I. The BLOOD and its components
II. The CIRCULATIONS
III. HEMODYNAMICS and fundamental pressure, resistance and flow inter-relationships
IV. The HEART as a highly coordinated, well regulated two-step "twin" pump
V. The VASCULATURE as a well regulated system of conduits
VI. BLOOD PRESSURE regulation
VII. Determinates of MYOCARDIAL WORK

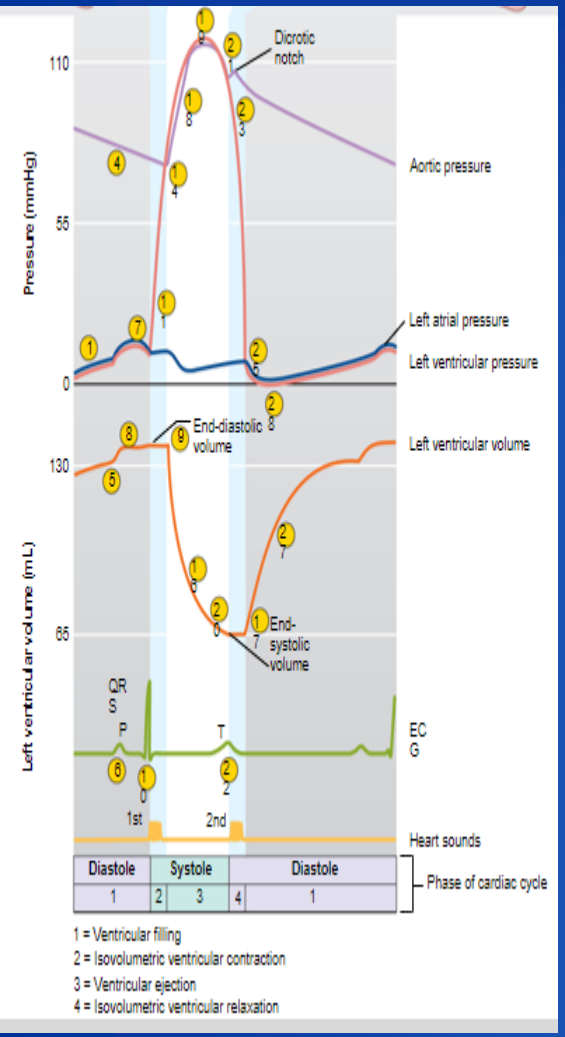
Read and study all of chapter 12

Independent studies:

BLOOD components

LIPOPROTEIN METABOLISM (Google Drive presentation)

Begin learning WELL the cardiac cycle NOW



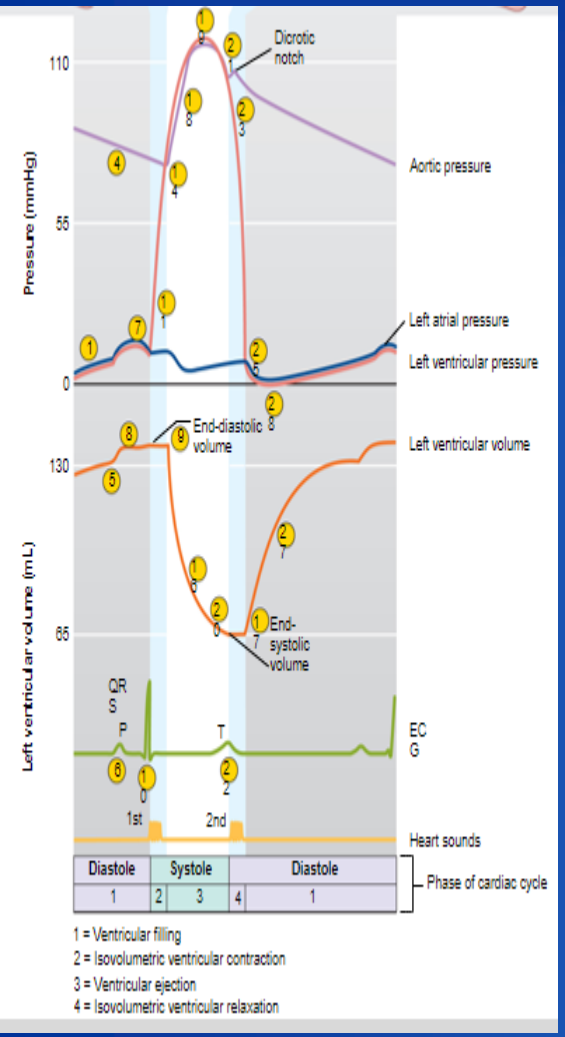
Learn, understand, draw and label Figure 12.22 (pg. 381)

Focus on interrelationships of ECG, ventricular pressure, aortic pressure, ventricular volume, valves opening and closing sequences, end-diastolic and end-systolic volumes & heart sounds

Identify ventricular systole and diastole phases, isovolumic contraction and relaxation phases, rapid and reduced ventricular ejection phases, and rapid and reduced ventricular filling phases

Understand completely the following sequence and how the steps below relate to Figure 12.22:

- **Membrane depolarization leads to ventricular muscle contraction**
- **Ventricular muscle contraction leads to ventricular pressure generation**
- **Ventricular pressure generation leads to ventricular blood ejection**
- **Ventricular blood ejection is a component of blood pressure**
- **Blood pressure drives blood flow**



Follow this order!!!

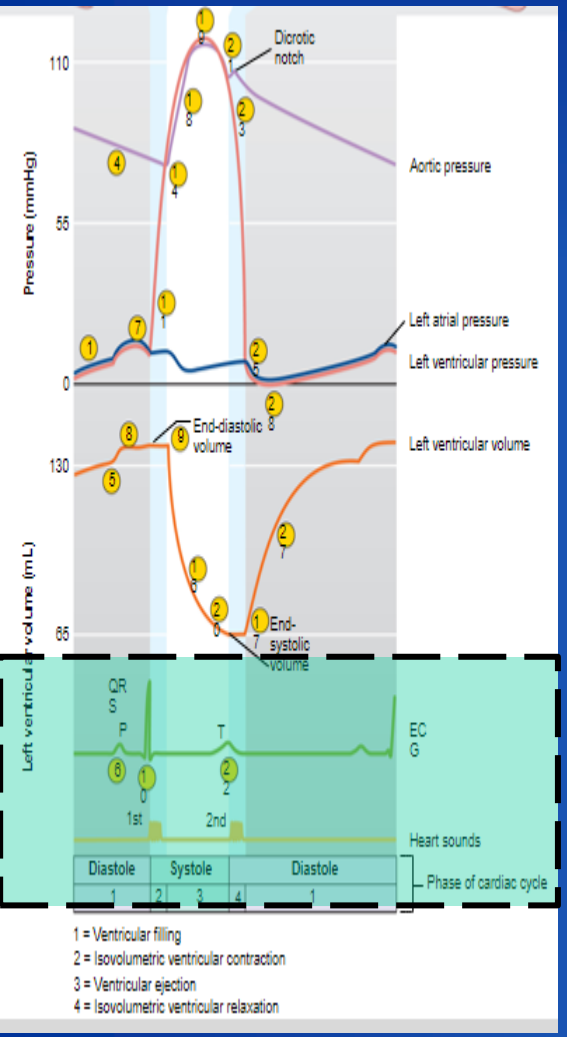
Ventricular membrane depolarization

Ventricular muscle contraction

Ventricular pressure generation

Ventricular blood ejection

Ventricular relaxation



Ventricular relaxation

Ventricular diastole; AV valve open, aortic valve closed
Blood flowing into ventricle
SA node depolarizes; ECG P wave; atrial depolarization
Atrial contraction; AV node depolarizes; PR internal

Ventricular membrane depolarization

ECG QRS complex; voltage-gated Na+ and L-type calcium channels open
Ca++ efflux from SR; Ca++ binds troponin; tropomyosin shifts

Ventricular muscle contraction

Myosin crossbridge binds thin filaments; crossbridge cycling

Ventricular pressure generation

As sarcomeres shorten, ventricular pressure increases and AV valve closes
Beginning of ventricular systole
Isovolumic contraction occurs as ventricular pressure increases

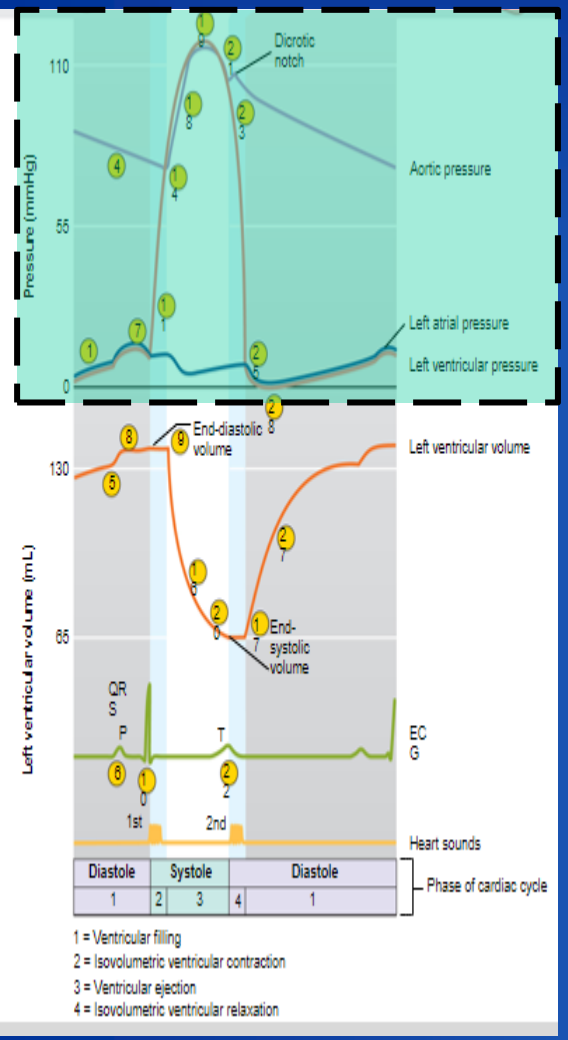
Ventricular blood ejection

When ventricular pressure exceeds aortic pressure aortic valve opens
Blood ejected into aorta
Ventricular blood volume decreases
ECG T wave indicates membrane repolarization

Ventricular relaxation

As ventricle relaxes, ventricular pressure drops faster than aortic pressure and aortic valve closes
Beginning of diastole and isovolumic relaxation
AV valve opens when ventricular pressure is lower than atrial pressure
Ventricle fills with blood

How do sympathetic and parasympathetic stimulation impact the elements of the cardiac cycle?



Ventricular relaxation

Ventricular diastole; AV valve open, aortic valve closed
Blood flowing into ventricle
SA node depolarizes; ECG P wave; atrial depolarization
Atrial contraction; AV node depolarizes; PR internal

Ventricular membrane depolarization

ECG QRS complex; voltage-gated Na+ and L-type calcium channels open
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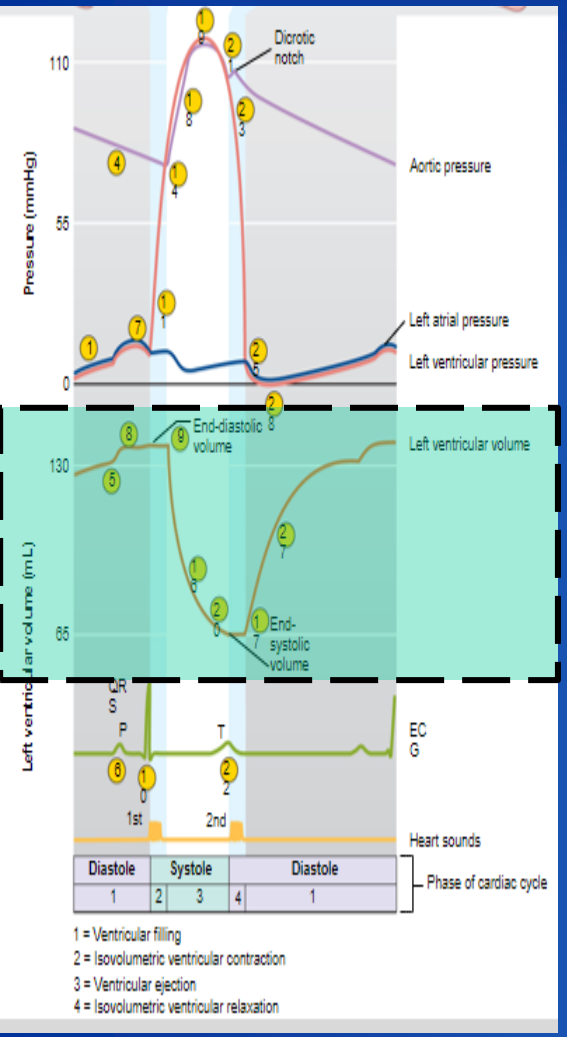
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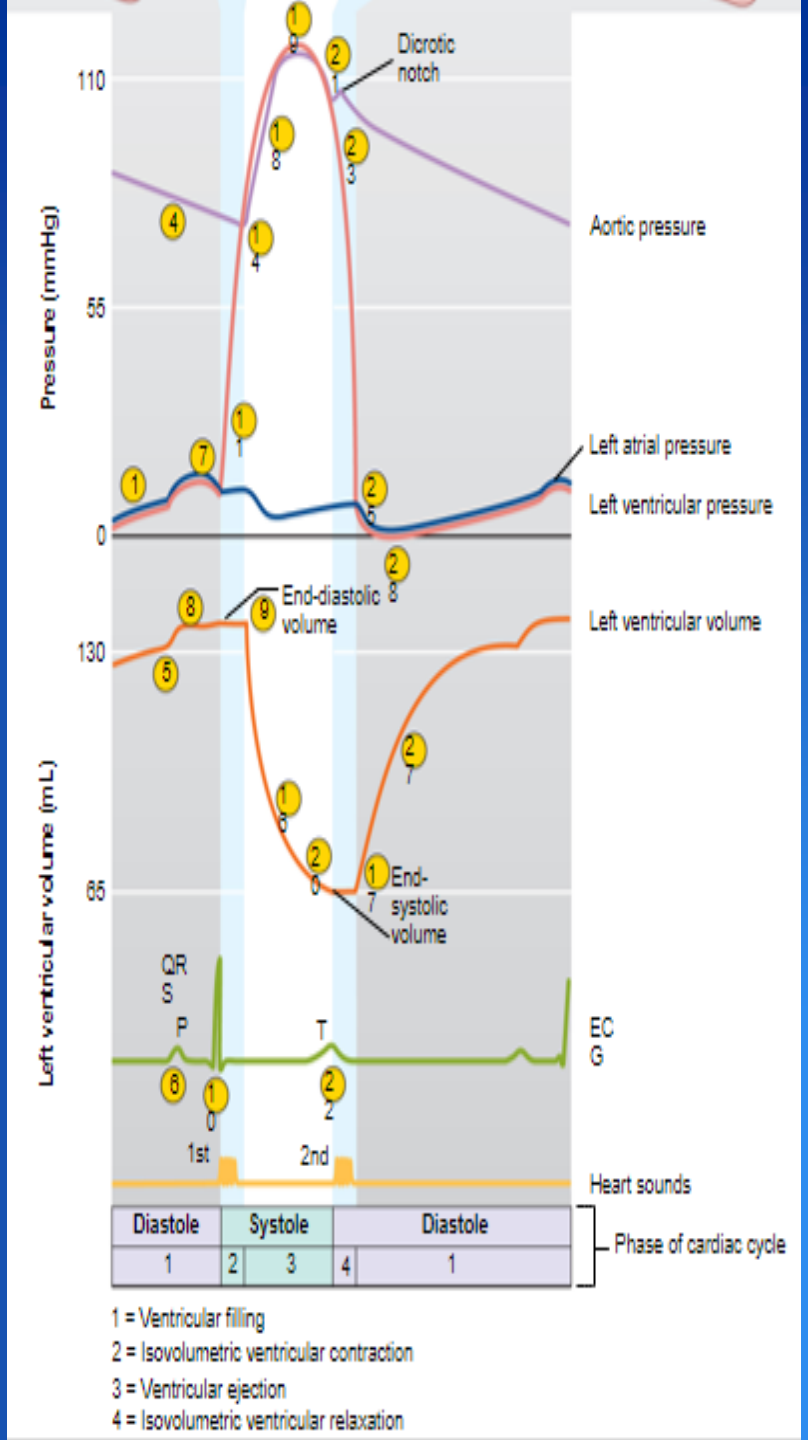
Ventricular blood ejection

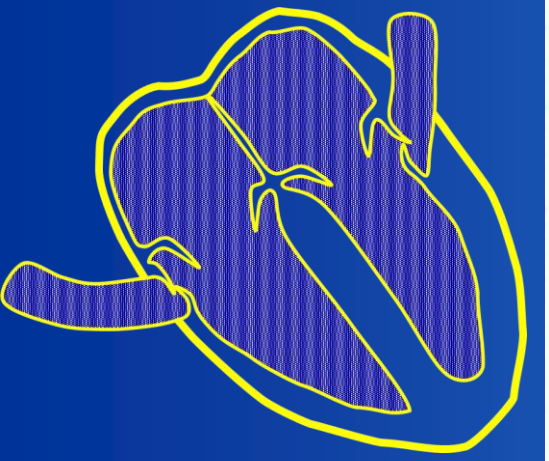
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Blood ejected into aorta
Ventricular blood volume decreases
ECG T wave indicates membrane repolarization

Ventricular relaxation

As ventricle relaxes, ventricular pressure drops faster than aortic pressure and aortic valve closes
Beginning of diastole and isovolumic relaxation
AV valve opens when ventricular pressure is lower than atrial pressure
Ventricle fills with blood

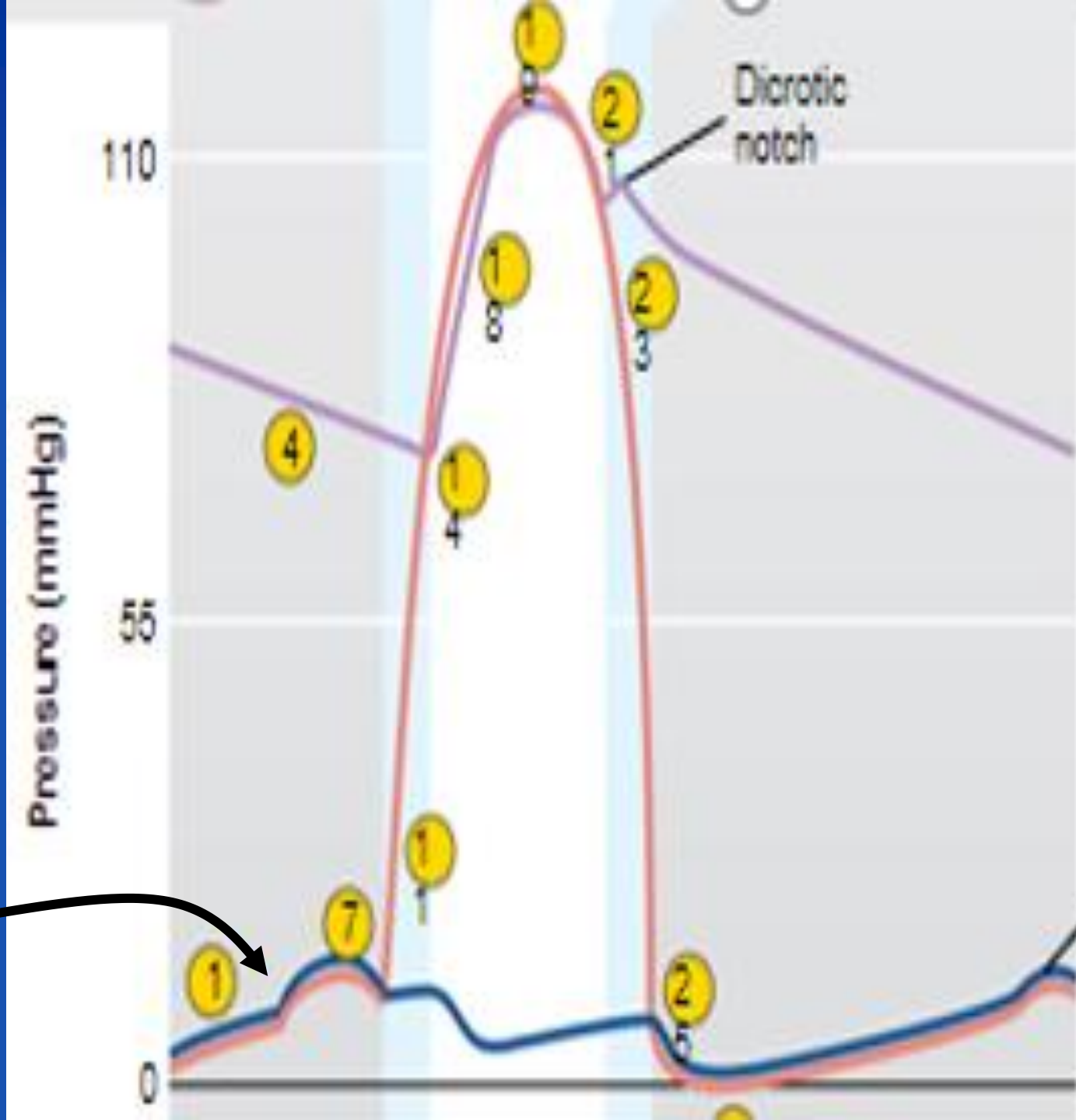
How do sympathetic and parasympathetic stimulation impact the elements of the cardiac cycle?

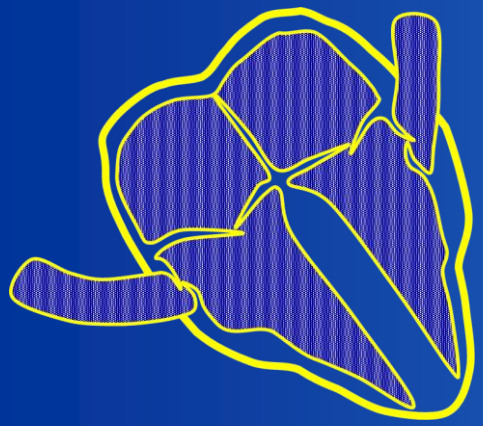

Dicrotic notch
110
55
0
Pressure (mmHg)
Aortic pressure
Left atrial pressure
Left ventricular pressure
End-diastolic volume
Left ventricular volume
130
65
Left ventricular volume (mL)
End-systolic volume
QRS
P
T
ECG
1st
2nd
Heart sounds
Diastole
Systole
Diastole
1
2
3
4
1
Phase of cardiac cycle
1 = Ventricular filling
2 = Isovolumetric ventricular contraction
3 = Ventricular ejection
4 = Isovolumetric ventricular relaxation



Aortic valve – closed
Mitral valve – open

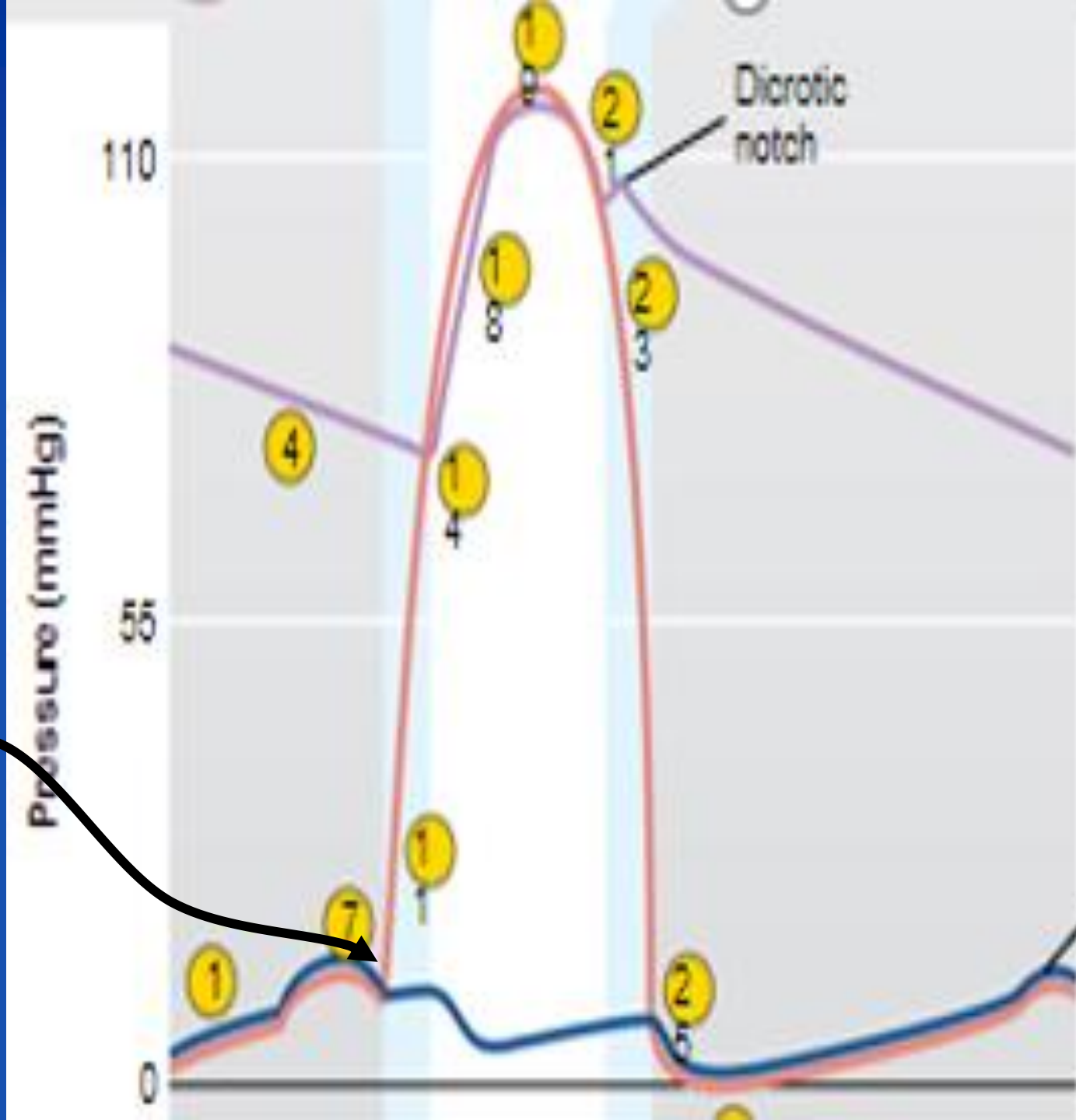
Diastolic filling phase

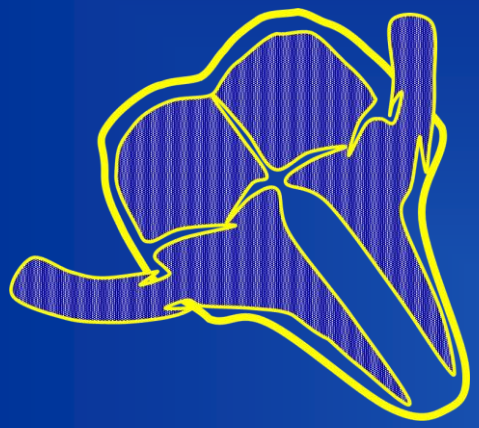




Aortic valve – closed
Mitral valve – closes

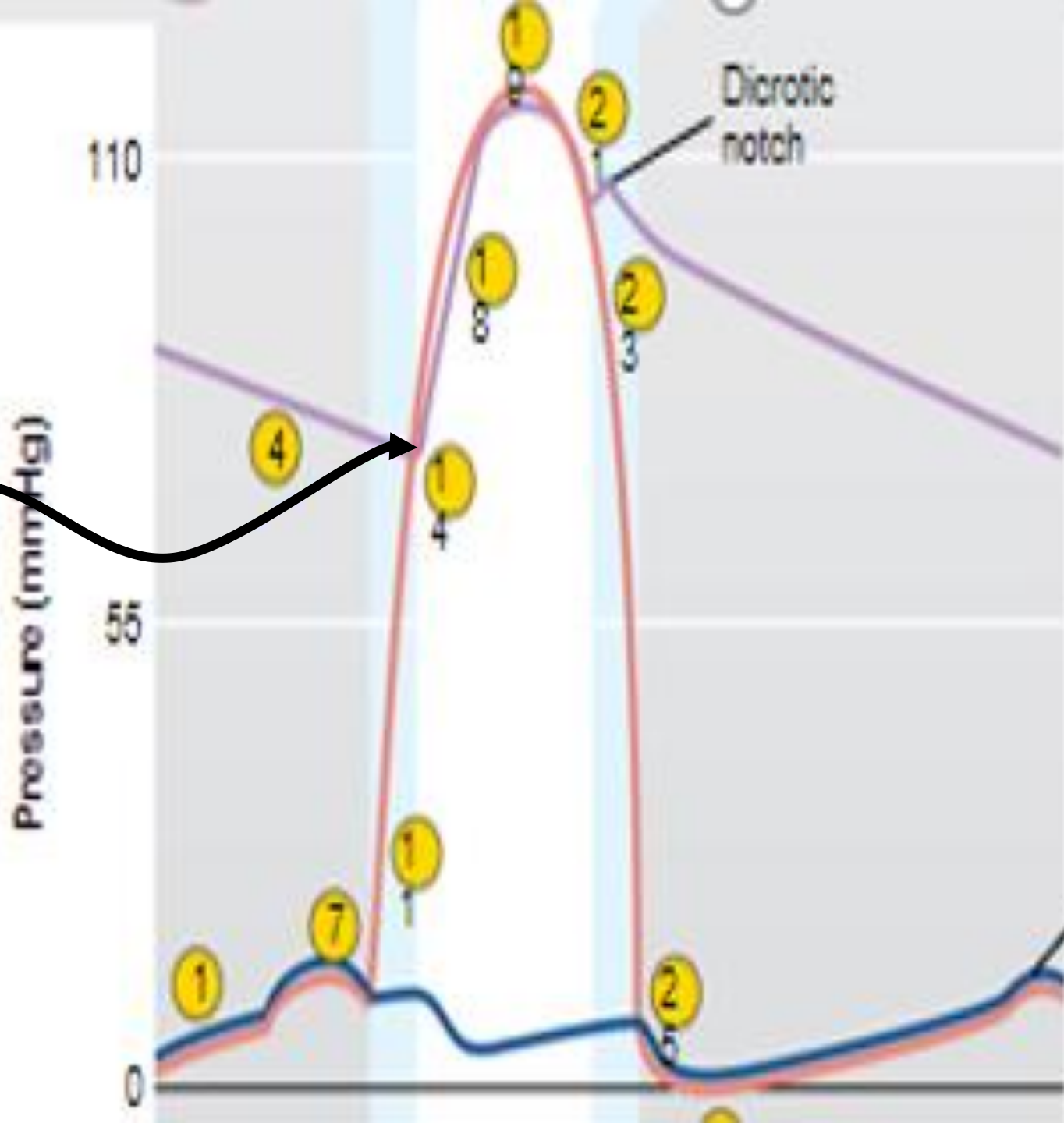
Systolic isovolumic contraction phase begins

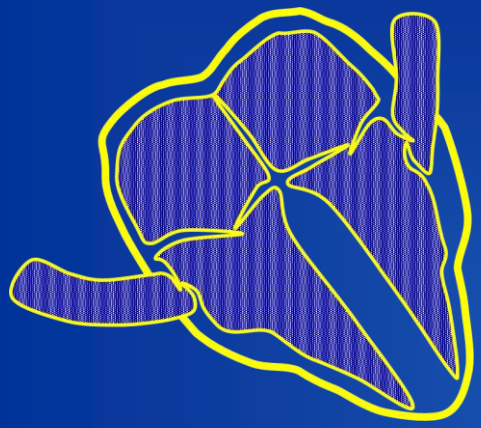




Aortic valve – opens
Mitral valve – closed

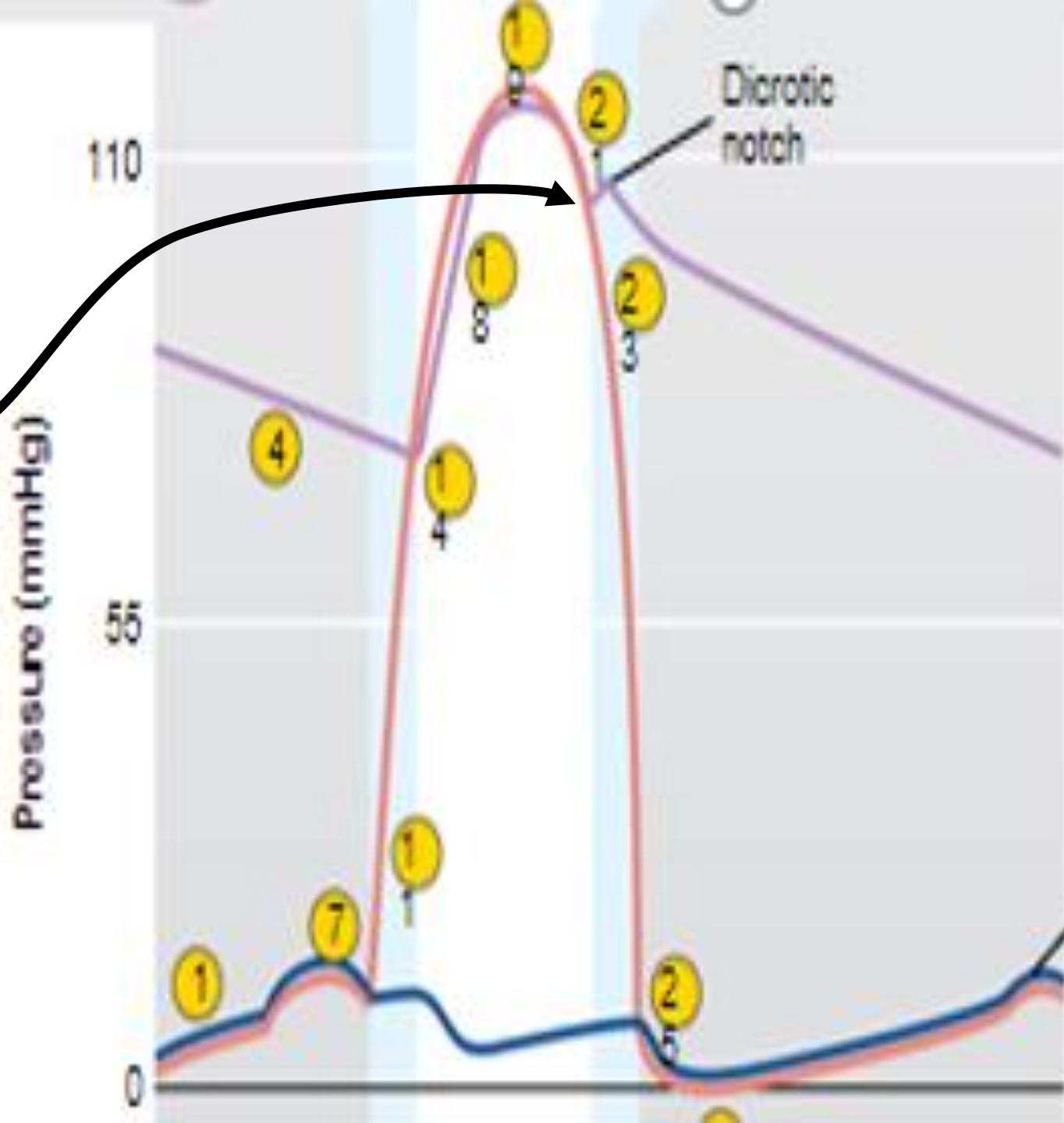
Systolic ejection phase begins

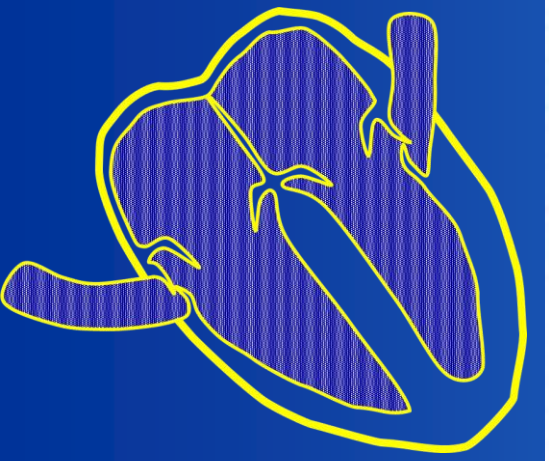




Aortic valve – closes
Mitral valve – closed

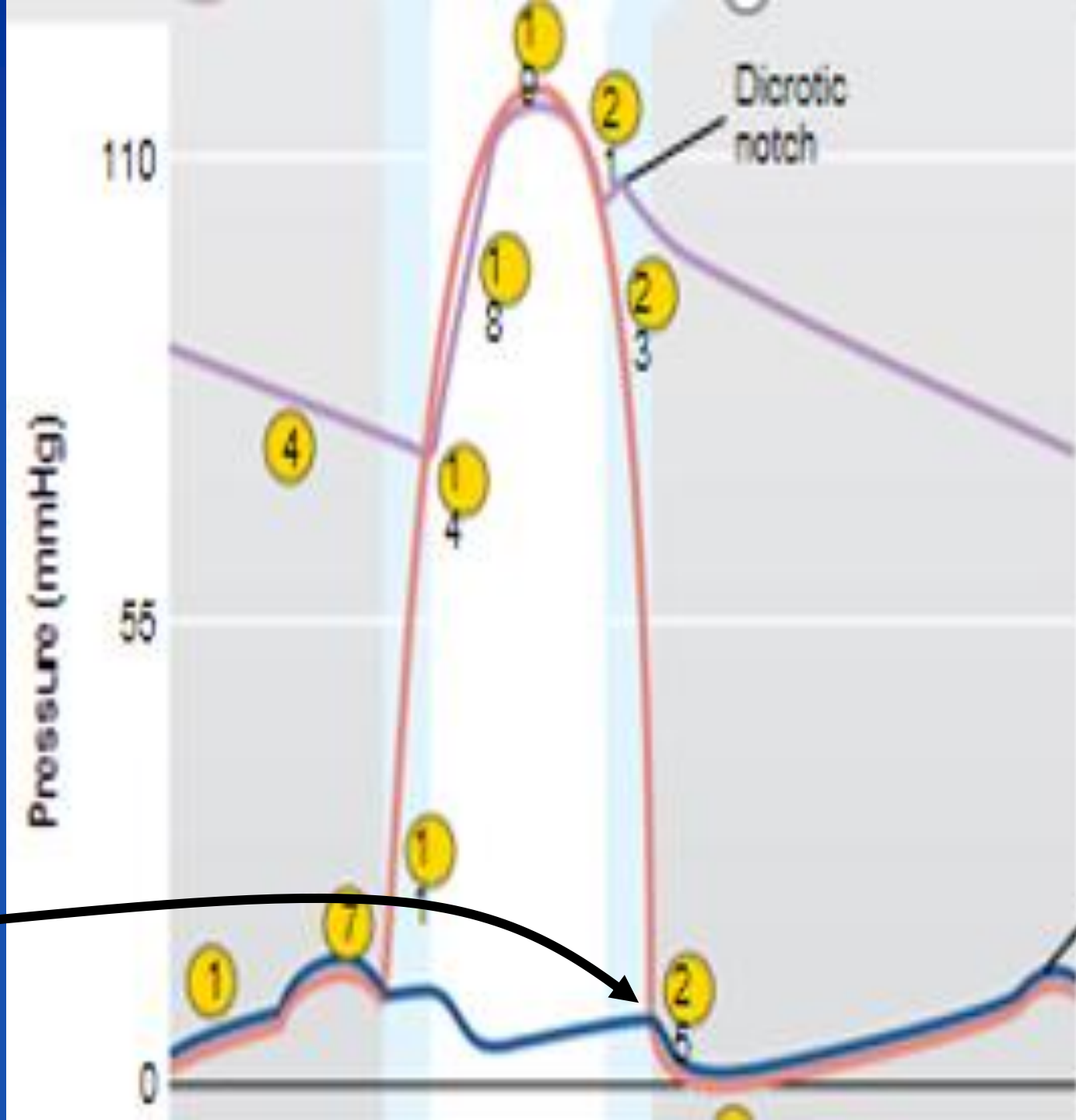
Diastolic isovolumic relaxation phase begins

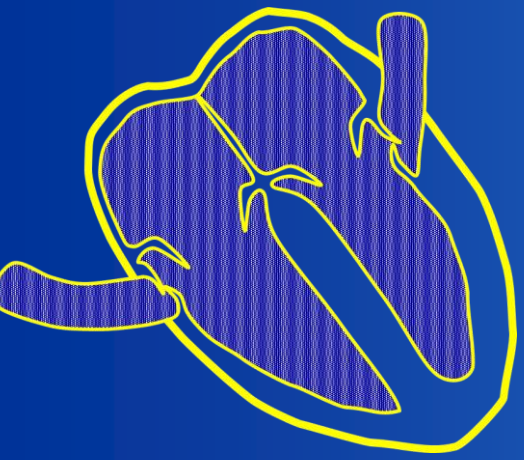


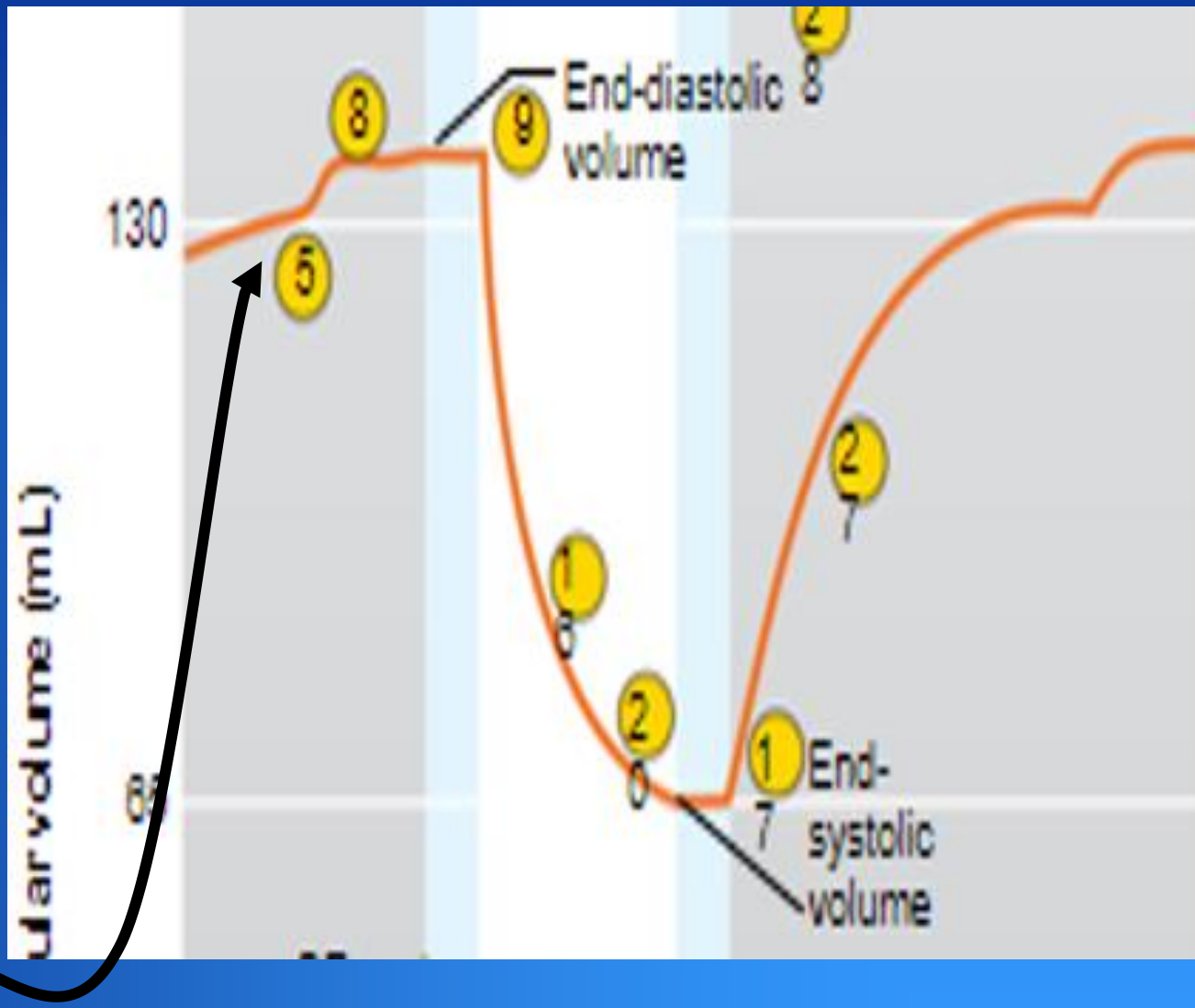


Aortic valve – closed
Mitral valve – opens

Diastolic filling phase begins

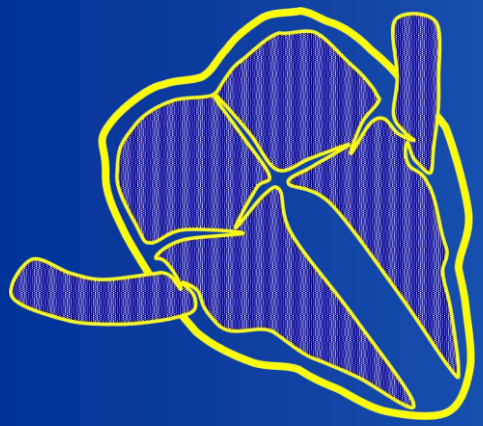






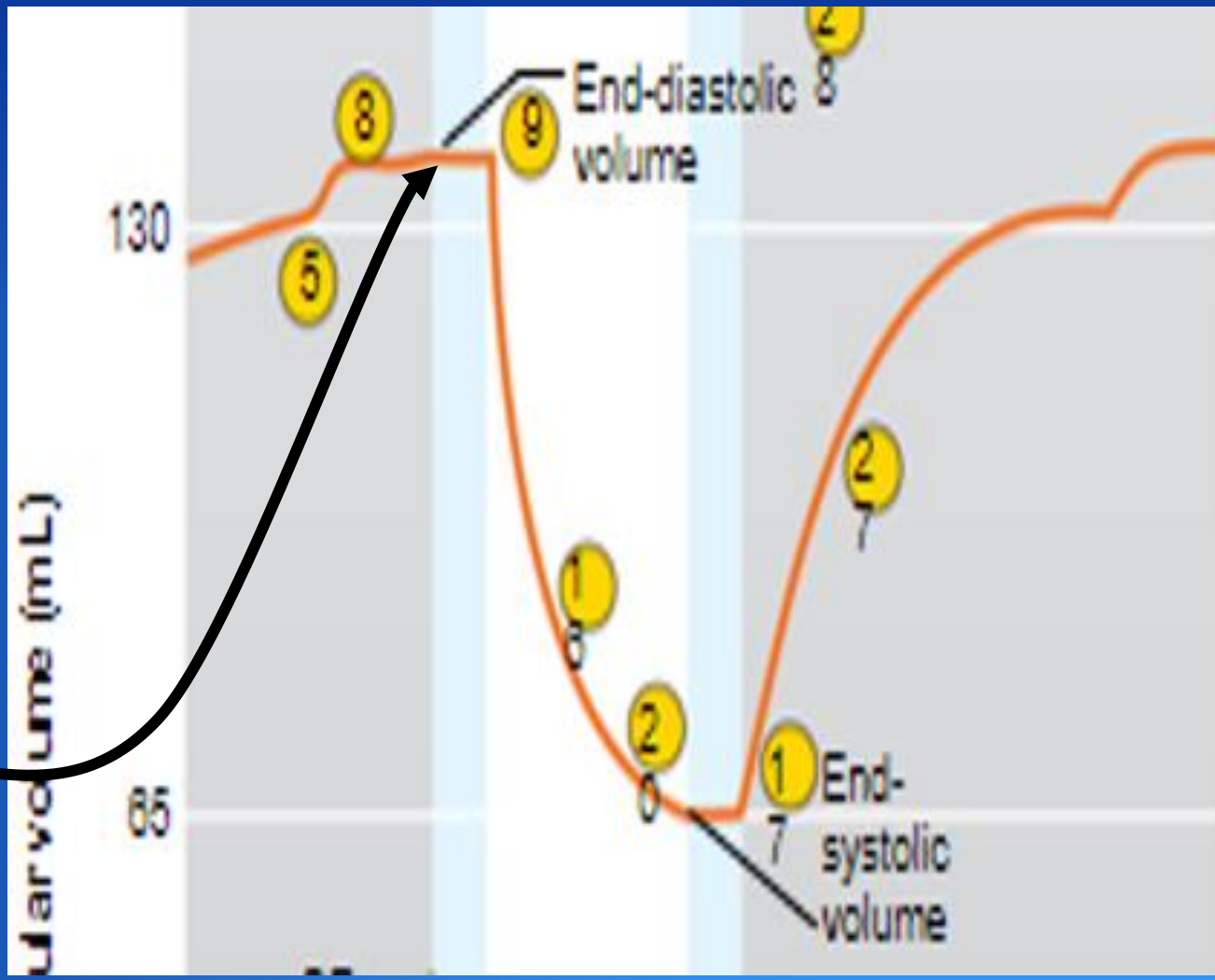
Aortic valve – closed
Mitral valve – open

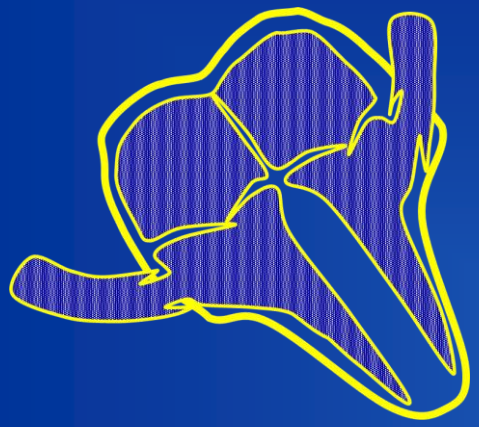
Diastolic filling phase



Aortic valve – closed
Mitral valve – closes

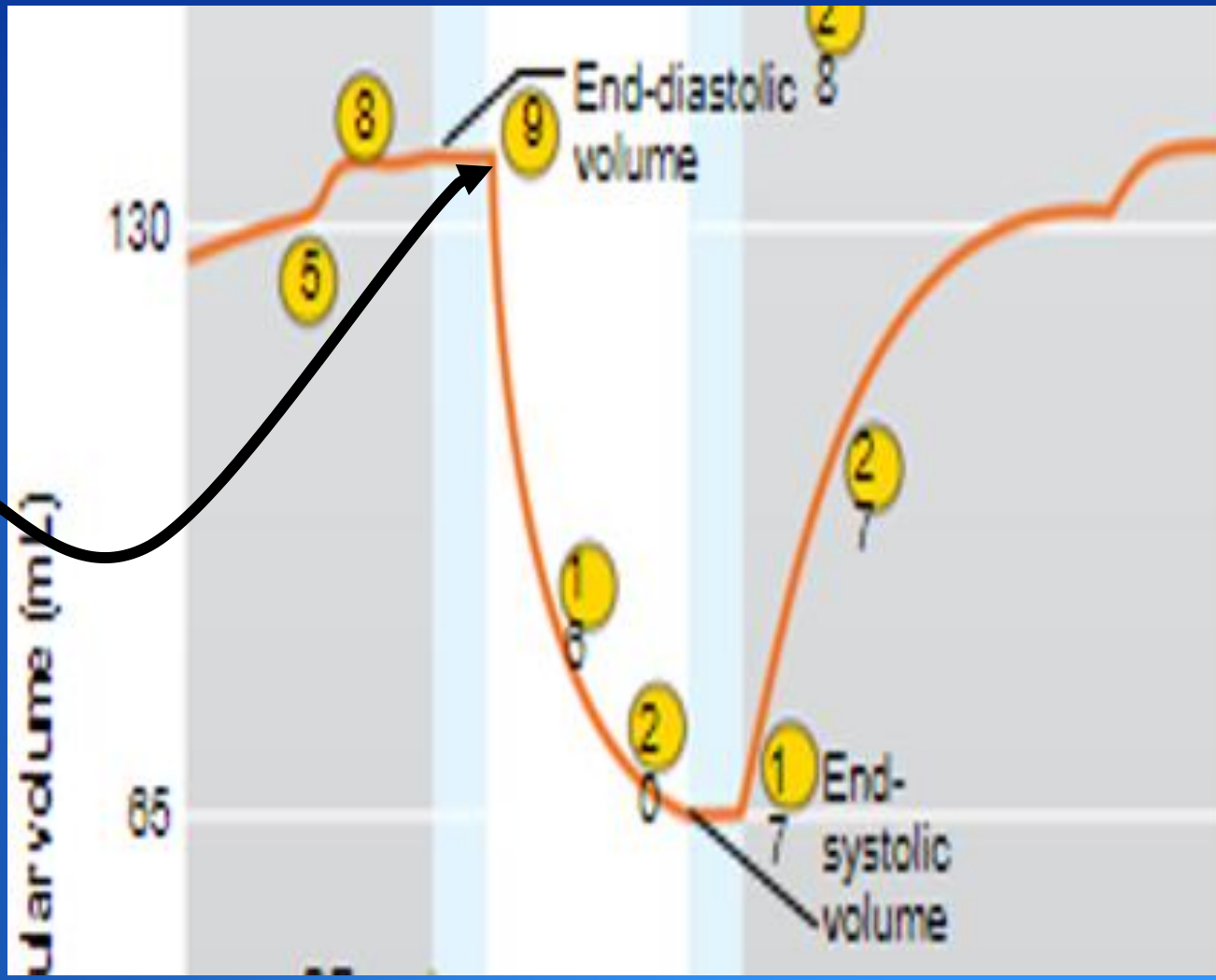
Systolic isovolumic contraction phase begins

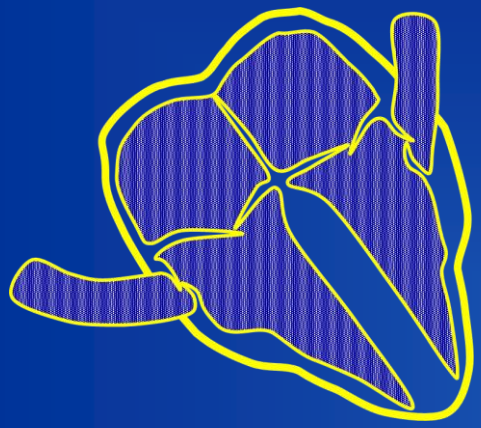




Aortic valve – opens
Mitral valve – closed

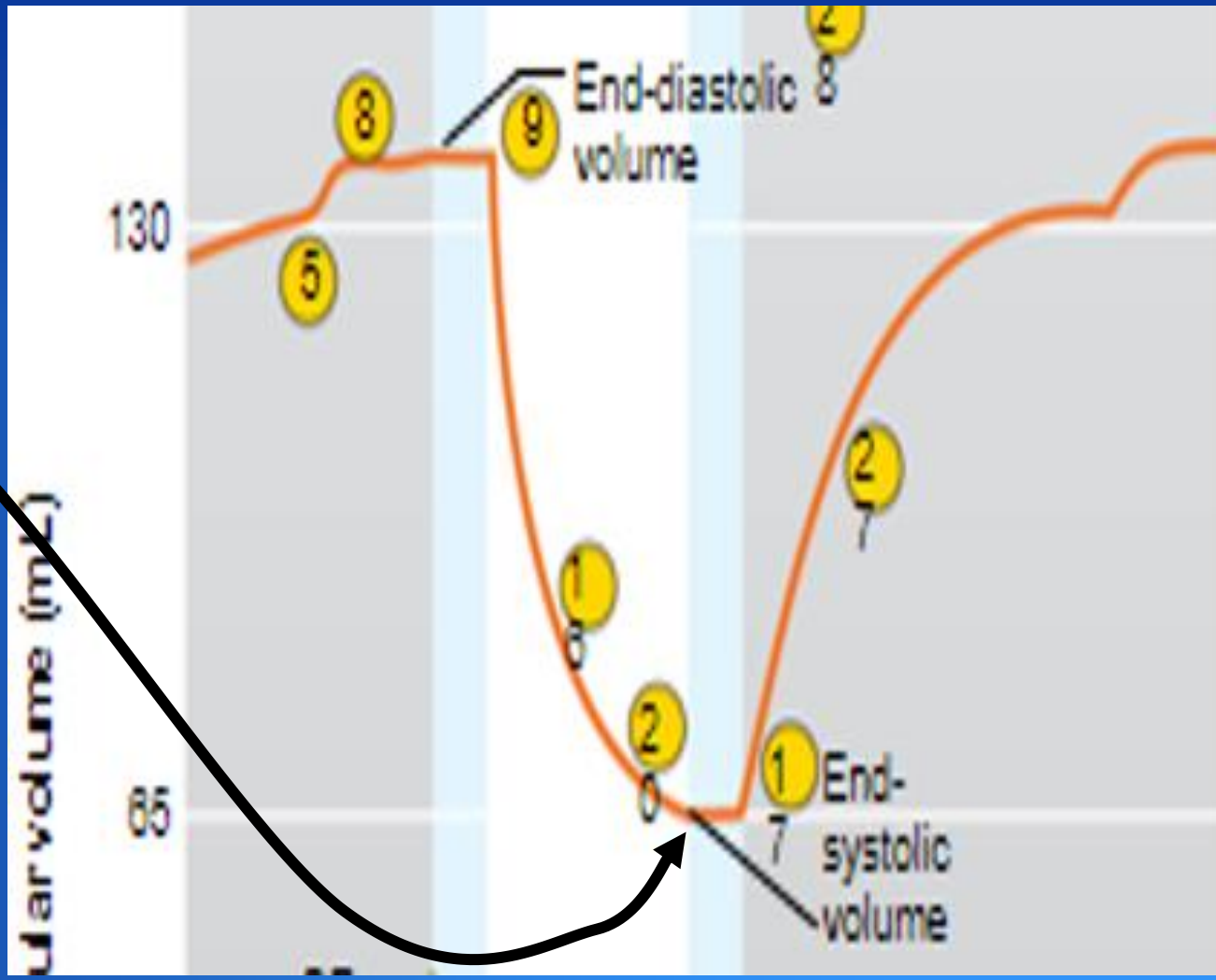
Systolic ejection phase begins

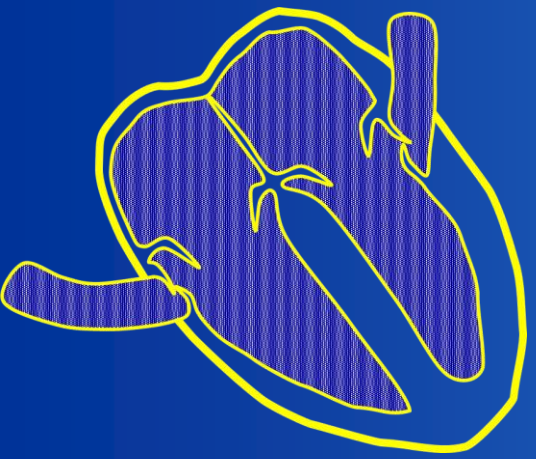


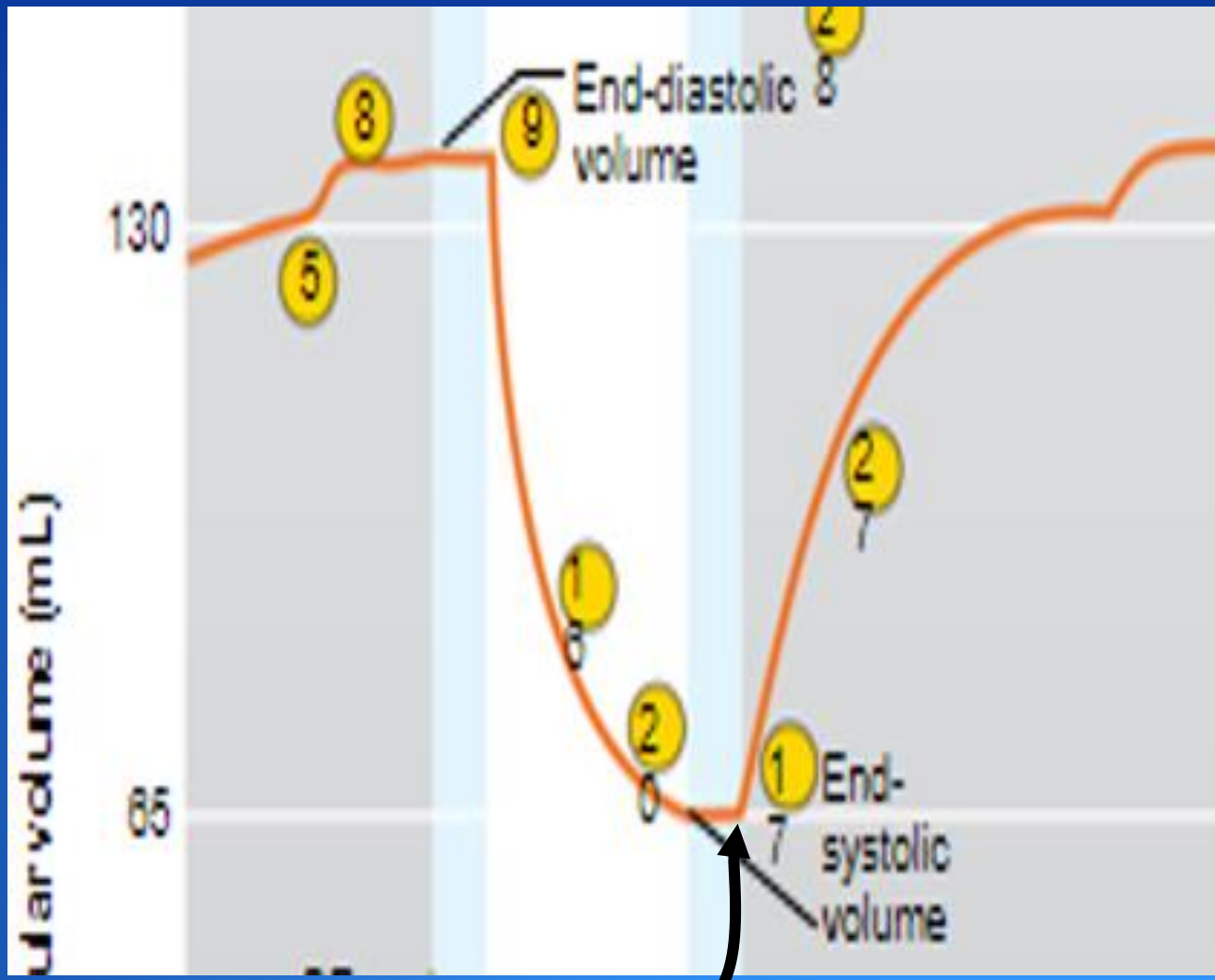


Aortic valve – closes
Mitral valve – closed

Diastolic isovolumic relaxation phase begins



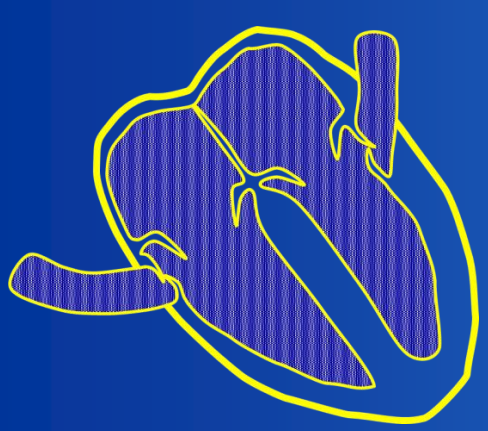


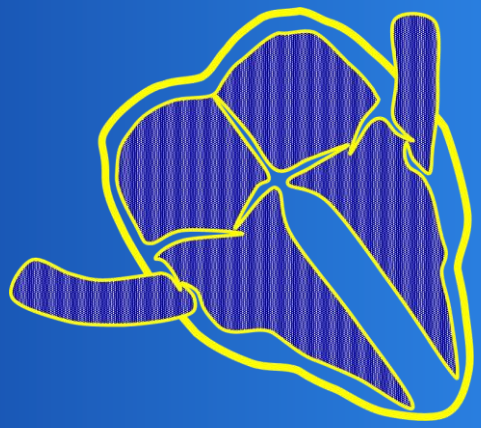


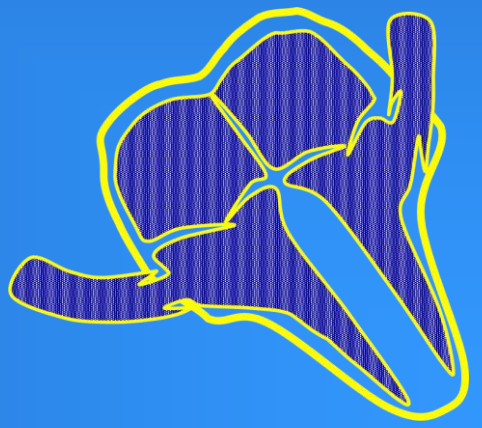
Aortic valve – closed
Mitral valve – opens

Diastolic filling phase begins


QR
S
P
T
6
1
2
0
2







Quiz 5 In-class quiz

Label and illustrate with correct sequencing the following for at least one full cardiac cycle of the left ventricle

- **ECG**
- **Left ventricular pressure**
- **Aortic pressure**
- **Left ventricular volume**
- **AV and aortic valve opening and closing time points**
- **End-diastolic and end-systolic time points**
- **Heart sounds 1 and 2**
- **Label the ventricular systole and diastole phases, isovolumic contraction and relaxation phases, rapid and reduced ventricular ejection phases, and rapid and reduced ventricular filling phases**

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VII. Determinates of MYOCARDIAL WORK

Read and study all of chapter 12

Independent studies:

BLOOD components

LIPOPROTEIN METABOLISM (Google Drive presentation)

I. The BLOOD and its components (*Independent study)*

A. Blood volume (~7%-8% of bodyweight)

B. Whole blood, plasma protein, plasma, serum and the hematocrit (ratio of RBC to whole blood volume)

C. Blood cells

1. RBC functions
2. Platelets functions
3. WBCs (leukocytes)

(1)Neutrophils
(2)Eosinophils
(3)Monocytes
(4)Macrophages
(5)Basophils
(6)Lymphocytes

} Know their primary functions

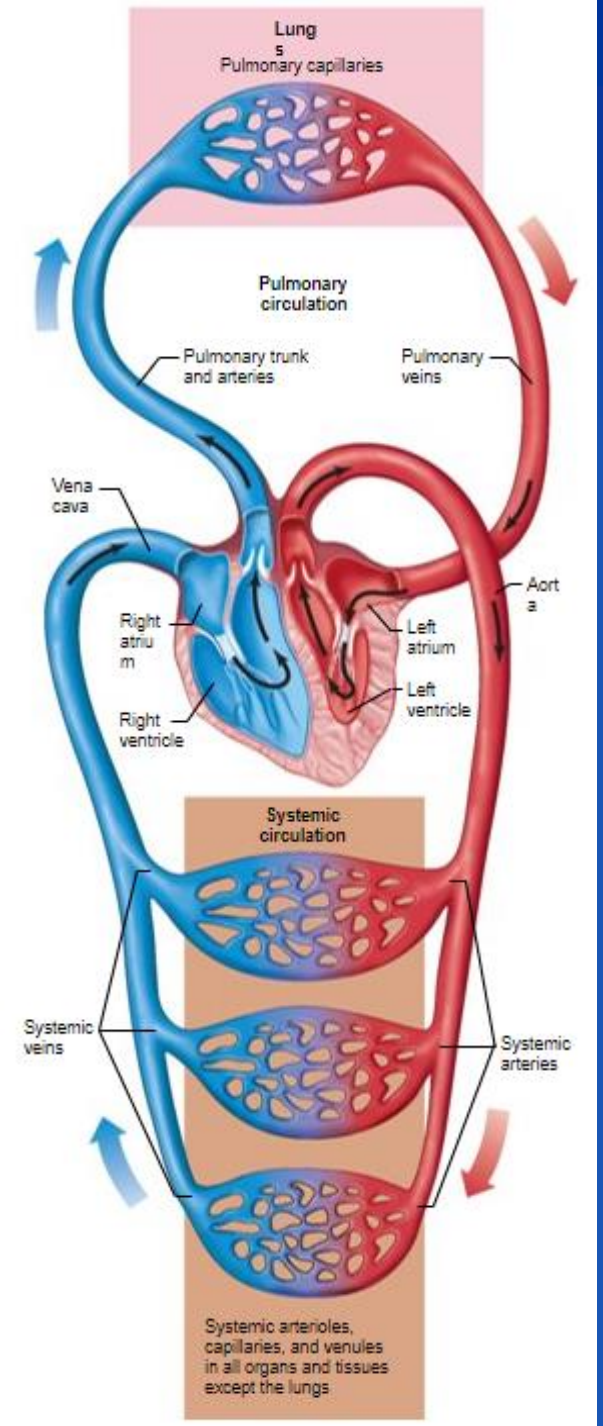
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Read and study all of chapter 12

Independent studies:

BLOOD components

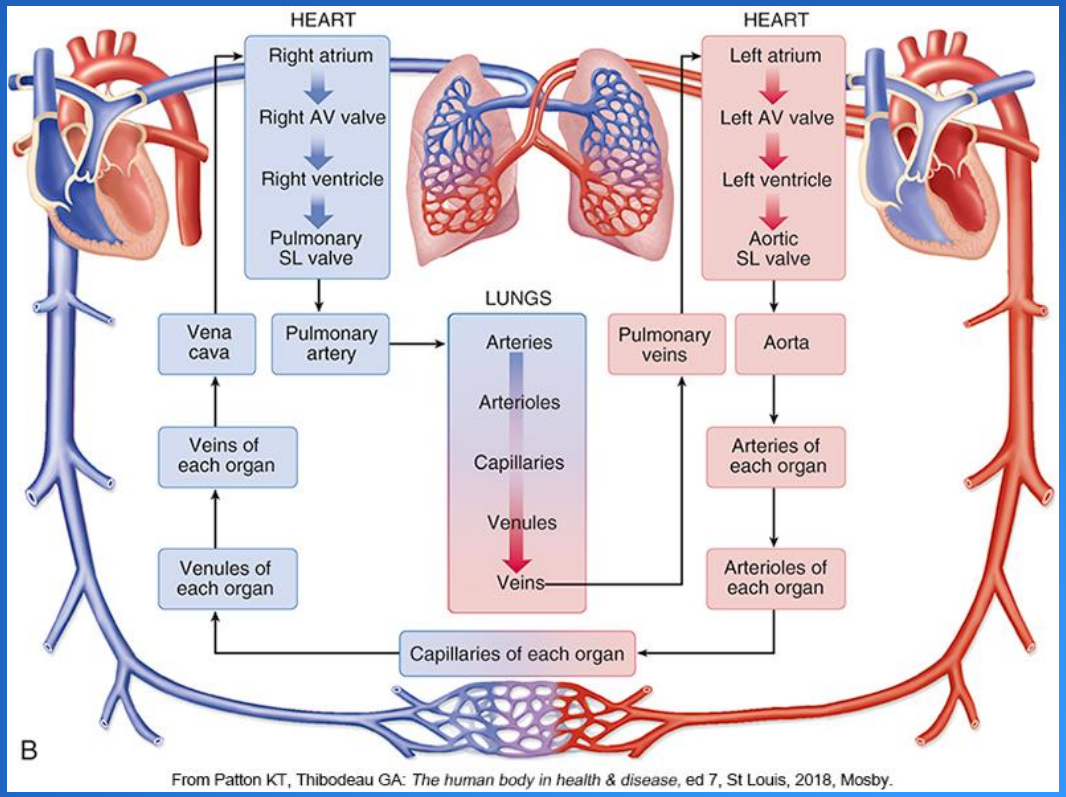
LIPOPROTEIN METABOLISM (Google Drive presentation)



The CIRCULATIONS (describe each of their pathways)

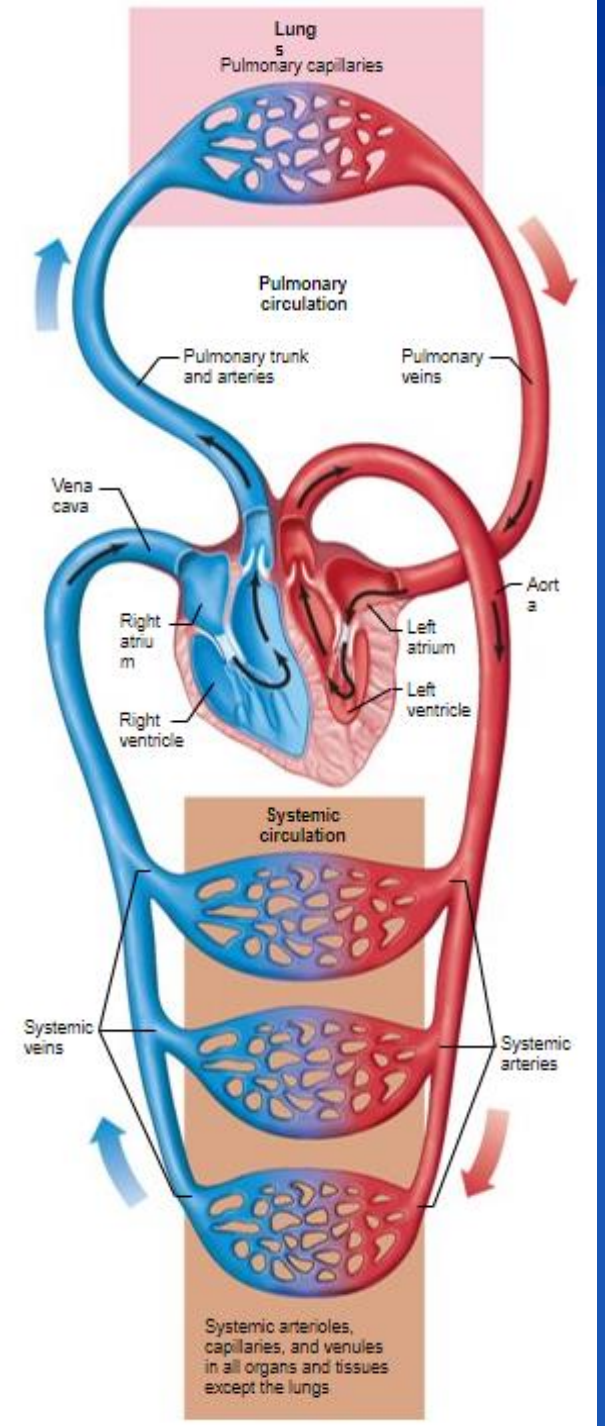
A. Systemic circulation (supplied by left heart)
   1. General system body circulation
   2. Coronary circulation

B. Pulmonary circulation (supplied by right heart)



From Patton KT, Thibodeau GA: *The human body in health & disease*, ed 7, St Louis, 2018, Mosby.

Figure 12.5



HEMODYNAMICS and fundamental pressure, resistance and flow inter-relationships

$$\text{Flow (Q)} = \frac{\text{Pressure1} - \text{Pressure2}}{\text{Resistance}} = \frac{\Delta P}{R}$$

Directly
Inversely

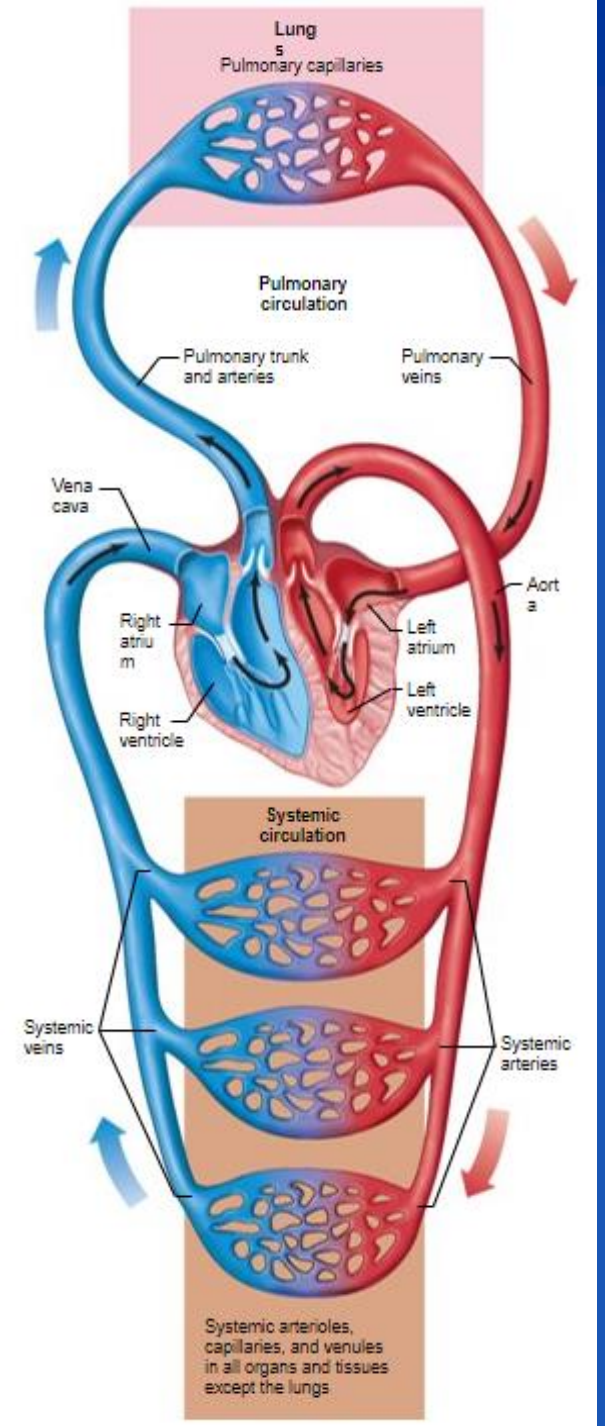
$$\Delta P = Q \times R$$

$$\text{MABP} - \text{RAP} = \text{CO} \times \text{TPR}$$

$$\text{MABP} = \text{CO} \times \text{TPR}$$

The amount of blood flow an organ or area receives is controlled by the resistance to the blood flow to the area.

Resistance is largely regulated by changing the radius of the blood vessels supplying an area.



HEMODYNAMICS and fundamental pressure, resistance and flow inter-relationships

Resistance is largely regulated by changing the radius of the blood vessels supplying an area.

$$\text{Resistance (R)} = \frac{8\,L\,\eta}{\pi\,r^4}$$

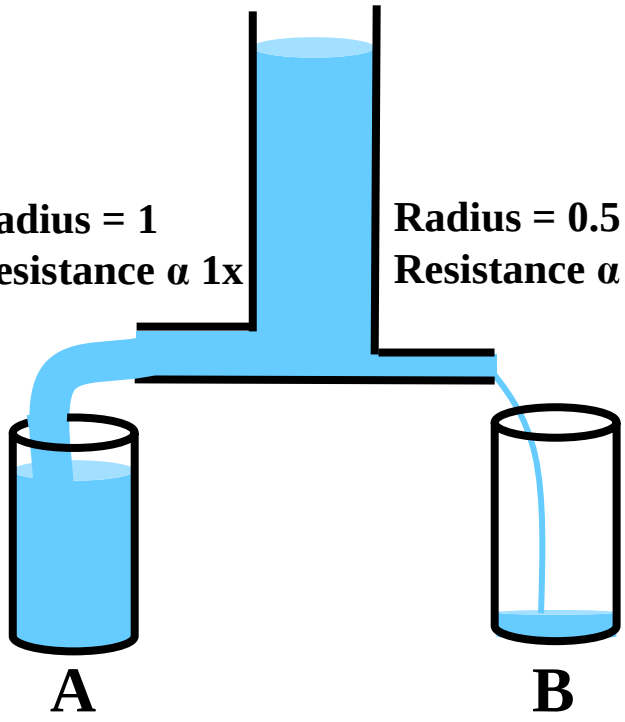
Where,

L = length of tube

η = viscosity of the fluid

r = radius of the tube

8π = constants



- B has 16x more resistance and 16x less flow than B
- B has only 6.25% of flow of A
- Small changes in radius leads to large changes in resistance and flow

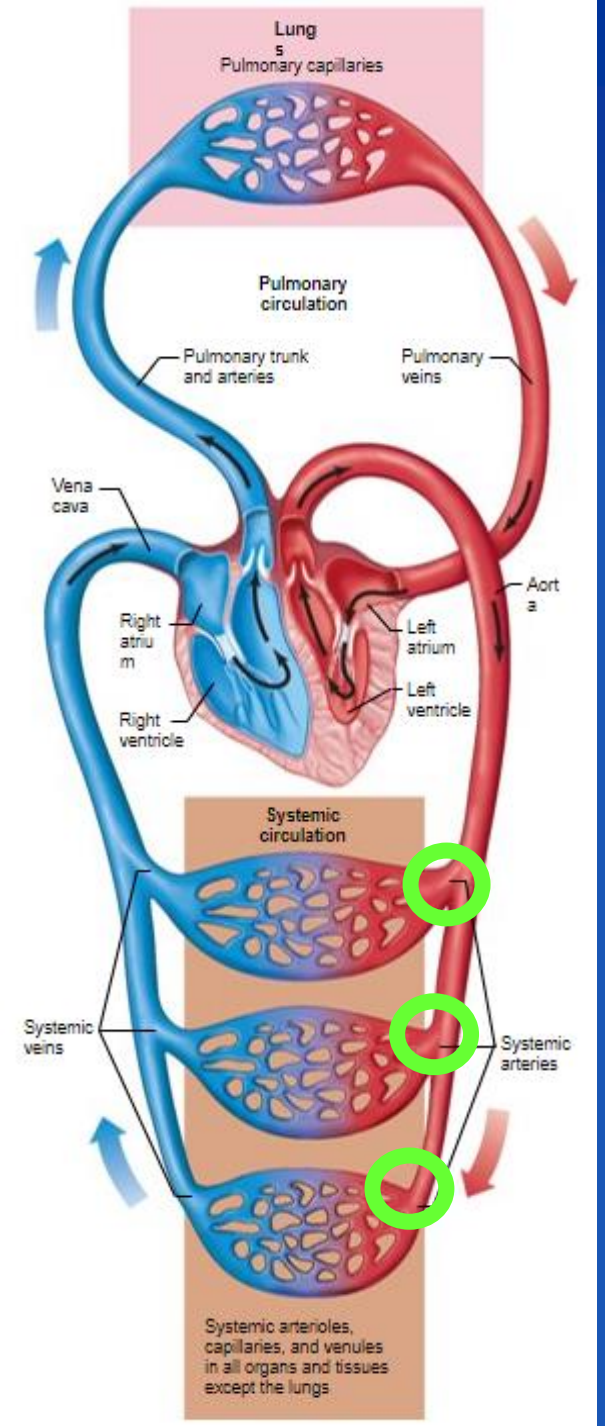
Redrawn and revised from Vander's Physiology 16 ed.

$$\text{Resistance (R)} = \frac{8\,L\,\eta}{\pi\,r^4}$$

*** Therefore, $R \alpha 1 / r^4$ ***

***Radius has a major impact on the regulation of resistance, and, therefore, radius has a major impact on the regulation of blood flow through a vessel and to an area.

***Controlling blood flow to an organ or area is regulated largely by vasoconstricting and vasodilating arterioles.



HEMODYNAMICS and fundamental pressure, resistance and flow inter-relationships

Resistance is largely regulated by changing the radius of the blood vessels supplying an area.

$$\text{Resistance (R)} = \frac{8\,L\,\eta}{\pi\,r^4}$$

Where,

L = length of tube
η = viscosity of the fluid
r = radius of the tube
8π = constants

Figure 32.1A

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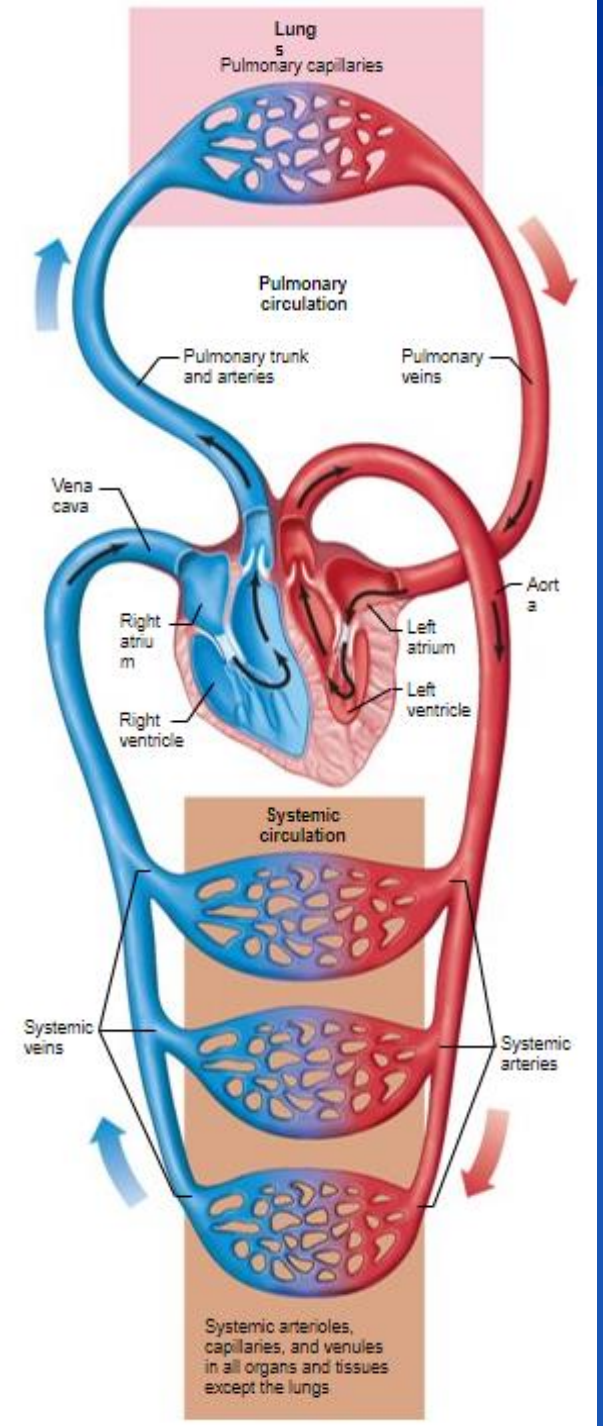
Read and study all of chapter 12

Independent studies:

BLOOD components

HEMOSTASIS

LIPOPROTEIN METABOLISM (Google Drive presentation)



The HEART is a highly coordinated, well regulated two-step "twin" pump

Coordination of the pumping action is made possible by:

1. **A specialized conduction system**
2. **Sets of one-way valves**



Figures 12.13 and 12.14

Depolarization

Muscle contraction

Pressure generation

Blood ejection





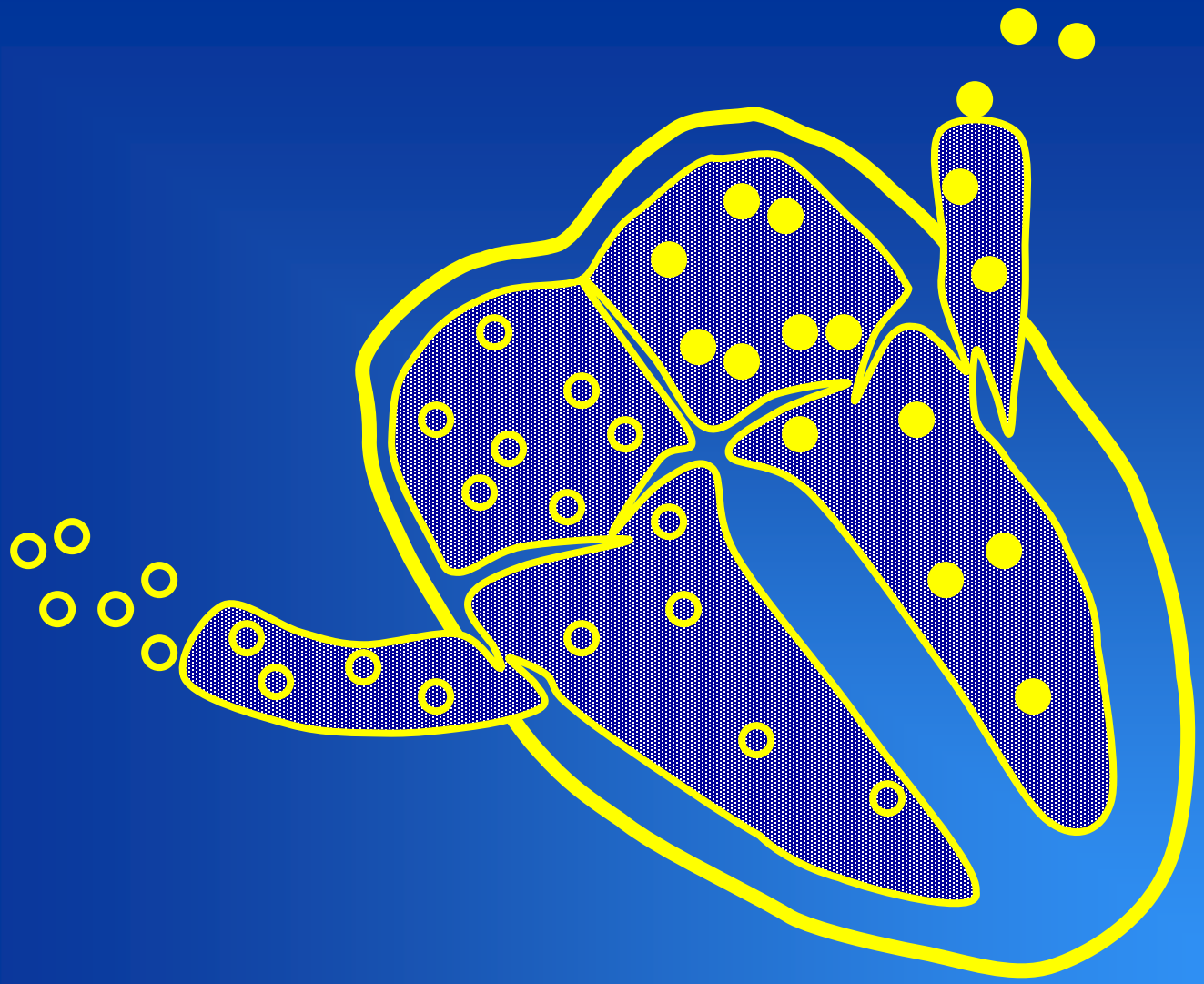
ECG elements

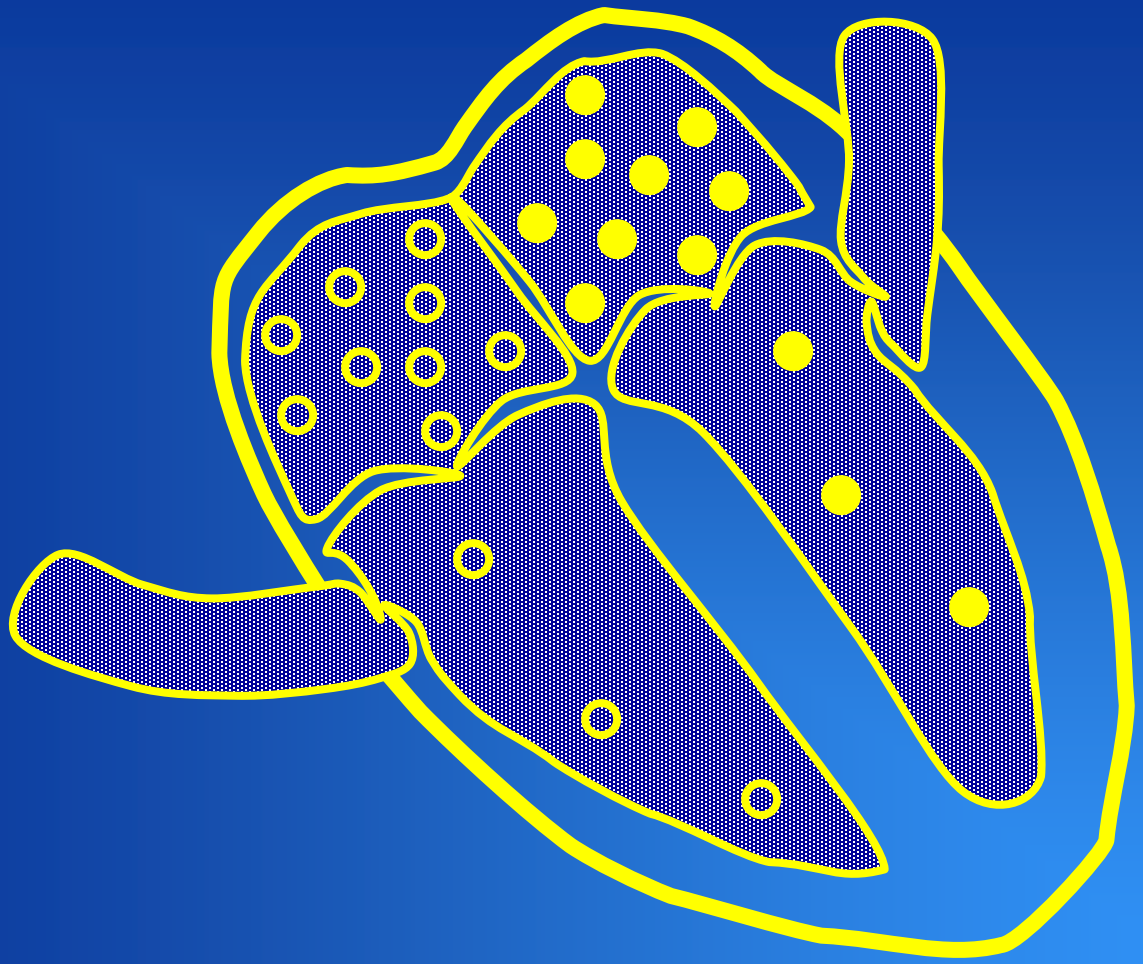
P wave – Atrial depolarization

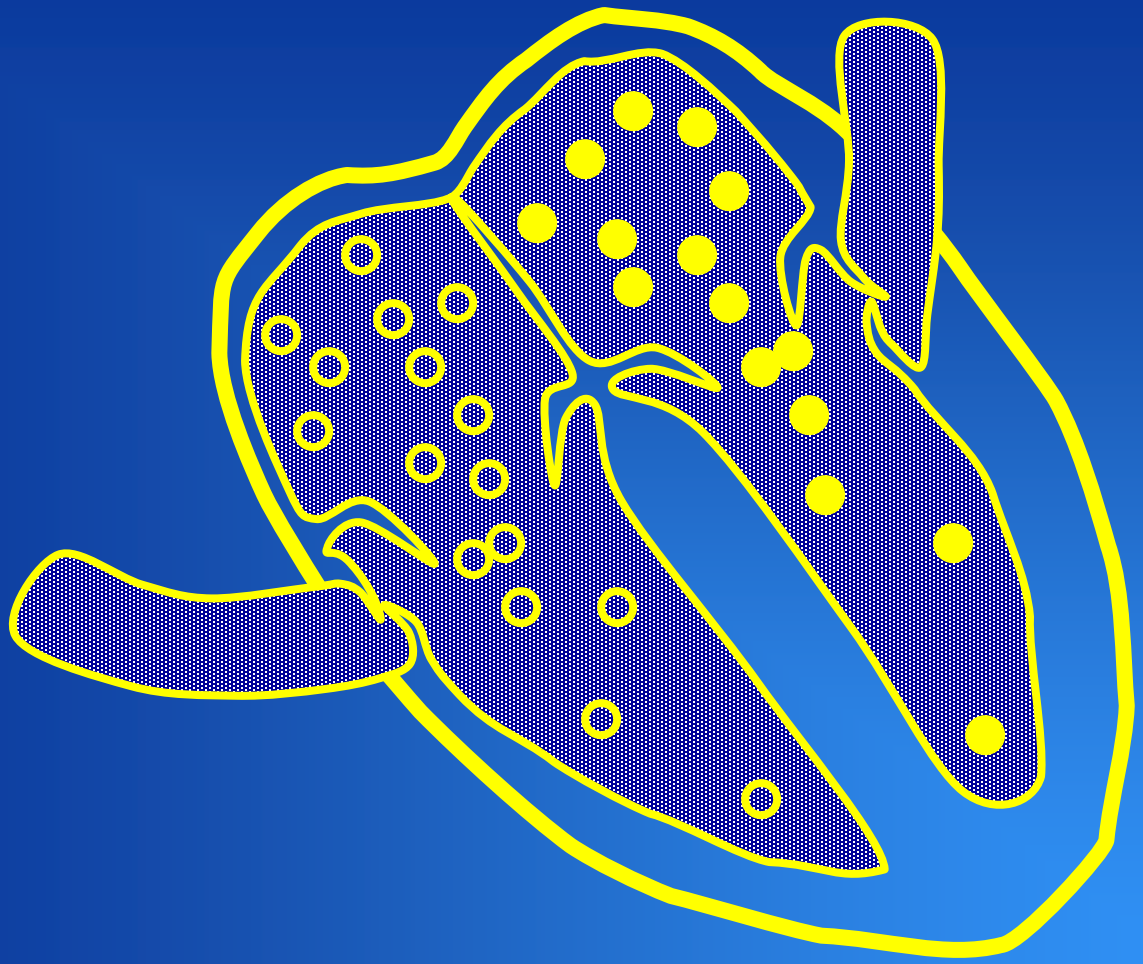
PR interval –AV node depolarization

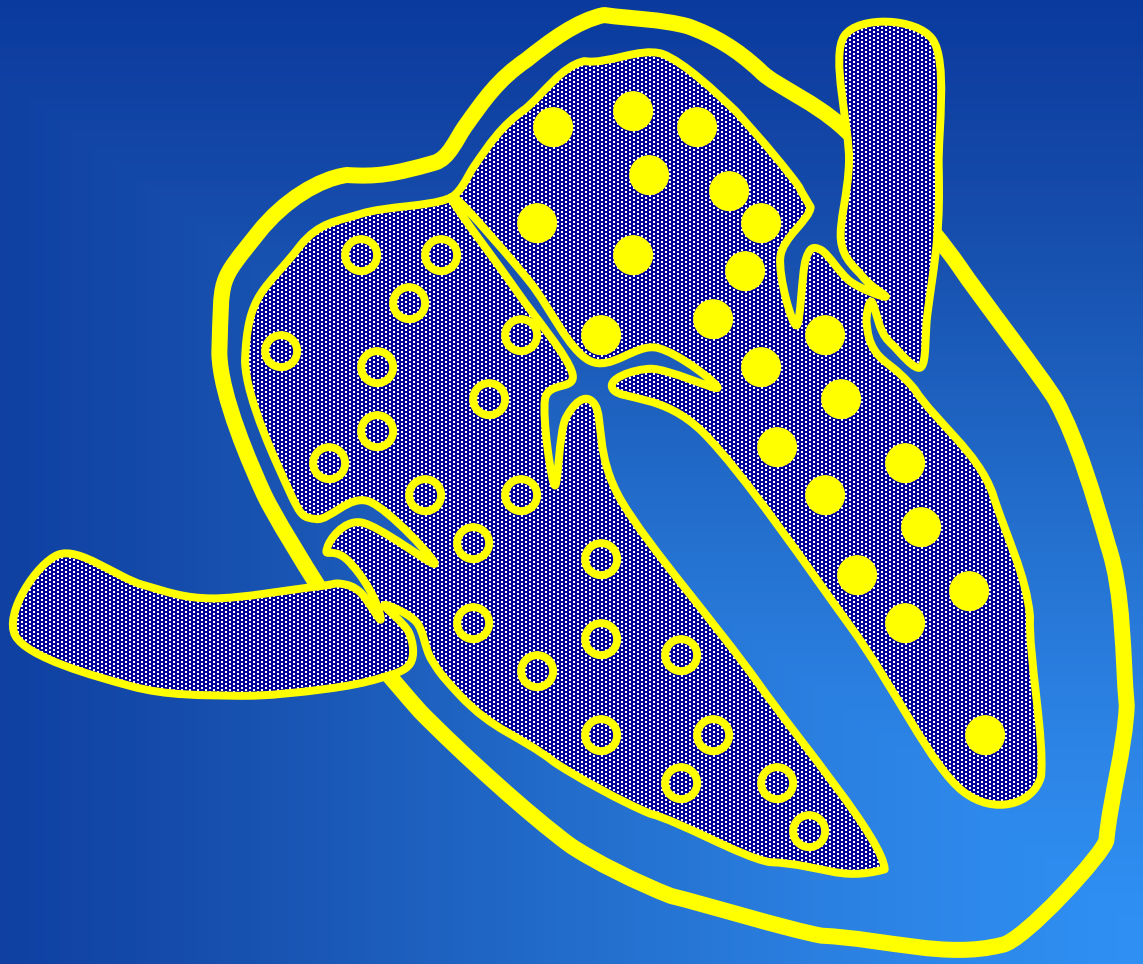
QRS complex –Septal and ventricular wall depolarization

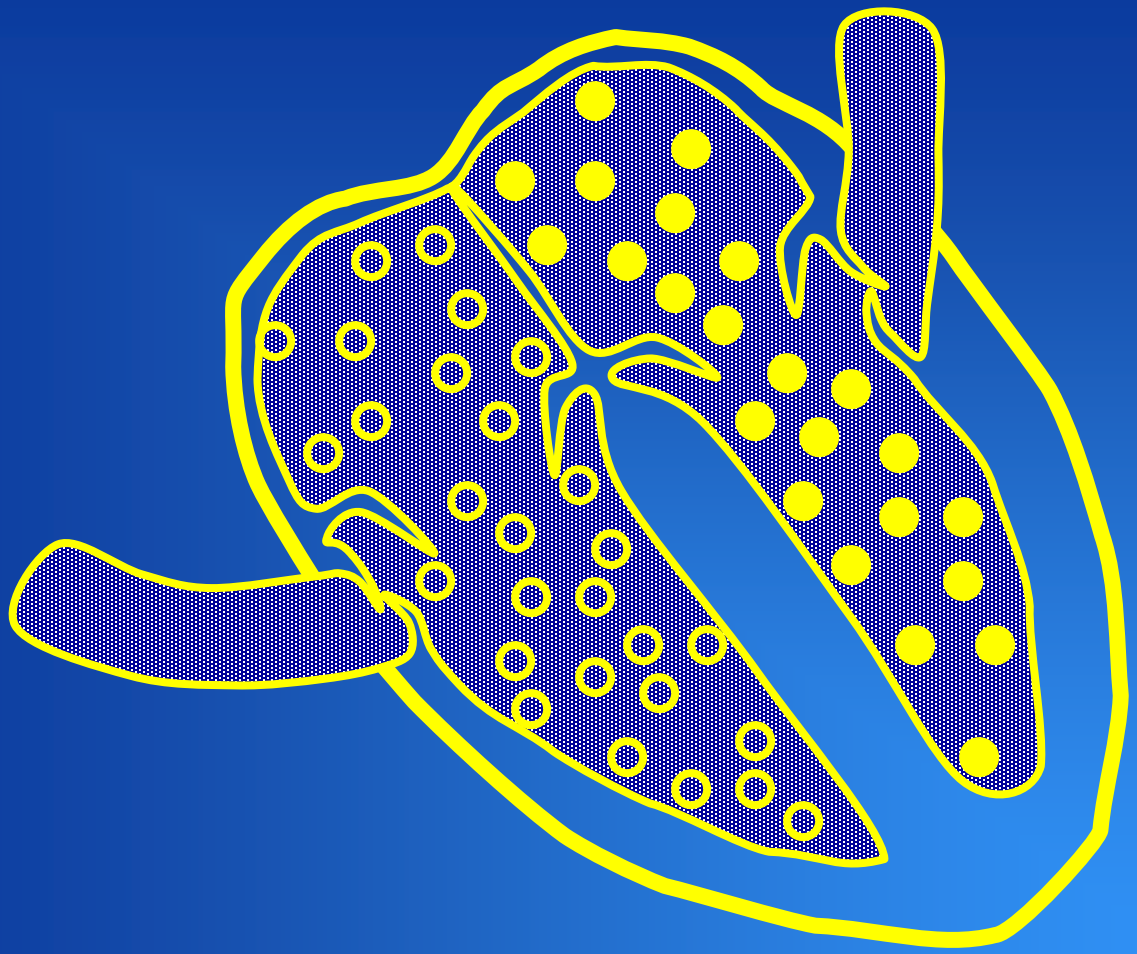
T wave –Ventricular repolarization

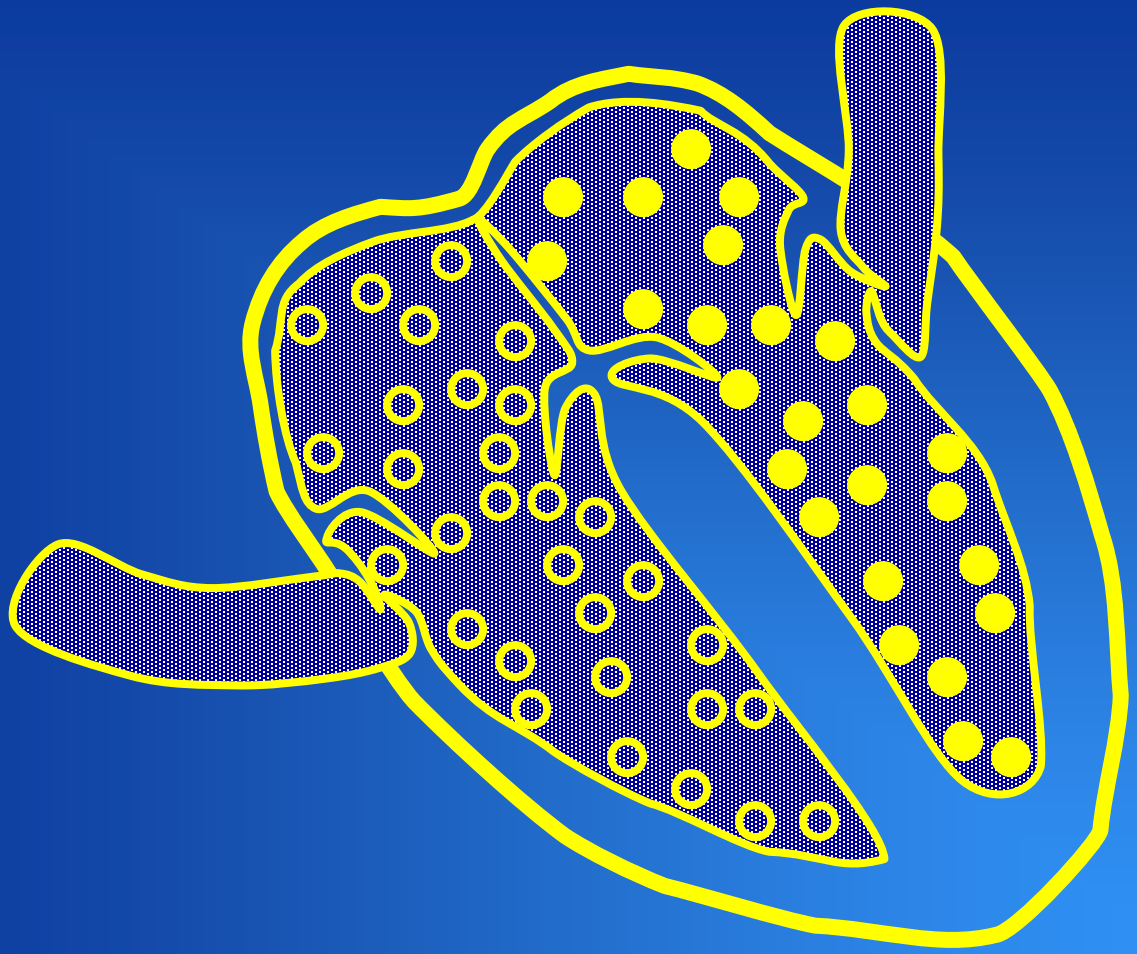


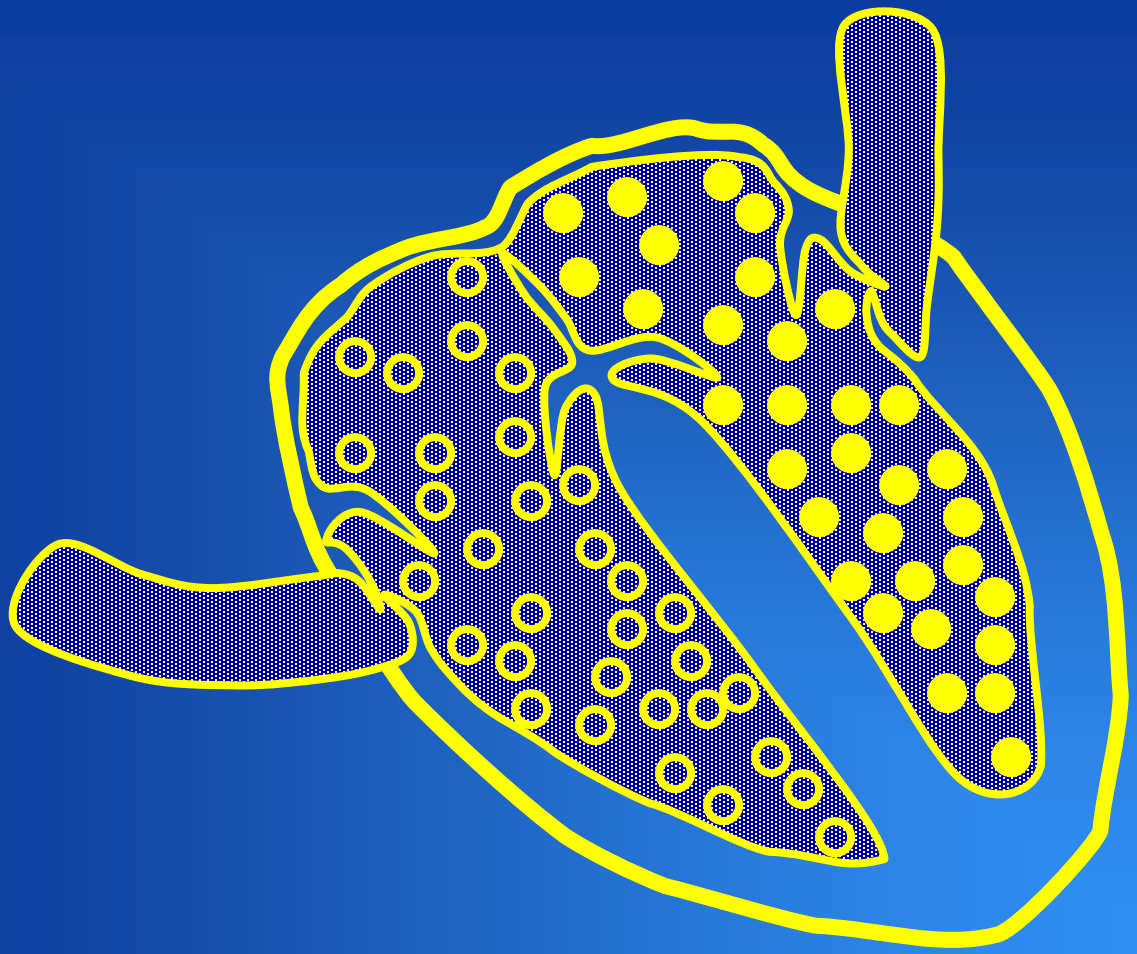


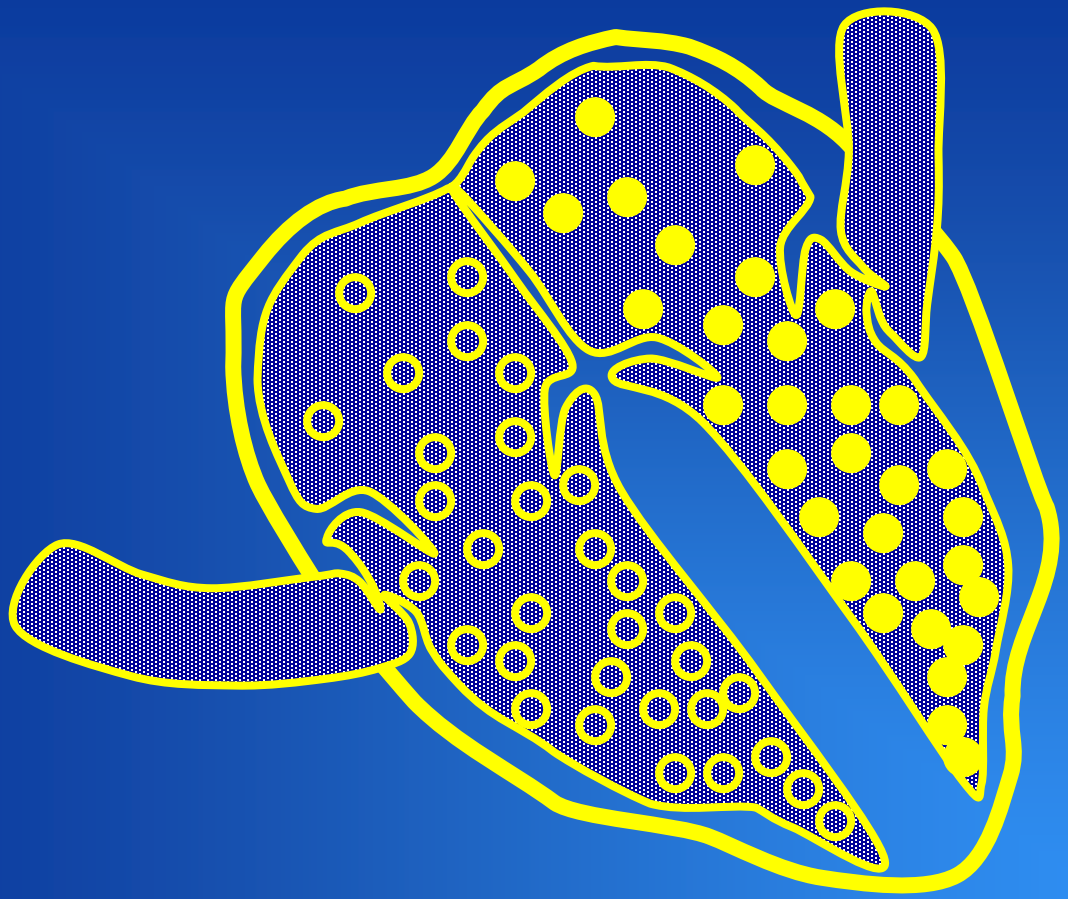


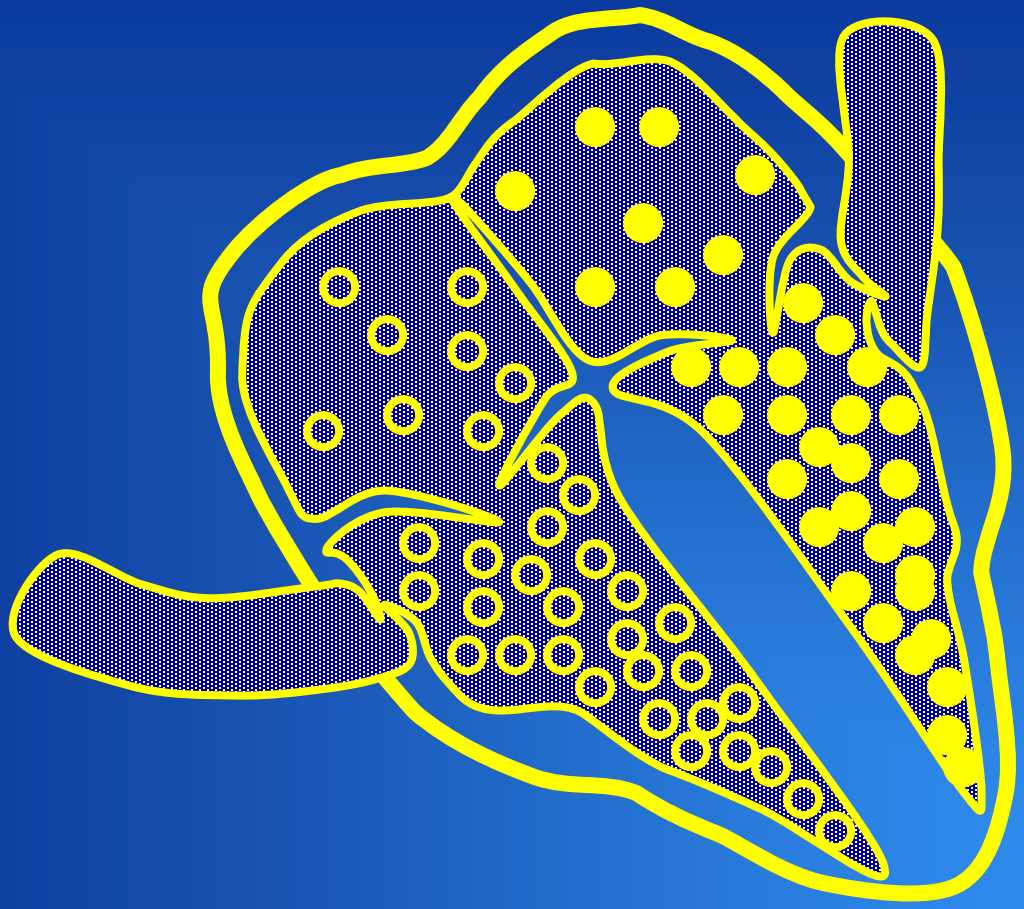


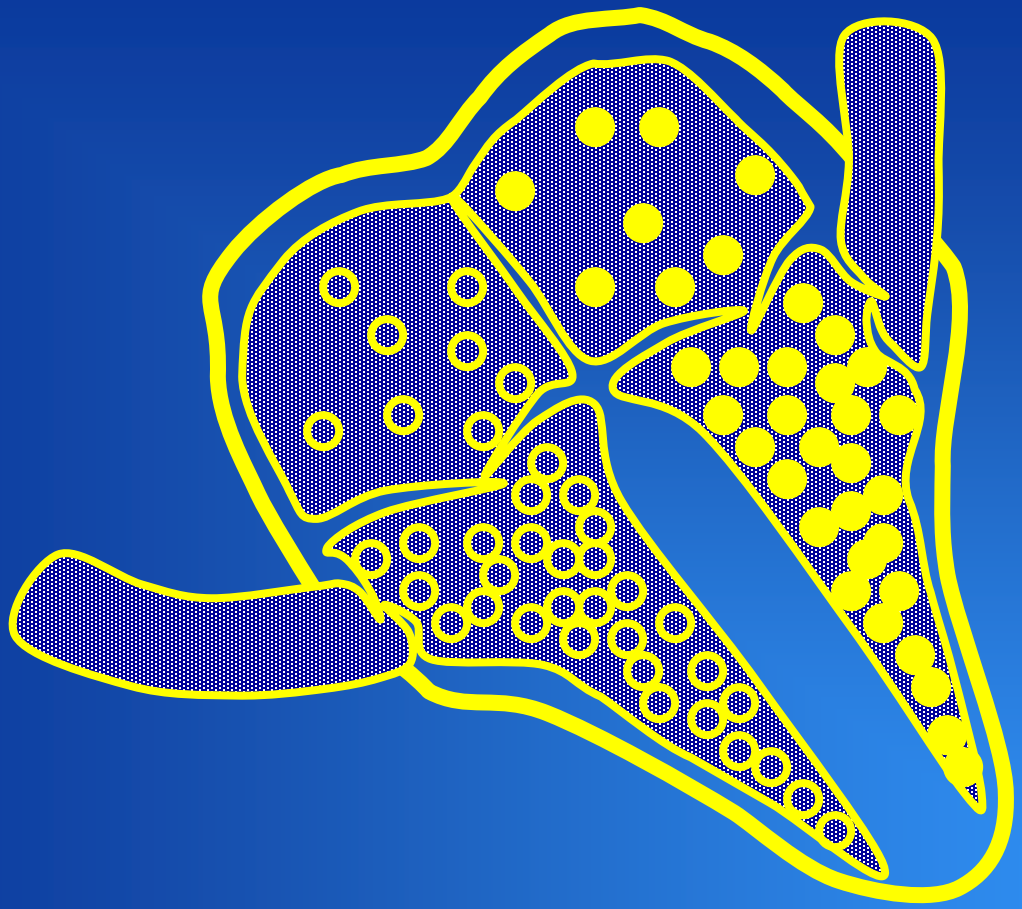




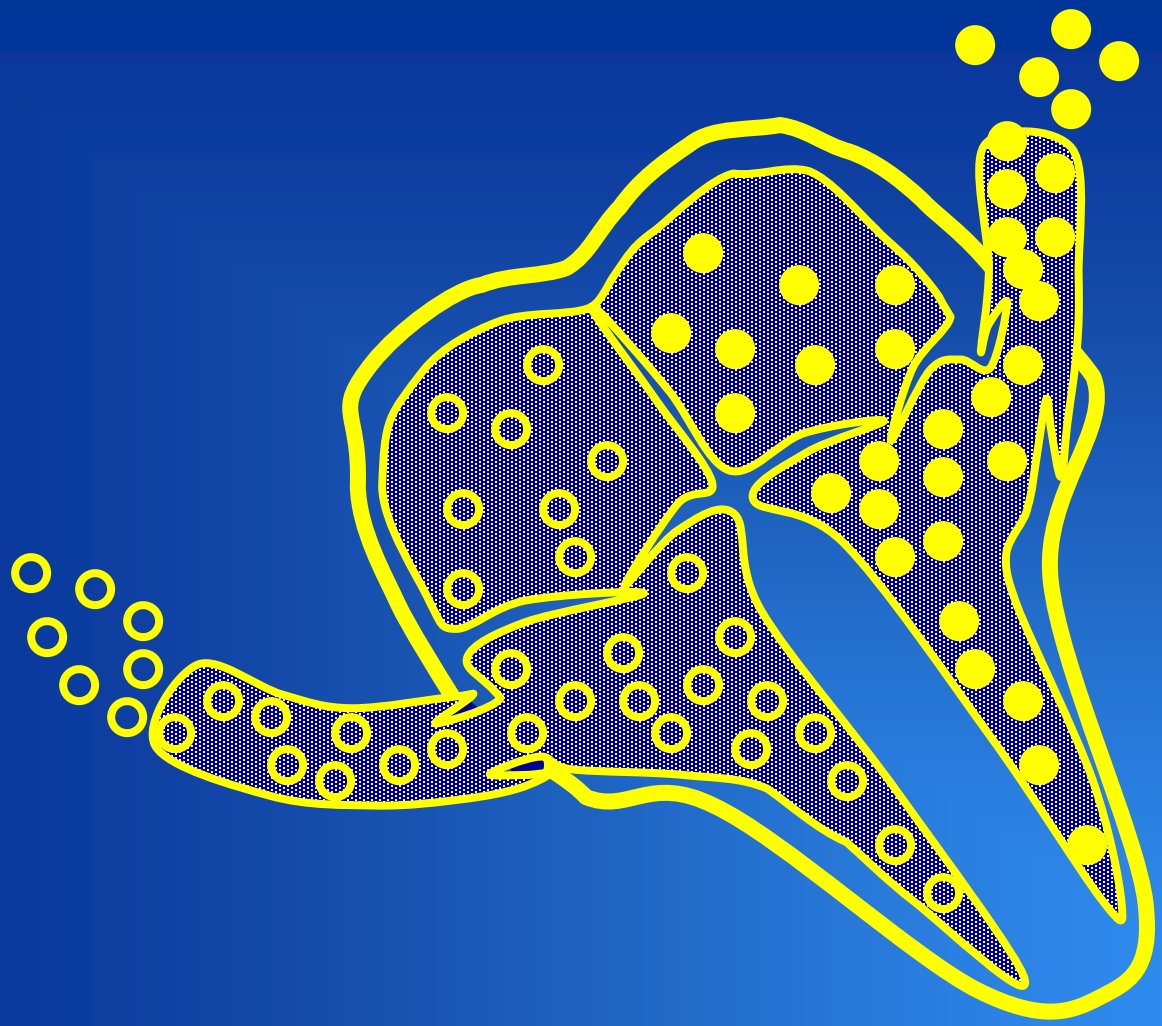


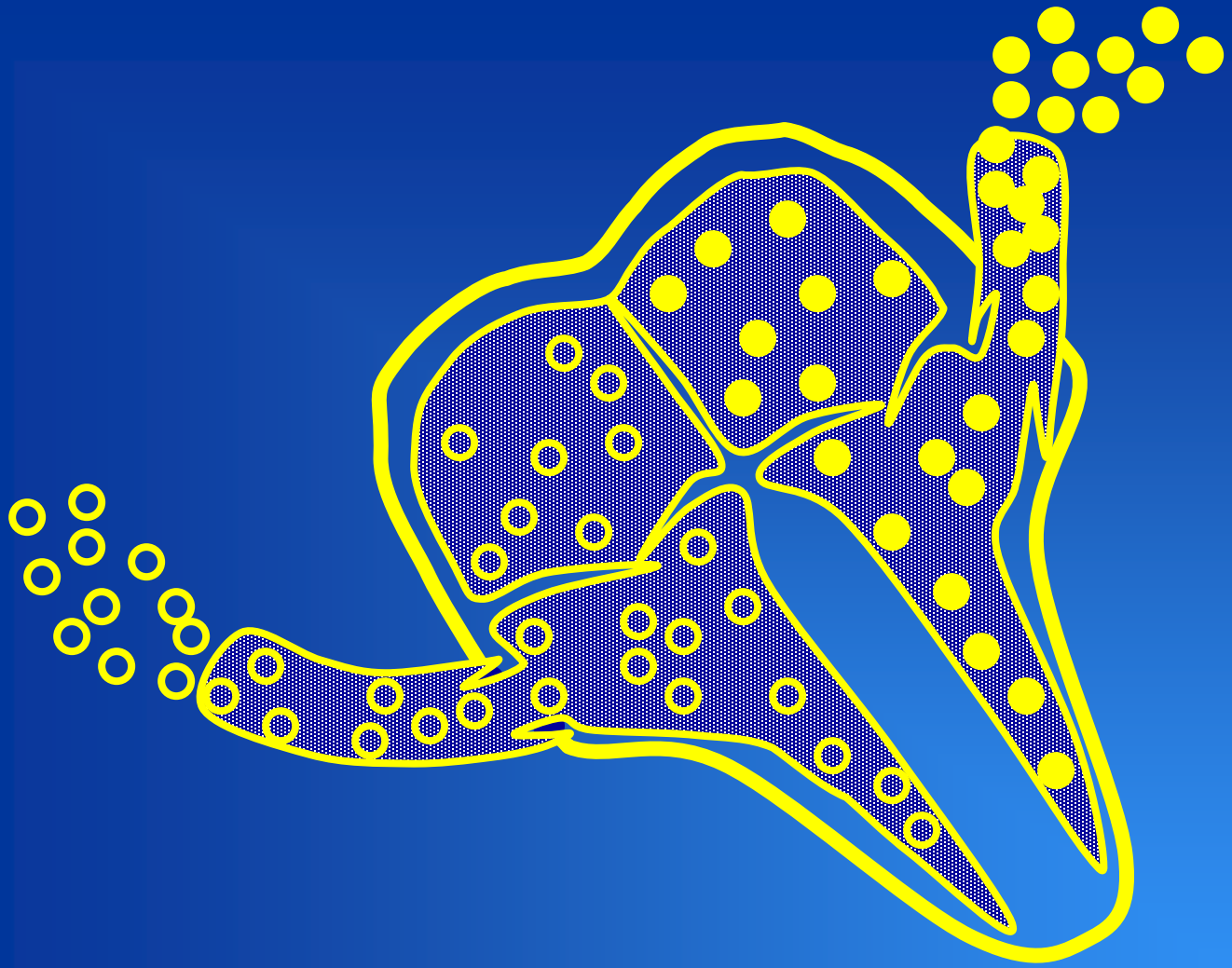


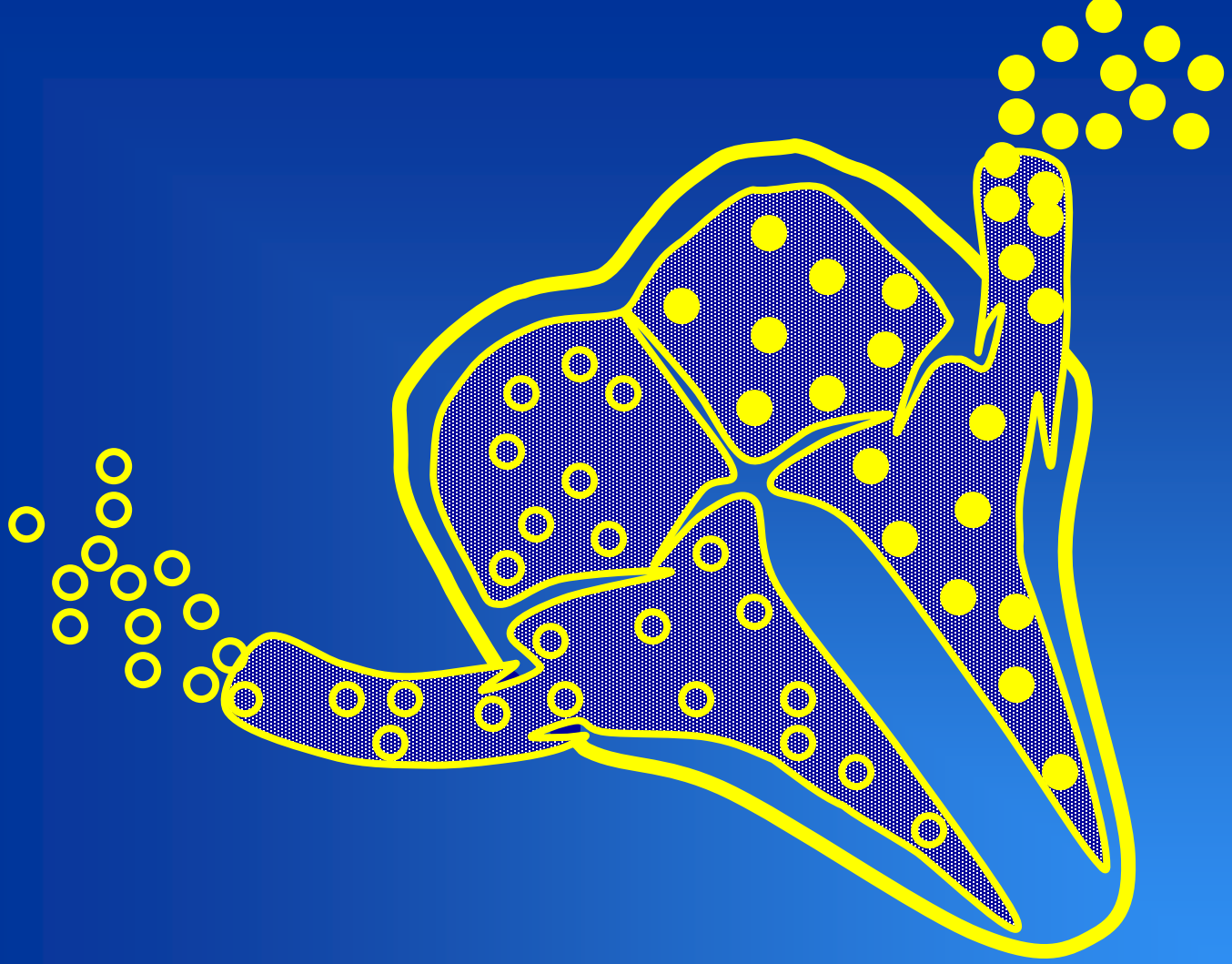


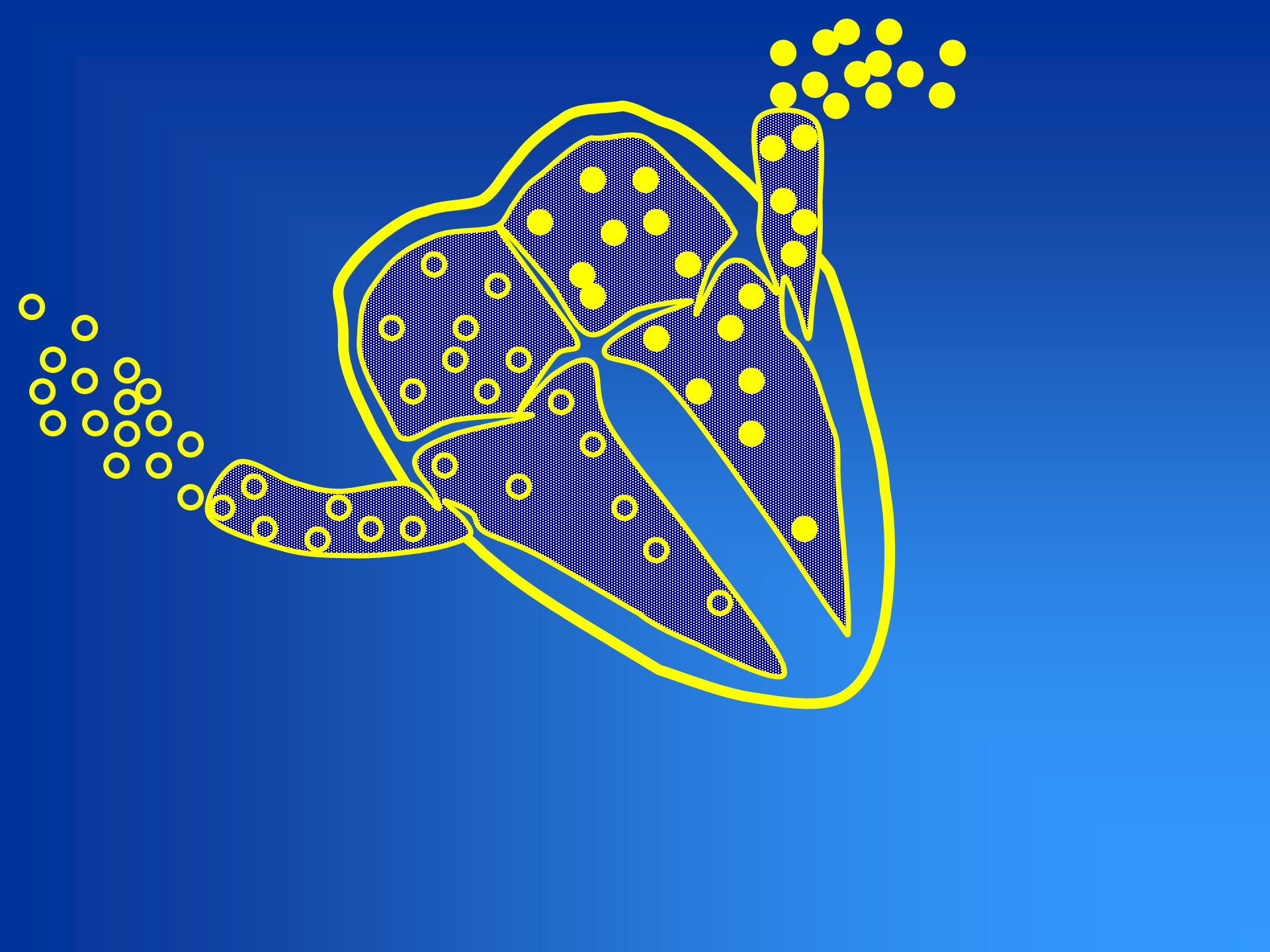


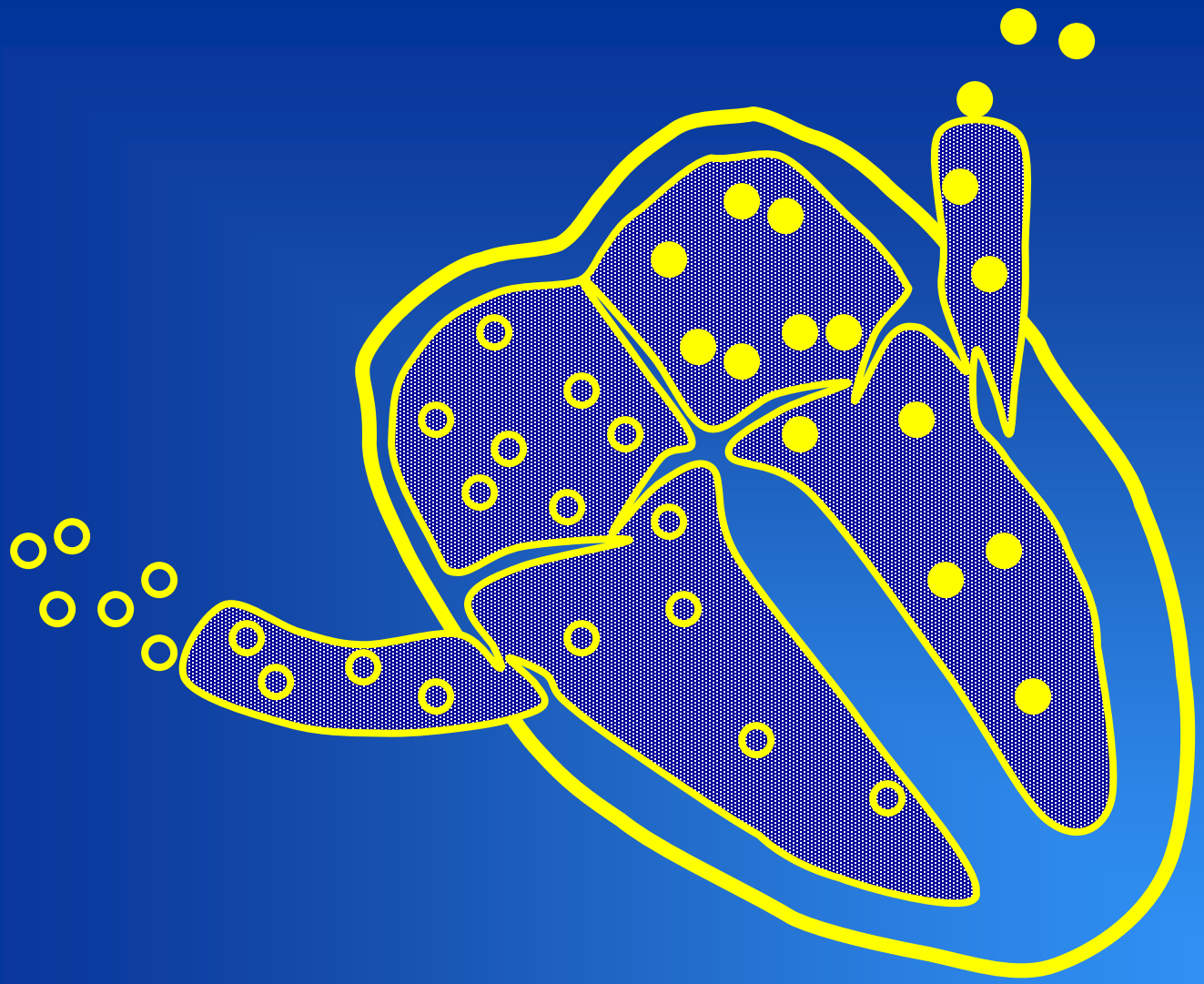


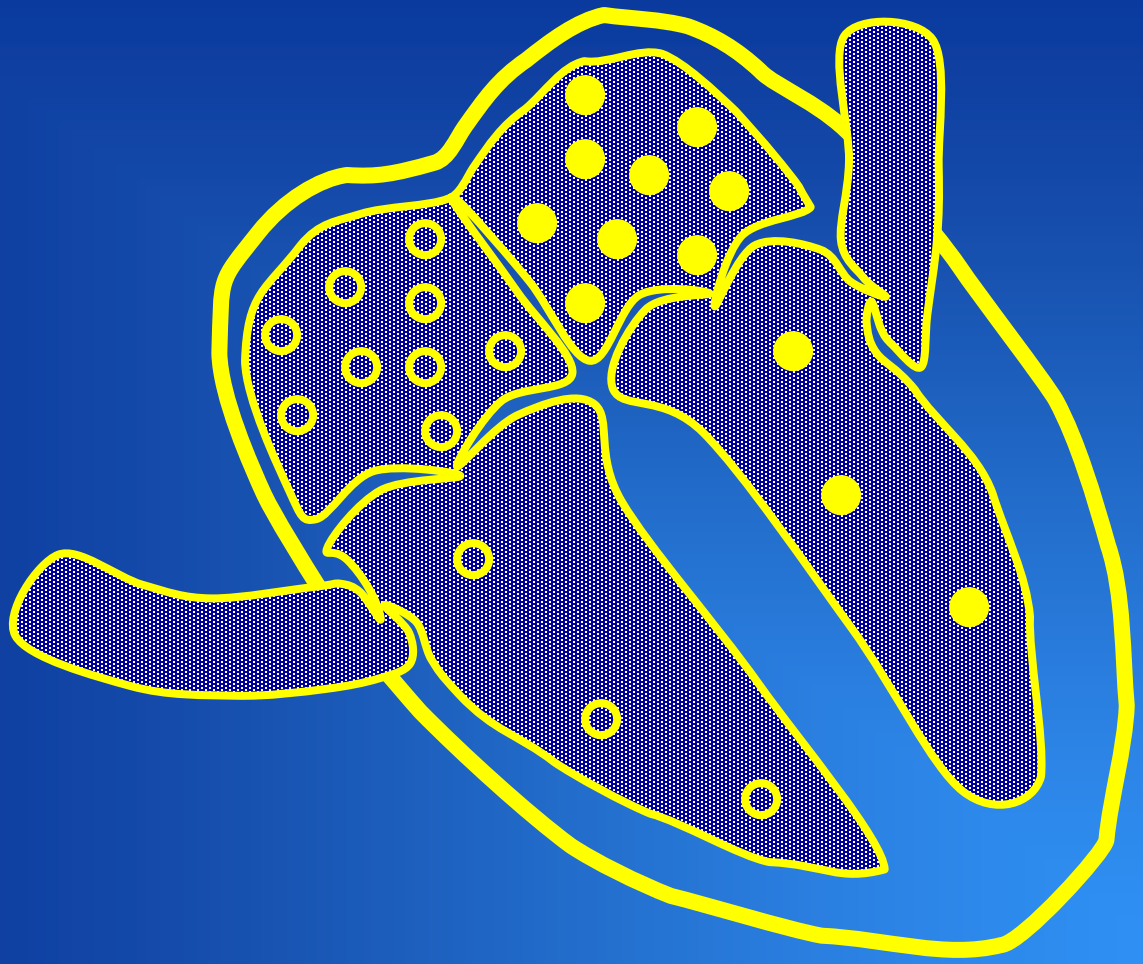


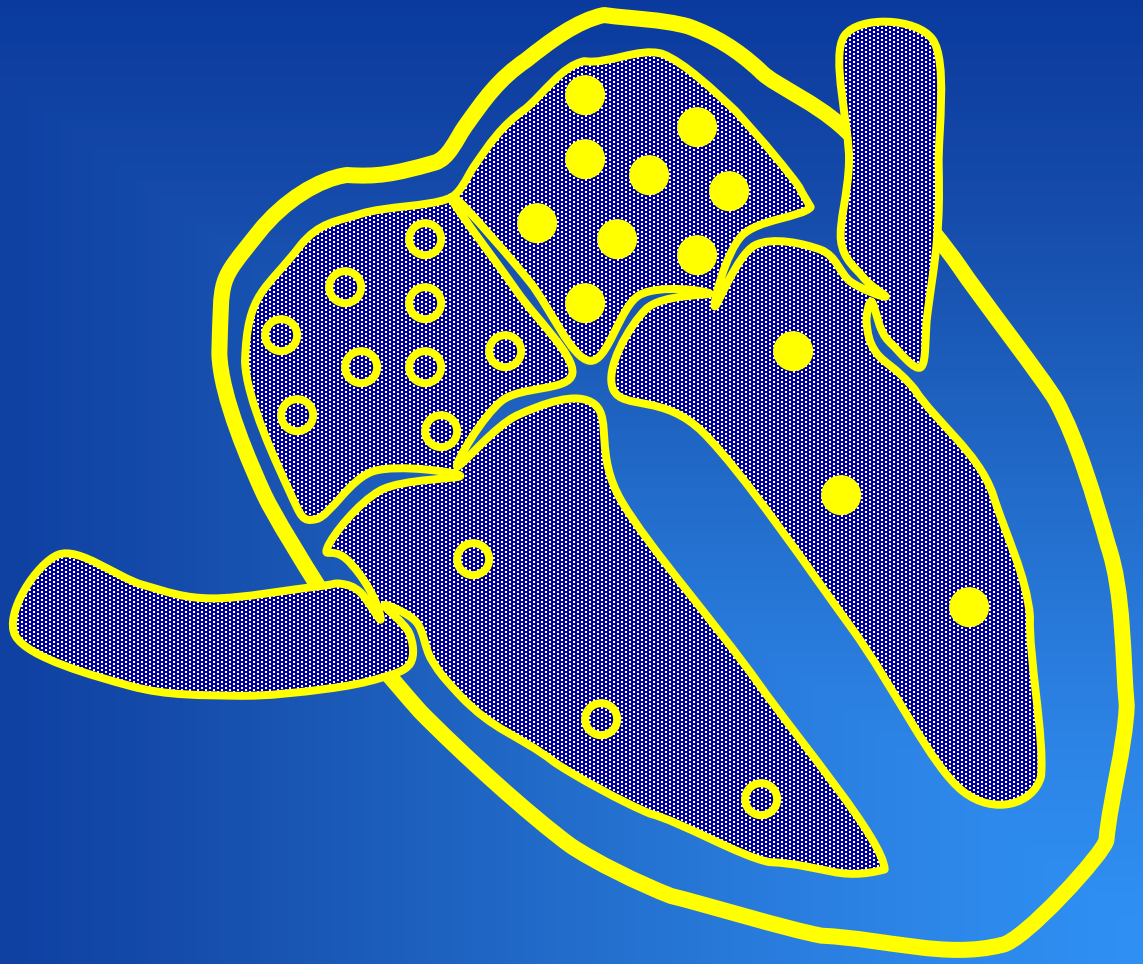


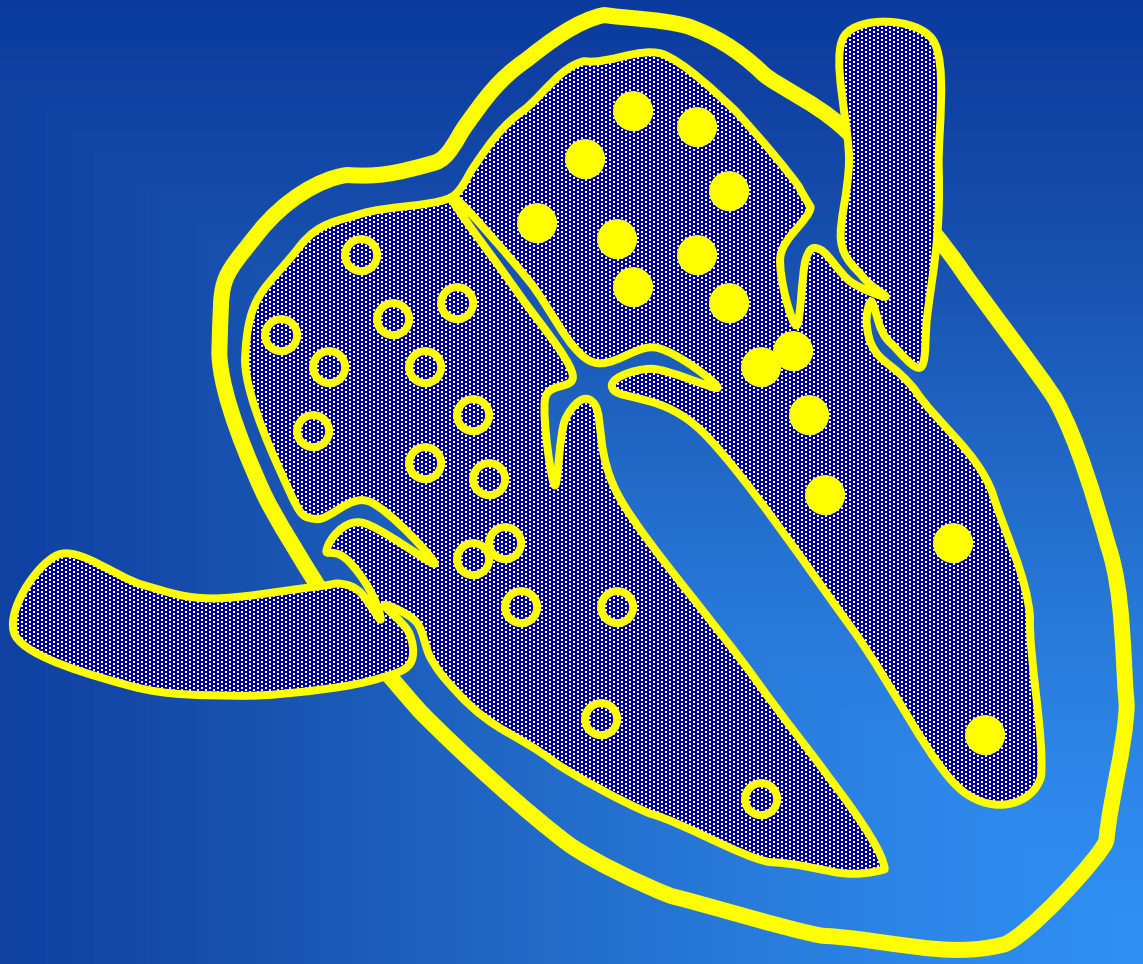


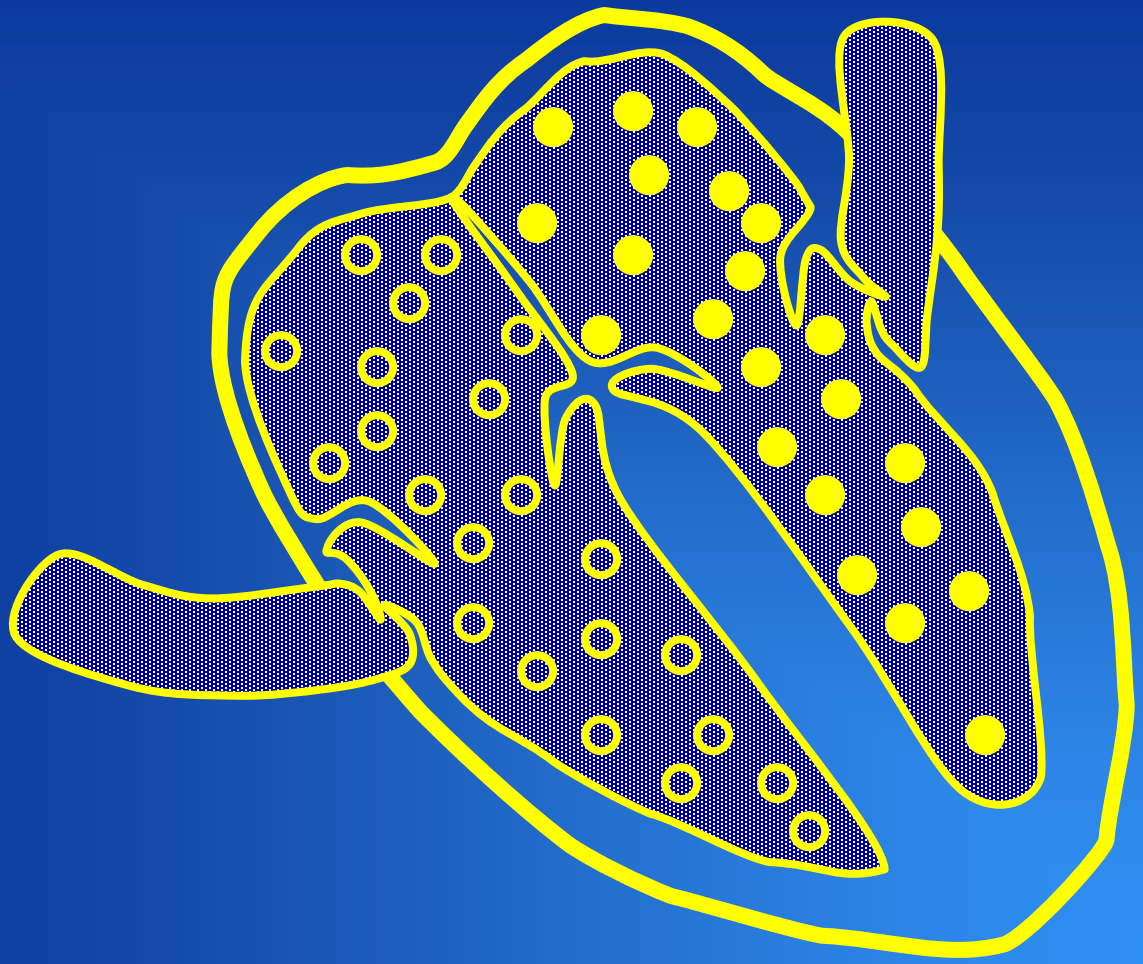


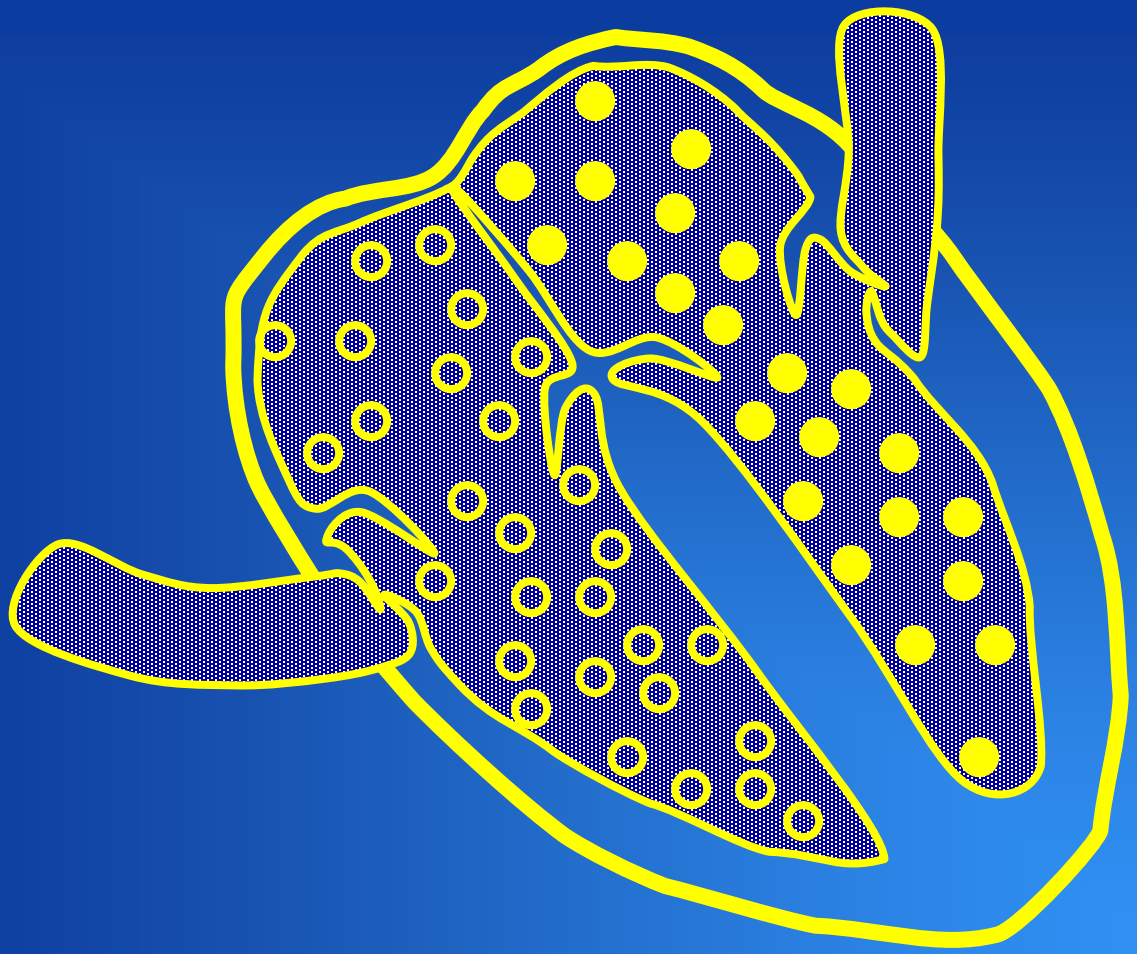


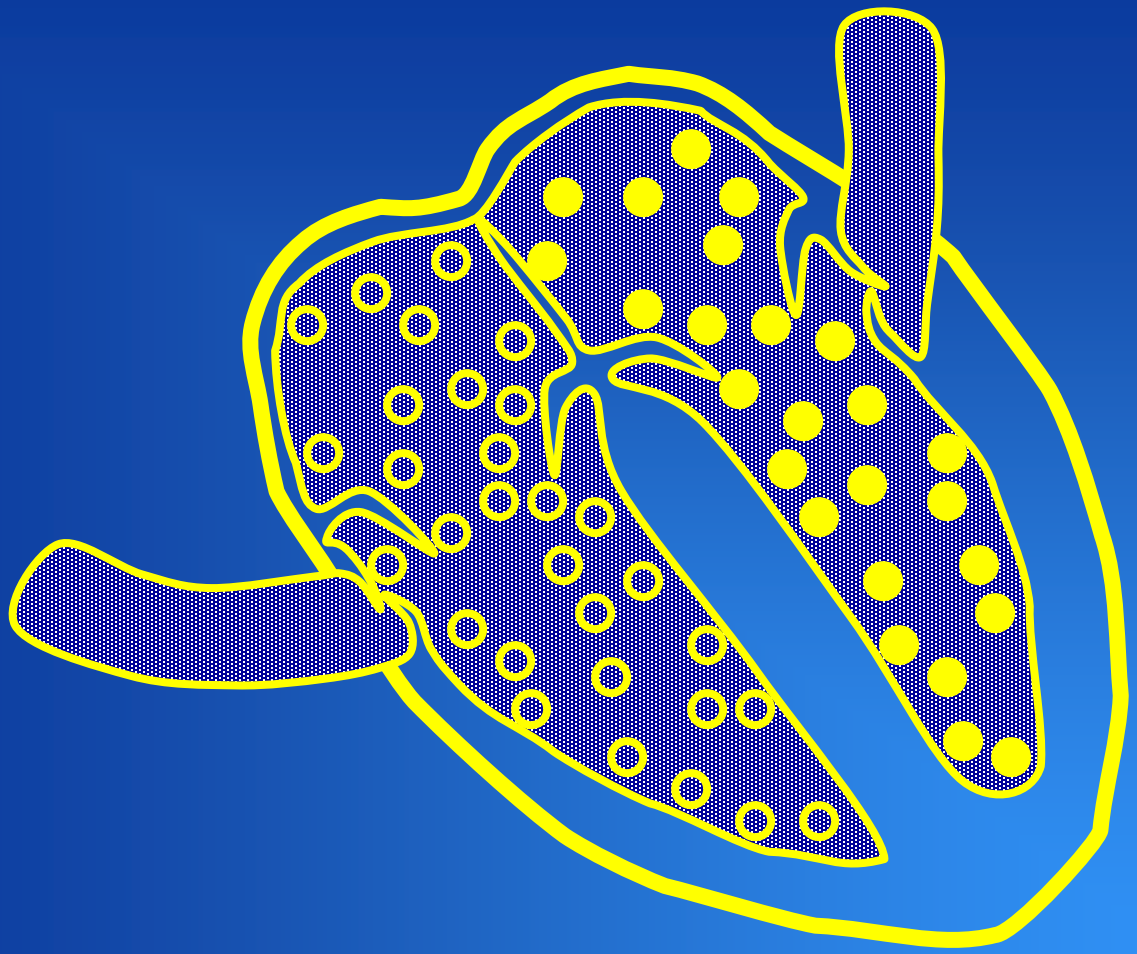


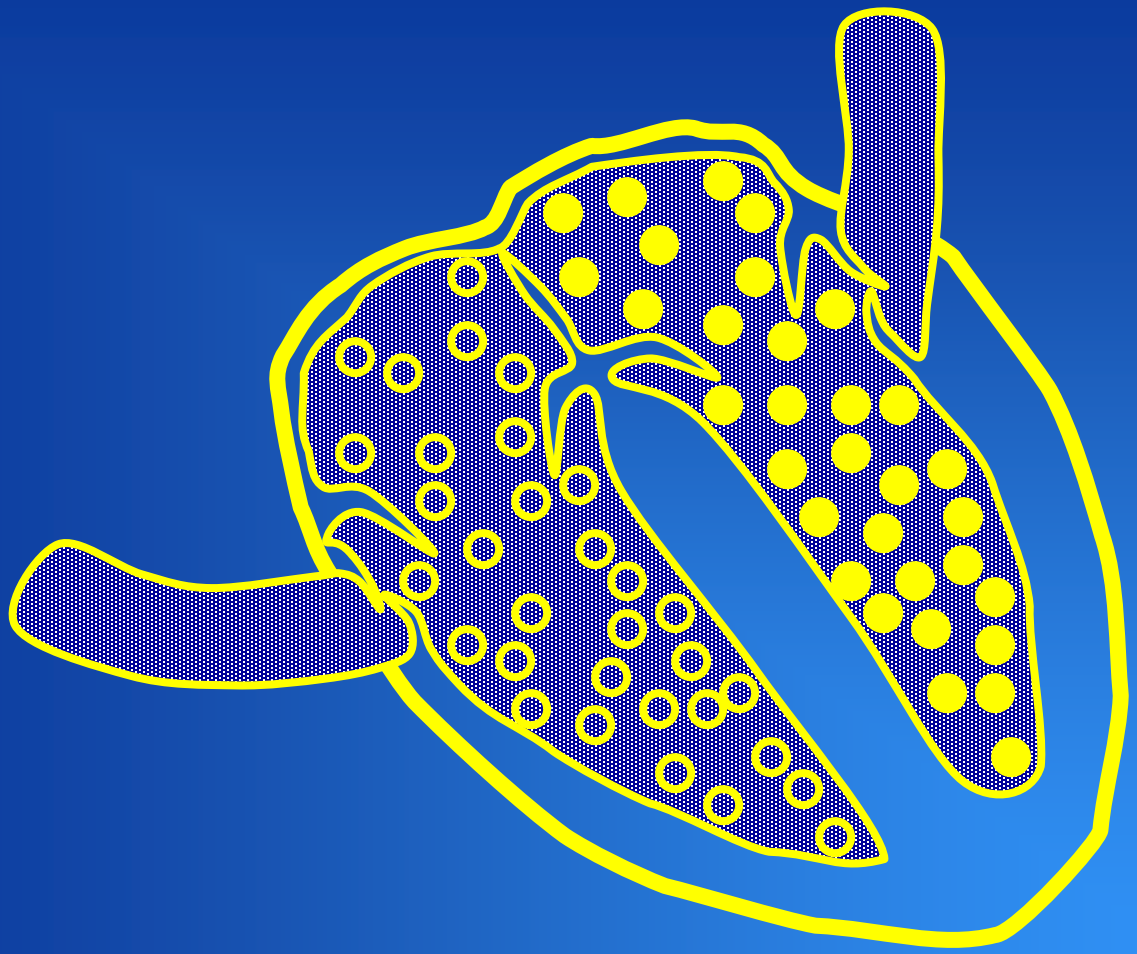


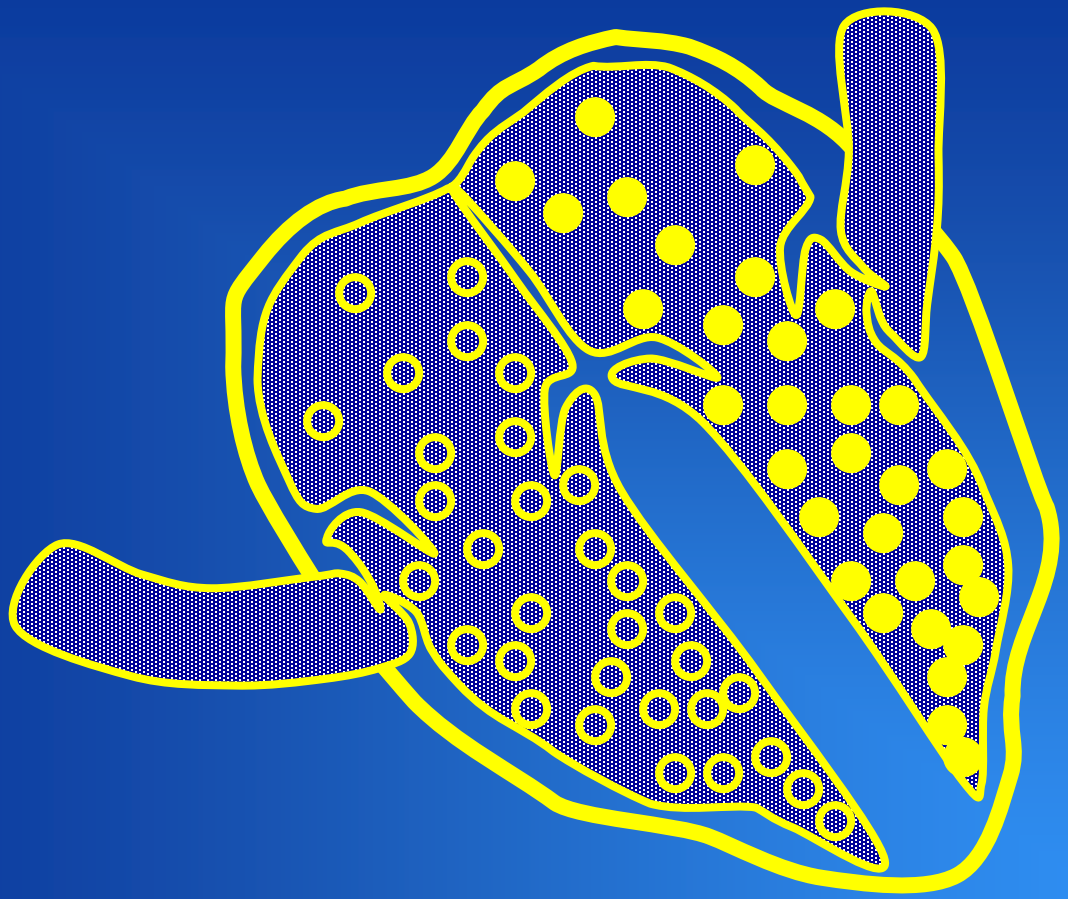


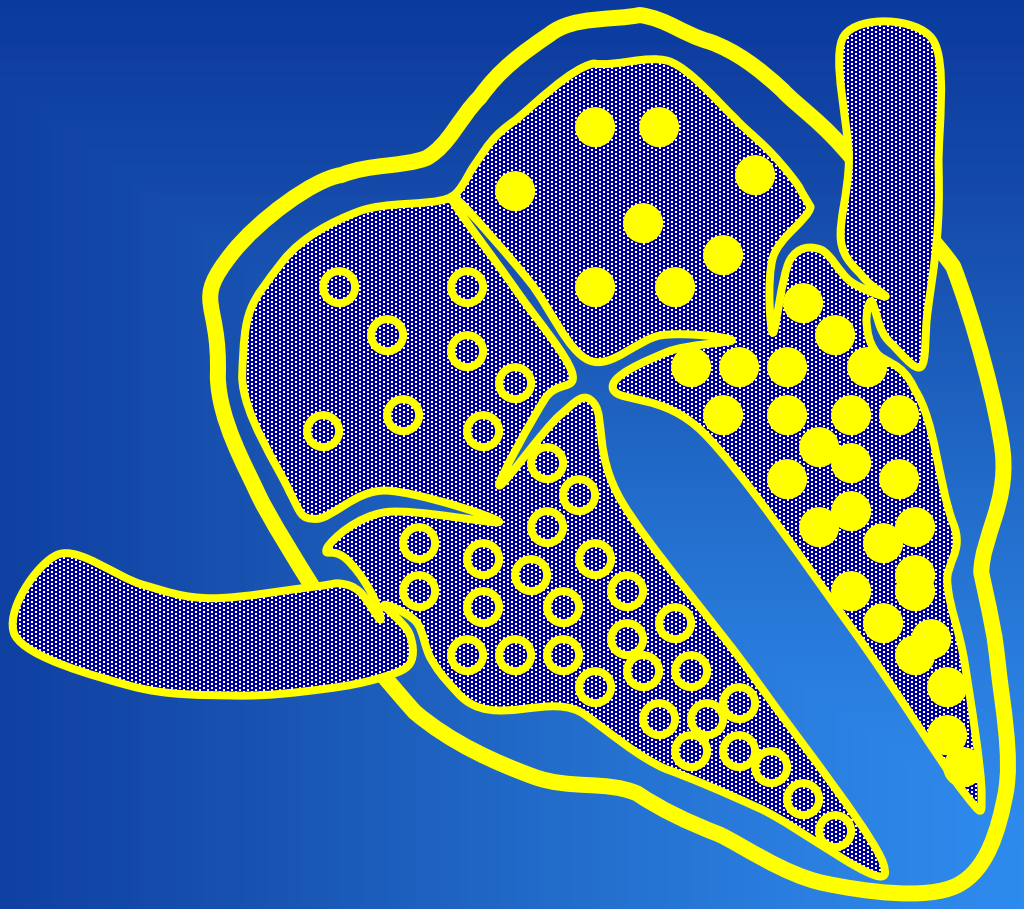


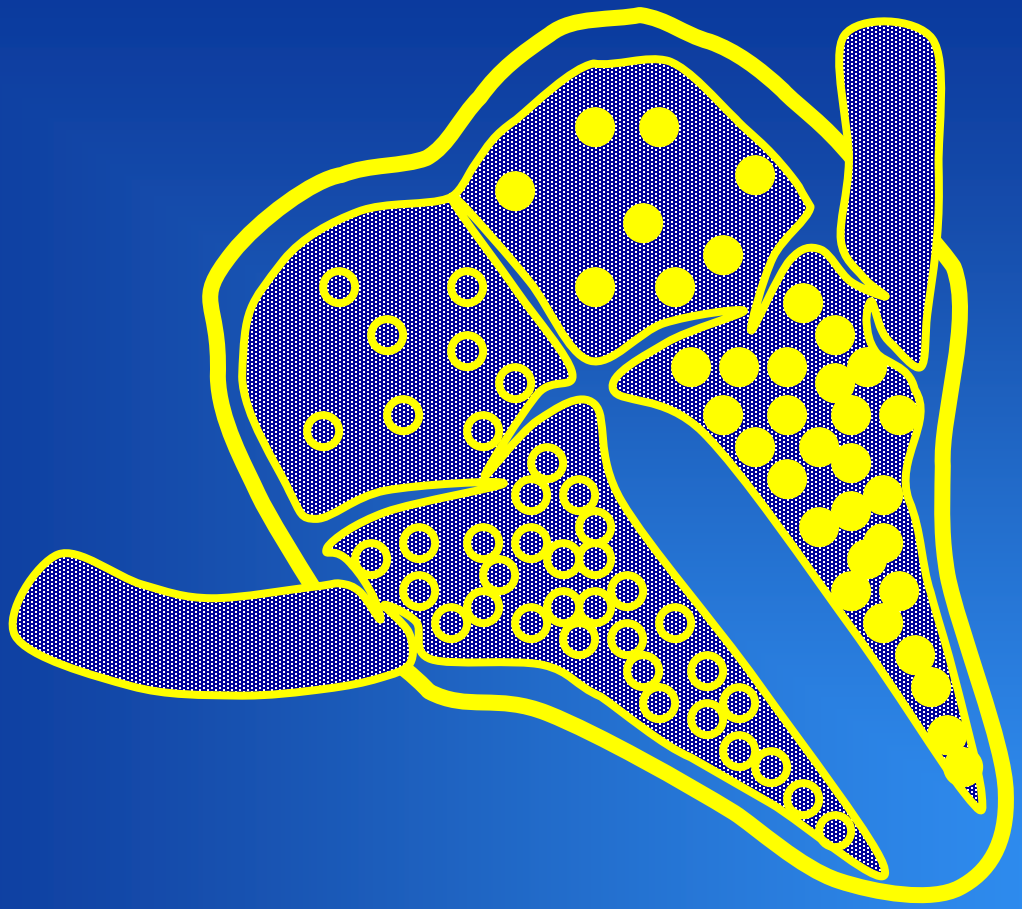




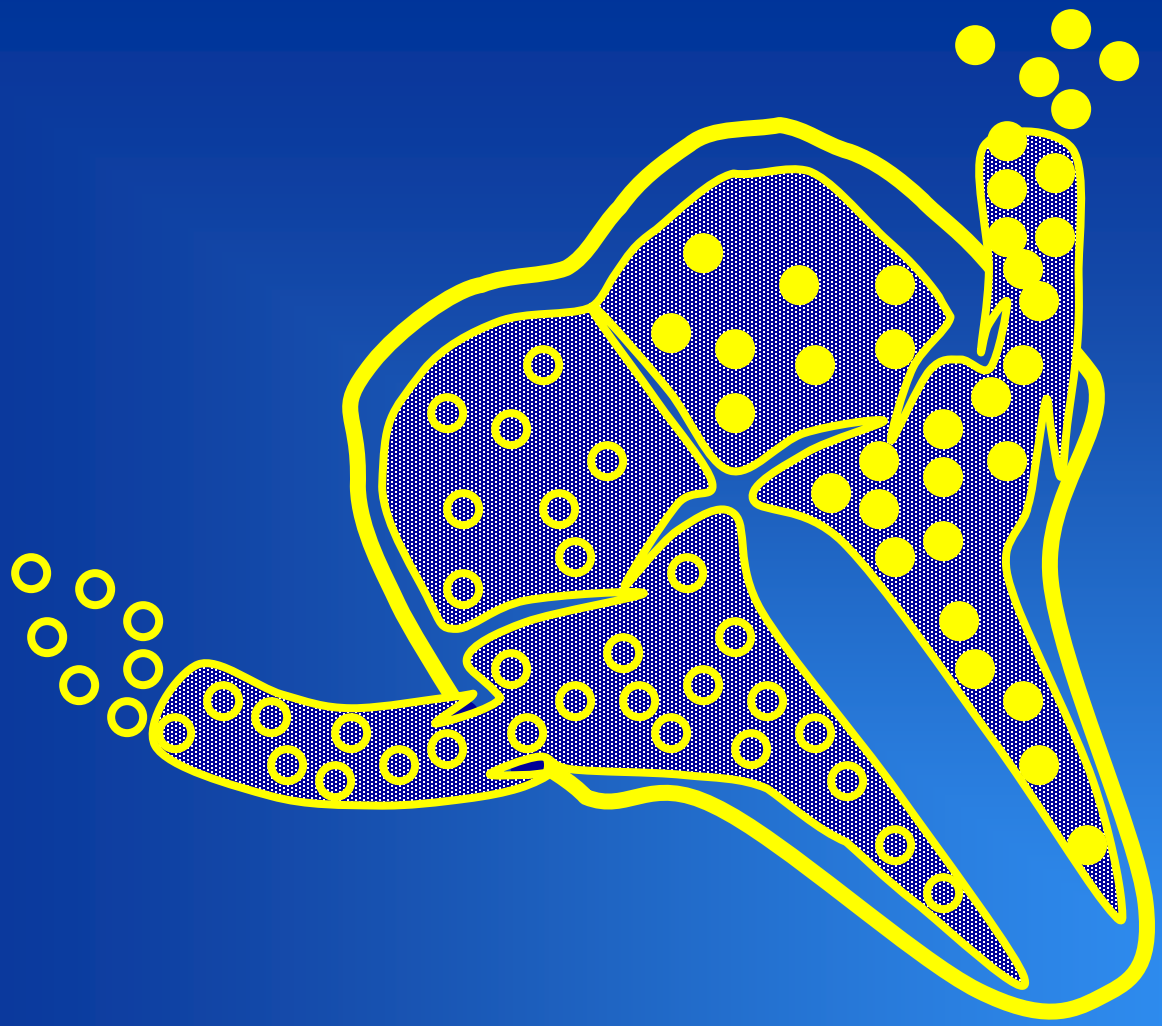


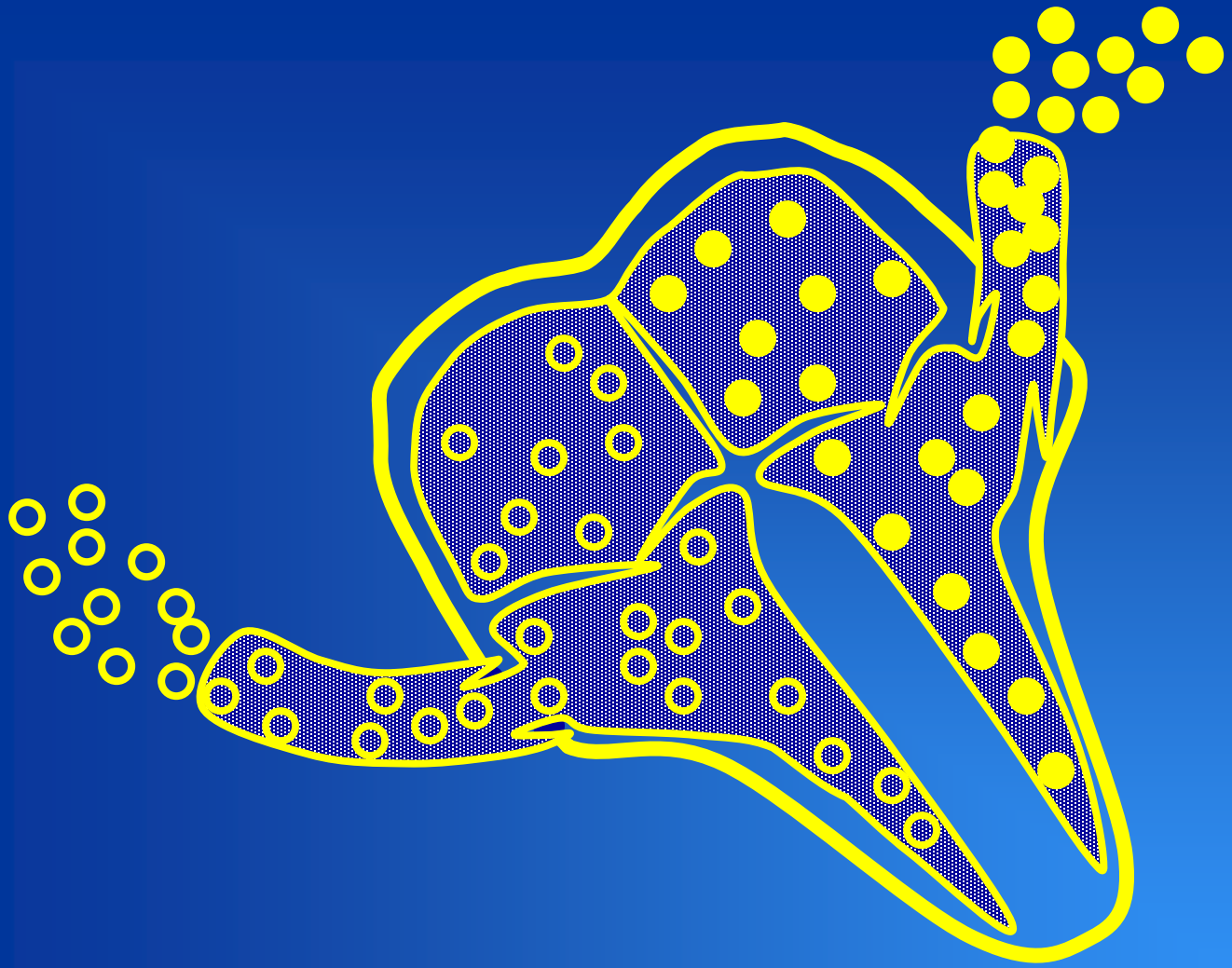


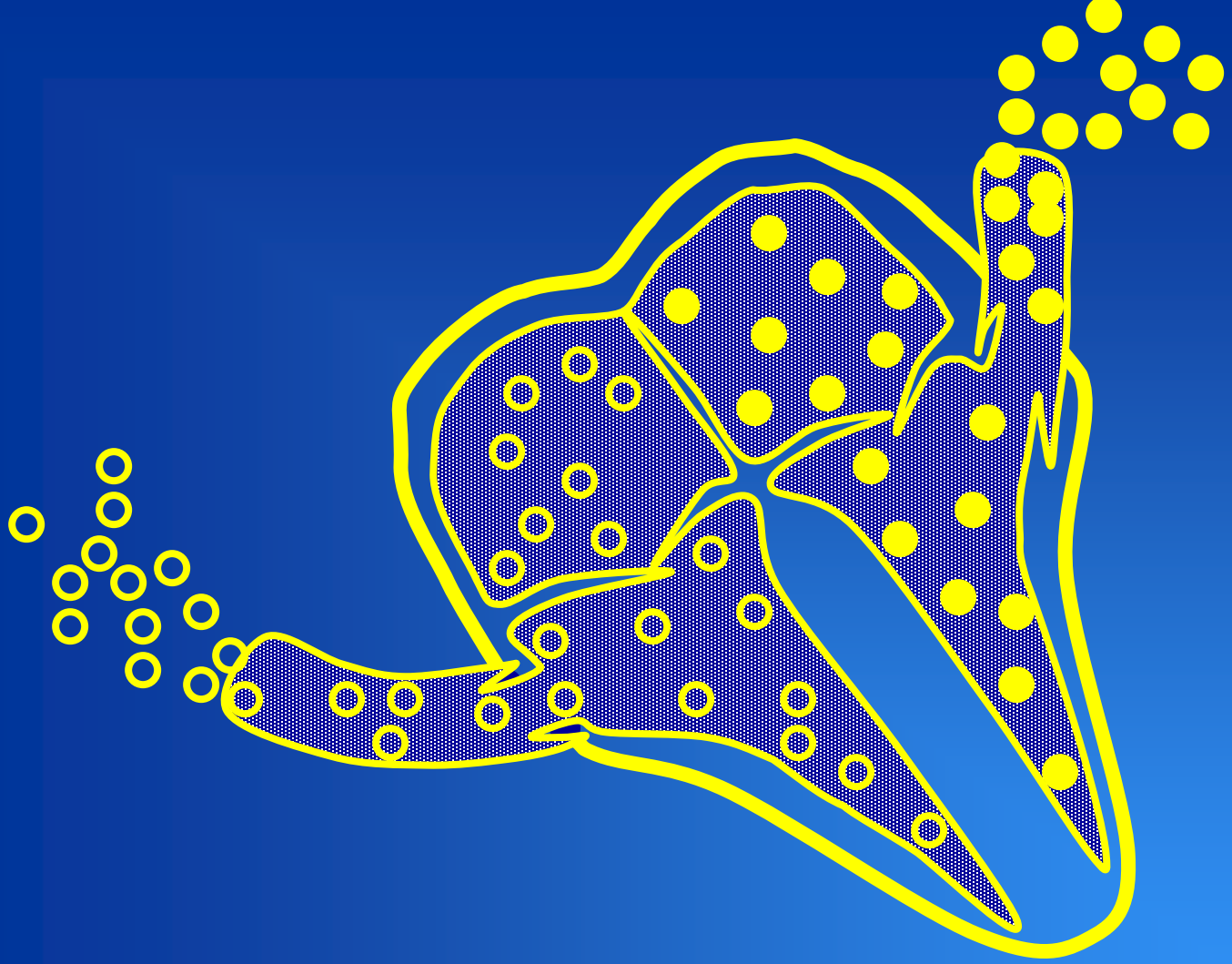


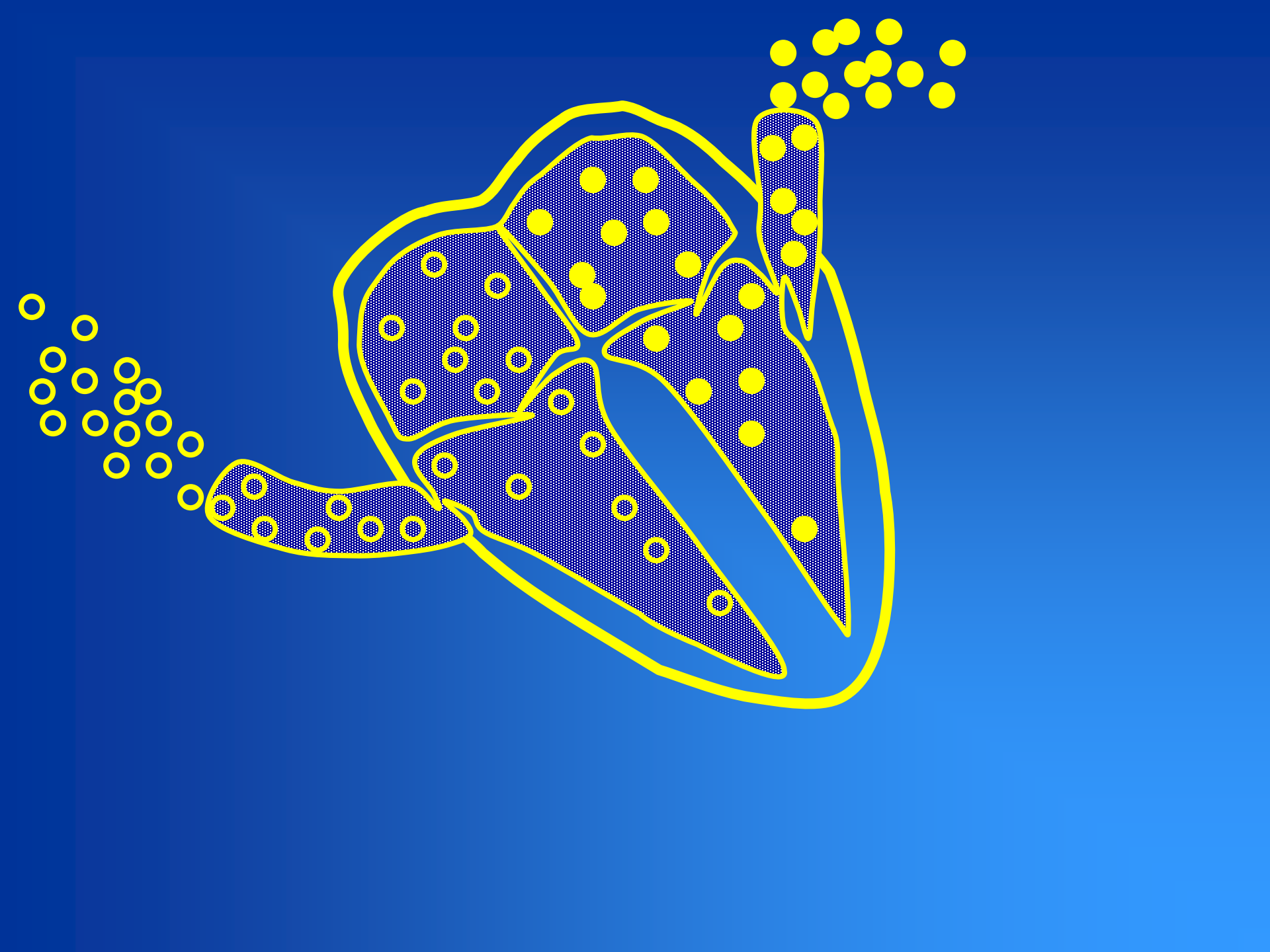


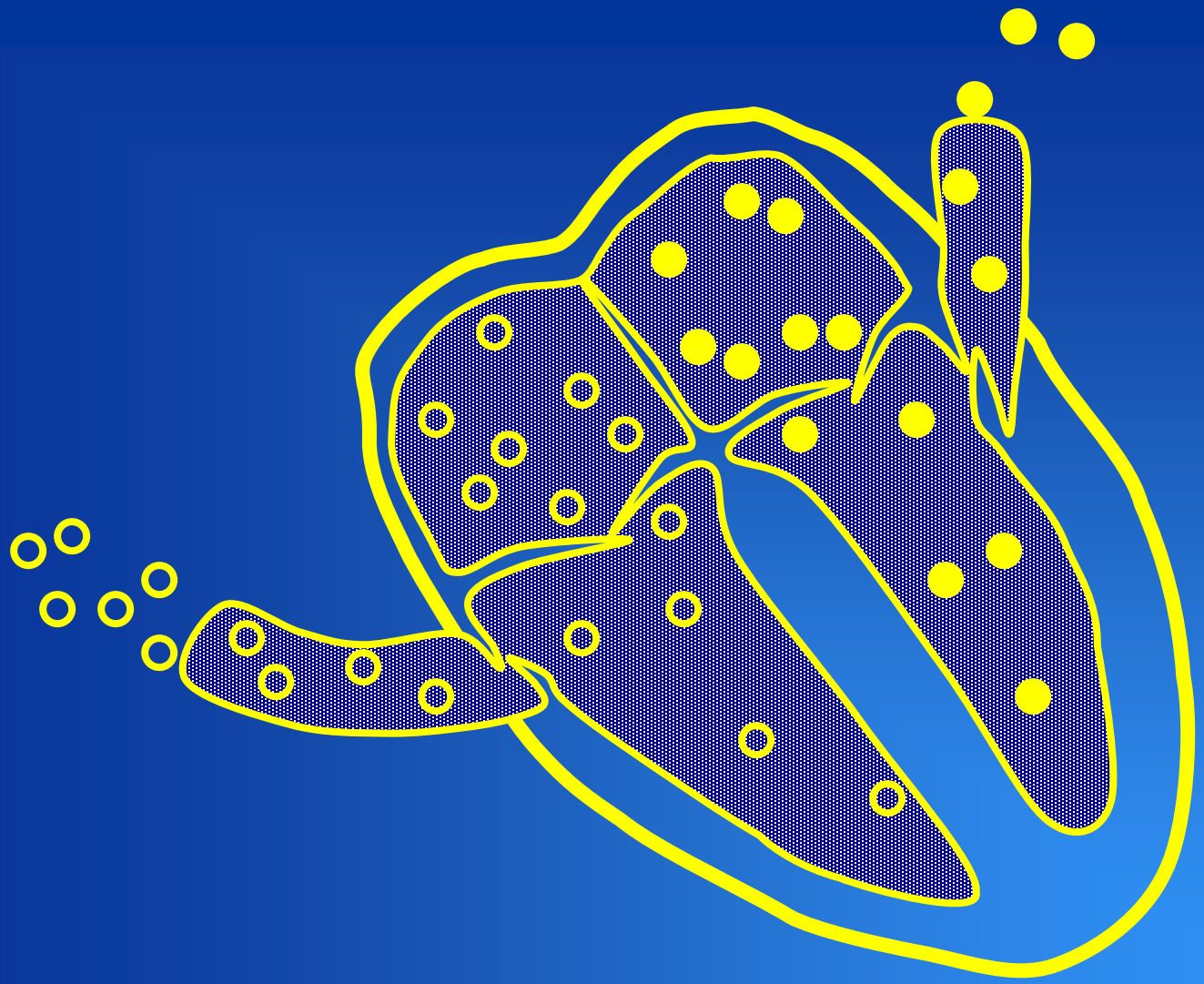












Depolarization

Muscle contraction

Pressure generation

Blood ejection





ECG elements

P wave – Atrial depolarization

PR interval –AV node depolarization

QRS complex –Septal and ventricular wall depolarization

T wave –Ventricular repolarization

SA node

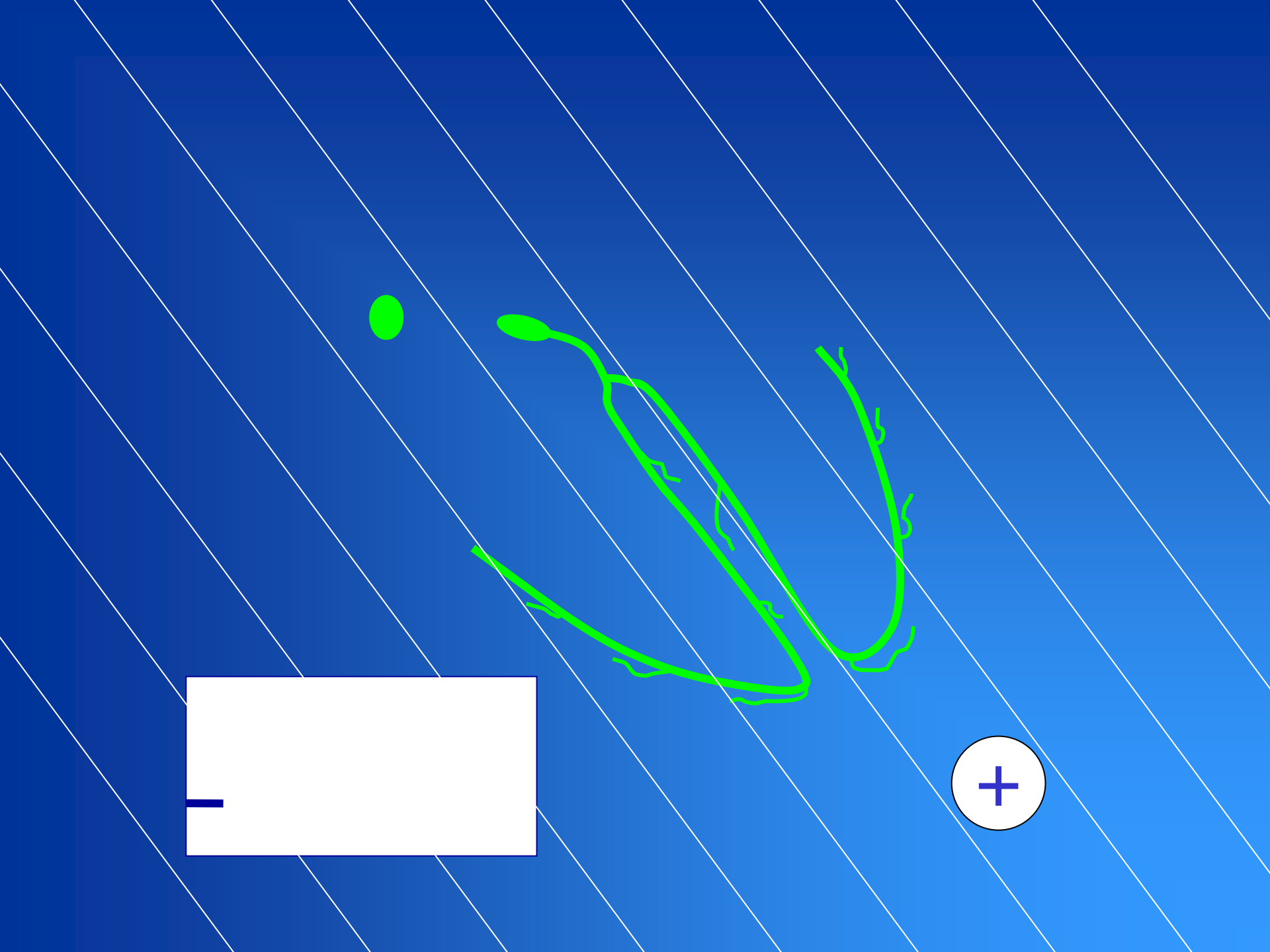
AV node

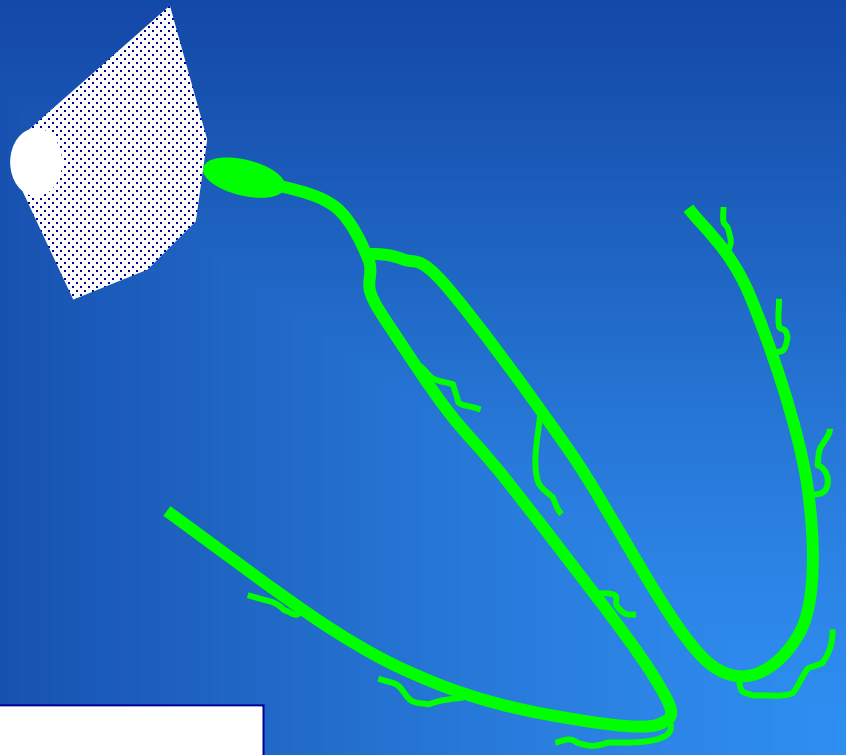


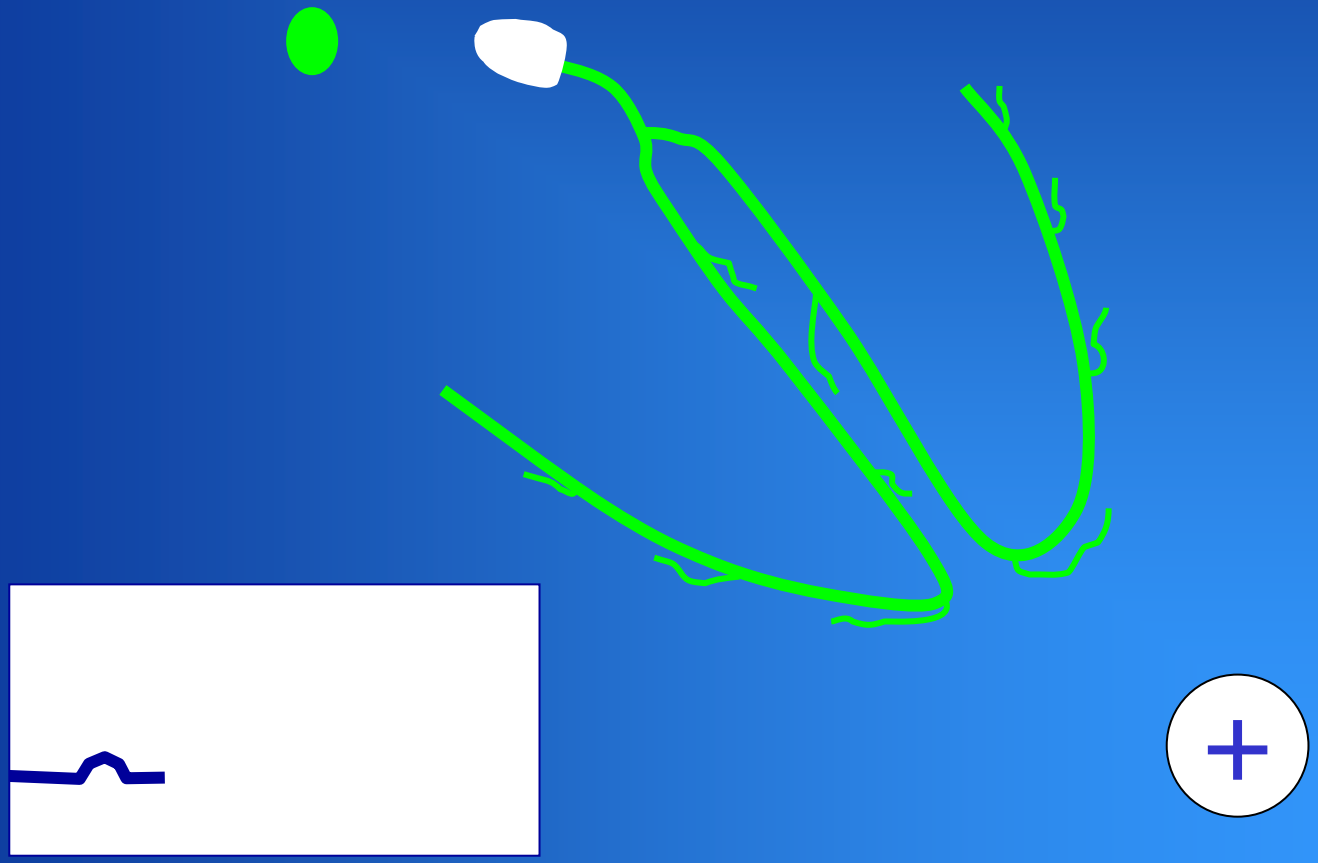
Bundle branches

Purkinje fibers

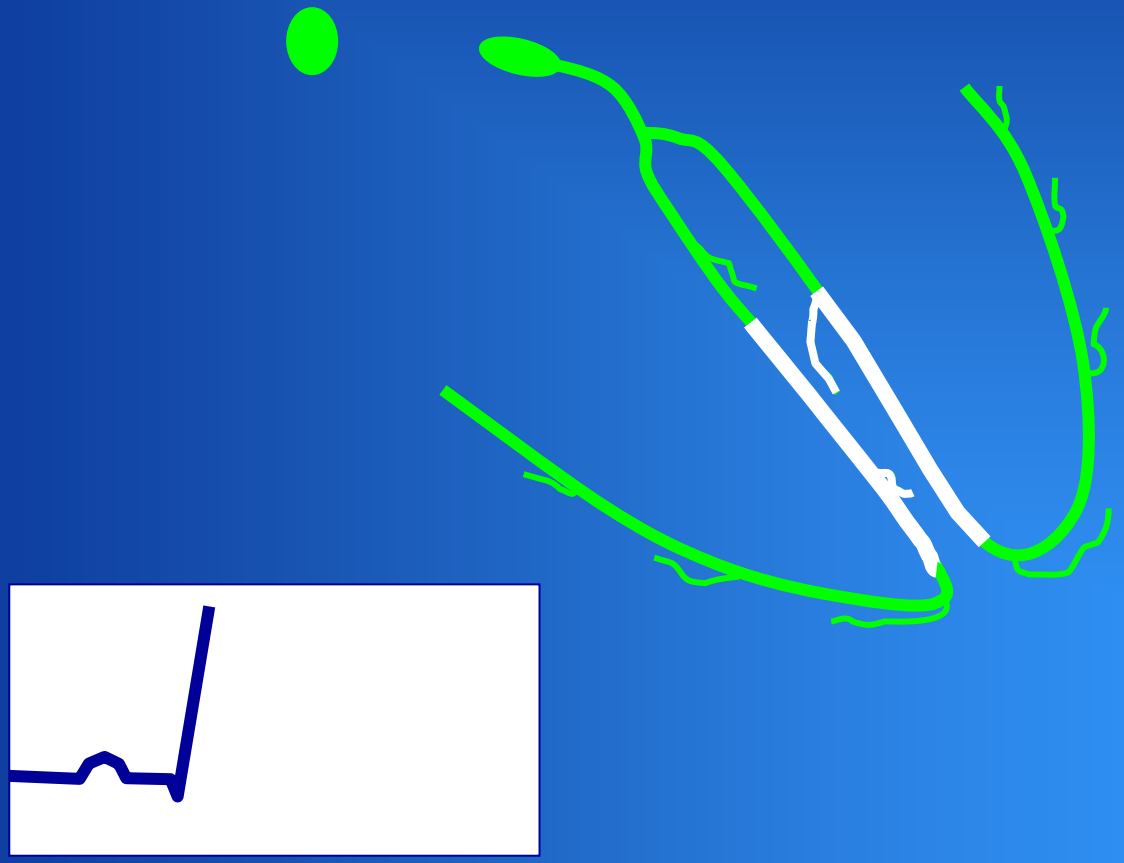
Ventricular myocytes



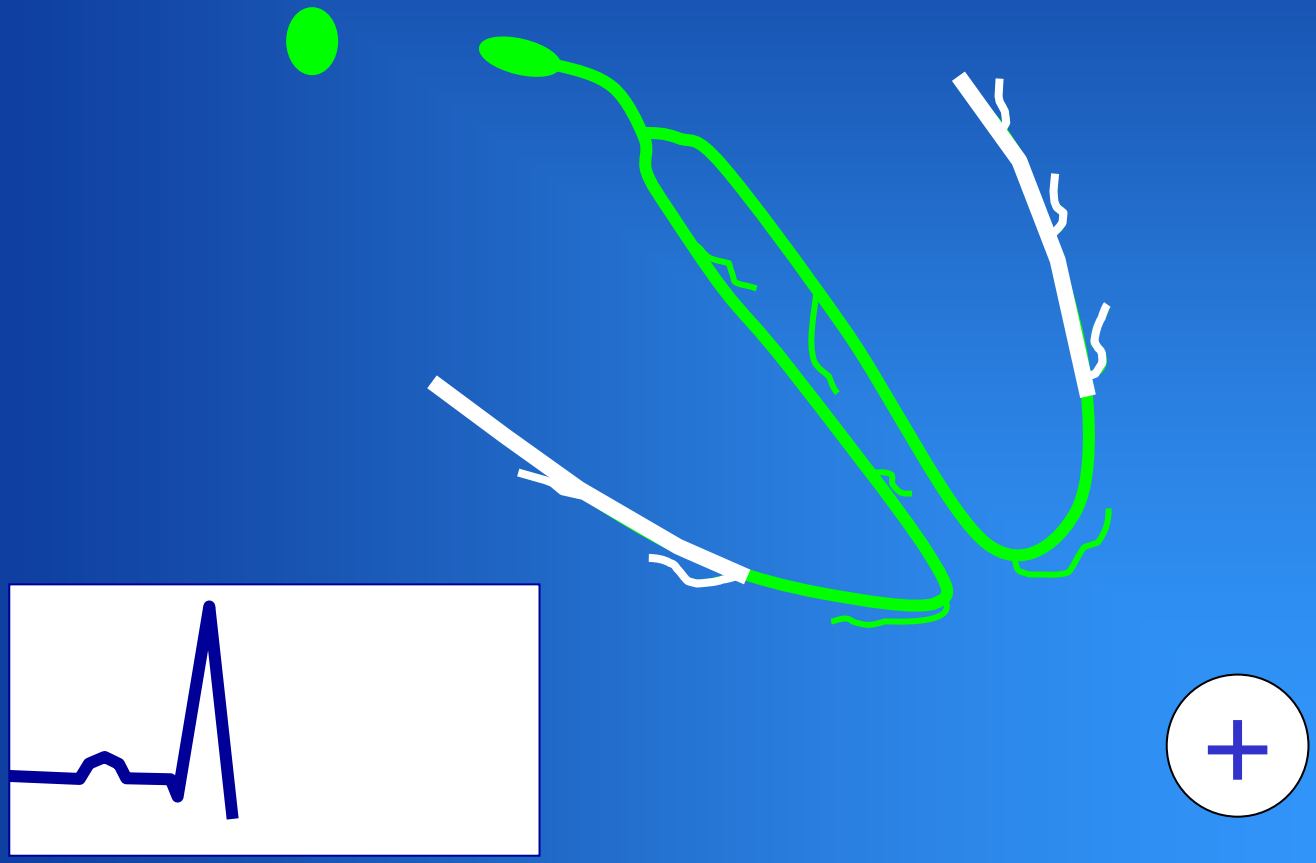


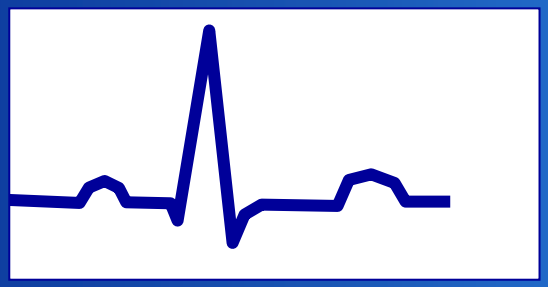














Depolarization

Muscle contraction

Pressure generation

Blood ejection





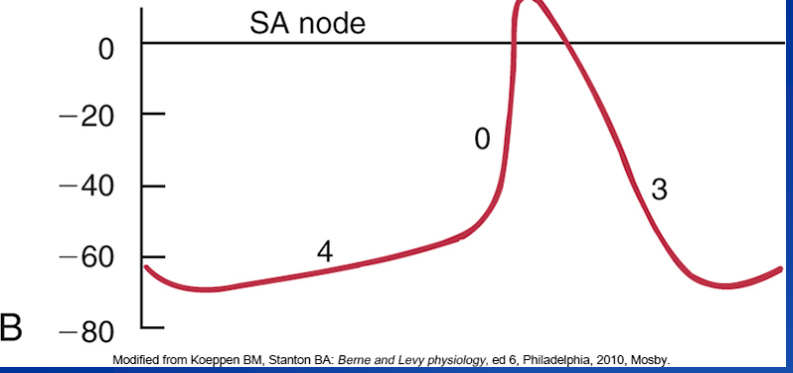
ECG elements

P wave – Atrial depolarization

PR interval –AV node depolarization

QRS complex –Septal and ventricular wall depolarization

T wave –Ventricular repolarization



Modified from Koeppen BM, Stanton BA: *Berne and Levy physiology*, ed 6, Philadelphia, 2010, Mosby.

SA node

(~90-100 depol/min)

AV node

(~40-60 depol/min)

Bundle branches

(~20-30 depol/min)

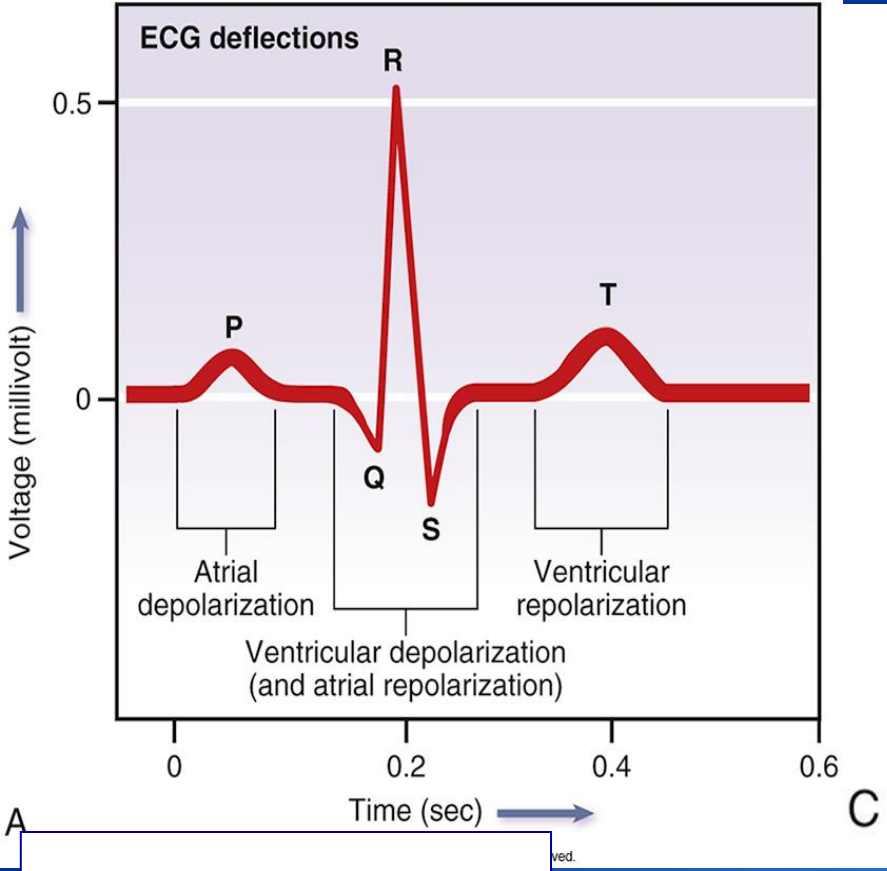
Purkinje fibers

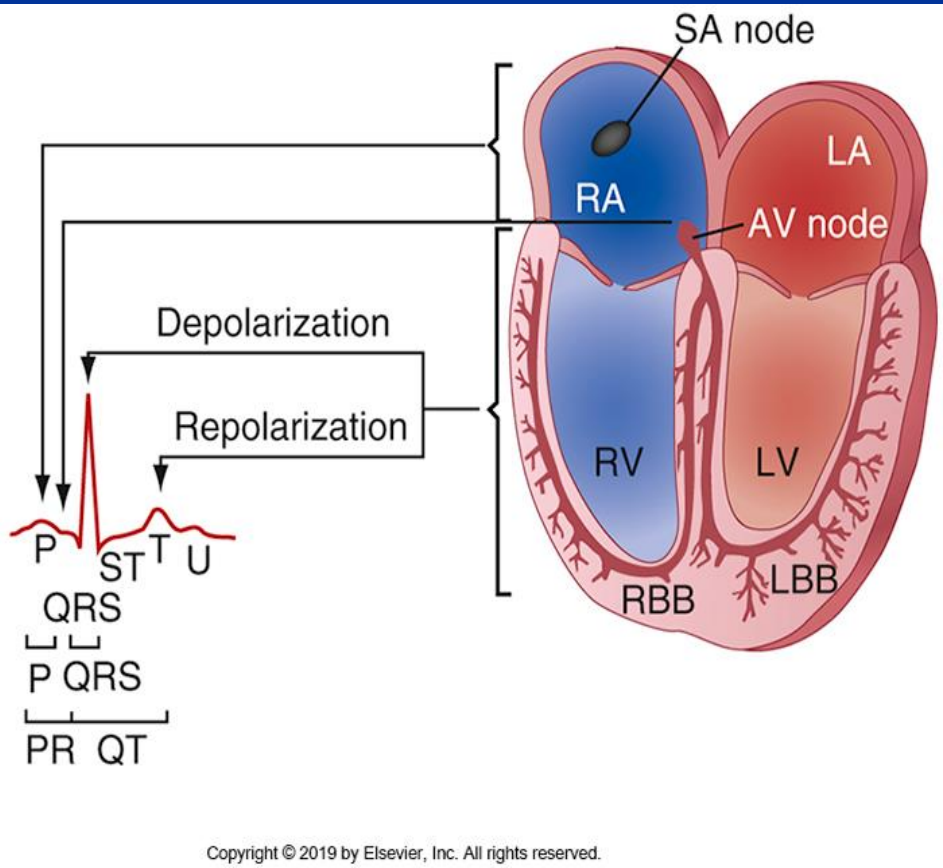
(~15-20 depol/min)



Ventricular myocytes

(~10 depol/min)





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ECG elements

- **P wave – Atrial depolarization**
- **PR interval –AV node depolarization**
- **QRS complex –Septal and ventricular wall depolarization**
- **T wave –Ventricular repolarization**

I. The BLOOD and its components
II. The CIRCULATIONS
III. HEMODYNAMICS and fundamental pressure, resistance and flow inter-relationships
IV. The HEART as a highly coordinated, well regulated two-step "twin" pump
V. The VASCULATURE as a well regulated system of conduits
VI. BLOOD PRESSURE regulation
VII. Determinates of MYOCARDIAL WORK

Read and study all of chapter 12

Independent studies:

BLOOD components

HEMOSTASIS

LIPOPROTEIN METABOLISM (Google Drive presentation)

CARDIAC OUTPUT (CO) and its regulation

1. CO = Heart rate (HR) X Stroke volume (SV)
    a) HR = number of beats (ventricular contractions) per minute (HR = beats/min)
    b) SV = volume of blood ejected from ventricle with each beat (SV = volume/beat)

2. HEART RATE (HR) and its regulation (chronotropy)

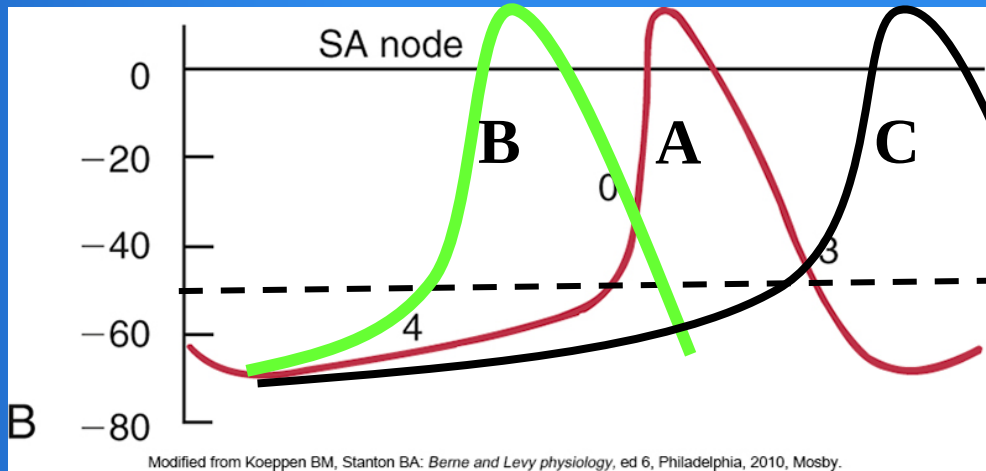
3. STROKE VOLUME and its regulation
    a) FRANK-STARLING mechanism
    b) CONTRACTILITY (ionotropy)
    c) AFTERLOAD

HEART RATE (HR) and its regulation (chronotropy) see Fig. 12.25

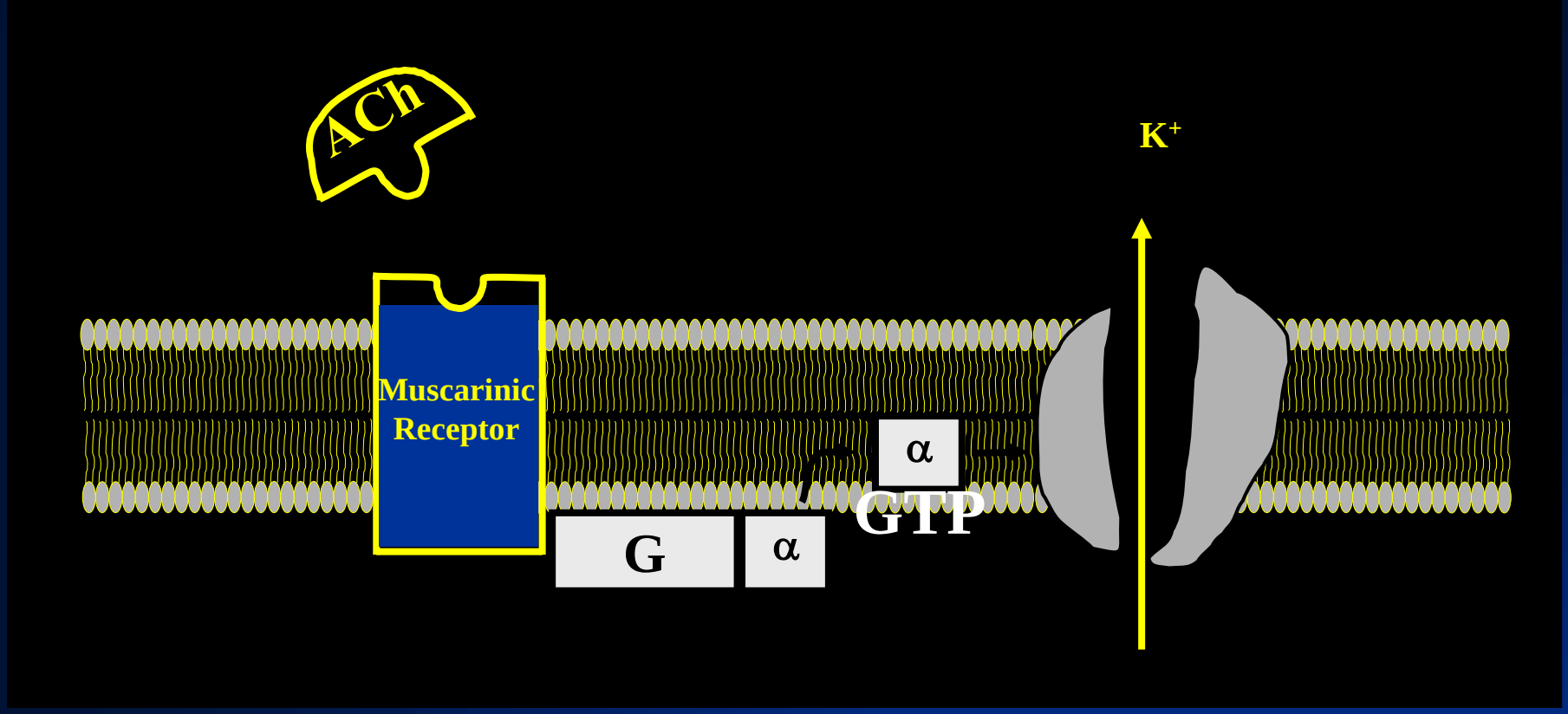
- Increasing PACEMAKER POTENTIAL depolarization rate increases HR
- Sympathetic nervous system (SNS) activation increases HR
- Via activation of SA node β1-adrenergic receptors by norepinephrine (SNS innervation) and epinephrine (released from adrenal medulla into blood stream)
- Decreasing Parasymp NS activation increases HR

- Decreasing PACEMAKER POTENTIAL depolarization rate decreases HR
- Parasympathetic nervous system (PNS) activation decreases HR via activation of SA node muscarinic receptors by acetylcholine (vagal innervation)
- Muscarinic receptor activation at the SA node initiates a G-protein-mediated signal transduction process that opens low capacity K+ channels leading to a slight K+ efflux and a decreased rate of the SA node pacemaker potential
- Decreasing Symp NS activation decreases HR

- When the membrane spontaneously depolarizes to 50 mV Action Pot begins
- A = Normal
- B = Sym stim & increased phase 4 pacemaker depolarization rate; or decreased Parasym
- C = Parasym stim & decreased phase 4 pacemaker depolarization rate; or decreased Symp

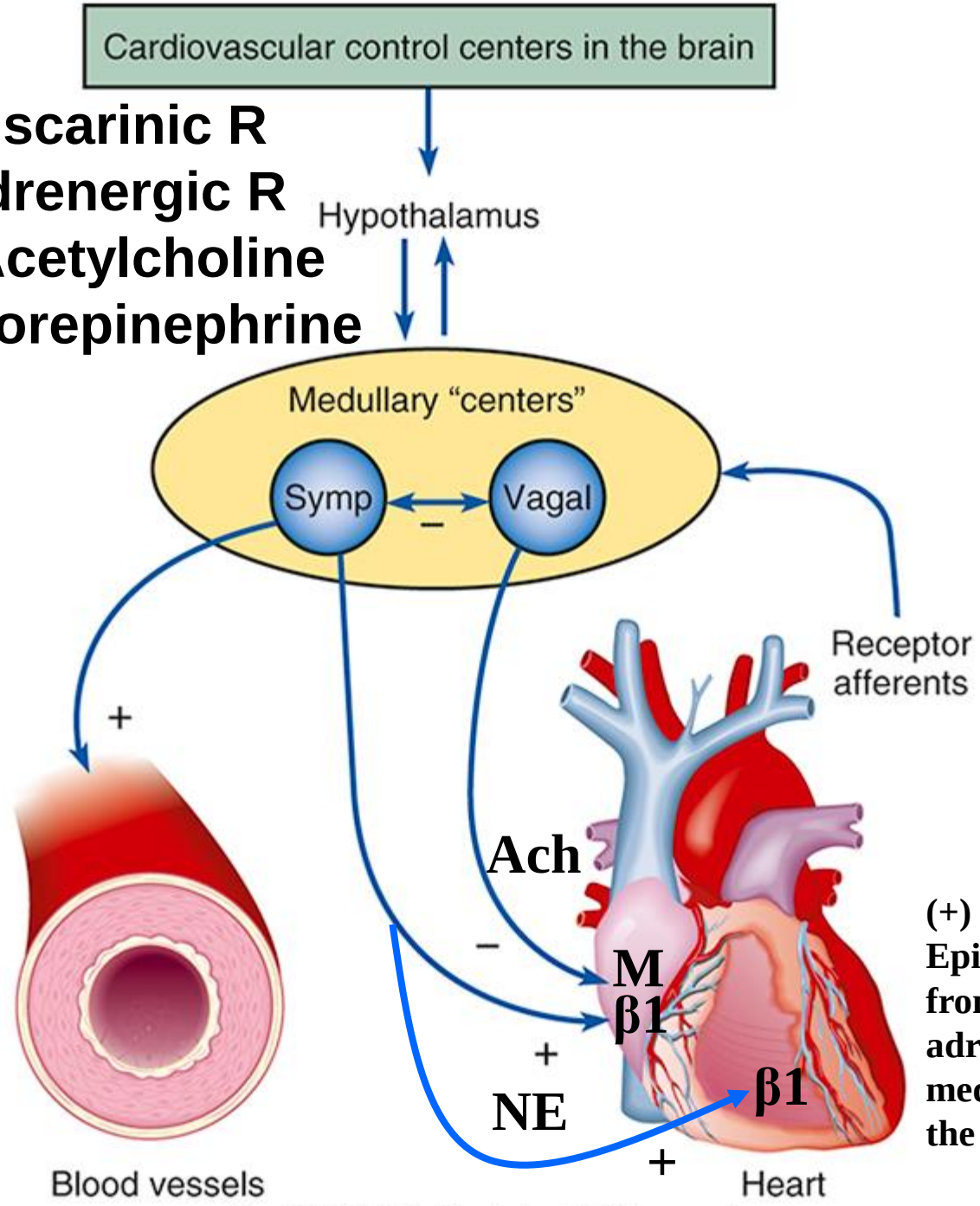


Modified from Koeppen BM, Stanton BA: *Berne and Levy physiology*, ed 6, Philadelphia, 2010, Mosby.



SIGNAL TRANSDUCTION VIA THE MUSCARINIC RECEPTOR IN THE HEART

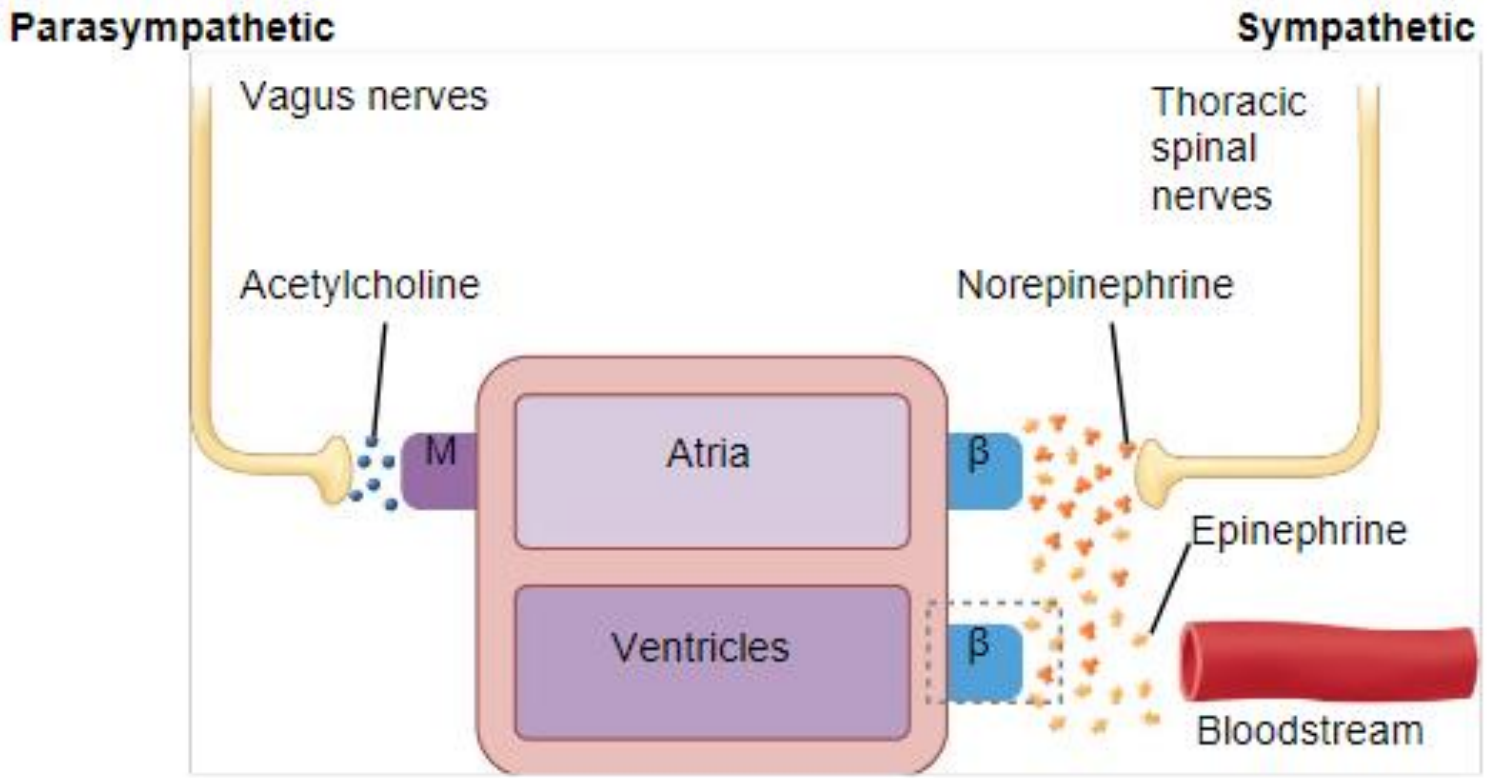
M = Muscarinic R
β1 = Adrenergic R
Ach =Acetylcholine
NE = Norepinephrine



(+) β1 Epinephrine from the adrenal medulla via the blood

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Fig 12.12


Parasympathetic
Sympathetic
Vagus nerves
Thoracic spinal nerves
Acetylcholine
Norepinephrine
M
Atria
β
Epinephrine
Ventricles
β
Bloodstream

CARDIAC OUTPUT (CO) and its regulation

1. CO = Heart rate (HR) X Stroke volume (SV)
    a) HR = number of beats (ventricular contractions) per minute (HR = beats/min)
    b) SV = volume of blood ejected from ventricle with each beat (SV = volume/beat)

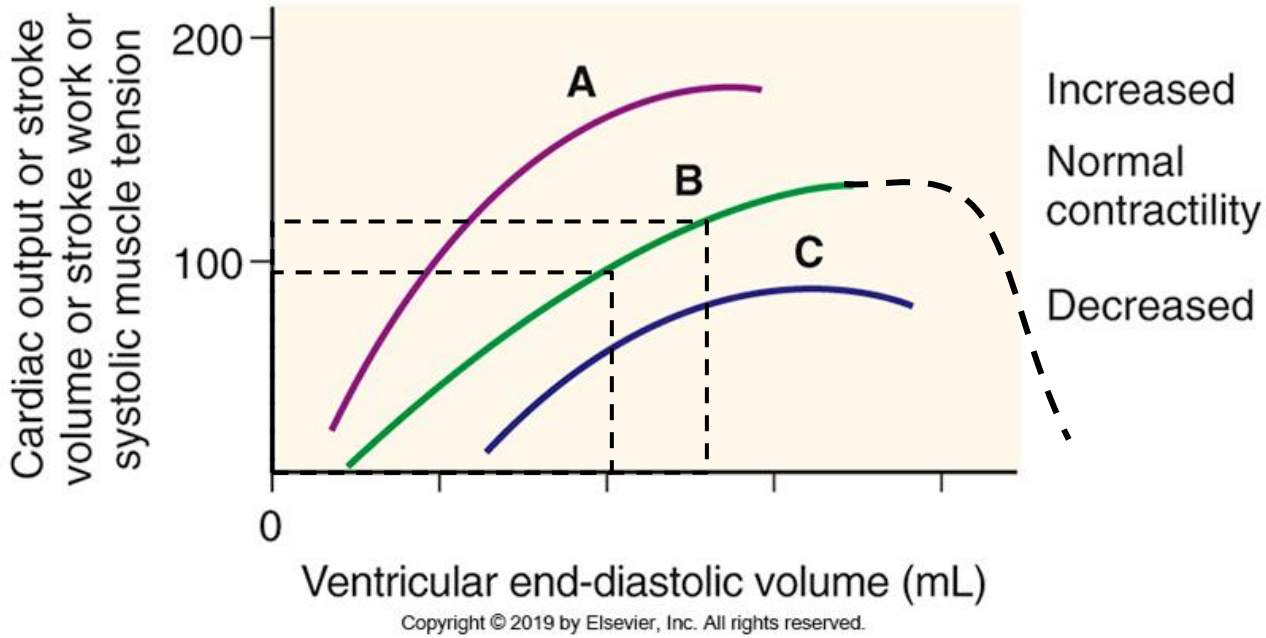
2. HEART RATE (HR) and its regulation (chronotropy)

3. STROKE VOLUME and its regulation
    a) FRANK-STARLING mechanism
    b) CONTRACTILITY (ionotropy)
    c) AFTERLOAD

STROKE VOLUME and its regulation

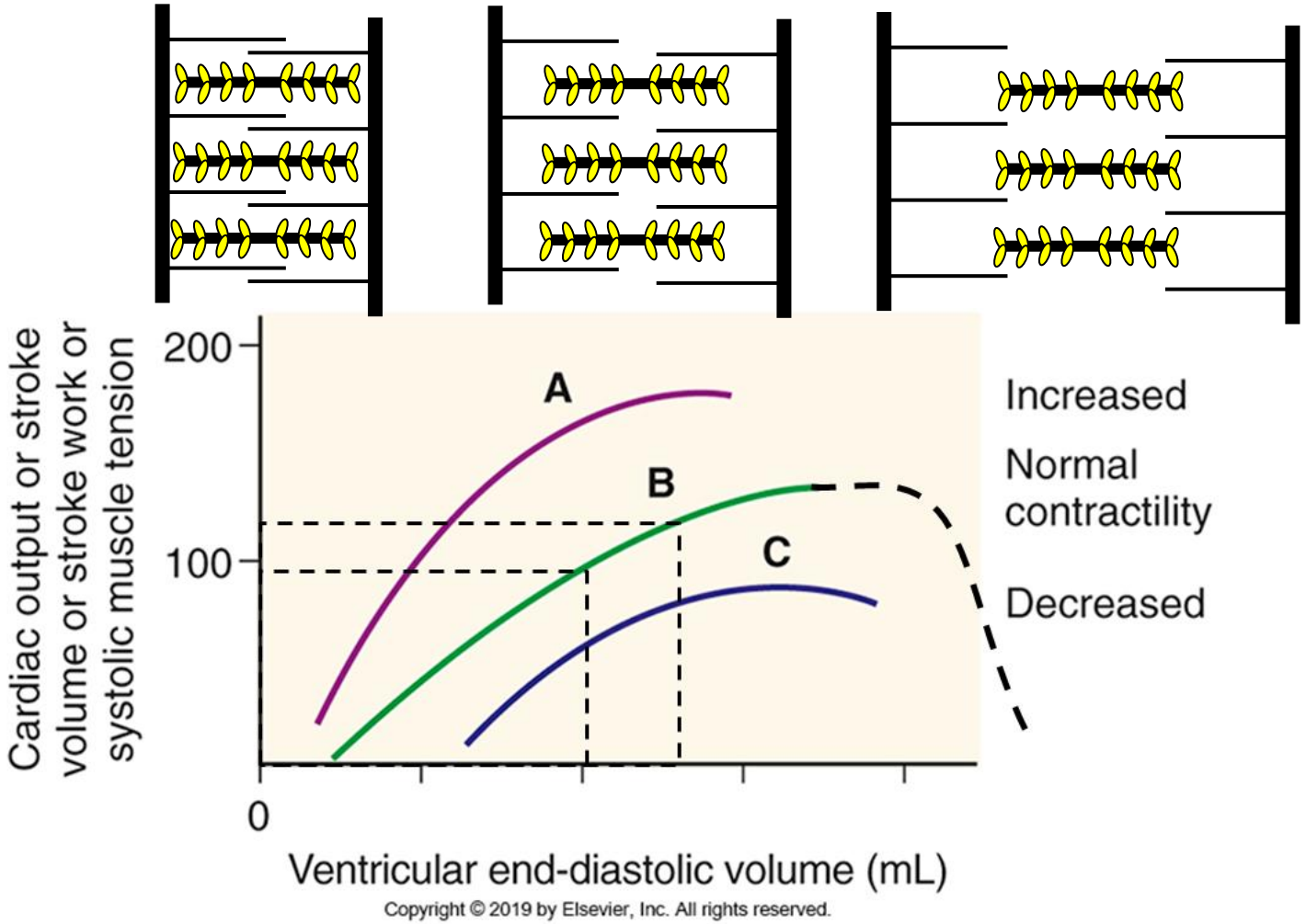
1. FRANK-STARLING mechanism

- Up to a point, as the ventricle fills with more blood, the SV will increase (figure 12.28)
- The volume of blood filling the ventricle just prior to contraction is the end-diastolic volume (EDV) or PRELOAD
- Mechanism is due to:
  - Better thick and thin filament overlap
  - Increased troponin sensitivity to Ca++
  - Increased Ca++ release from sarcoplasmic reticulum



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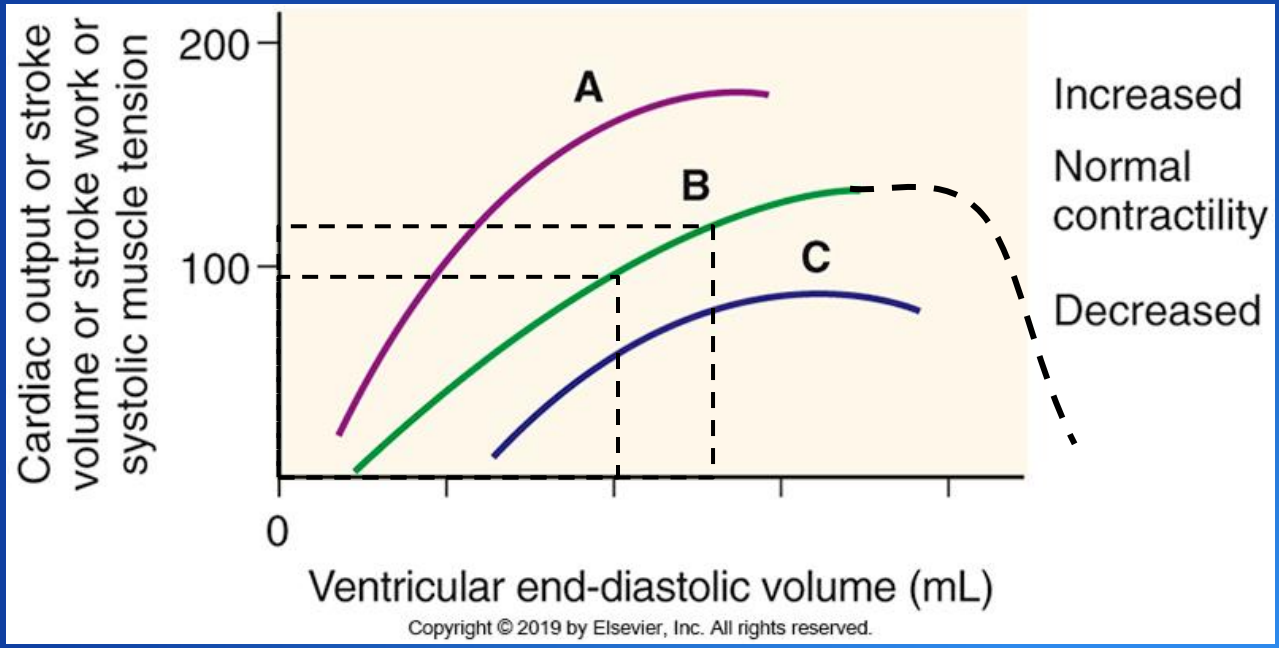
- **Up to a point, increased ventricular filling leads to better cross-bridge overlap; more cross-bridges allow for increased tension development and increased stroke volume**
- **Over-filling the ventricle, which happens in certain types of heart failure, decreases cross-bridge over lap and stroke volume; called decompensation)**



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STROKE VOLUME and its regulation

1. FRANK-STARLING mechanism



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Examples of what affects ventricular EDV

↑ Venoconstriction (↓ Venous compliance)

↑ Skeletal mms contraction pump

↑ Respiratory pump

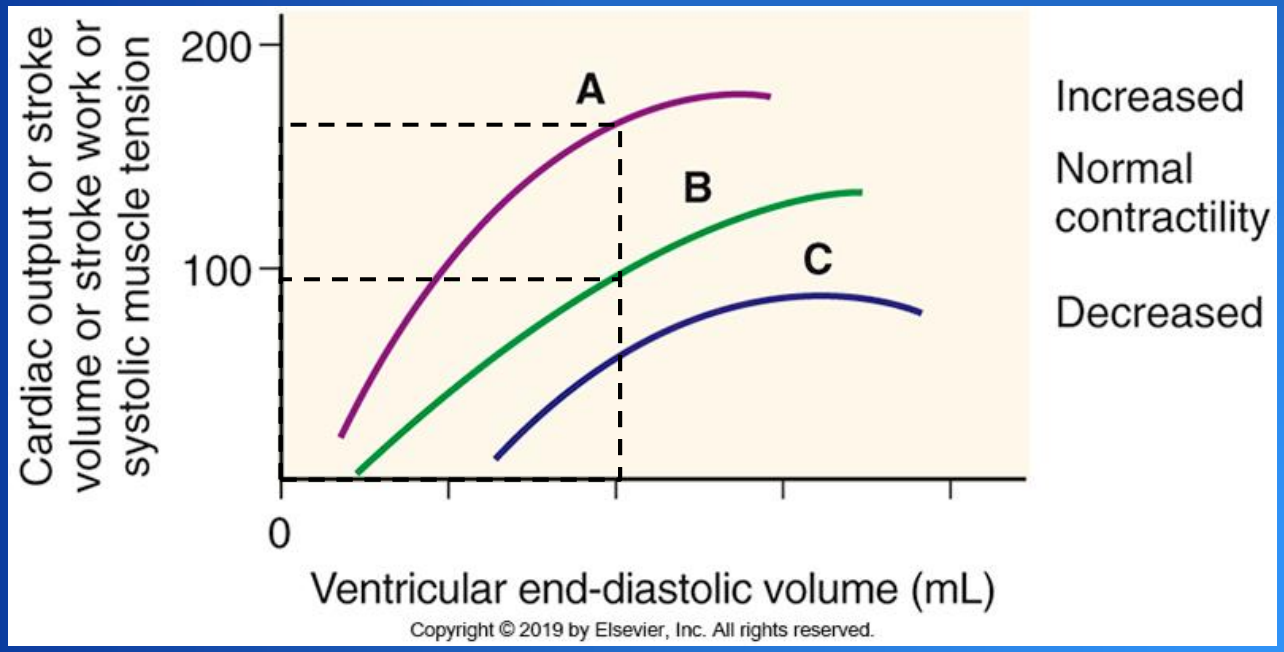
↑ Blood or plasma volume

→ ↑ Venous return → ↑ Ventricular filling → ↑ EDV → ↑ SV

STROKE VOLUME and its regulation

2. CONTRACTILITY (ionotropy)

- **An increase in the strength of contraction for any end-diastolic volume (EDV) (figure 12.28) (Note that rates of force development AND relaxation both increase)**



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STROKE VOLUME and its regulation

2. CONTRACTILITY (ionotropy)

- An increase in the strength of contraction for any end-diastolic volume (EDV) (figure 12.28)
- Increases in contractility increase EJECTION FRACTION
- Ejection fraction (EF) is the ratio of stoke volume (SV) to EDV

Normal		Increased Contractility	
EDV	135 ml	EDV	140 ml
ESV	65 ml	ESV	20 ml
SV	70 ml	SV	120 ml
EF = SV/EDV		EF = SV/EDV	
EF = 70/135 = 52%		EF = 120/140 = 86%	

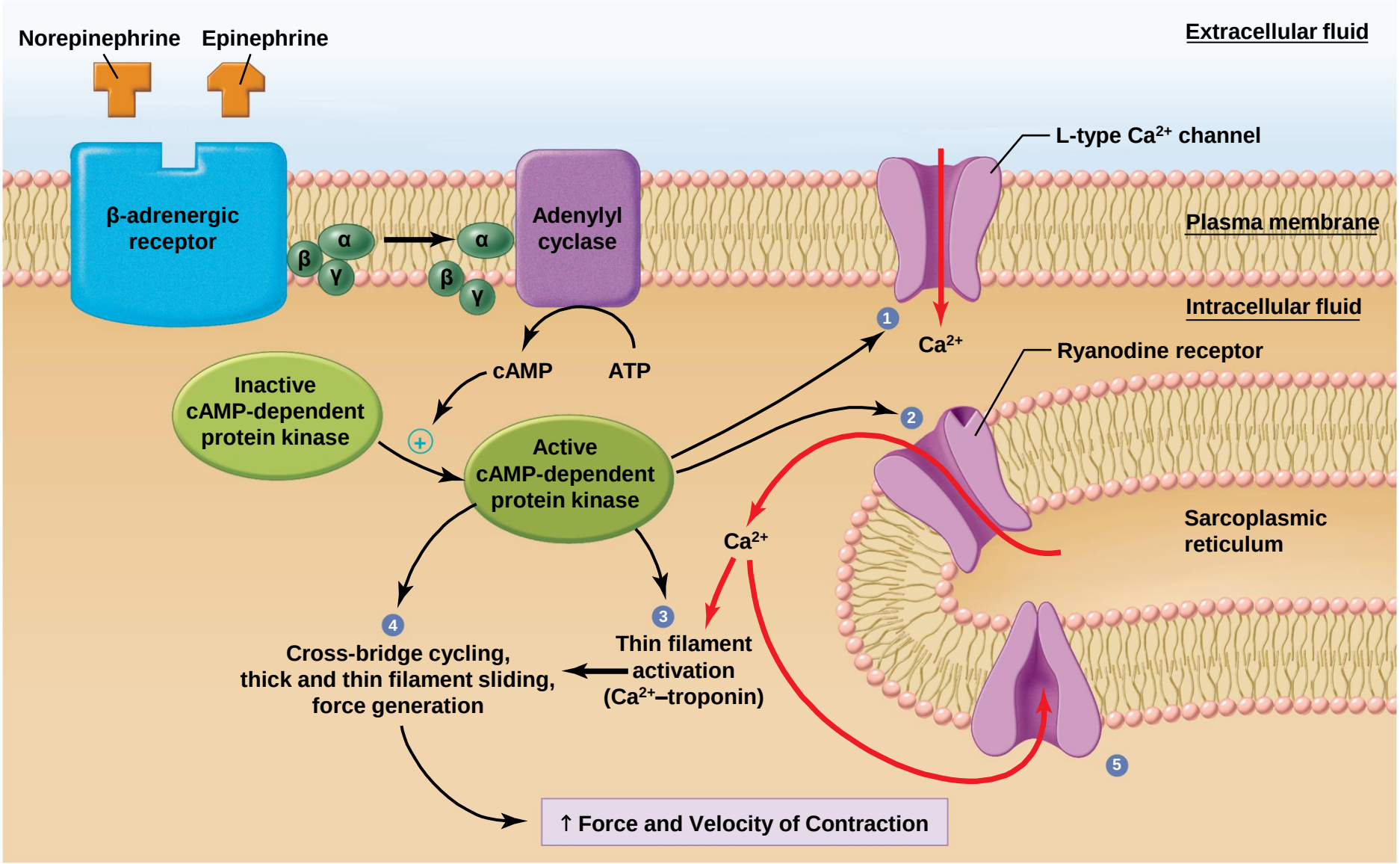
Normal resting EJ% ~ 50-70%

- Sympathetic activation of β1-adrenergic receptors on ventricular myocytes increase contractility by actions involving calcium at
  - Plasma membrane L-type Ca++ channels
  - Sarcoplasmic reticulum ryanodine receptor
  - Troponin
  - Myosin cross-bridges
  - Sarcoplasmic reticulum Ca++-ATPase pumps (speeds relaxations)

- Increased contractility strengthens and increases the velocity of ventricular contractions and increases the velocity of ventricular relaxations

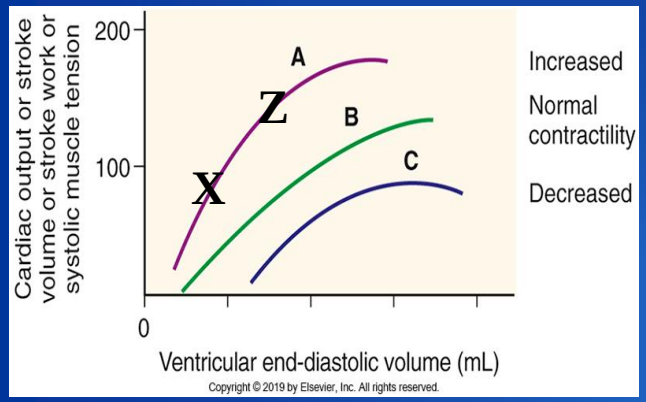
Fig. 12.29

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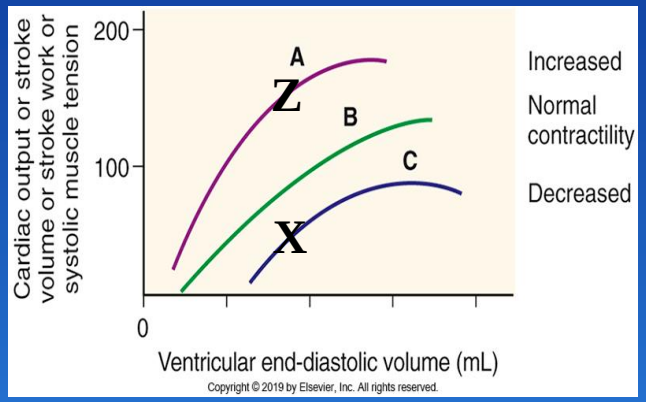


STROKE VOLUME and its regulation

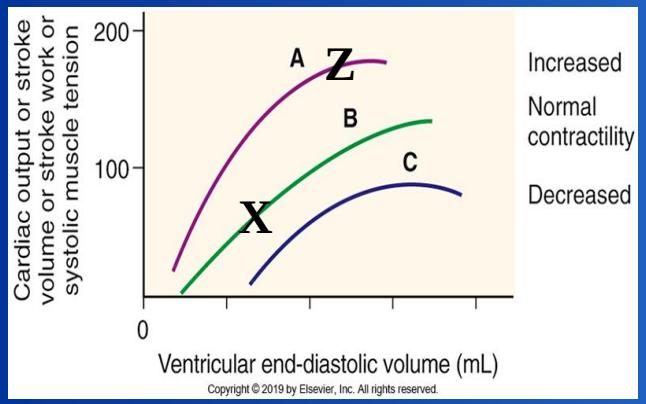
EXAMPLES



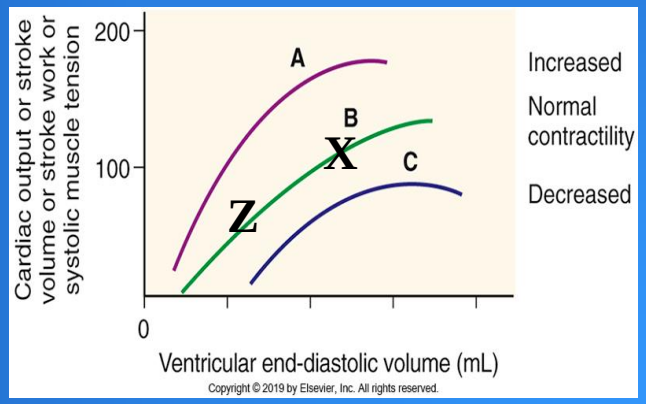
Moving from X to Z along line A is due to F-S mechanism



Moving from X to Z while EDV is unchanged is due to contractility



Moving from X to Z is due to both increased EDV (F-S mech) and increased contractility



Moving from X to Z is due to decreased ventricular filling (F-S mechanism)

CARDIAC OUTPUT (CO) and its regulation

1. CO = Heart rate (HR) X Stroke volume (SV)
    a) HR = number of beats (ventricular contractions) per minute (HR = beats/min)
    b) SV = volume of blood ejected from ventricle with each beat (SV = volume/beat)

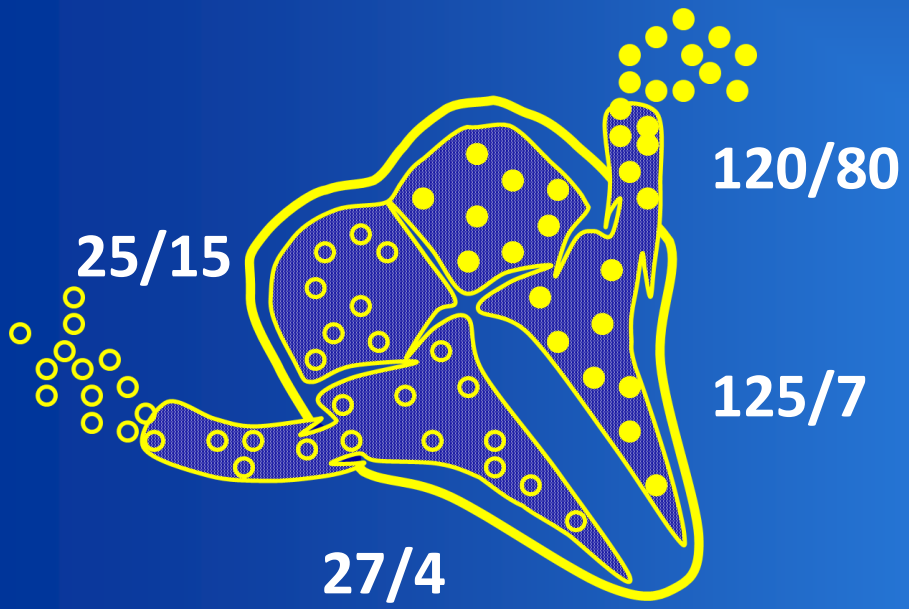
2. HEART RATE (HR) and its regulation (chronotropy)

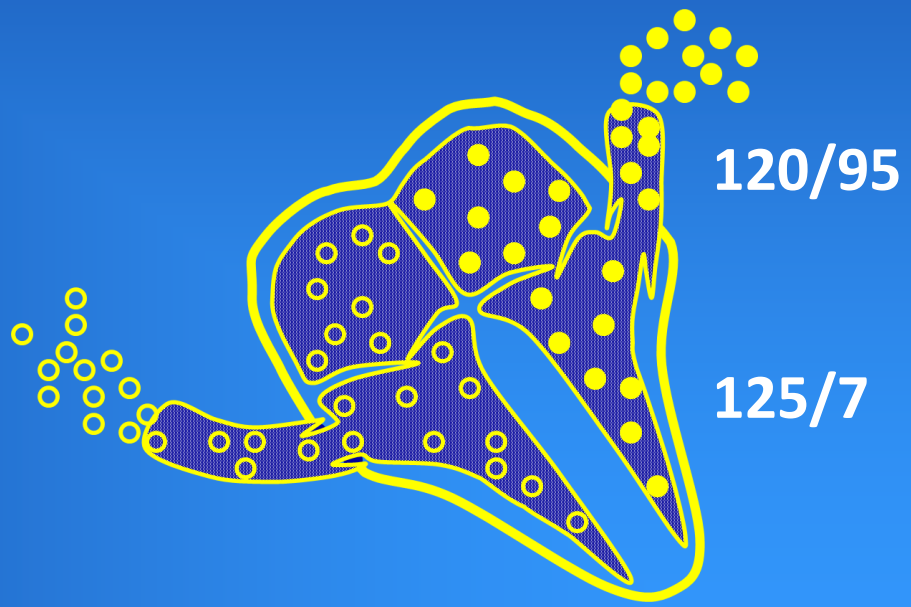
3. STROKE VOLUME and its regulation
    a) FRANK-STARLING mechanism
    b) CONTRACTILITY (ionotropy)
    c) AFTERLOAD

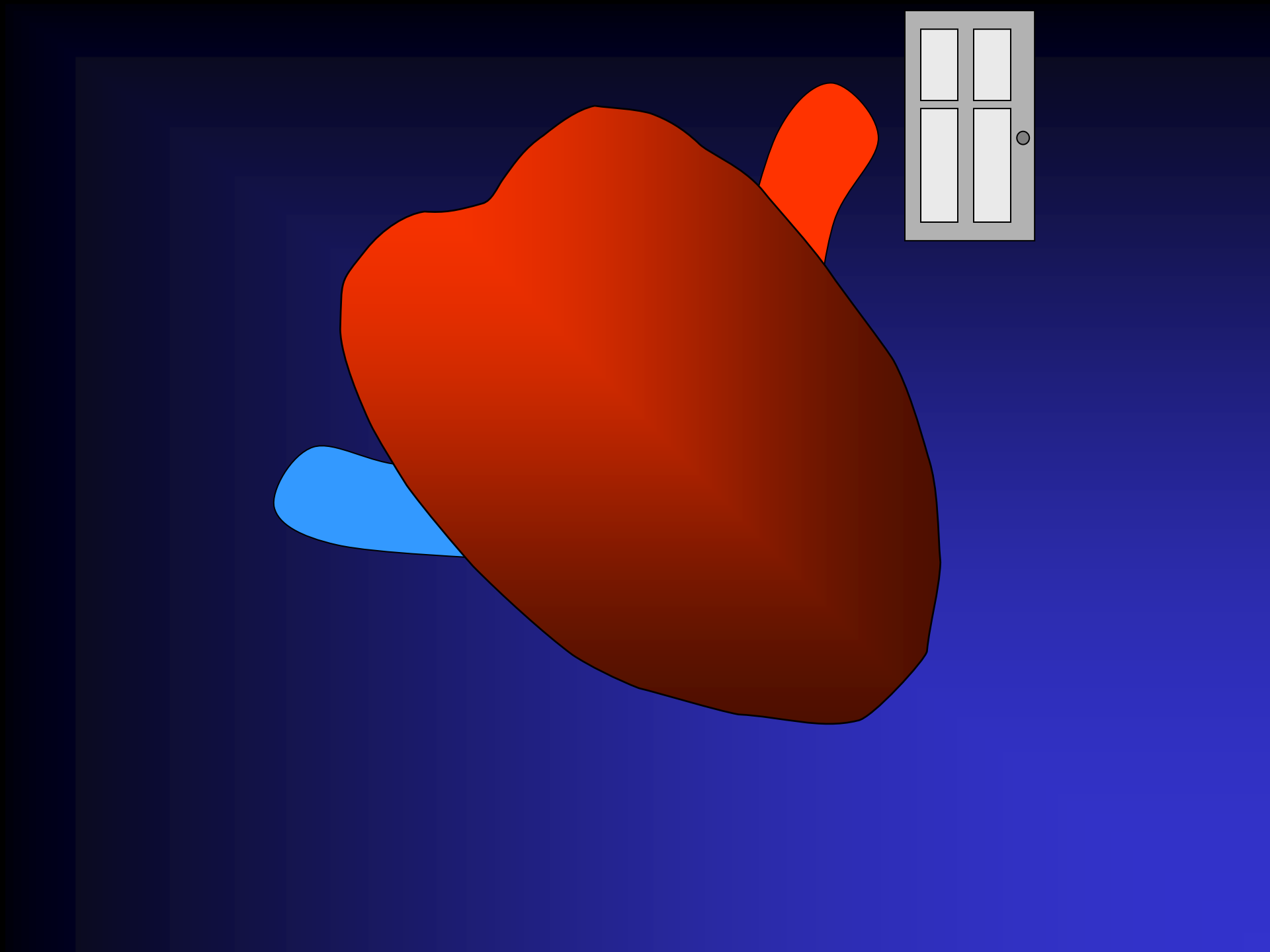
STROKE VOLUME and its regulation

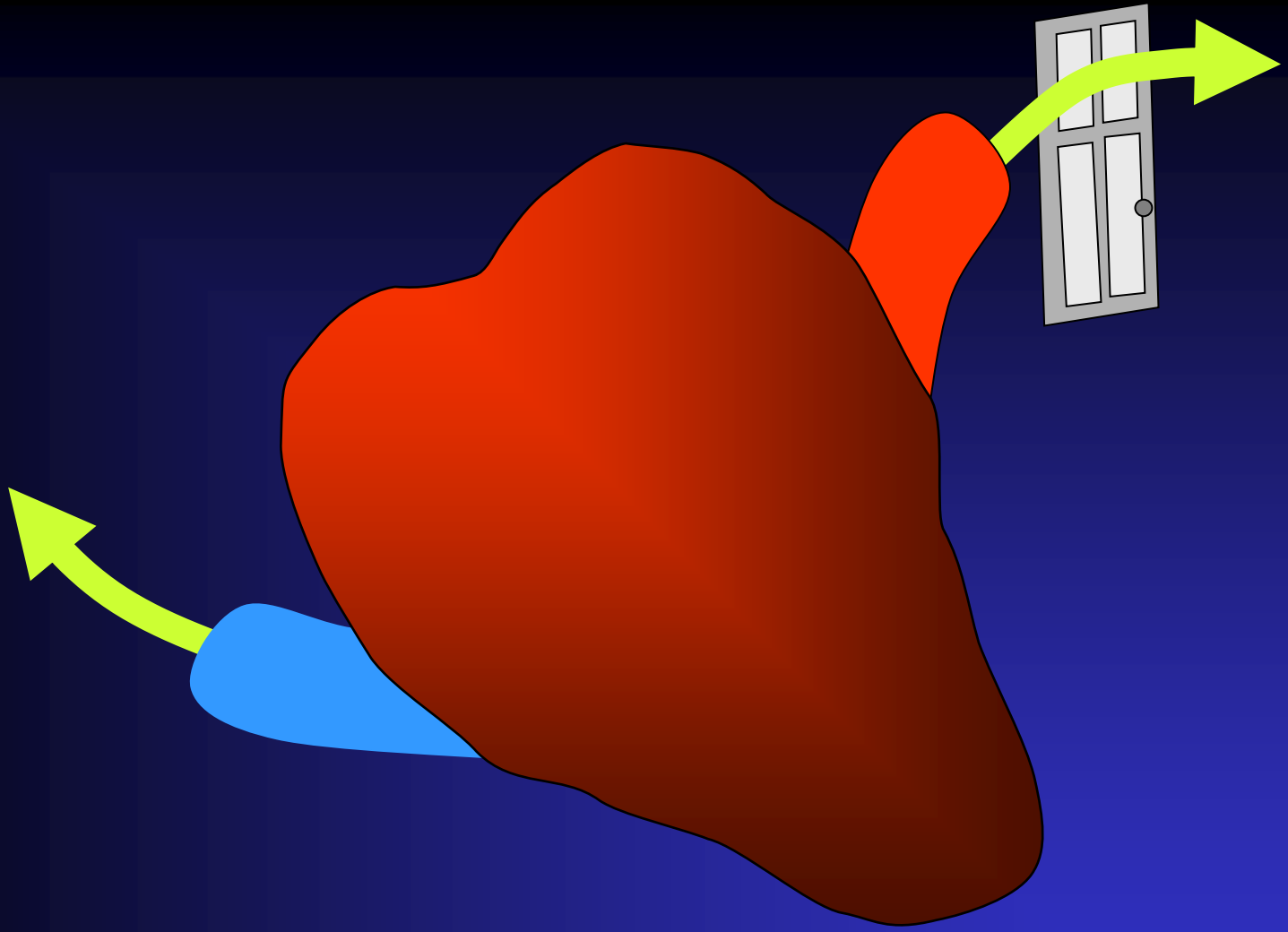
c. AFTERLOAD

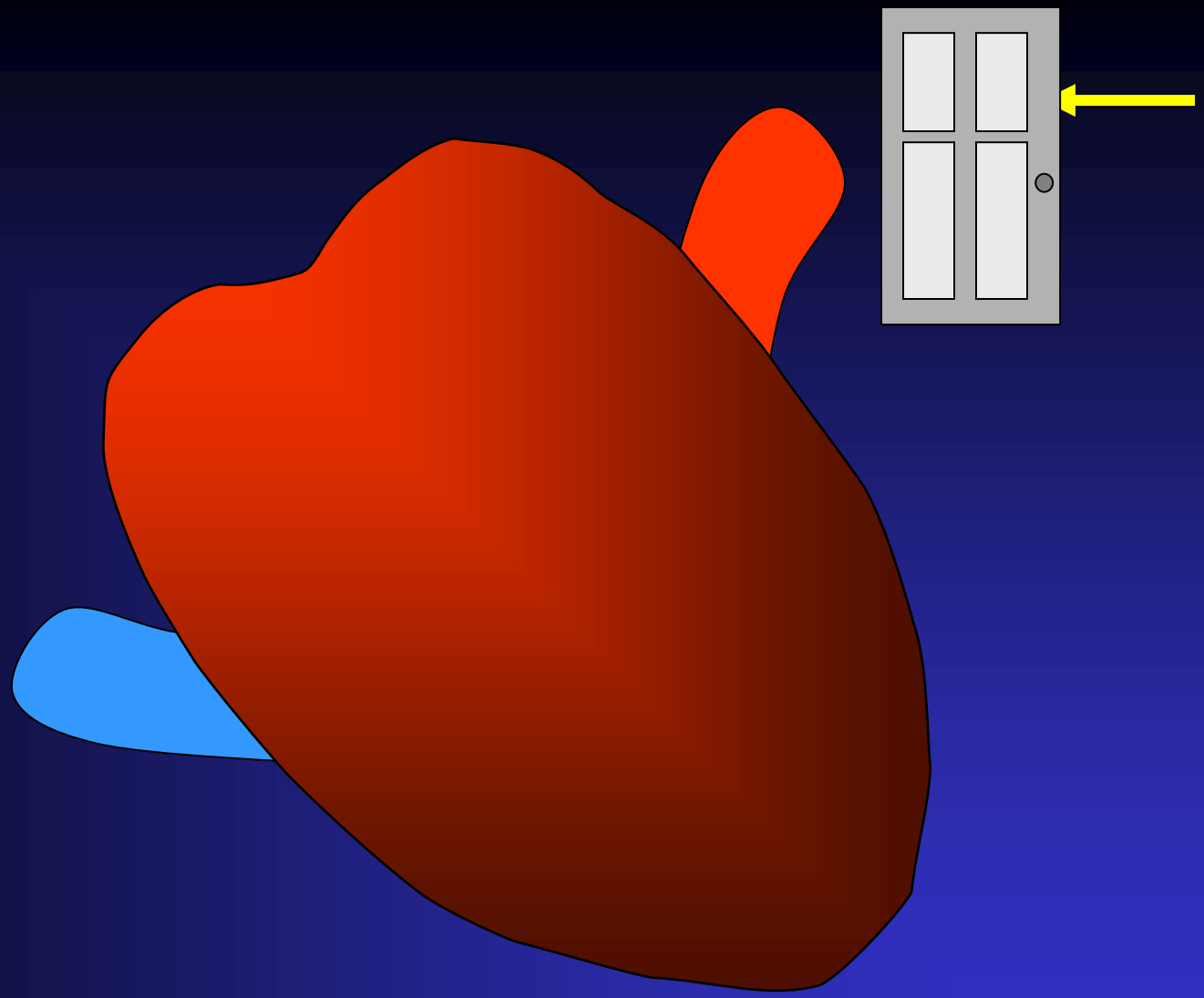
- Afterload is aortic and pulmonic blood pressure and represents the load the ventricles experience after initiating contraction to eject blood
- More energy used to overcome the afterload or blood pressure, the less energy available for blood ejection ***
- Increased afterload decreases ejection volume (decreased stroke volume)

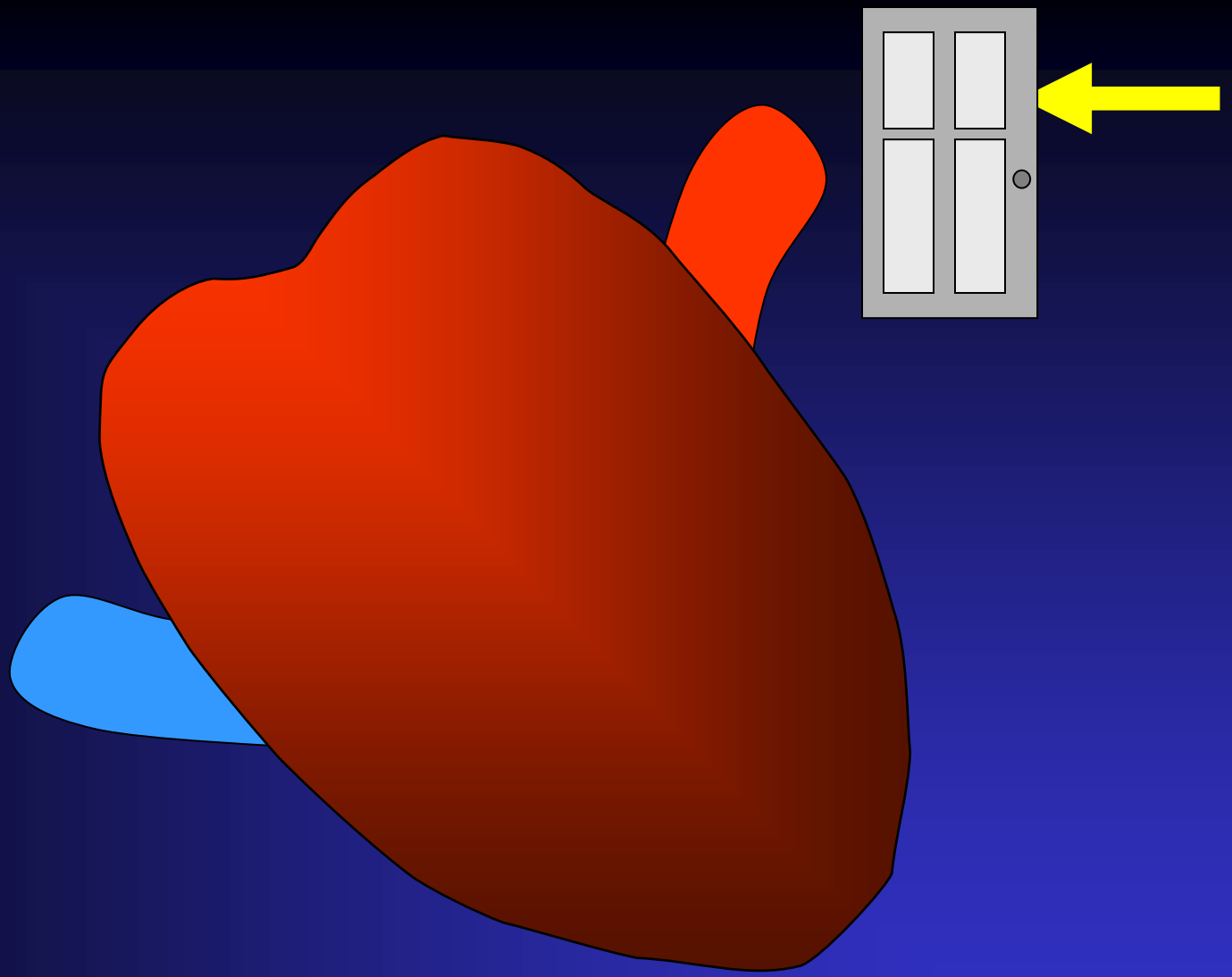


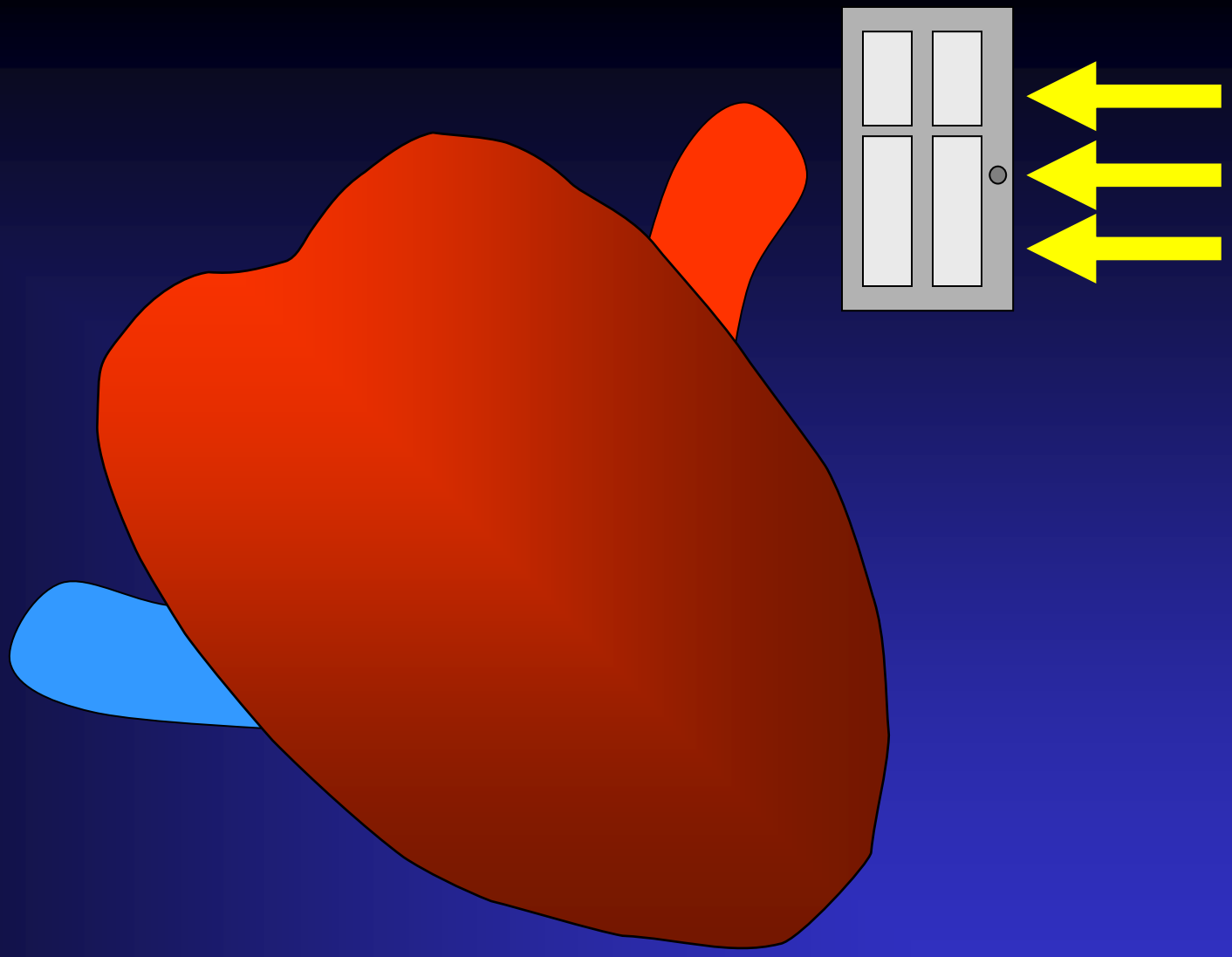




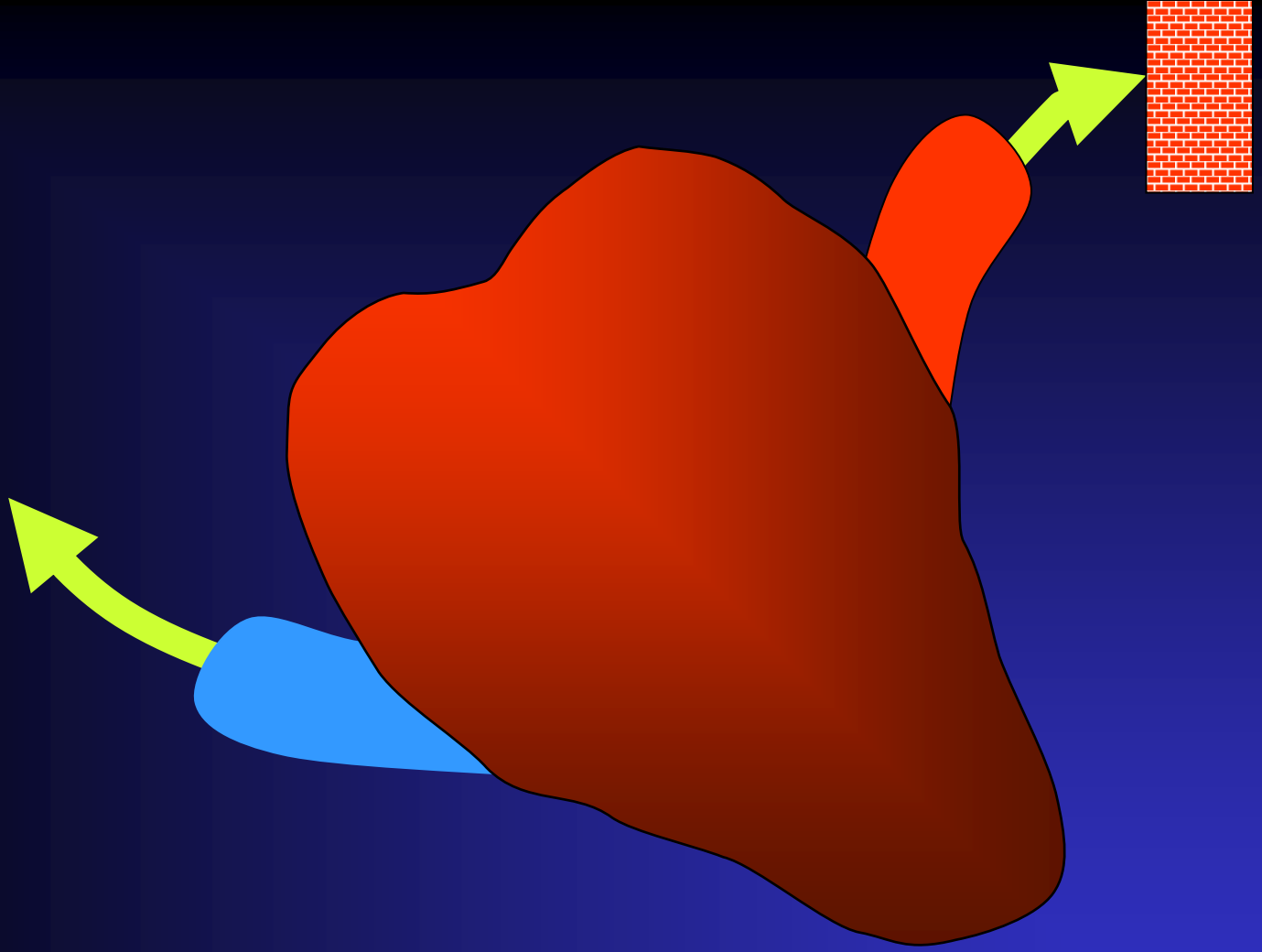








HYPERTENSION is an abnormally high AFTER-LOAD (arterial blood pressure) that greatly increases myocardial work and O2 demand



HYPERTENSION is an abnormally high AFTER-LOAD (arterial blood pressure) that greatly increases myocardial work and O2 demand

CARDIAC OUTPUT (CO) and its regulation

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2. HEART RATE (HR) and its regulation (chronotropy)

3. STROKE VOLUME and its regulation
    a) FRANK-STARLING mechanism
    b) CONTRACTILITY
    c) AFTERLOAD

I. The BLOOD and its components
II. The CIRCULATIONS
III. HEMODYNAMICS and fundamental pressure, resistance and flow inter-relationships
IV. The HEART as a highly coordinated, well regulated two-step "twin" pump
V. The VASCULATURE as a well regulated system of conduits
VI. BLOOD PRESSURE regulation
VII. Determinates of MYOCARDIAL WORK

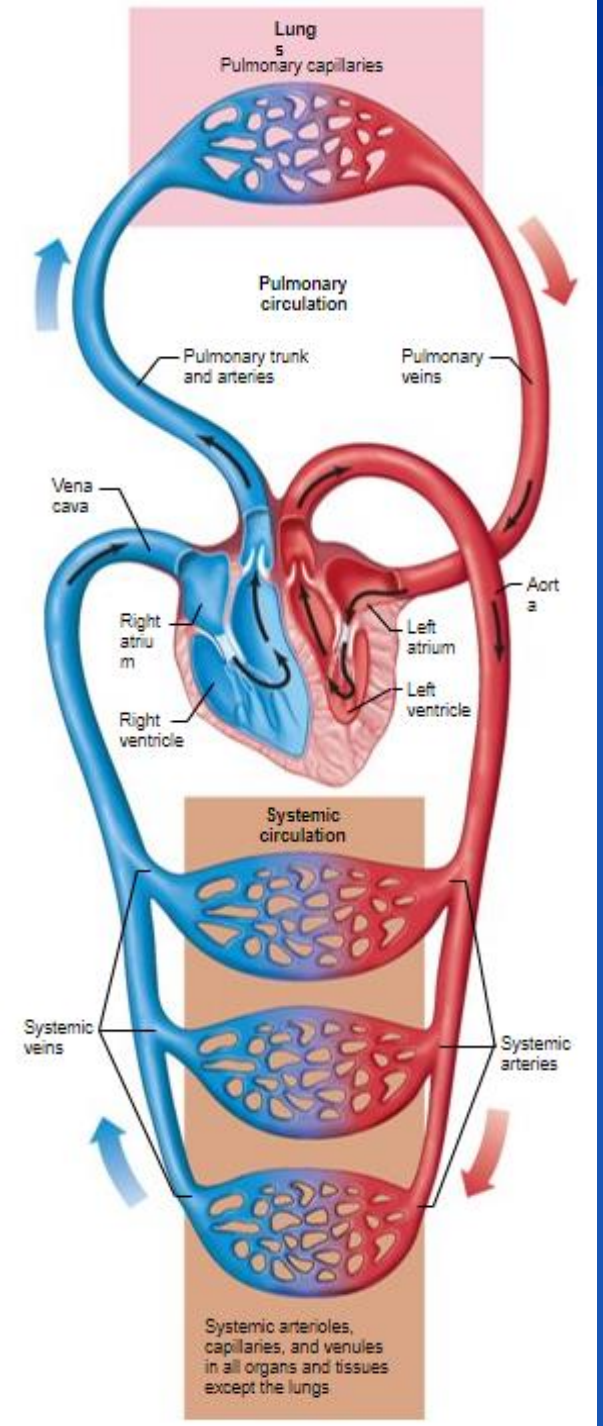
Read and study all of chapter 12

Independent studies:

BLOOD components

HEMOSTASIS

LIPOPROTEIN METABOLISM (Google Drive presentation)



The VASCULATURE as a well-regulated system of conduits

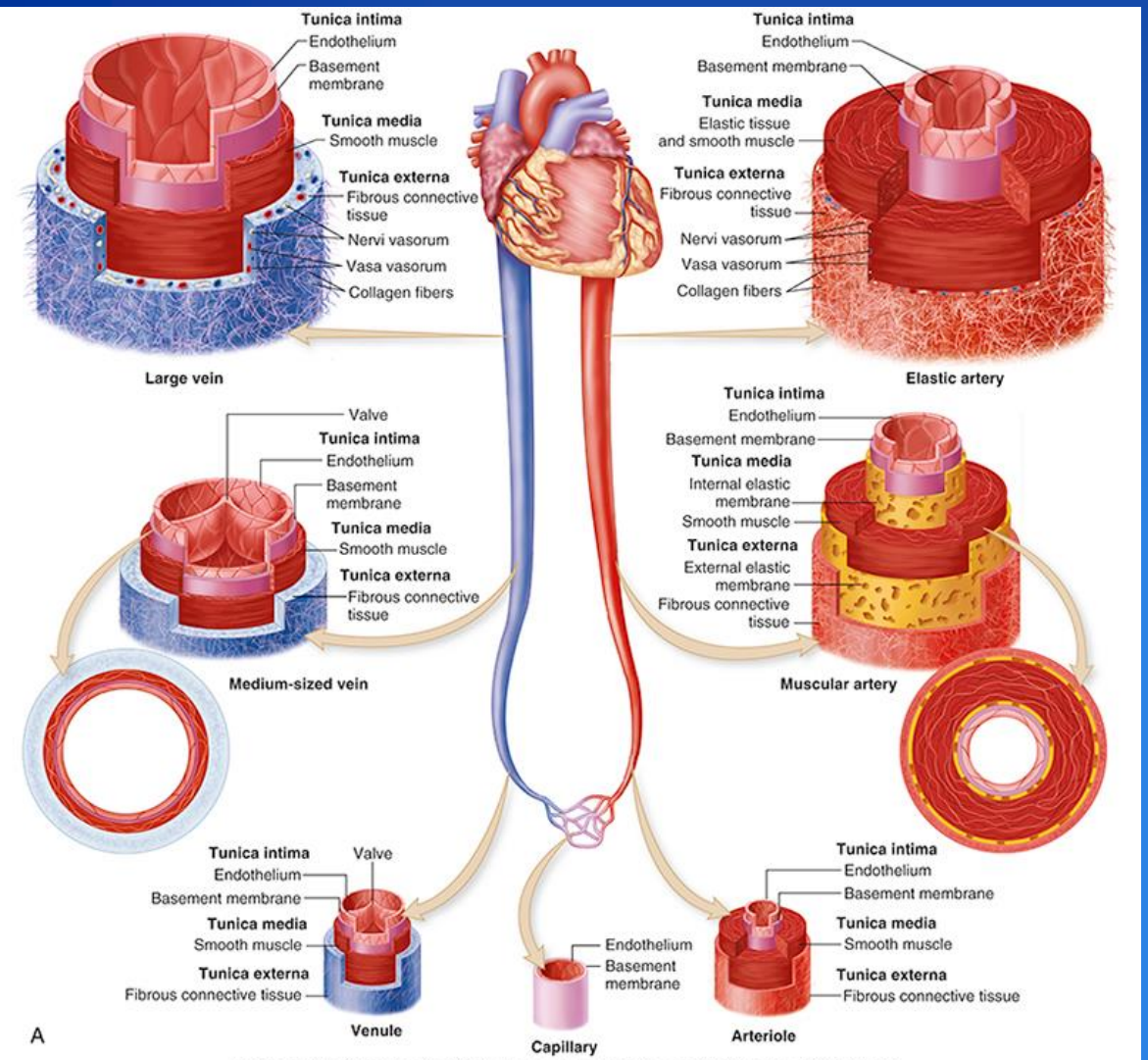
- Overall structural characteristics of the vascular system (study figure 12.31)
- Structural characteristics lead to the following (simplistic but largely accurate) statement:

  "Arteries store blood pressure while veins store volume."
- Blood flows down its pressure gradient, from an area of high pressure (e.g., left ventricle) to an area of low pressure (e.g., right atrium)
- Arterioles are central to the distribution of blood flow
- Regulation of arteriole vasoconstriction and vasodilation determines blood flow to an area

Figure 12.5

The VASCULATURE as a well-regulated system of conduits

- Overall structural characteristics of the vascular system (study figure 12.31)
- Structural characteristics lead to the follow (simplistic but largely accurate) statement:
  "Arteries store blood pressure while veins store volume."



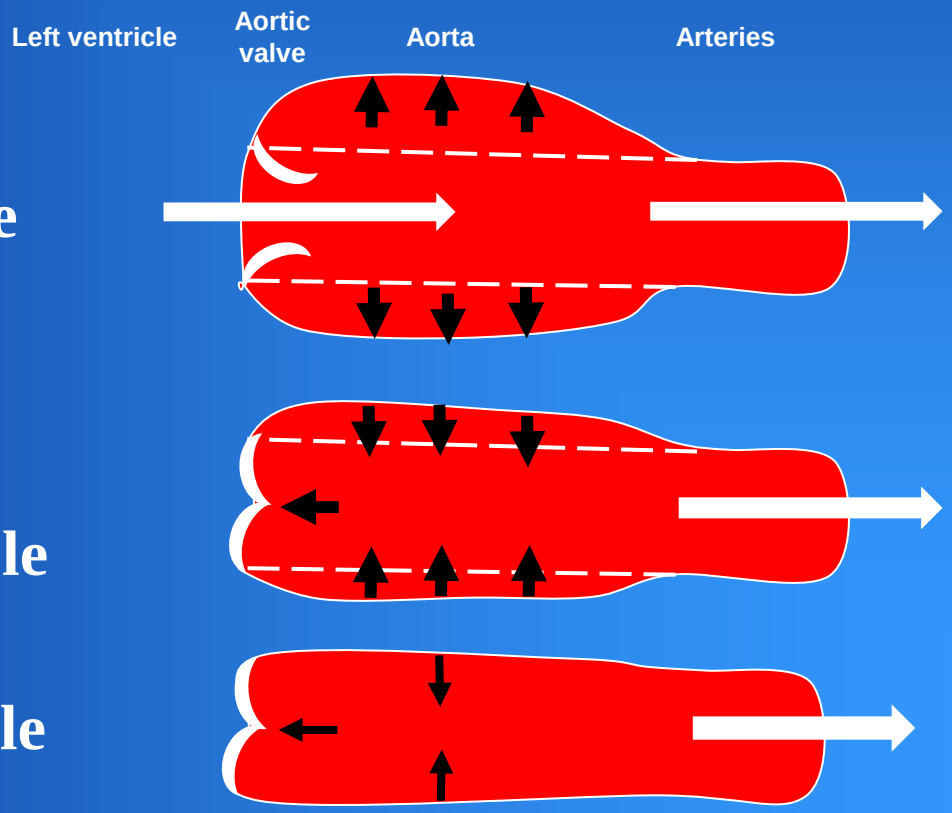
From Patton KT, Thibodeau GA: *Anatomy & physiology*, ed 9, St Louis, 2016, Mosby.

- Different areas of the vasculature have different compliances; COMPLIANCE pertains to yielding to pressure
- Compliance = Δ Volume/Δ Pressure = ΔV/ΔP
- Example: Blowing lightly (i.e., low pressure) into a very thin walled balloon results in a large change in balloon volume; the thin walled balloon is highly compliant. A thick walled balloon requires you to blood much harder ((i.e., higher pressure) to achieve the same balloon volume; the thick walled balloon has low compliance.

The VASCULATURE as a well-regulated system of conduits

ARTERIES

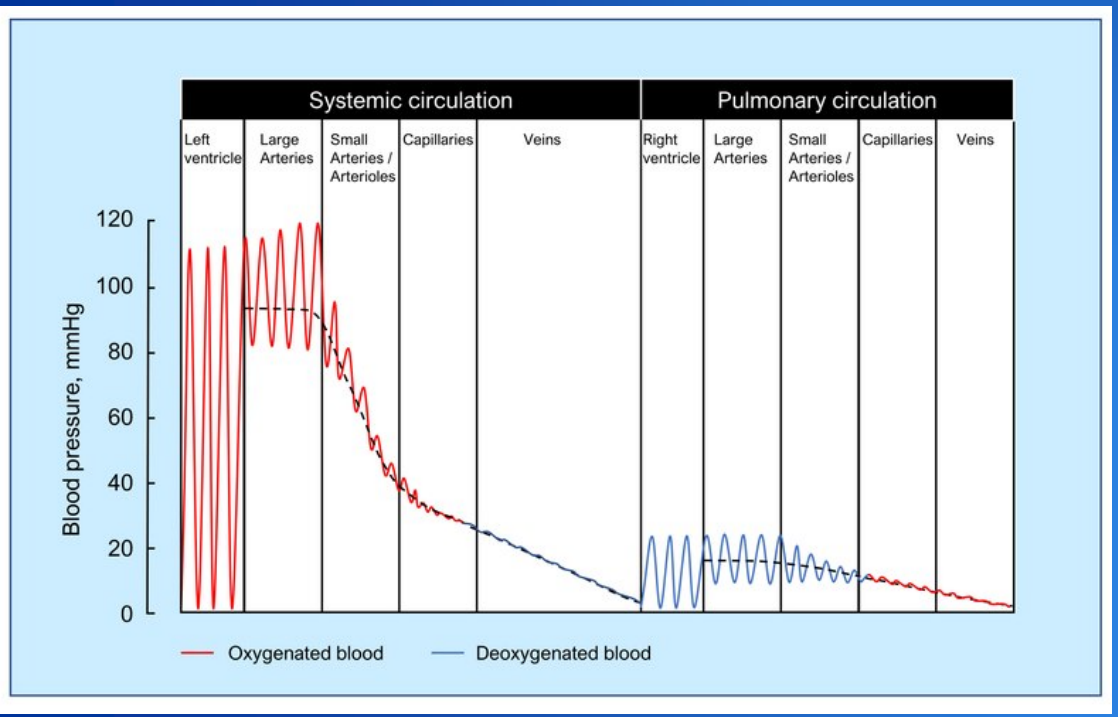
- Elastic tubes with relatively large radii (therefore, low resistance)
- Act as "pressure reservoirs" (they store pressure)
- As blood is ejected from the left ventricle during ventricular systole, the elastic aortic walls are distended outward while some blood moves down the aorta. During ventricular diastole, the distended elastic walls of the aorta recoil inward, keeping the blood pressurized. A closed aortic valve ensures that blood keeps moving down its pressure gradient down the vasculature tree during diastole.

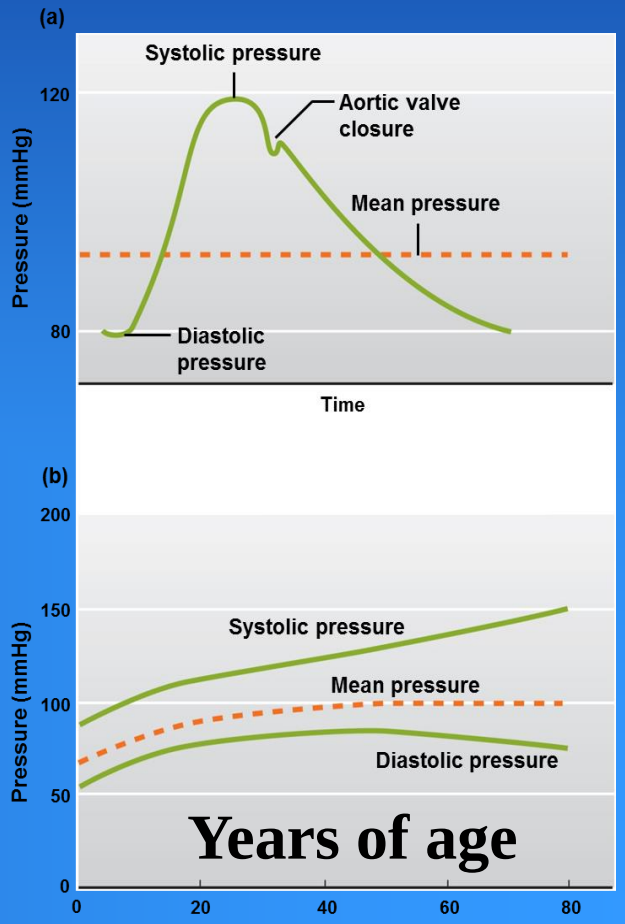


The VASCULATURE as a well-regulated system of conduits

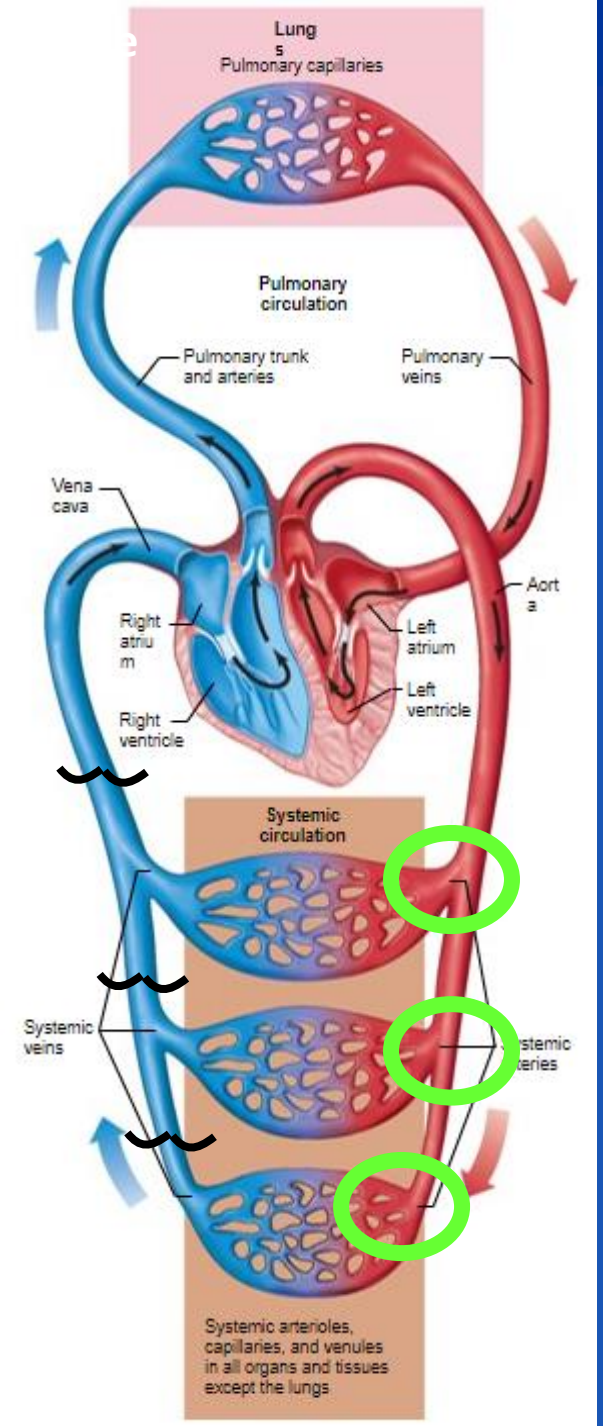
Blood pressures in the vascular tree

- SYSTOLIC PRESSURE (SP) – maximum arterial pressure occurring during ventricular systole
- DIASTOLE PRESSURE (DP) – minimum arterial pressure occurring at the end of ventricular diastole
- PULSE PRESSURE(PP) – the difference between SP and DP
- PP = SP –DP A person with a systolic blood pressure of 120 mmHg and a diastolic pressure of 80 mmHg has a pulse pressure of 40 mmHg
  - ARTERIOSCLEROSIS is the progressive loss of arterial compliance (arteries are becoming stiffer, less elastic with age or disease progression
  - Pulse pressure is a diagnostic indictor of arteriosclerosis





https://commons.wikimedia.org/wiki/File:Circulation_pressures_v1.tif



VASCULATURE as a well-regulated system of conduits

ARTERIOLES AND DISTRIBUTION OF BLOOD FLOW

1. Distribution of blood flow to different organs
2. Arteriole function is primary determinate of mean arterial blood pressure
3. The radius of a tube is the primary determinate of resistance and flow through a tube
4. Regulation of arteriole radius to an organ, is the primary determinate for resistance to flow to an organ
5. Because $R \alpha 1 / r^4$, vasoconstriction increases resistance and decreases flow to an organ. Vasodilation does the opposite.

The VASCULATURE as a well-regulated system of conduits

- **ARTERIOLES**

  Regulation of arteriole resistance

  a) Whether resistance is due to local or extrinsic controls, in almost all cases the controls are regulating resistance by impacting arteriole smooth muscle cytosolic calcium or the effects of calcium; signals that increase arterial smooth muscle calcium or it effects increase contraction and vasoconstriction, and vice versa.

  b) Many medications regulate arteriole resistance by directly regulating smooth muscle cytosolic calcium or by activating or blocking the control described below.

  c) LOCAL ARTERIOLE CONTROLS

  (1) Local controls almost always predominate over extrinsic (neural and hormonal) controls

  (2) METABOLIC PRODUCTS

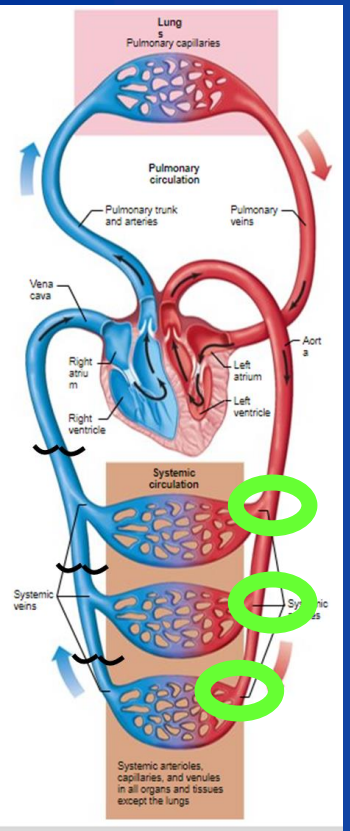
  (a) More metabolically active tissue requires more blood flow

  (b) Arteriole smooth muscle is highly sensitive to many metabolites

  (c) Locally produced metabolites are usually vasodilating

  (3) MYOGENIC RESPONSE

  (a) If BP to a tissue is high, the BP may distend the arterial walls outward, stretching the arteriole smooth muscle. Some arterioles react with a reflex activation and vasoconstriction. This increases resistance flow to maintain a constant blood flow in spite of the increased BP. This also occurs in reverse to some degree in some arterioles.



The VASCULATURE as a well-regulated system of conduits

- **ARTERIOLES**

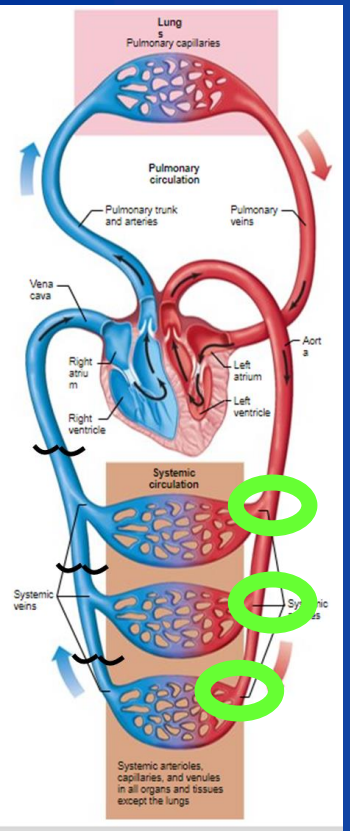
  Regulation of arteriole resistance

  d) EXTRINSIC ARTERIOLE CONTROLS

  (1) Sympathetic nervous system activation

  (a) Many arteriole smooth muscle beds are richly populated by α1-adrenergic receptors. During sympathetic stimulation, norepinephrine from sympathetic neurons and epinephrine from the adrenal medulla bind the α1-adrenergic receptors it initiate an IP3 signal transduction process to increase smooth muscle calcium, vasoconstriction and decreased blood flow to the area.

  (b) Exception: Some arterioles supplying skeletal muscles have β2-adrenergic receptors. Epinephrine has high affinity to β2-adrenergic receptors and will bind them to initiate a cAMP response that results in vasodilation and increased blood flow to the skeletal muscle.



(2) Hormonal control

(a) Angiotensin II is a powerful vasoconstrictor; Angiotensin II binding to its receptor on arteriole smooth muscle activates an IP3/Ca++ signaling pathway.

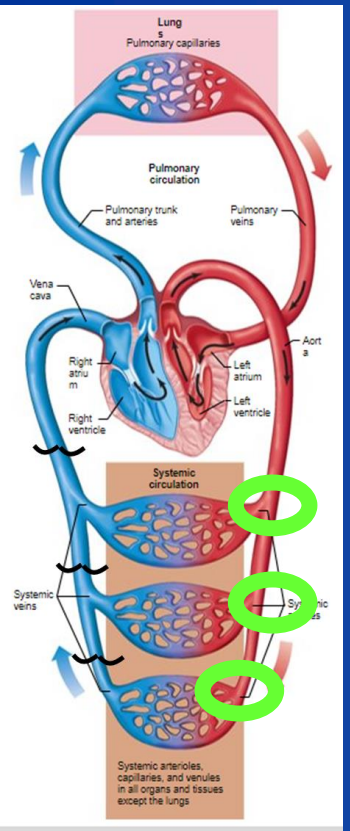
(b) Vasopressin is a vasoconstrictor released by the posterior pituitary. Vasopressin receptors found on arterioles also activate an IP3/Ca++ signaling pathway.

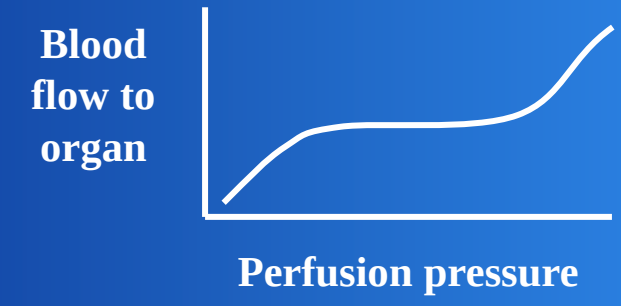
(c) Atrial natriuretic peptide is a vasodilator

(3) Others – Neurons that release nitric oxide

The VASCULATURE as a well-regulated system of conduits

- **ARTERIOLES**
  6. **Examples of dynamic flow regulation responses**
     a) **Active hyperemia**
        (1) The more metabolically active a tissue is the more blood flow it requires
        (2) Metabolically active tissue produces more vasodilating local factors which increase blood flow to the area
        (3) Therefore, blood flow is regulated by tissue metabolic activity
     b) **Reactive hyperemia**
        (1) At times, blood flow to an area can be occluded
        (2) Vasodilating local factors accumulate causing relaxation of arterial smooth muscle
        (3) If blood flow is restored, there is a strong compensatory increase in blood flow
     c) **Flow autoregulation**
        (1) Several organs (brain, kidney, etc.) regulate blood flow by regulating resistance such that flow is near constant for a wide range of PERFUSION PRESSURES
        (2) If perfusion pressure rises, resistance to flow increases
        (3) If perfusion pressure drops, resistance to flow decreases





- **MEAN ARTERIAL BLOOD PRESSURE (MAP)**
- **MAP is the average BP over the entire cardiac cycle**
- **Because diastole lasts twice as long as systole, diastole BP has twice to impact on MAP as SP**

MAP = DP + 1/3(PP)

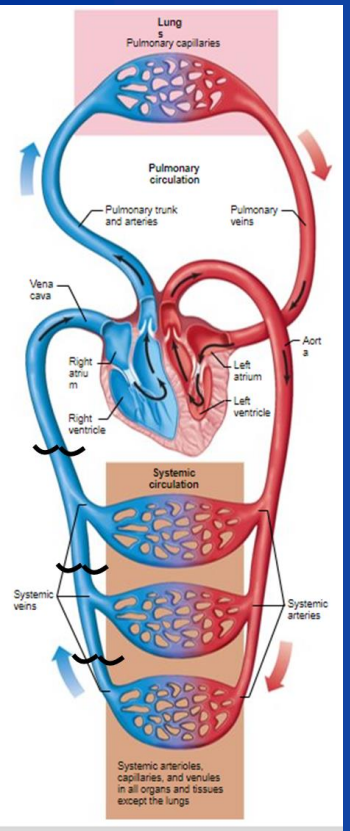
A person with a systolic blood pressure of 120 mmHg and a diastolic pressure of 80 mmHg has a pulse pressure of 40 mmHg.

Therefore, MAP = 80 + 1/3(40) = 93 mmHg

- **MEAN ARTERIAL (Aortic) BLOOD PRESSURE (MABP)**

MABP = DP + 1/3(PP) e.g., 120/80

MABP = 80 + 1/3(40) = 93 mmHg



Flow = Δ Pressure / Resistance

Flow = (Aortic pressure – Right atrial pressure) / Resistance

Flow = (MABP – 0) / Resistance = MABP/Resistance

Therefore, Flow = MABP/Resistance

Rearrange equation to:

MABP = Flow x Resistance

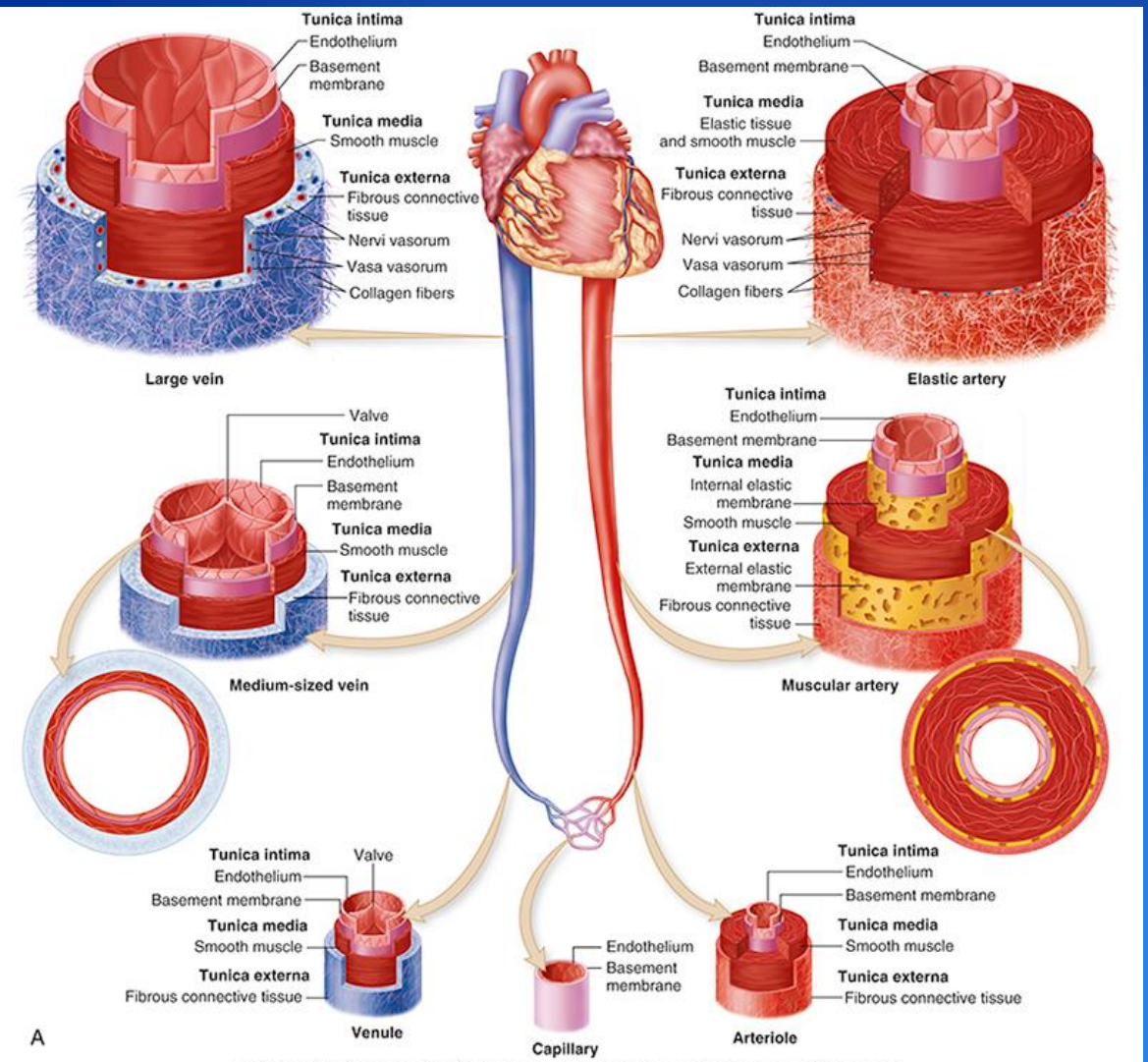
Therefore, for systemic blood flow:

MABP = Cardiac output x Total peripheral resistance

MABP = (HR x SV) x Total peripheral resistance

The VASCULATURE as a well-regulated system of conduits

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- Structural characteristics lead to the follow (simplistic but largely accurate) statement:
  "Arteries store blood pressure while veins store volume."

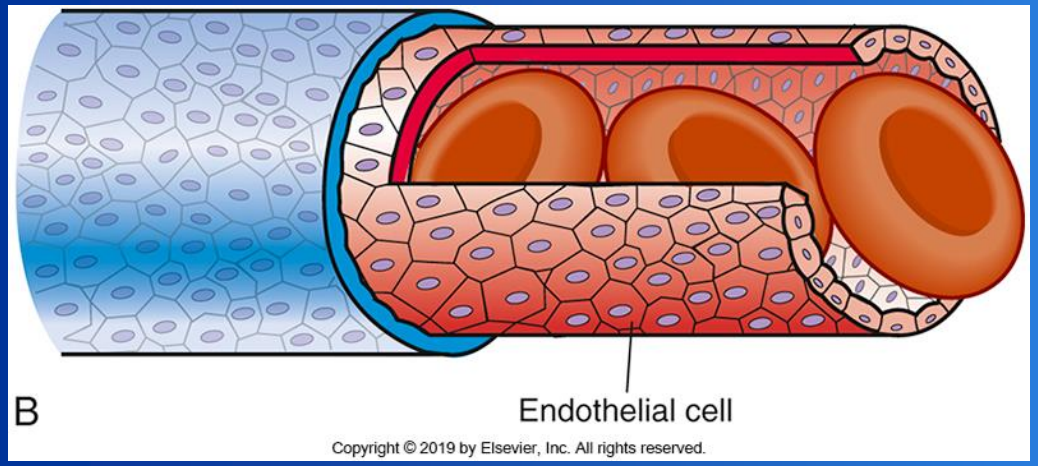


From Patton KT, Thibodeau GA: *Anatomy & physiology*, ed 9, St Louis, 2016, Mosby.

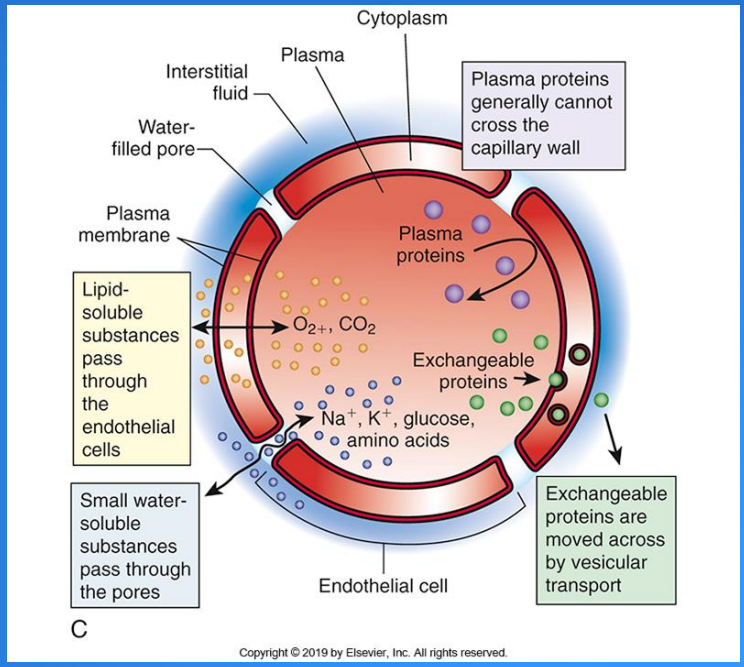
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CAPILLARIES

- **Site of transfer of gases, nutrients, cell secretions and metabolic end products**
- **Structure (fig. 12.31)**
  - Single layer of endothelial cells (ENDOTHELIUM)
  - Adjacent endothelial cells are attached loosely to one another by water-filled intercellular clefts
    - Brain – very tight clefts, limited transport
    - Liver – very loose clefts, increased transport
  - Some capillaries also have endocytotic and exocytotic vesicles



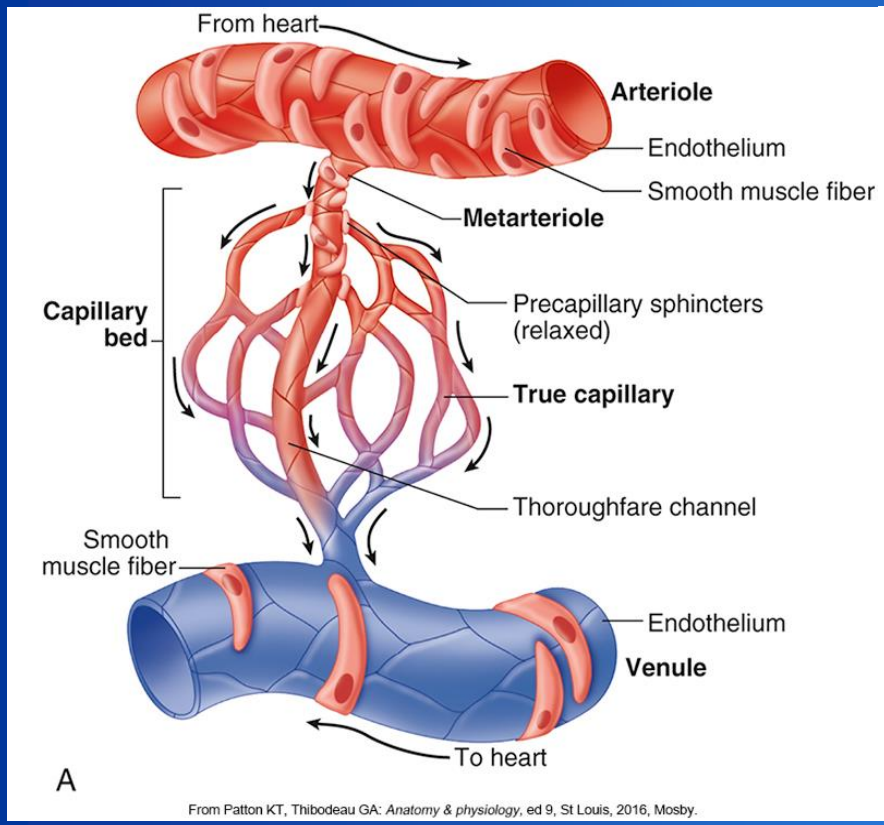
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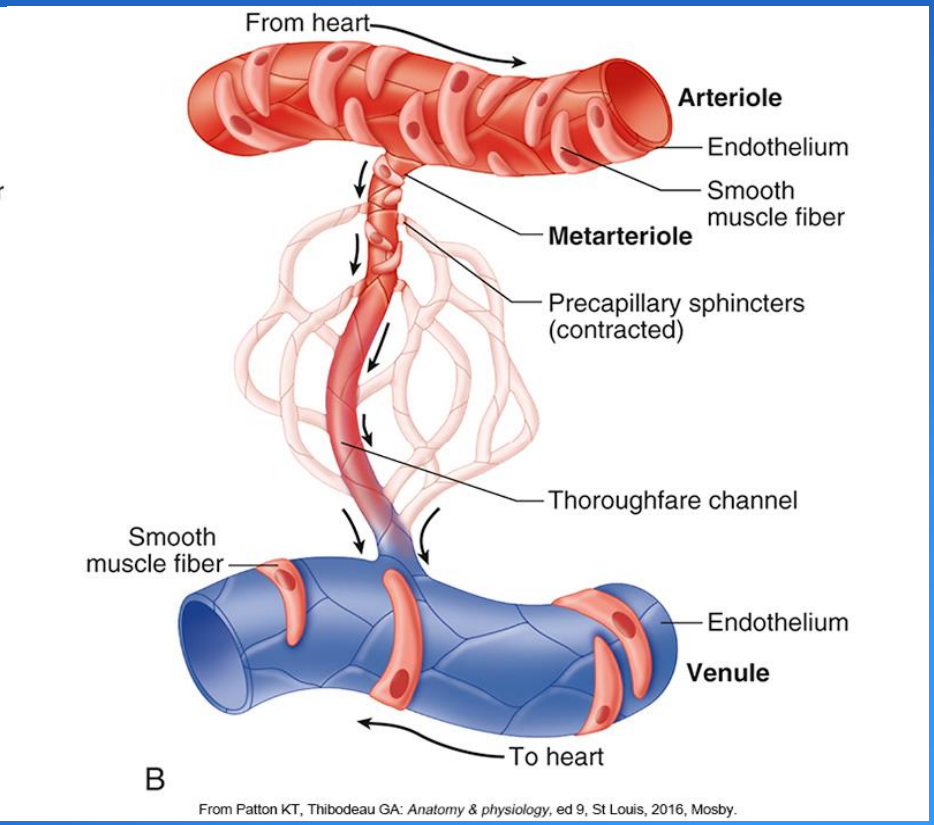


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CAPILLARIES

- **Capillary networks (fig 12.31)**
  - Blood flow in a capillary bed is greatly regulated by controlling resistance in the supplying arteriole
  - Precapillary sphincters – regulate blood entering capillaries according to local metabolic mediators
  - Metarterioles – connect arterioles to veins in some tissues to regulate capillary blood flow



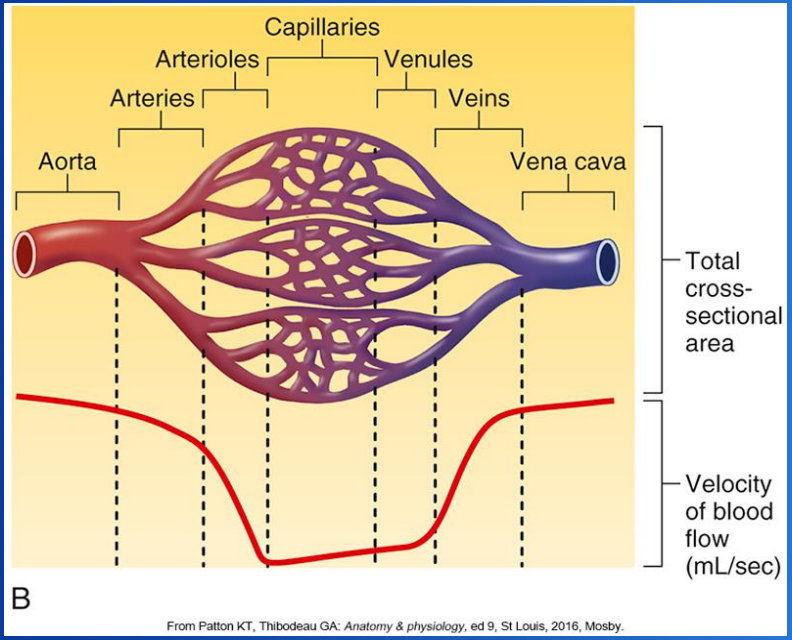


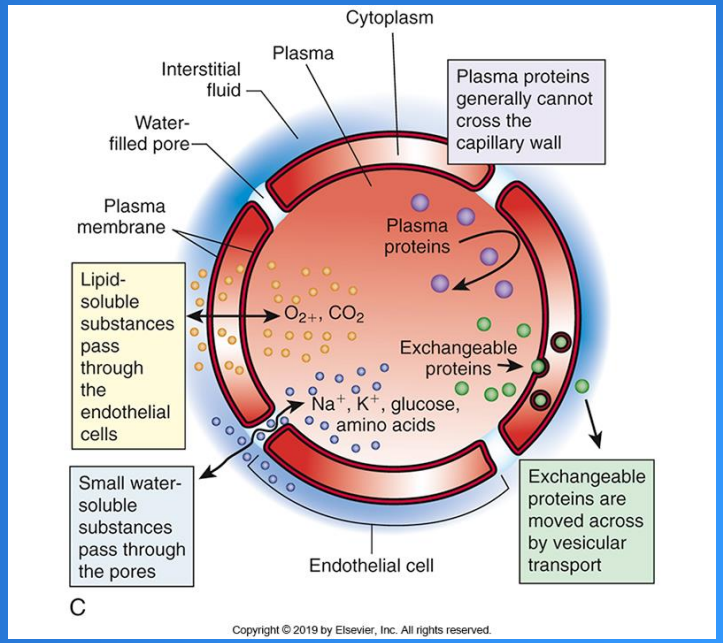
CAPILLARIES

- Cross-sectional area of blood vessels and velocity of blood flow
- As cross-sectional area increases, velocity decreases (fig. 32.24b)
- Velocity in capillaries is very low, making it ideal for diffusion (fig. 32.22c)
- Capillaries constitute a huge cross-sectional surface area, making them ideal for diffusion; recall Fick's Law of Diffusion

$$\text{Diffusion rate} = \frac{[\text{Surface area} \times (\text{Diffusivity}/\sqrt{\text{MW}}) \times \text{Concentration gradient}]}{\text{Thickness of barrier}}$$

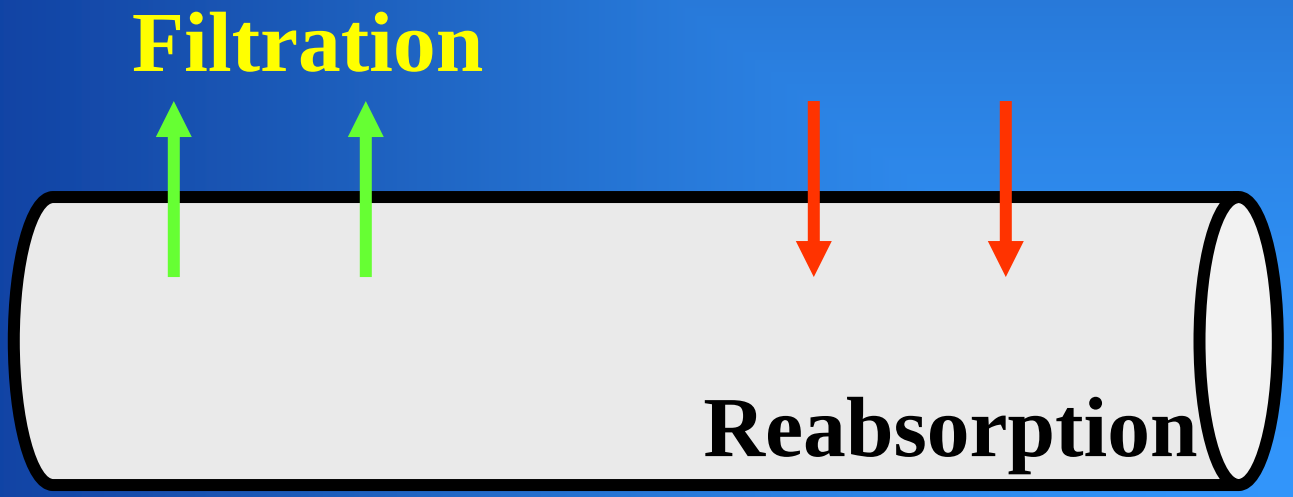
- *"In all capillaries, excluding the brain, diffusion is the only important means by which movement of nutrients, oxygen and metabolic end products occurs across the capillary wall."* Vander's Physiology 16th ed
- Mediated transport is important in the brain





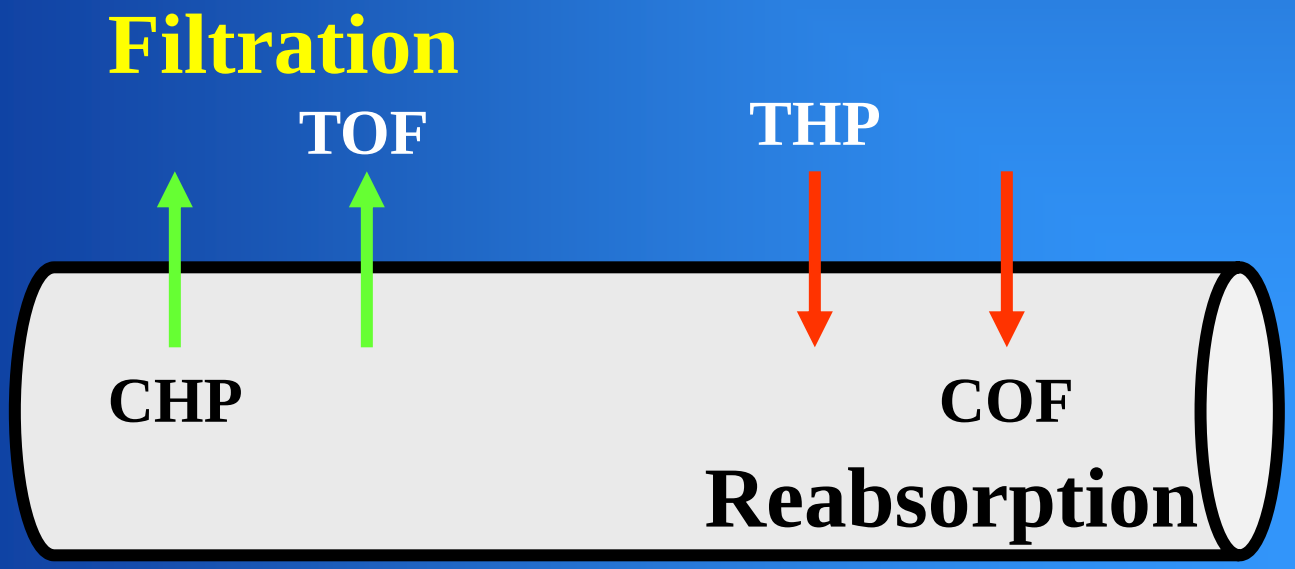
CAPILLARIES

- CAPILLARY FLUID DYNAMICS
  - Filtration and reabsorption across the capillary by Starling forces
  - Fluid is exchanged between the plasma and the intestinal fluid across the capillary
    - FILTRATION = movement of fluid out of the plasma into the interstitium
    - REABSORPTION = movement of fluid out of interstitium and into the plasma



CAPILLARIES

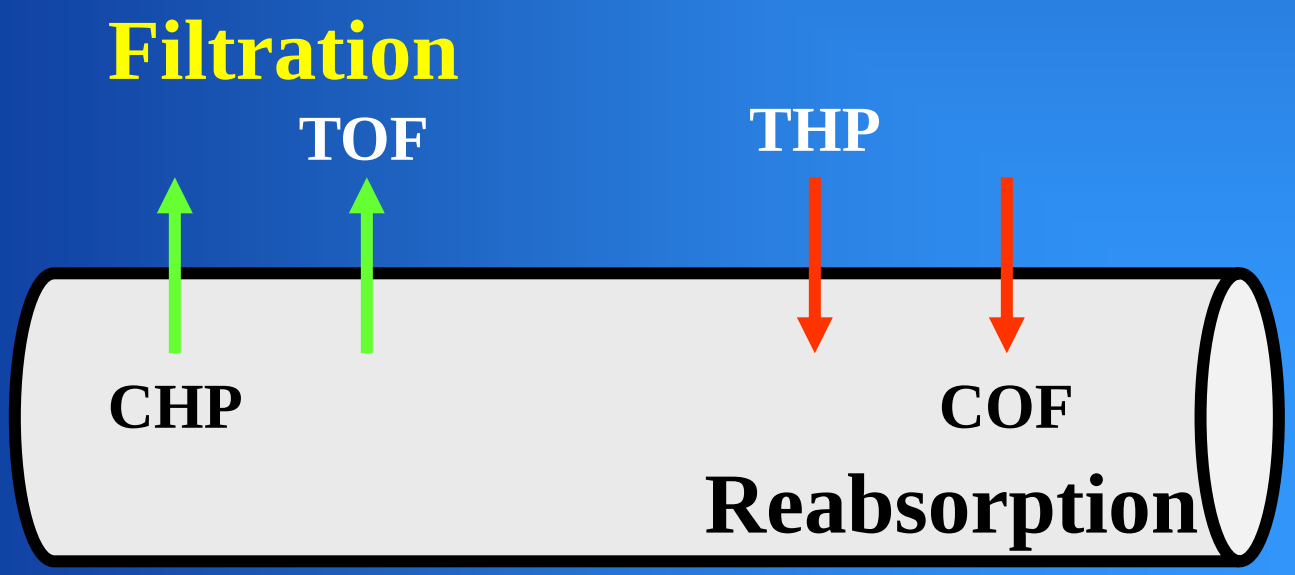
- CAPILLARY FLUID DYNAMICS
  - Two forces favor filtration
    - Capillary hydrostatic (blood) pressure (CHP) “pushes” fluid out of the capillary into the interstitium
    - Tissue (interstitial) osmotic force (TOF) “pulls” fluid out of the capillary into the interstitium
  - Two forces favor reabsorption
    - Capillary osmotic force (COF) “pulls” fluid out of the interstitium into the capillary
    - Tissue (interstitial) hydrostatic (blood) pressure (THP) “pushes” fluid out of the interstitium into the capillary



Proteins produce oncotic/ osmotic forces

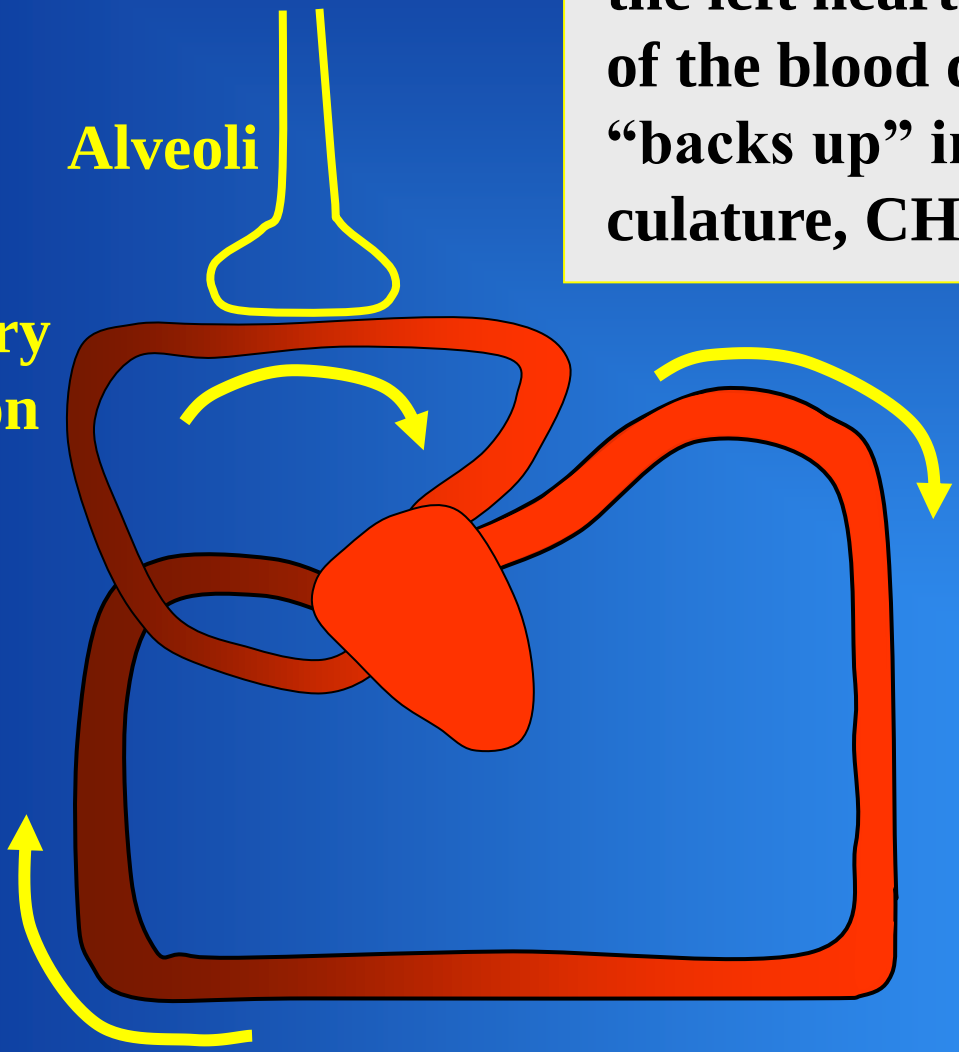
CAPILLARIES

- **CAPILLARY FLUID DYNAMICS**
  - **Net fluid movement across the capillary is determined the balance between filtration and reabsorption**
  - **Net fluid movement = Filtration forces – Reabsorption forces**
  - **Net fluid movement = (CHP + TOF) – (THP + COF)**
  - **Examples**
    - **If blood pressure drops significantly, CHP is reduced and fluid moves from the interstitium to the plasma to help support blood pressure**
    - **If an area becomes infected and inflamed, plasma proteins move into the inflamed tissue which increases the TOF and draws water into the tissue (swelling)**
    - **EDEMA – abnormal accumulation of fluid in the interstitium**
      - **Example of left heart (ventricular) failure**

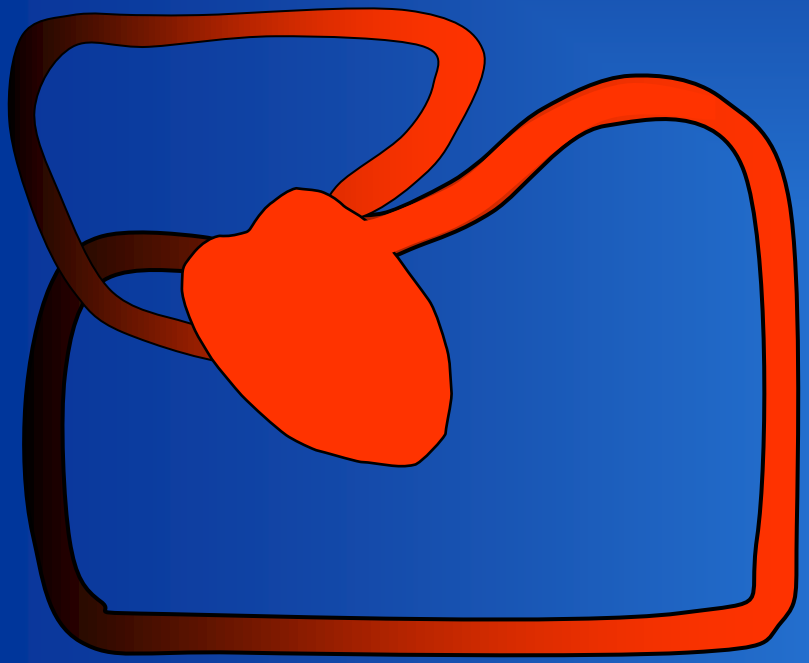


Proteins produce oncotic/ osmotic forces

Although there are several causes of pulmonary edema, one of the most common is left heart failure; the left heart is unable to pump all of the blood delivered to it; blood "backs up" in the pulmonary vasculature, CHP ↑ & edema results



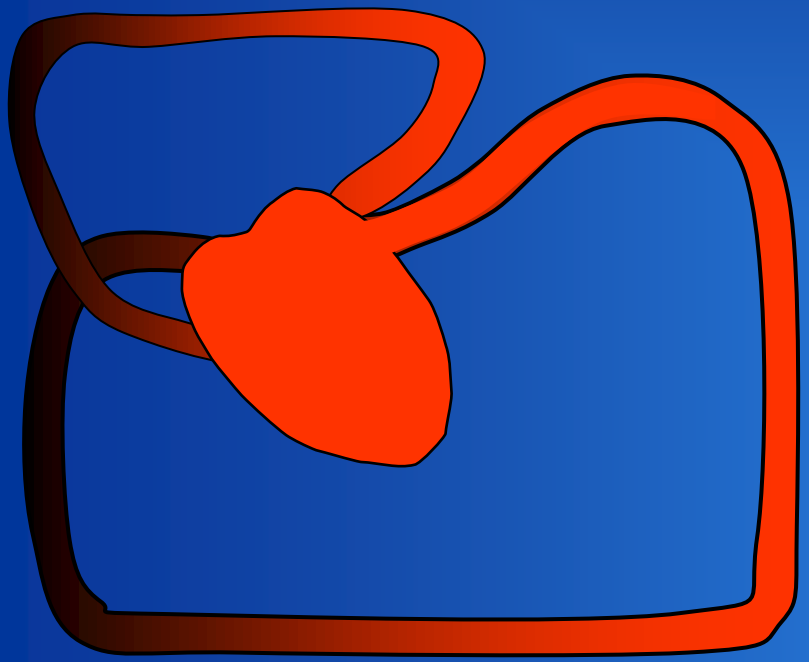
CONGESTIVE HEART FAILURE



Over a period of time:

Left Cardiac Output must equal Right Cardiac Output

CONGESTIVE HEART FAILURE

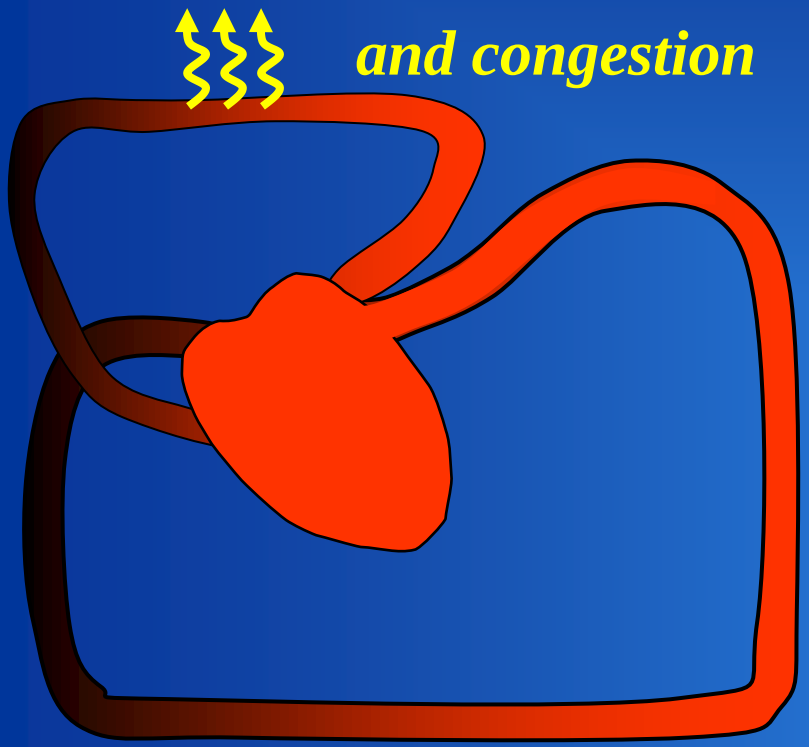


But what if Right CO is greater than Left CO by one drop of blood (0.1 ml) for each heart beat?

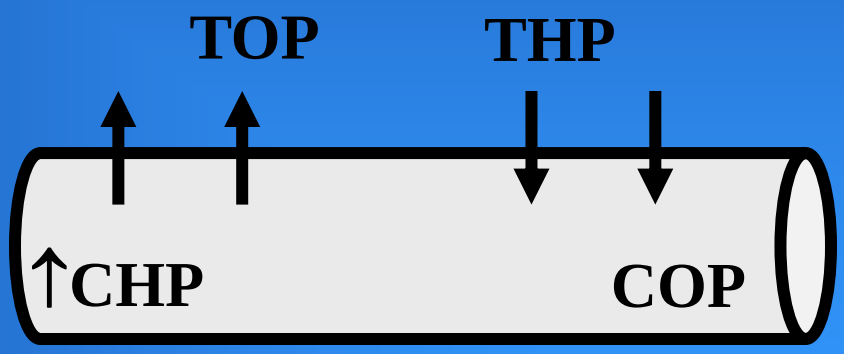
0.1 ml/beat * 80 beats/min =

8 ml of blood would back up in the pulmonary vasculature every minute

CONGESTIVE HEART FAILURE



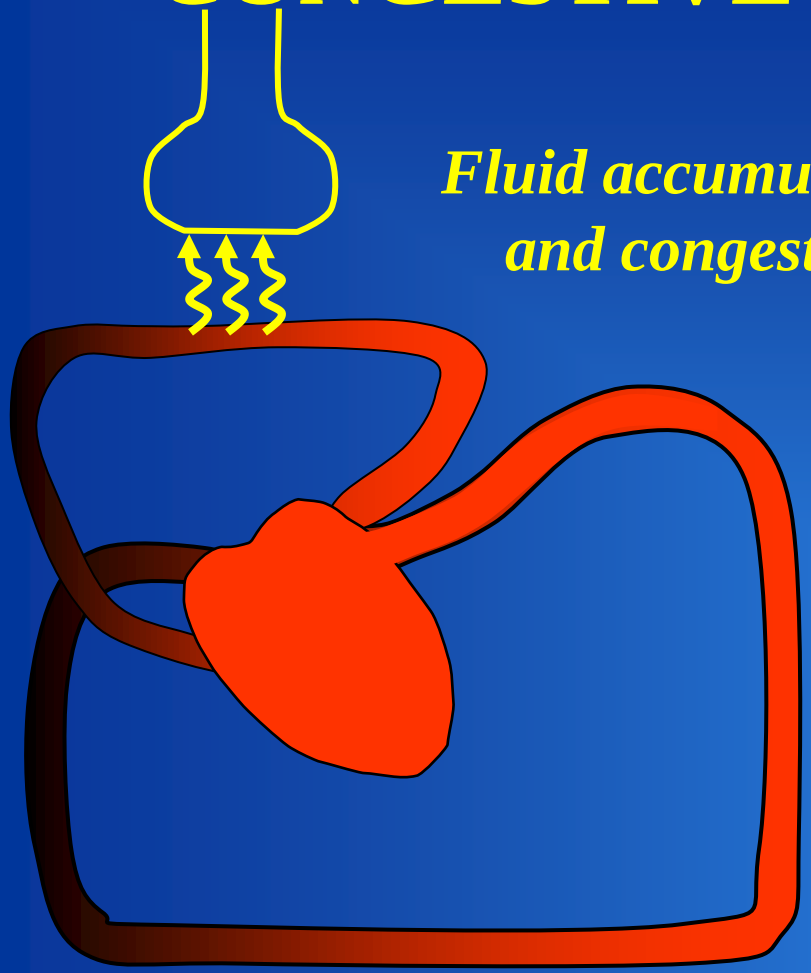
But what if Right CO is greater than Left CO by one drop of blood (0.1 ml) for each heart beat?



PULMONARY EDEMA!

Diffusion impairment!!!

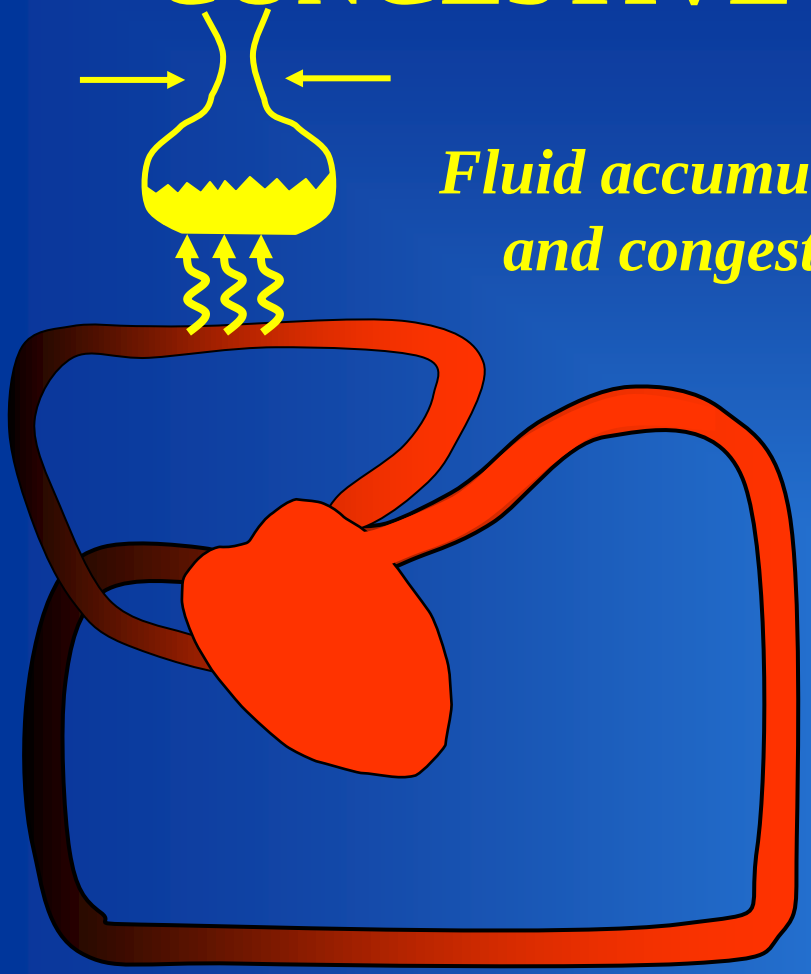
CONGESTIVE HEART FAILURE



Fluid accumulation and congestion

PULMONARY EDEMA!

CONGESTIVE HEART FAILURE



Fluid accumulation and congestion

Left Heart Failure

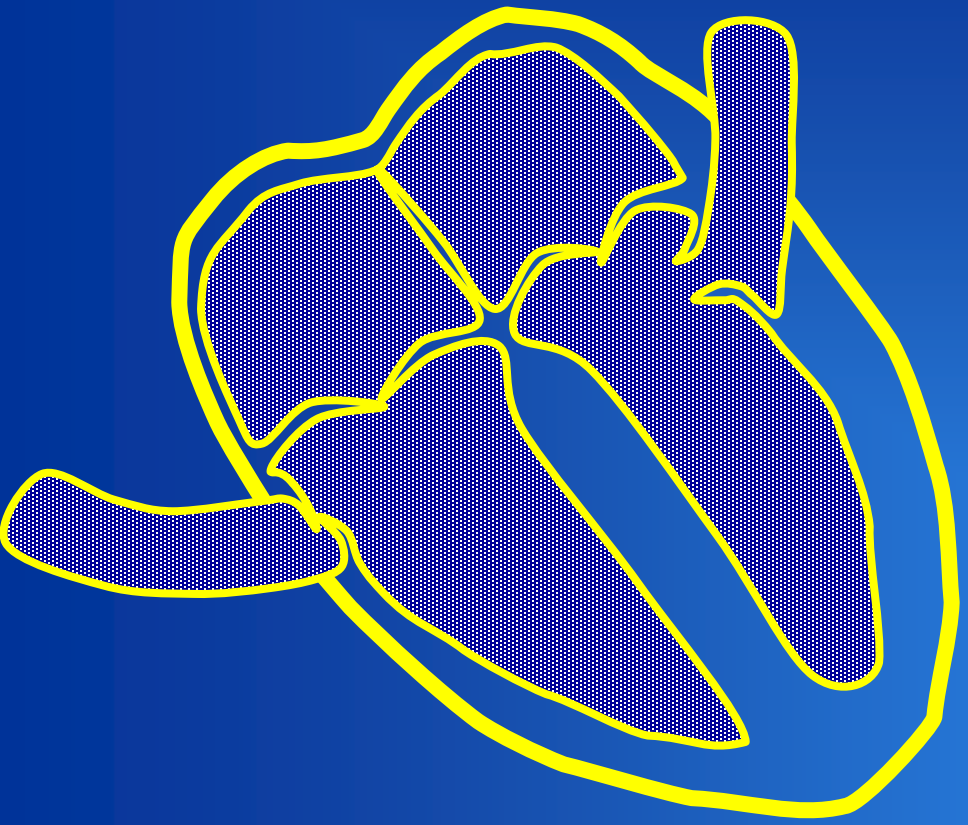
- **Right CO > Left CO**
- **Pulmonary edema**
- **Dyspnea**
- **Causes:**

  Hypertension

  Weakened left ventricle

  CAD, MI, myopathy

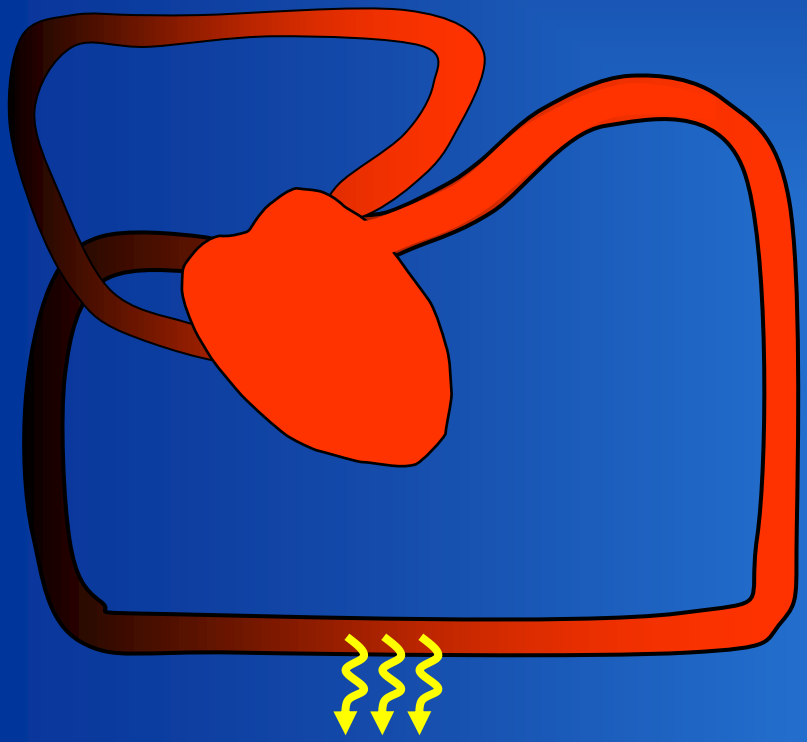
CONGESTIVE HEART FAILURE



Left Heart Failure

- Right CO > Left CO
- Pulmonary edema
- Dyspnea
- Causes:
  - Hypertension
  - Weakened left ventricle
    - *CAD, MI, myopathy*
  - Valve disease
    - *Stenosis*
    - *Regurgitation*

CONGESTIVE HEART FAILURE

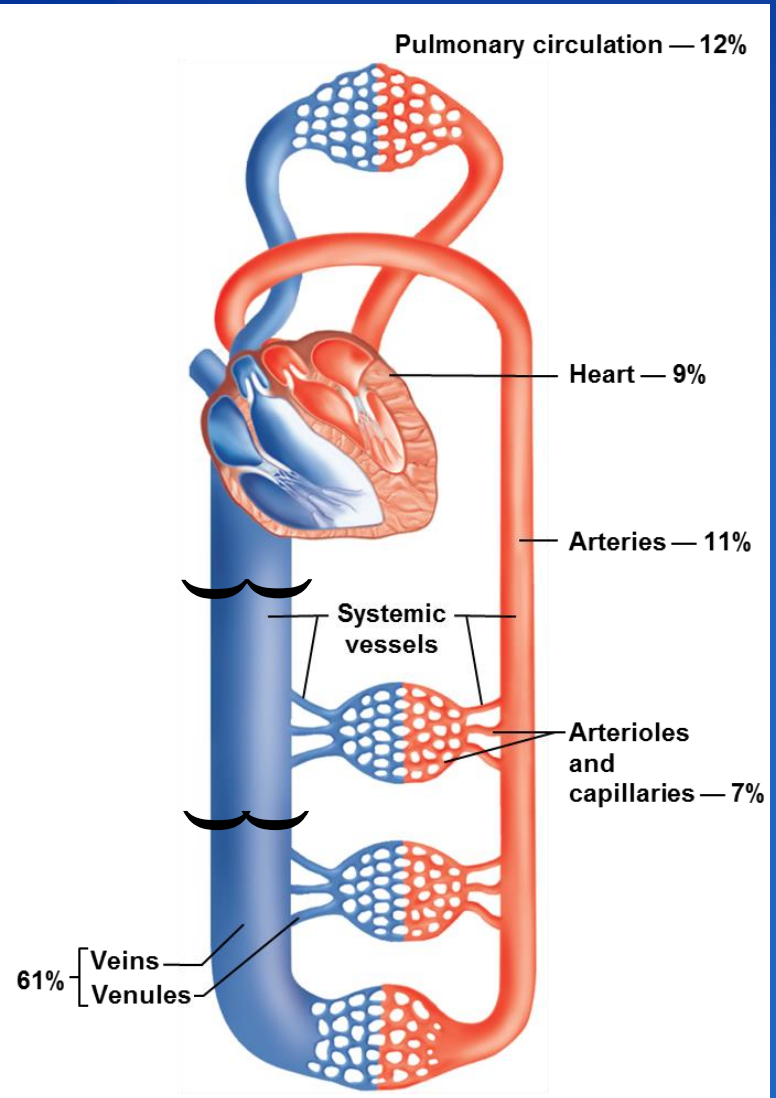


Dependent edema

Right Heart Failure

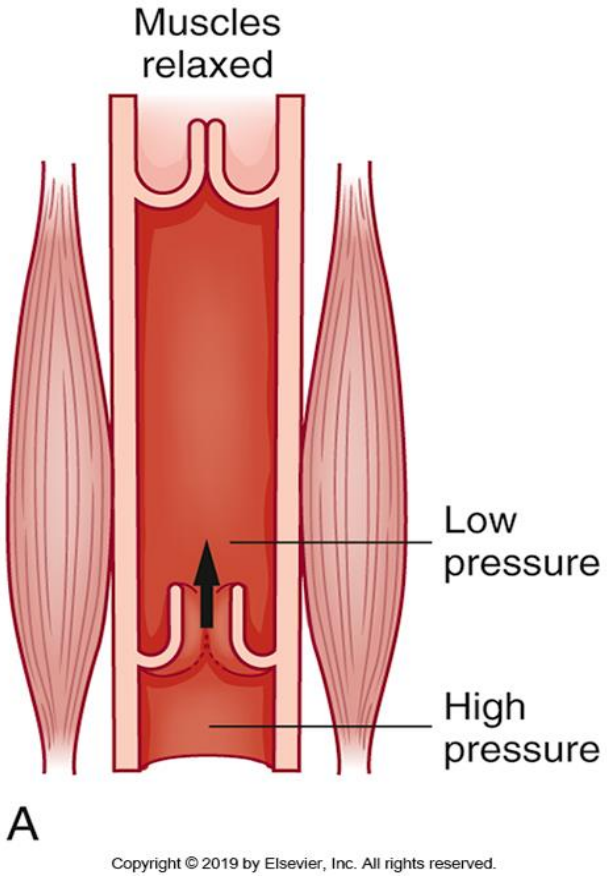
- Left CO > Right CO
- Peripheral, dependent edema
- Causes:
  - Pulmonary hypertension
  - Valve disease

VEINS

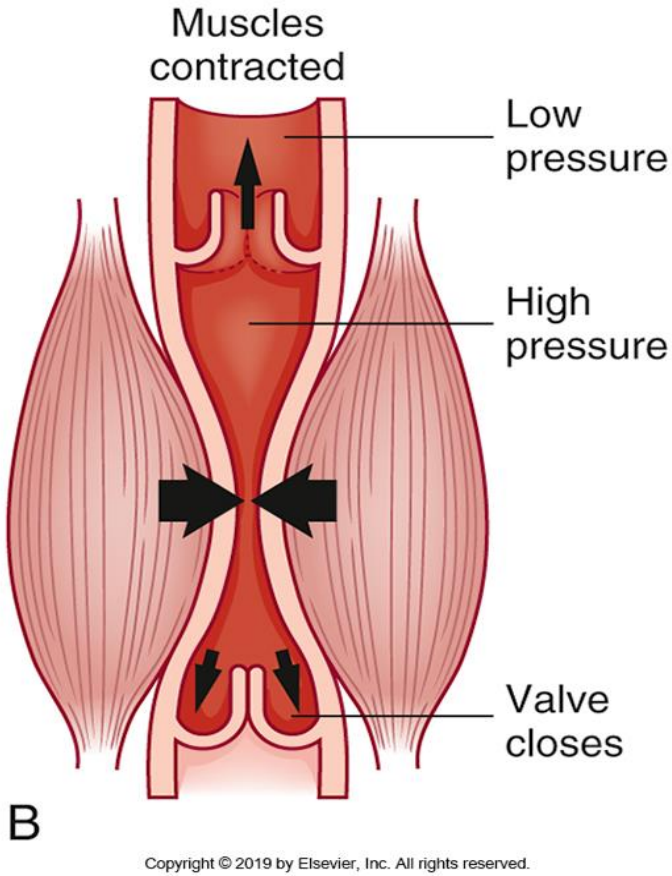


- Recall that "arteries store blood pressure while veins store volume."
- Veins function as a "volume reservoir (fig. 12.47 Vander's Physiology 16th ed)
- Mean venous pressure is usually very low (~10 mmHg); very low resistance
- Venous system has very high compliance; a small change in pressure leads to a large change in volume;
- Compliance = Δ Volume/Δ Pressure
- Regulation of venous return
  - Venous return plays a vital role in regulating ventricular filling, stroke volume and cardiac output
  - Sympathetic nervous system activity
  - Skeletal muscle pump (fig 12.48)
  - Respiratory pump

Skeletal muscle pump (see fig. 12.48)

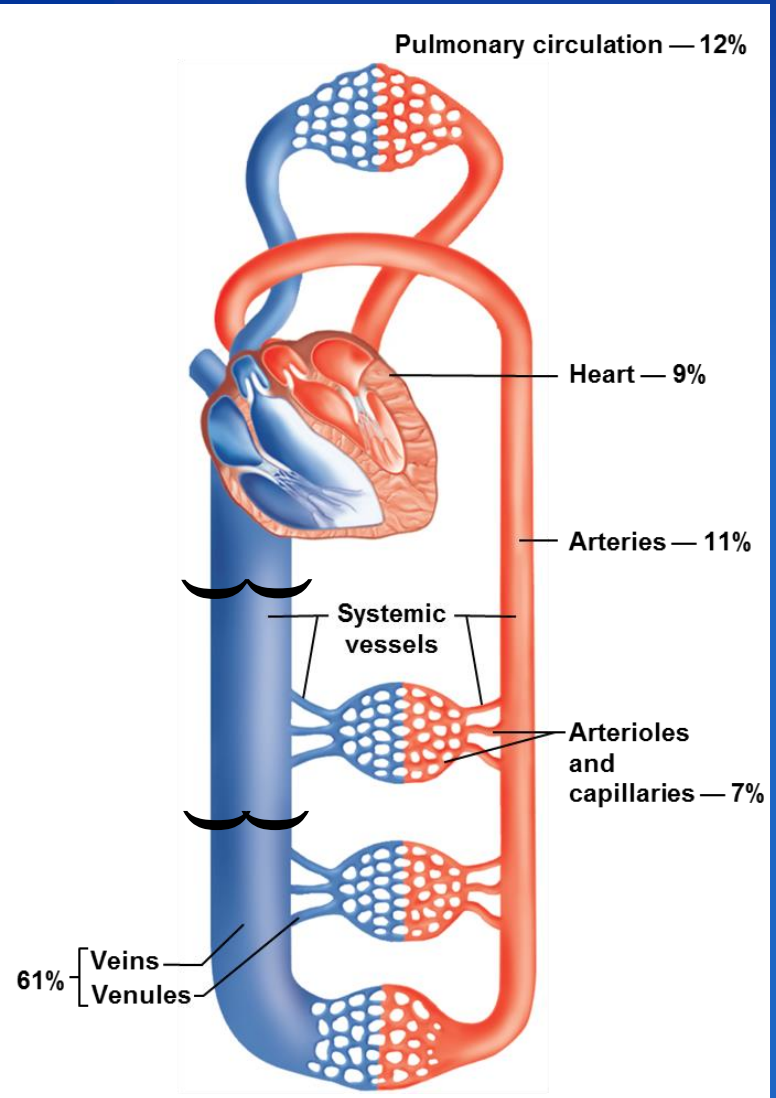


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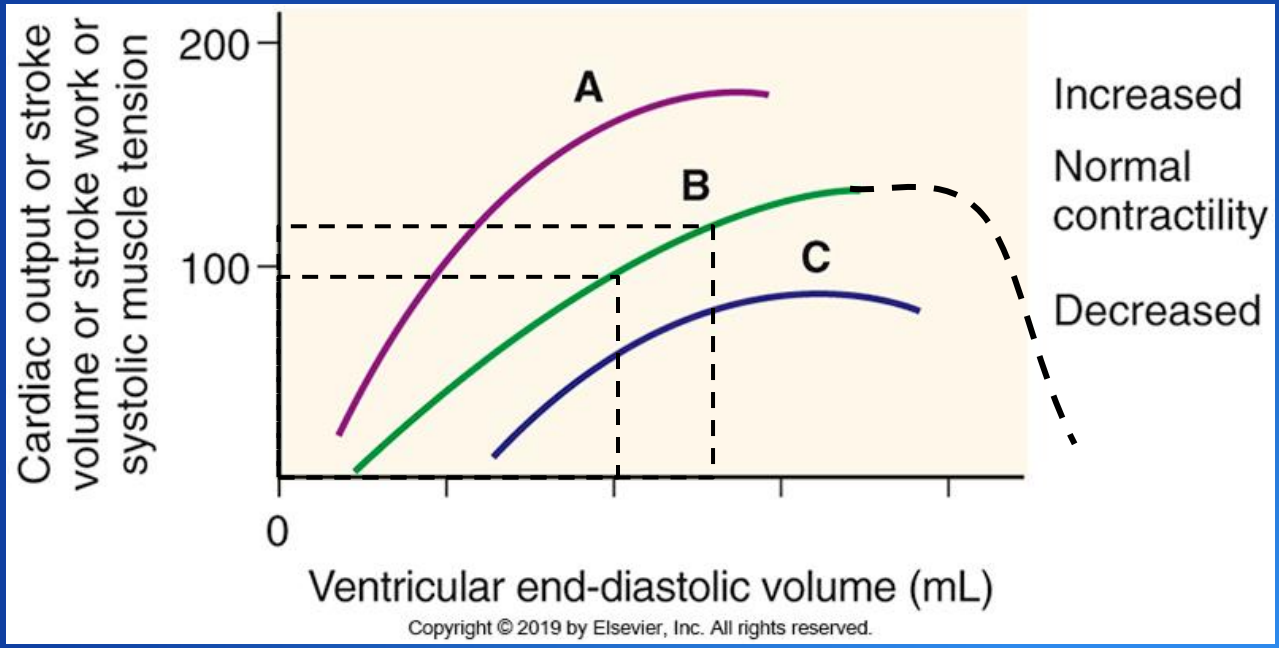
VEINS



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  - Respiratory pump

STROKE VOLUME and its regulation

1. FRANK-STARLING mechanism



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Examples of what affects ventricular EDV

↑ Venoconstriction (↓ Venous compliance)

↑ Skeletal mms contraction pump

↑ Respiratory pump

↑ Blood or plasma volume

→ ↑ Venous return → ↑ Ventricular filling → ↑ EDV → ↑ SV

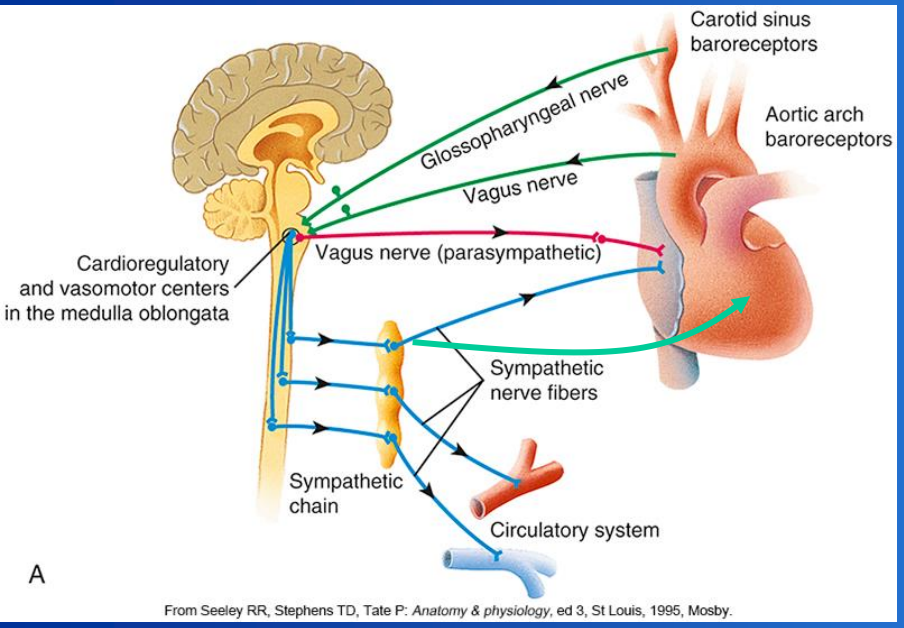
BLOOD PRESSURE REGULATION

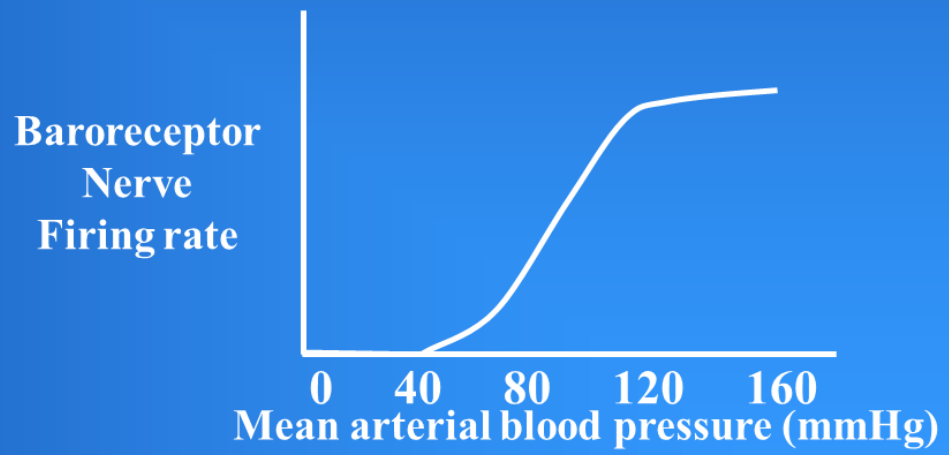
- Recall that for systemic blood flow:

  MABP = Cardiac output x Total peripheral resistance

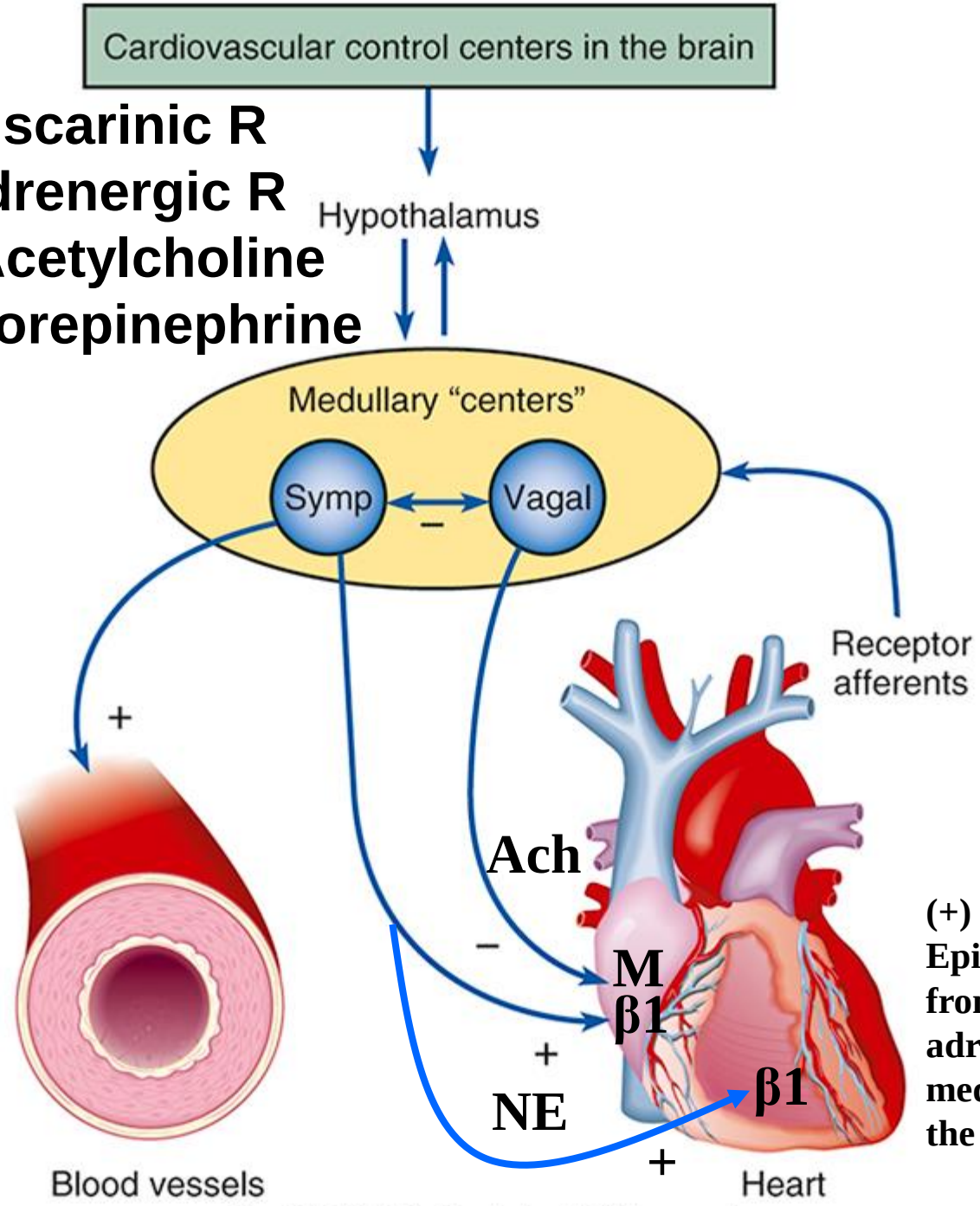
  MABP = CO x TPR or MABP = (HR x SV) x TPR

- Blood pressure sensors
  - ARTERIAL BARORECEPTORS (mechanoreceptors that sense arterial blood pressure) AND AUTONOMIC NERVOUS SYSTEM REGULATION
  - Decreased MABP decreases baroreceptor nerve firing rate to the brainstem centers, which increase sympathetic nervous system activation and decreases parasympathetic nervous system activation



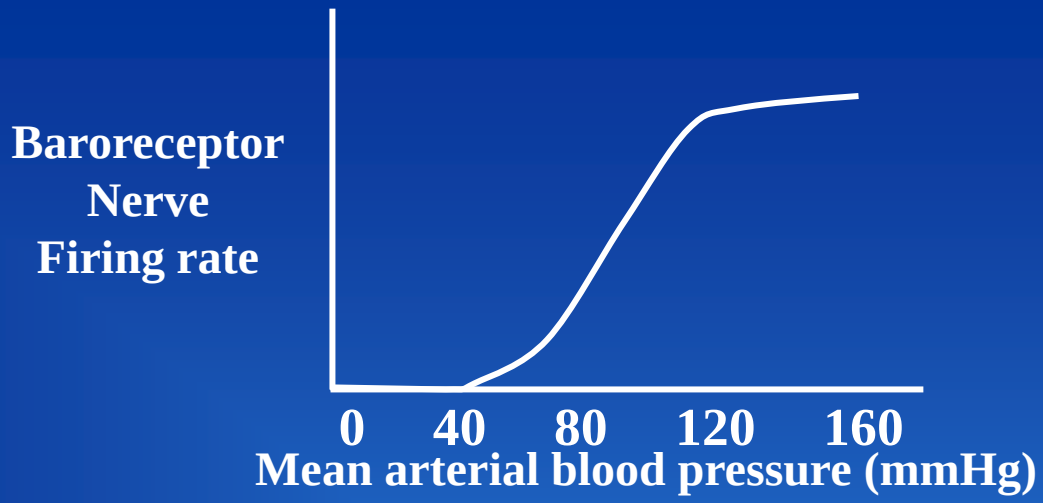


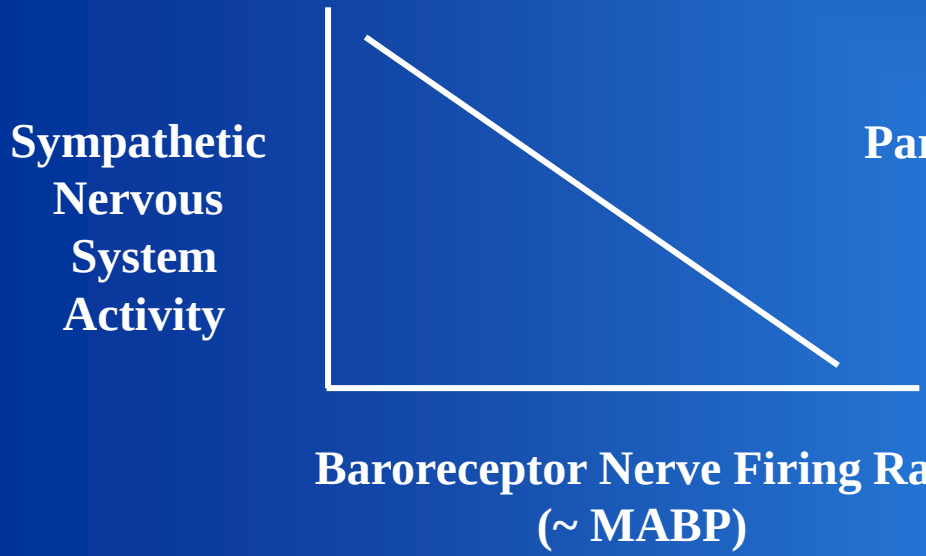
M = Muscarinic R
β1 = Adrenergic R
Ach =Acetylcholine
NE = Norepinephrine

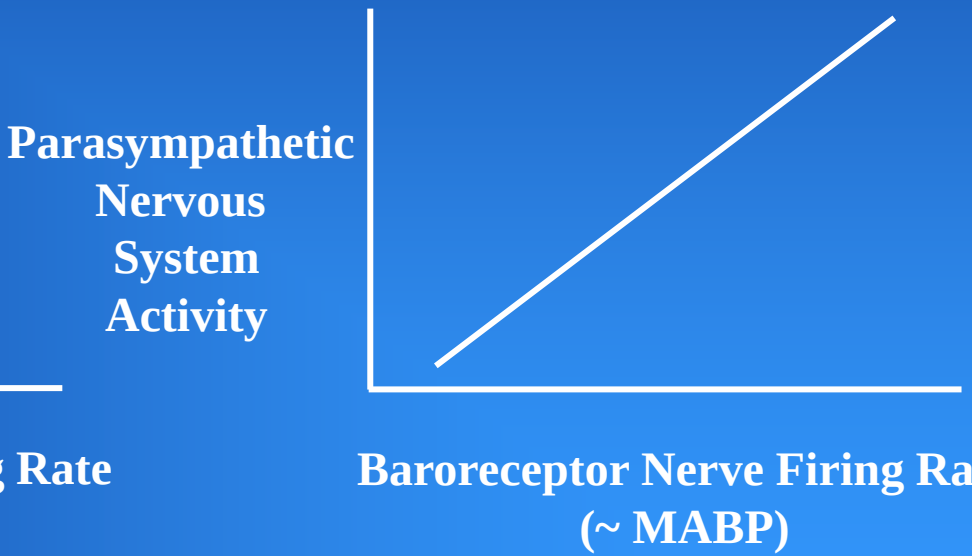


(+) β1 Epinephrine from the adrenal medulla via the blood

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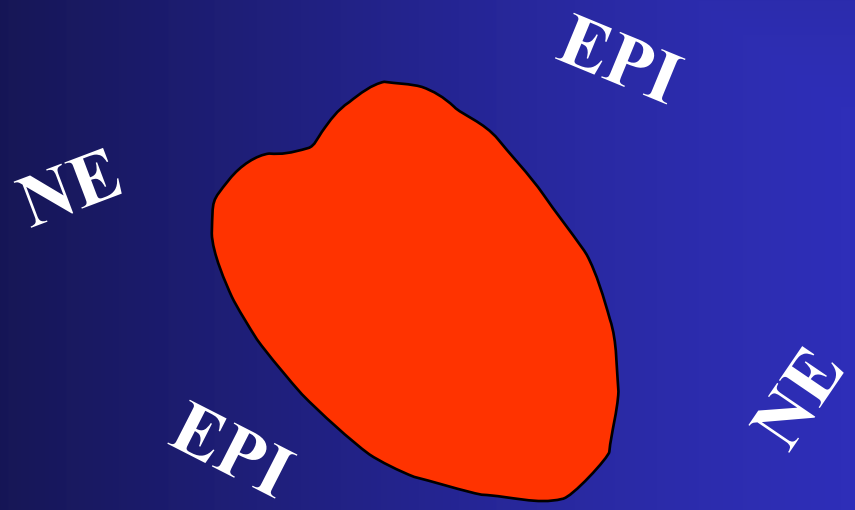
- As MABP increases, SNS activity decreases and PNA increases
- As MABP decreases, SNS activity increases and PNA decreases
- Baroreceptors can reset their set points

*MABP = (HR * SV) * TPR*

Heart rate

Sympathetic nervous system activity

↑SNS → ↑HR → ↑CO (via β_1-adrenergic receptors)



*MABP = (HR * SV) * TPR*

Heart rate

Sympathetic nervous system activity

↑SNS → ↑HR → ↑CO (via β_1-adrenergic receptors)

Parasympathetic nervous system activity

↓PNS → ↑HR → ↑CO

*MABP = (HR * SV) * TPR*

Stroke volume

Contractility (ventricular ejection force)

Preload (venous return and ventricular filling)

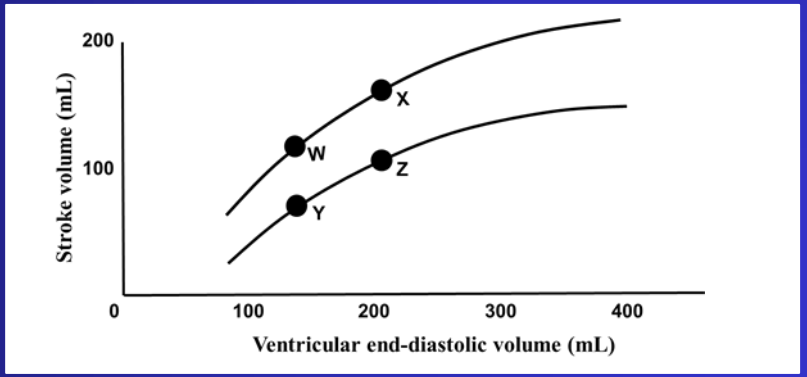
*MABP = (HR * SV) * TPR*

Stroke volume

Contractility (ventricular ejection force)

↑SNS → ↑ejection force → ↑SV → ↑CO
(via β_1-adrenergic receptors)

Epinephrine → ↑ejection force → ↑SV → ↑CO
(via β_1-adrenergic receptors)



*MABP = (HR * SV) * TPR*

Stroke volume

Preload (venous return and ventricular filling)

*MABP = (HR * SV) * TPR*

Stroke volume

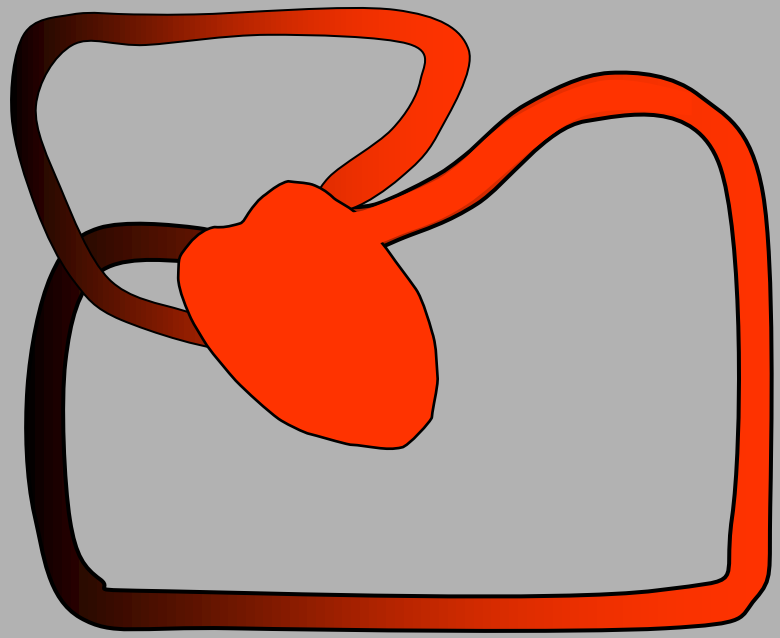
Preload (venous return and ventricular filling)

Venous compliance

↓Compliance → ↑VR → ↑SV

Plasma or blood volume

↑H_2O retention → ↑VR → ↑SV



*MABP = (HR * SV) * TPR*

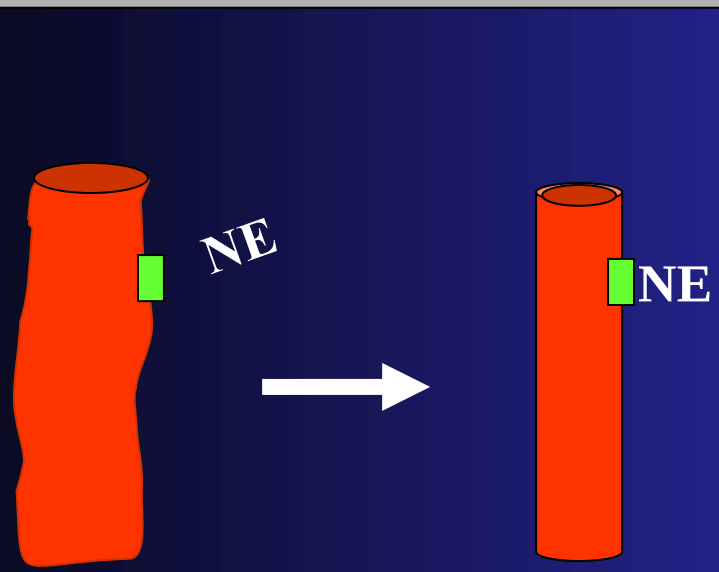
Stroke volume

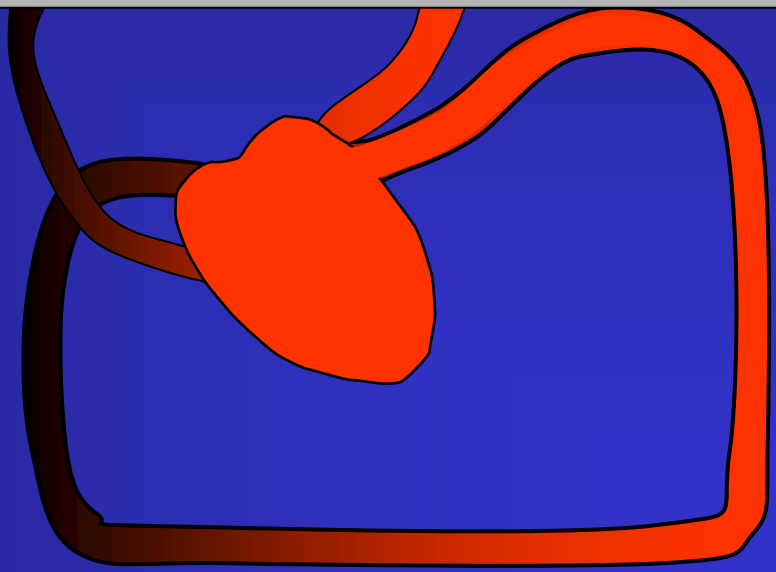
Preload (venous return and ventricular filling)

Venous compliance (α_1-adrenergic & AII receptors)

↑SNS → ↓Compliance → ↑VR → ↑Filling → ↑CO

↑AII → ↓Compliance → ↑VR → ↑Filling → ↑CO





*MABP = (HR * SV) * TPR*

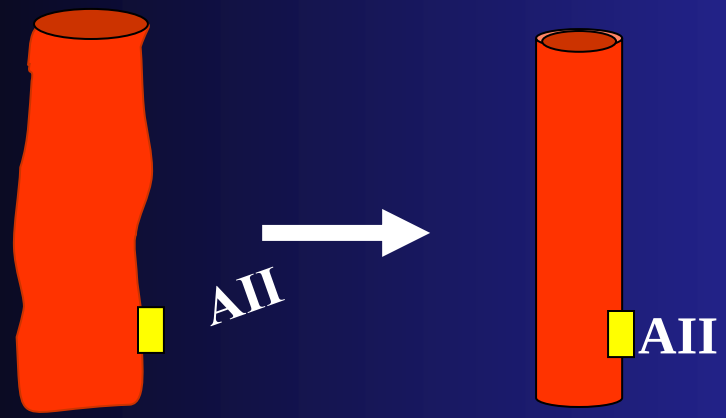
Stroke volume

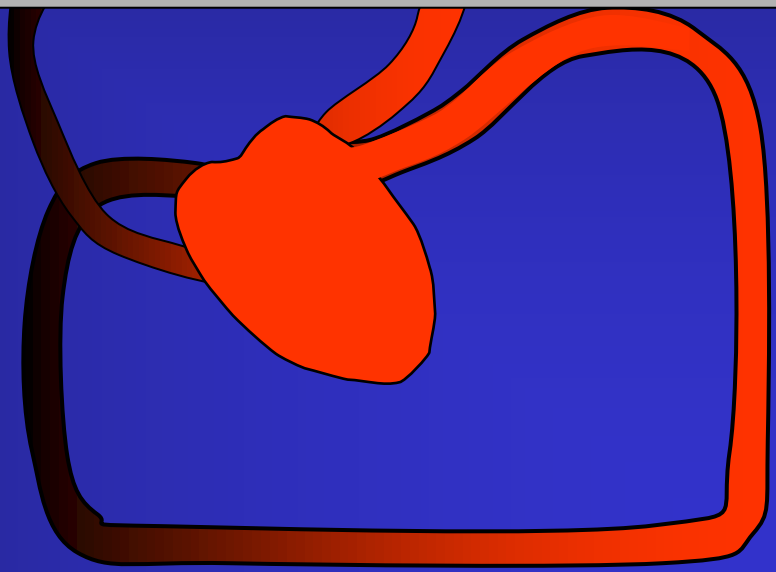
Preload (venous return and ventricular filling)

Venous compliance (α_1-adrenergic & AII receptors)

↑SNS → ↓Compliance → ↑VR → ↑Filling → ↑CO

↑AII → ↓Compliance → ↑VR → ↑Filling → ↑CO





*MABP = (HR * SV) * TPR*

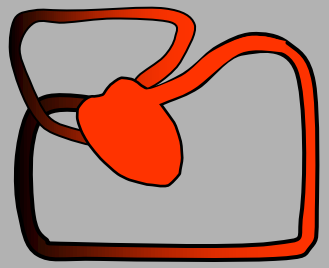
Stroke volume

Preload (venous return and ventricular filling)

Venous compliance (α_1-adrenergic & AII receptors)

↑SNS → ↓Compliance → ↑VR → ↑Filling → ↑CO

↑AII → ↓Compliance → ↑VR → ↑Filling → ↑CO



Plasma or blood volume

Water volume (aldosterone or ADH)

↑H_2O retention → ↑VR → ↑Filling → ↑CO

Sodium intake or retention

↑Na+ → ↑H_2O retention → ↑VR → ↑Filling → ↑CO

MABP = (HR x SV) x TPR

Autonomic NS

SymNS
(β1-receptors)
ParasymNS
(muscarinic receptors)

Contractility

SymNS
(β1-receptors)

Preload

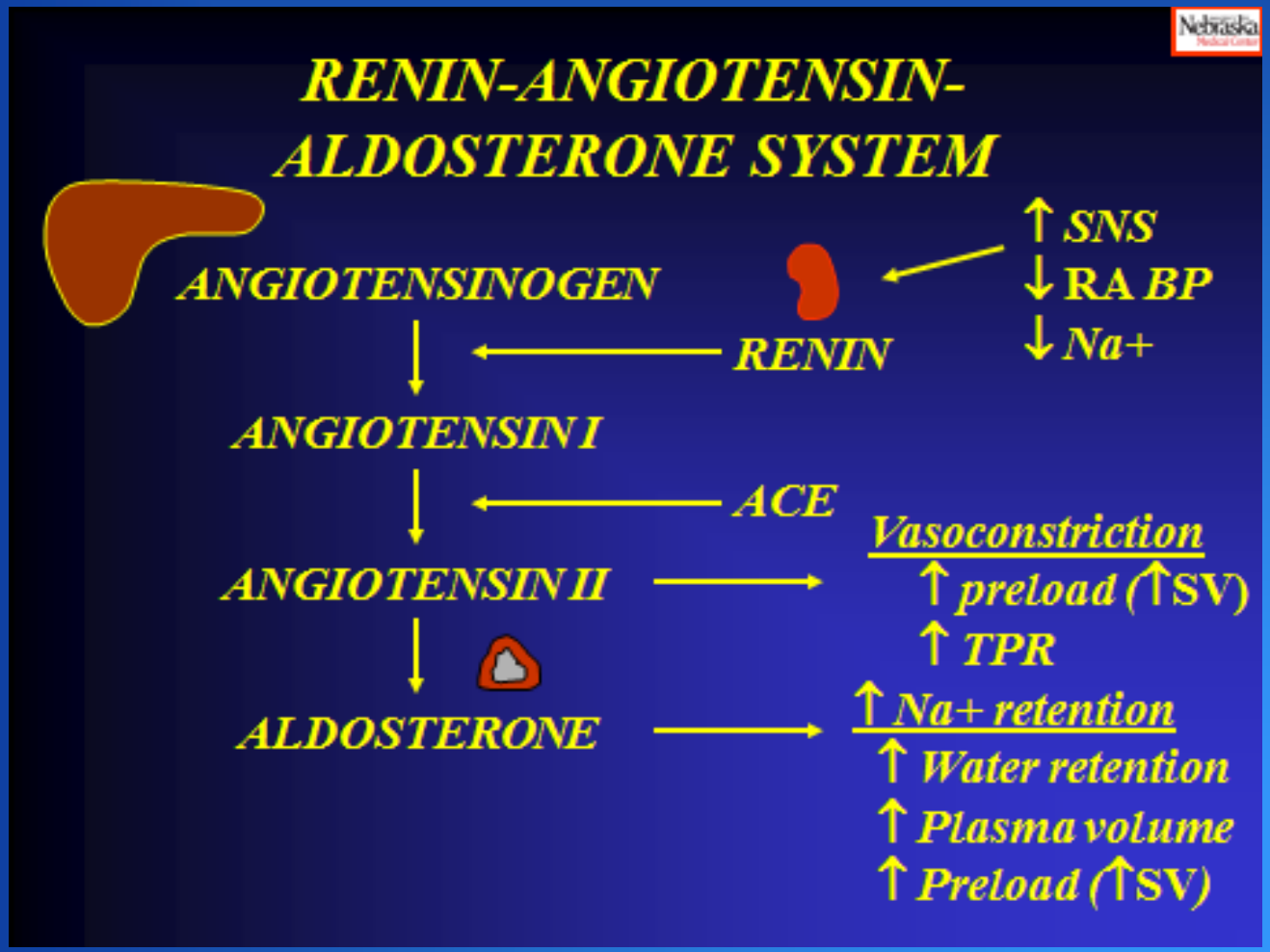
Venous compliance
SymNS (α1-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction

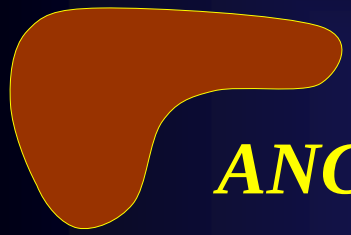
SymNS
(α1 -receptors)
Angiotensin II
(AII receptors)
Vasopressin
(V_{1A} receptors)

BLOOD PRESSURE REGULATION

- Blood pressure sensors
  - RENAL AFFERENT ARTERIOLE BARORECEPTORS
  - Decreased renal afferent arteriole blood pressure causes the activation of the renin-angiotensin-aldosterone system



RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



ANGIOTENSINOGEN



↓ ← RENIN

ANGIOTENSIN I

↓ ← ACE

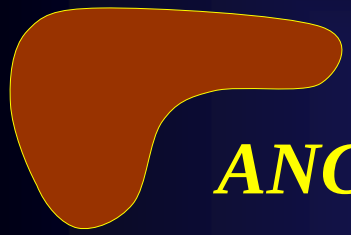
ANGIOTENSIN II → Vasoconstriction
↑ preload (↑SV)
↑ TPR

↓



ALDOSTERONE

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



ANGIOTENSINOGEN



RENIN

ANGIOTENSIN I

ACE

ANGIOTENSIN II



ALDOSTERONE

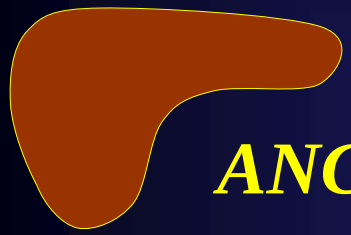
↑ Na+ retention

↑ Water retention

↑ Plasma volume

↑ Preload (↑SV)

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



↑ SNS

↓ RA BP

↓ Na+



ANGIOTENSINOGEN

RENIN

ANGIOTENSIN I

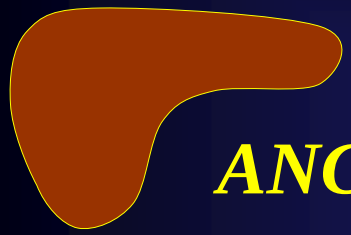
ACE

ANGIOTENSIN II



ALDOSTERONE

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



↑ SNS
↓ RA BP
↓ Na+

ANGIOTENSINOGEN



RENIN

ANGIOTENSIN I

ACE

ANGIOTENSIN II

<u>Vasoconstriction</u>
↑ preload (↑SV)
↑ TPR

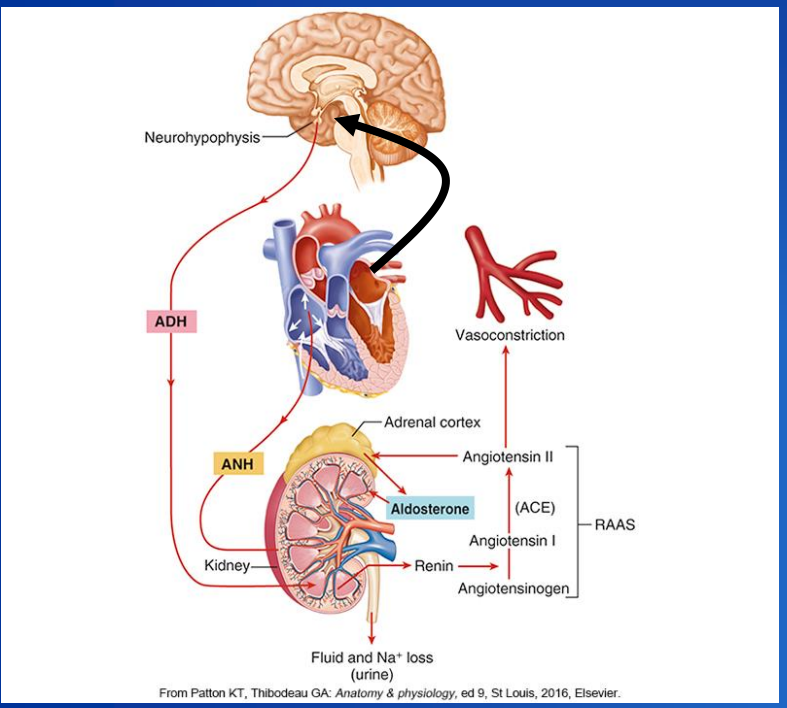


ALDOSTERONE

<u>↑ Na+ retention</u>
↑ Water retention
↑ Plasma volume
↑ Preload (↑SV)

BLOOD PRESSURE REGULATION

- Blood pressure sensors
  - ATRIAL (and other) BARORECEPTORS AND VASOPRESSIN RELEASE



From Patton KT, Thibodeau GA: *Anatomy & physiology*, ed 9, St Louis, 2016, Elsevier.

- If venous return is increased, baroreceptor activation located in the atria increases; this increases neural activation to the hypothalamus, causing a reduction in the release of vasopressin; this reduces vasopressin-induced vasoconstriction, thus reducing blood pressure and venous return
- If venous return is decreased, baroreceptor activation located in the atria decreases; this reduces neural activation to the hypothalamus, causing an increase in the release of vasopressin; this increases vasopressin-induced vasoconstriction, thus increasing blood pressure and venous return

*MABP = (HR * SV) * TPR*

Total peripheral resistance

Arterial vasoconstriction

SNS and α_1-adrenergic receptors

Renin-Angio system and AII receptors

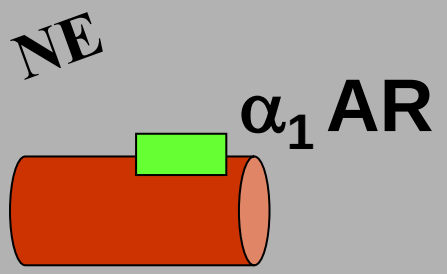
*MABP = (HR * SV) * TPR*

Total peripheral resistance

Arterial vasoconstriction

SNS and α_1-adrenergic receptors

$\uparrow$SNS $\rightarrow$ vasoconstriction $\rightarrow$ $\uparrow$TPR



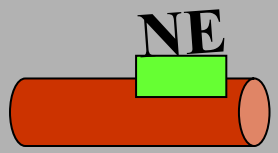
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Arterial vasoconstriction

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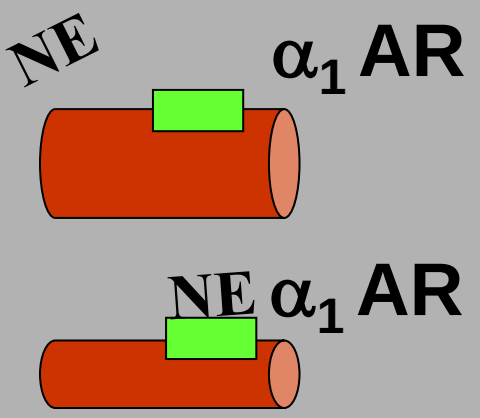
↑SNS → vasoconstriction→ ↑TPR



*MABP = (HR * SV) * TPR*

Total peripheral resistance

Arterial vasoconstriction



SNS and α_1-adrenergic receptors

$\uparrow$SNS $\rightarrow$ vasoconstriction$\rightarrow$ $\uparrow$TPR

*MABP = (HR * SV) * TPR*

Total peripheral resistance

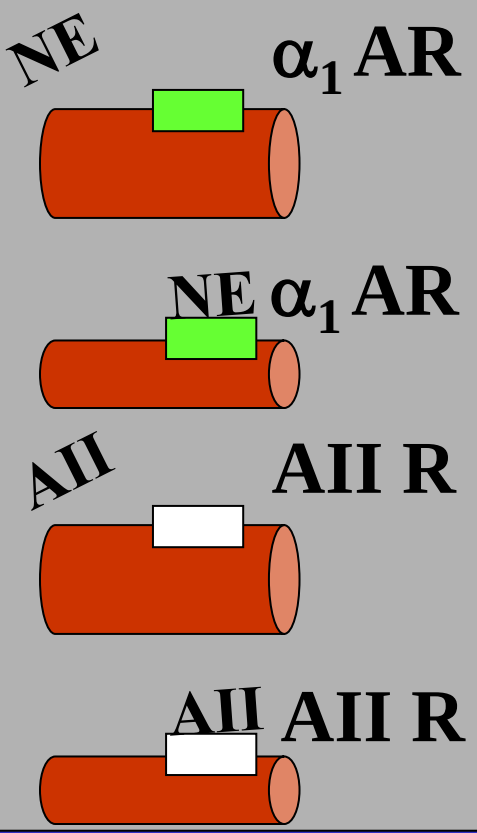
Arterial vasoconstriction

SNS and α_1-adrenergic receptors

↑SNS → vasoconstriction→ ↑TPR

Renin-Angio system and AII receptors

↑ AII → vasoconstriction→ ↑TPR



*MABP = (HR * SV) * TPR*

Total peripheral resistance

Arterial vasoconstriction

SNS and α_1-adrenergic receptors

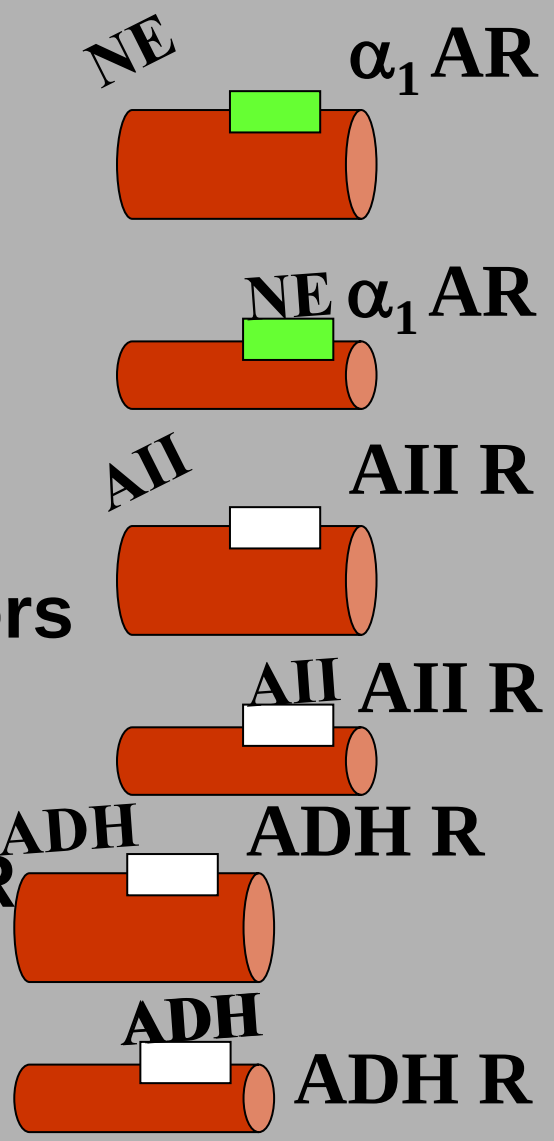
↑SNS → vasoconstriction→ ↑TPR

Renin-Angio system and AII receptors

↑ AII → vasoconstriction→ ↑TPR

Vasopressin/ADH and specific ADHR

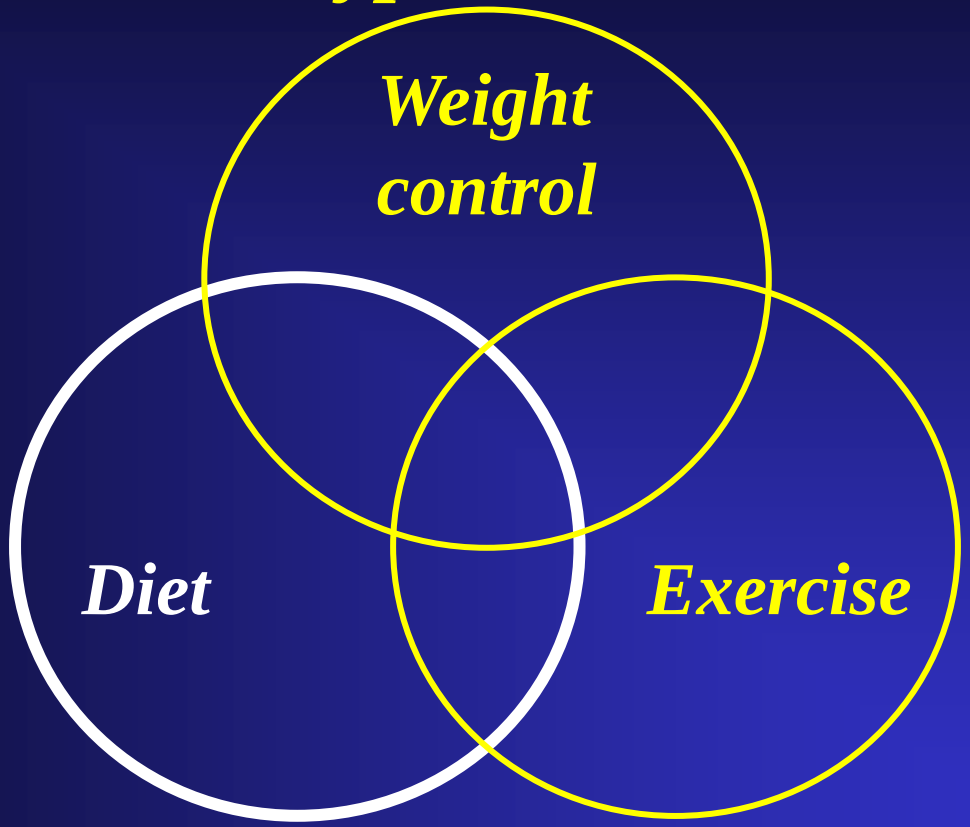
↑ ADH → vasoconstriction→ ↑TPR



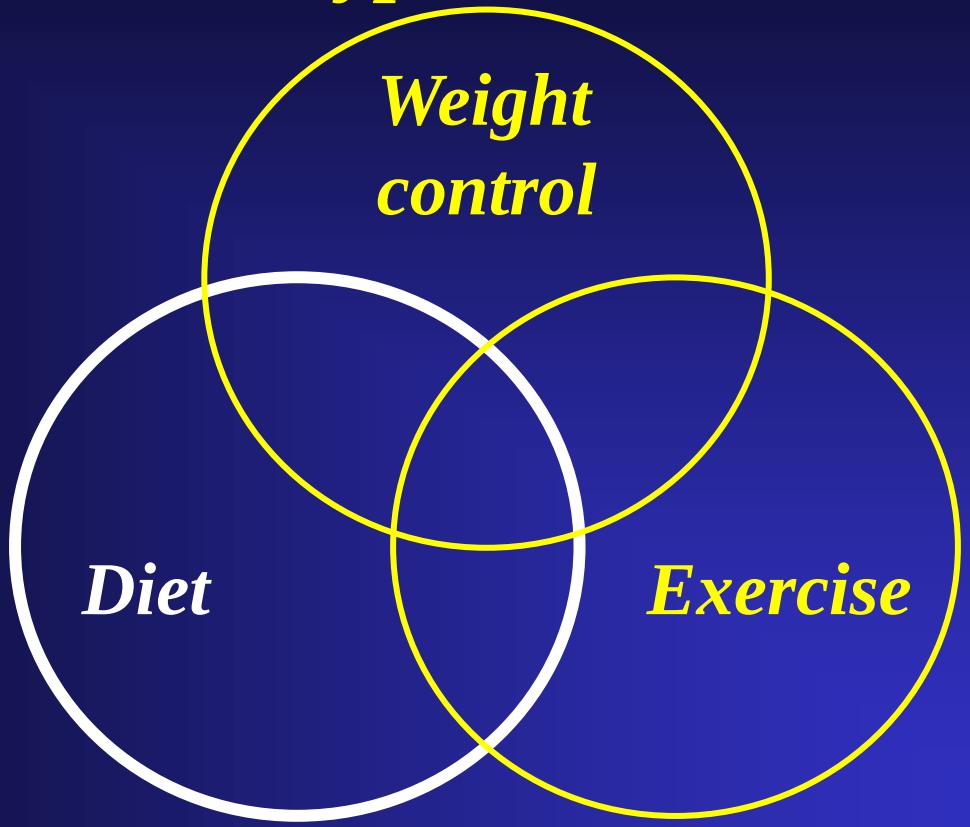
Non-pharmacological Management of Hypertension

DASH Diet

Rich in fruit, vegetables, low fat dairy products; reduced red meat and fat; (high in K+, Ca++, Mg++)



Non-pharmacological Management of Hypertension



DASH Diet

Rich in fruit, vegetables, low fat dairy products; reduced red meat and fat; (high in K+, Ca++, Mg++)

Na+ intake

The New England
Journal of Medicine

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VOLUME 336 APRIL 17, 1997 NUMBER 16

A CLINICAL TRIAL OF THE EFFECTS OF DIETARY PATTERNS ON BLOOD PRESSURE

LAWRENCE J. APPEL, M.D., M.P.H., THOMAS J. MOORE, M.D., EVA OBARZANEK, PH.D., WILLIAM M. VOLLMER, PH.D., LAURA P. SVETKEY, M.D., M.H.S., FRANK M. SACKS, M.D., GEORGE A. BRAY, M.D., THOMAS M. VOGT, M.D., M.P.H., JEFFREY A. CUTLER, M.D., MARLENE M. WINDHAUSER, PH.D., R.D., PAO-HWA LIN, PH.D., AND NJERI KARANJA, PH.D., FOR THE DASH COLLABORATIVE RESEARCH GROUP*

The DASH diet:

+ + fruits and vegetables

\- decreased meat, sweets, Na+and saturated fat

+ + lowfat dairy, nuts

+ + K+, Ca++, Mg++, arginine

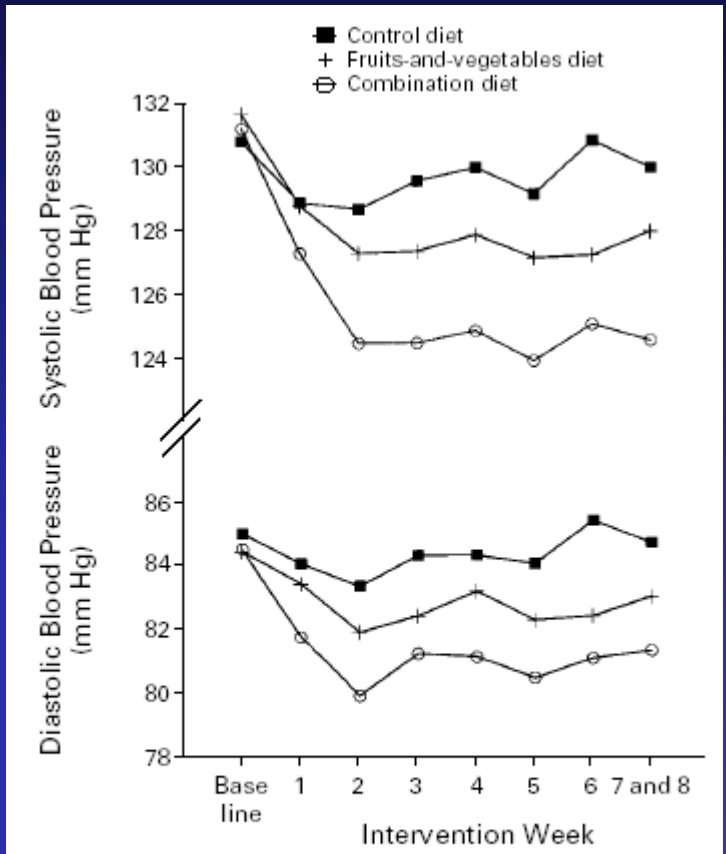


Figure 1. Mean Systolic and Diastolic Blood Pressures at Base Line and during Each Intervention Week, According to Diet, for 379 Subjects with Complete Sets of Weekly Blood-Pressure Measurements.

The New England

Journal of

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VOLUME 336 APRIL

A CLINICAL TRIAL OF THE EF
ON BLOOD

LAWRENCE J. APPEL, M.D., M.P.H., THOMAS J. MOORE, M
LAURA P. SVETKEY, M.D., M.H.S., FRANK M. SACKS, M.D.
JEFFREY A. CUTLER, M.D., MARLENE M. WINDHAUSER, PH.
FOR THE DASH COLLABO

The DASH diet:

+ + fruits and

- decreased

and satura

+ + lowfat dai

+ + K+, Ca++,

arginine

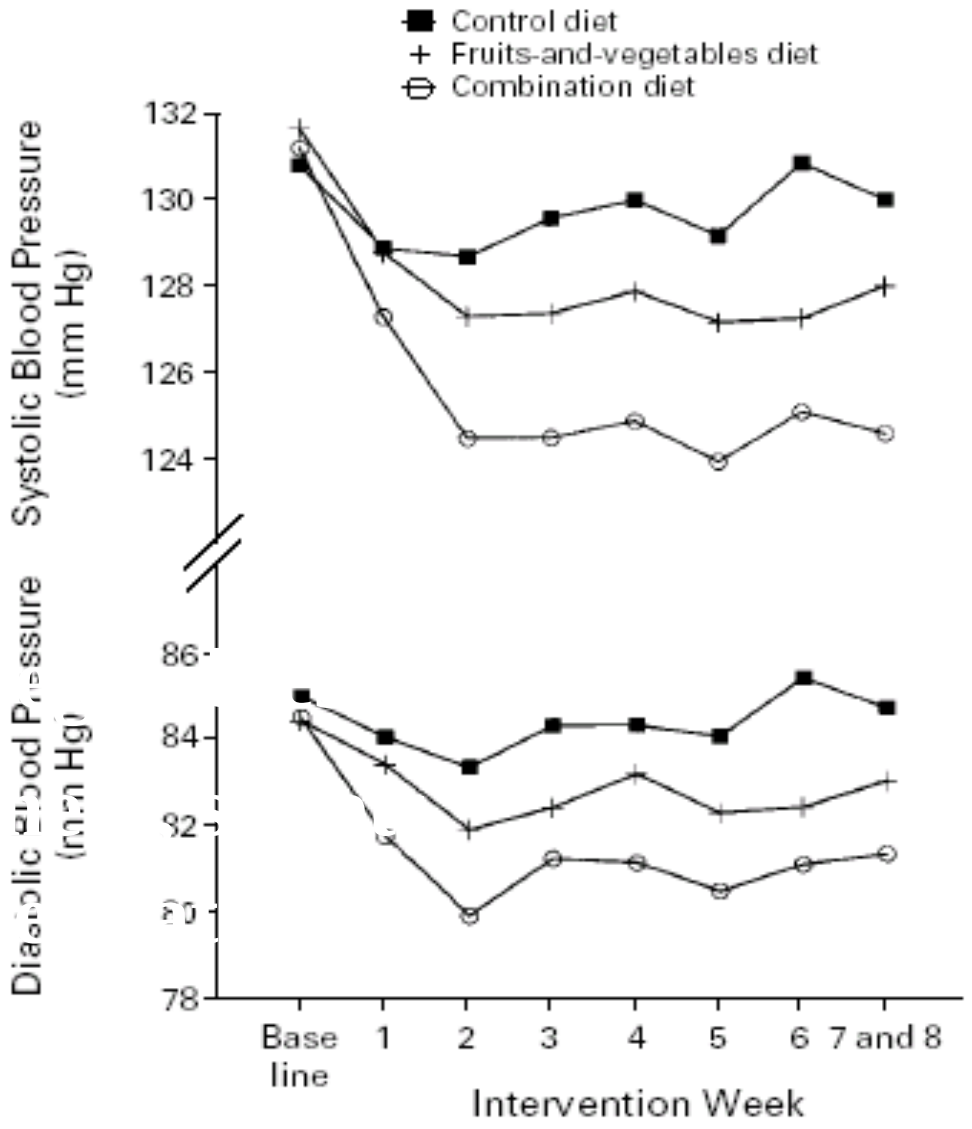
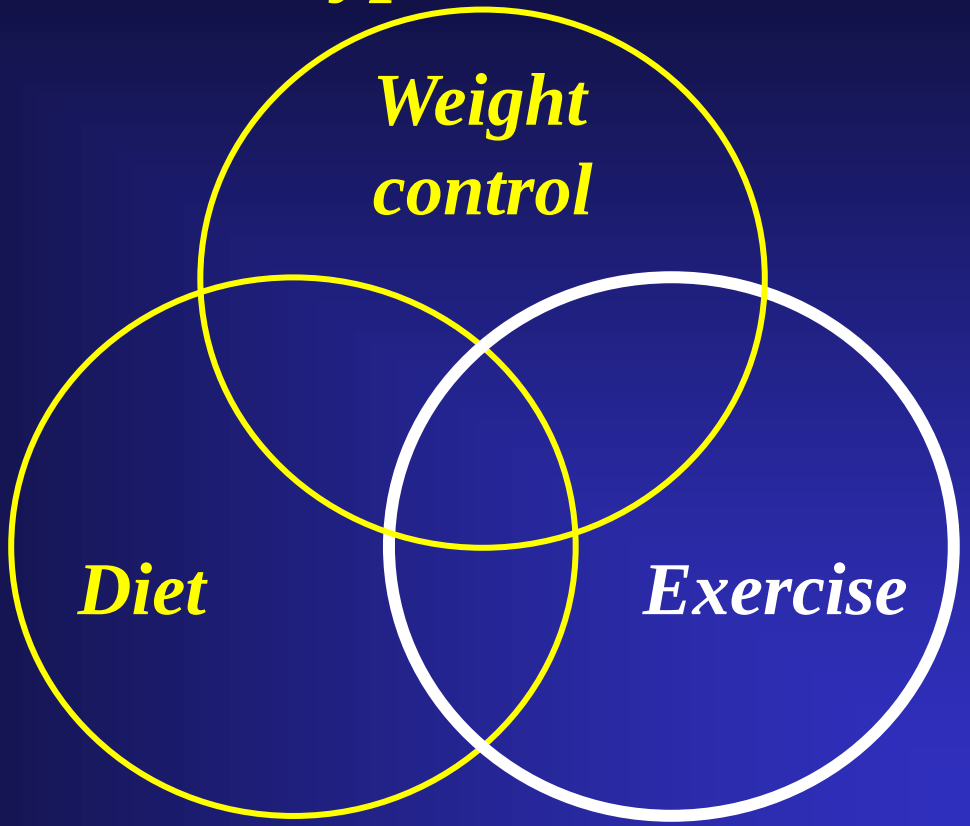


Figure 1. Mean Systolic and Diastolic Blood Pressures at Base Line and during Each Intervention Week, According to Diet, for 379 Subjects with Complete Sets of Weekly Blood-Pressure Measurements.

Non-pharmacological Management of Hypertension



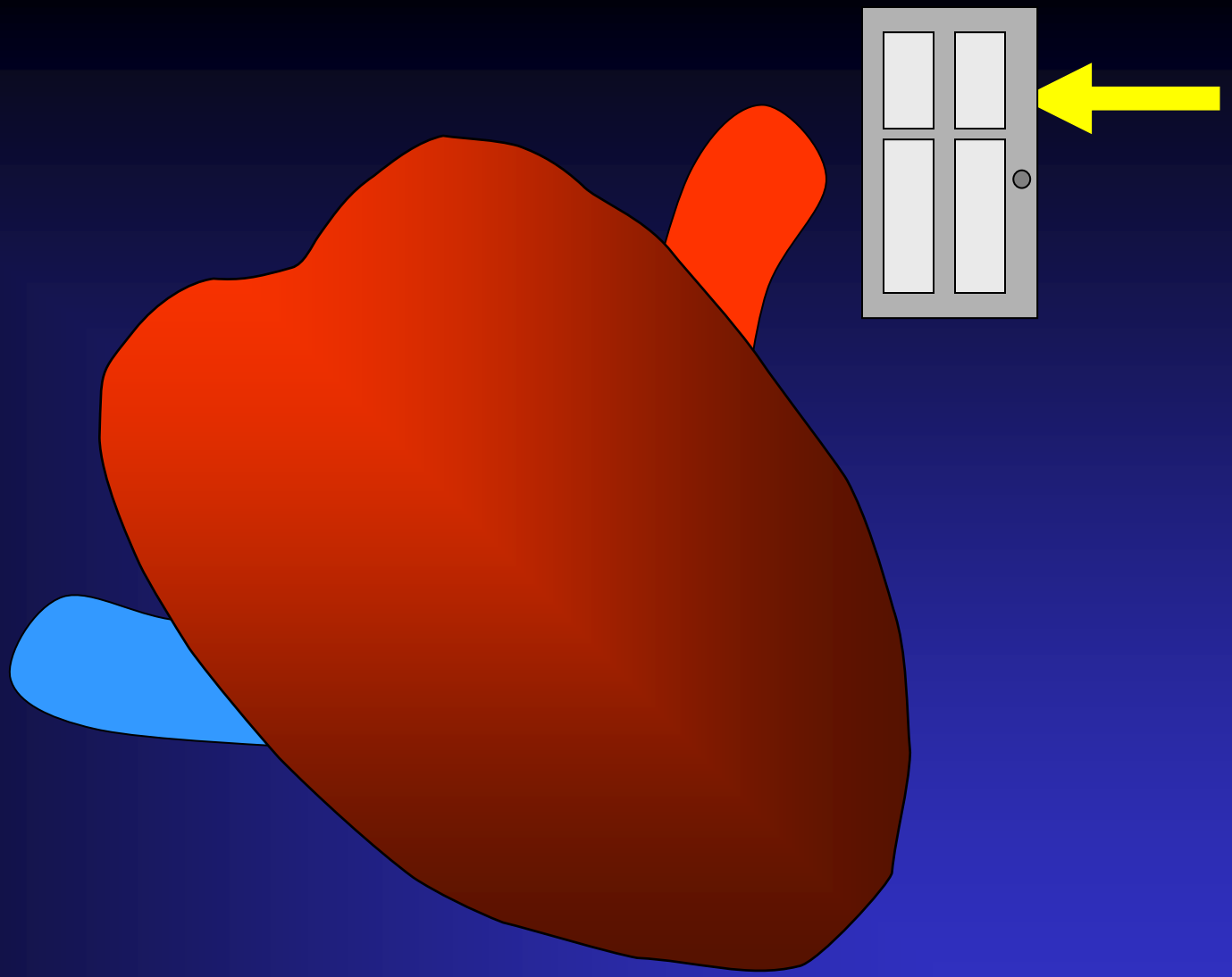
30 min most days of the week; acute and chronic BP lowering effects; Post-exercise hypotension phase

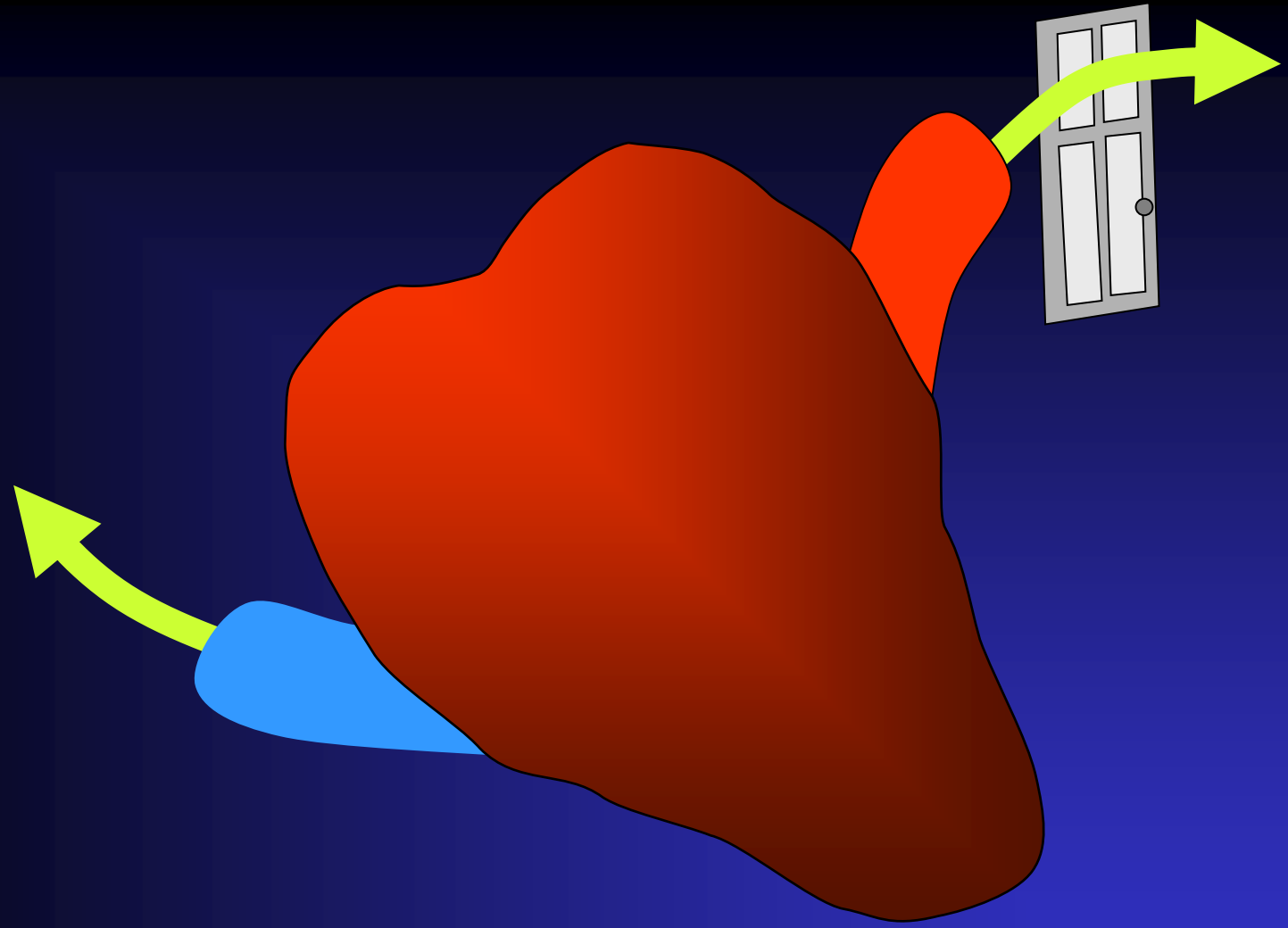
Acute delayed BP effects

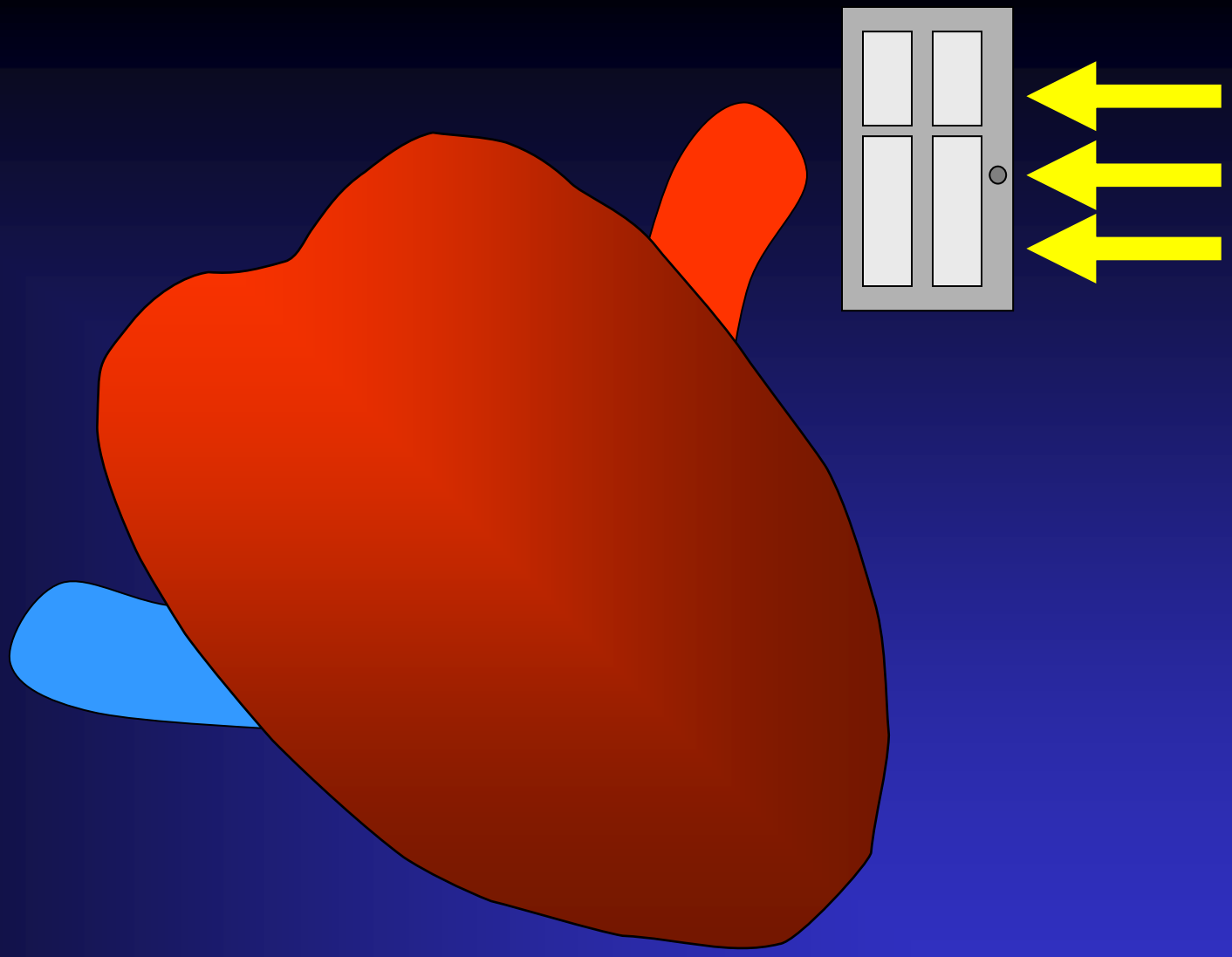
Pharmacological Management of Hypertension

CLASSES OF ANTIHYPERTENSIVE MEDICATIONS

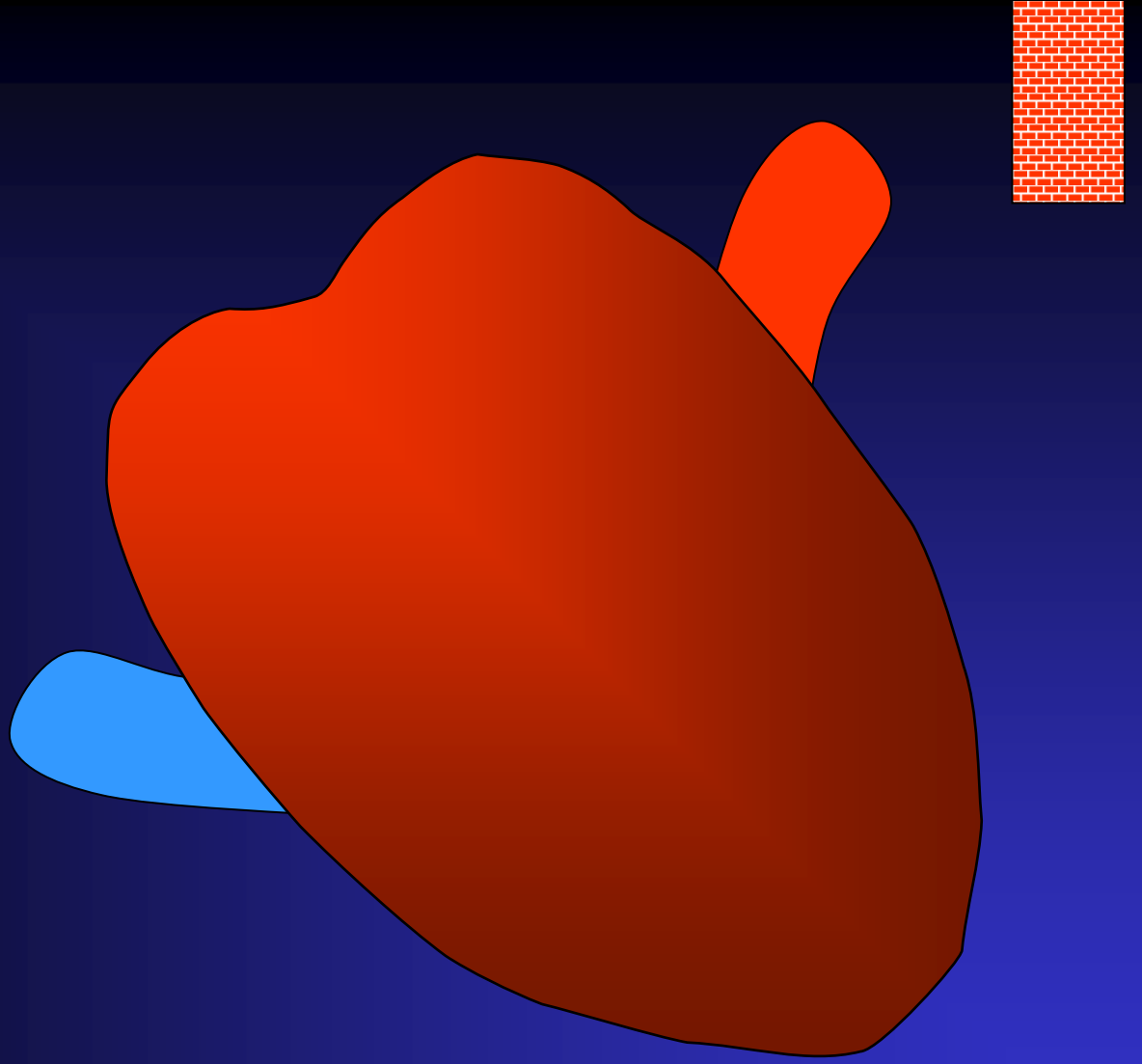
1. Diuretics
2. β-blockers
3. ACE inhibitors
4. Angiotensin II receptor antagonists (ARB)
5. Aldosterone receptor blockers
6. Calcium channel blockers (CCB)
7. α-blockers
8. Direct vasodilators







HYPERTENSION is an abnormally high AFTER-LOAD (arterial blood pressure) that greatly increases myocardial work and O2 demand



HYPERTENSION is an abnormally high AFTER-LOAD (arterial blood pressure) that greatly increases myocardial work and O2 demand

MABP = (HR x SV) x TPR

Autonomic NS

SymNS
(β1-receptors)
ParasymNS
(muscarinic receptors)

Contractility

SymNS
(β1-receptors)

Preload

Venous compliance
SymNS (α1-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction

SymNS
(α1 -receptors)
Angiotensin II
(AII receptors)
Vasopressin
(V_{1A} receptors)

Pharmacological Management of Hypertension

*MABP = (HR * SV) * TPR*

SymNS (*β1-receptors*)

Contractility
SymNS (*β1-receptors*)

Preload
Venous compliance
SymNS (α1-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α1 -receptors)
Angiotensin II
(AII receptors)

DIURETICS

-azide
-amide
-ide

Pharmacological Management of Hypertension

*MABP = (HR * SV) * TPR*

SymNS
(β1-receptors)

Contractility
SymNS
(β1-receptors)

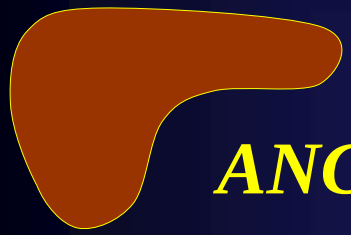
Preload
Venous compliance
SymNS (α1-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo
(β1 AR)

Arterial vaso-constriction
SymNS
(α1 -receptors)
Angiotensin II
(AII receptors)

β-BLOCKERS

-olol

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



↑ SNS
(β1 AR)

ANGIOTENSINOGEN

RENIN

ANGIOTENSIN I

ANGIOTENSIN II → Vasoconstriction

ALDOSTERONE → Na+ and water retention

β-BLOCKERS

Pharmacological Management of Hypertension

*MABP = (HR * SV) * TPR*

SymNS
(β-receptors)

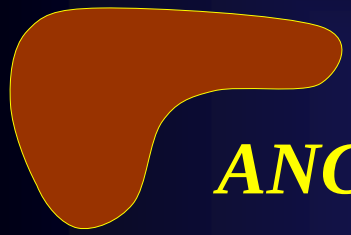
Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

ACE INHIBITORS **-pril**

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



ANGIOTENSINOGEN

↓

ANGIOTENSIN I

↓ ⟵// ***ACE*** ← ***ACE INHIBITORS***

ANGIOTENSIN II ⟶// ***Vasoconstriction***

↓

ALDOSTERONE ⟶// ***Na+ and water retention***

Pharmacological Management of Hypertension

*MABP = (HR * SV) * TPR*

SymNS
(β-receptors)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

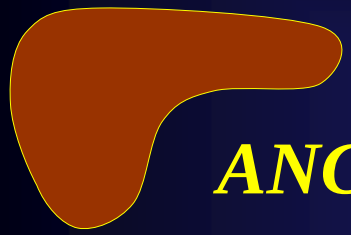
Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

ANGIOTENSION II RECEPTOR BLOCKERS

ARBS

-artan

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



ANGIOTENSINOGEN

↓

ANGIOTENSIN I

↓

ANGIOTENSIN II ⟶ *Vasoconstriction*

↓

ALDOSTERONE ⟶ *Na+ and water retention*

ANGIO II RECEPTOR BLOCKERS

Pharmacological Management of Hypertension

*MABP = (HR * SV) * TPR*

SymNS
(β1-receptors)

Contractility
SymNS
(β1-receptors)

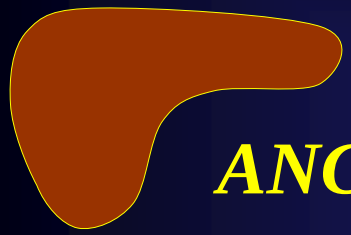
Preload
Venous compliance
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Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α1 -receptors)
Angiotensin II
(AII receptors)

ALDOSTERONE-RECEPTOR BLOCKERS

-one

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM



↑ SNS

ANGIOTENSINOGEN

RENIN

ANGIOTENSIN I

ACE

ANGIOTENSIN II → Vasoconstriction

ALDOSTERONE →// Na+ and water retention

ALDOSTERONE-RECEPTOR BLOCKERS

Pharmacological Management of Hypertension

*MABP = (HR * SV) * TPR*

SymNS
(β1-receptors)

Contractility
SymNS
(β1-receptors)

Preload
Venous compliance
SymNS (α1-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α1 -receptors)
Angiotensin II
(AII receptors)

CALCIUM CHANNEL BLOCKERS

-ipine

Pharmacological Management of Hypertension

*MABP = (HR * SV) * TPR*

SymNS
(β1-receptors)

Contractility
SymNS
(β1-receptors)

Preload
Venous compliance
SymNS (α1-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α1 -receptors)
Angiotensin II
(AII receptors)

α BLOCKERS

-zosin

Management of Acute Angina or Hypertensive Crisis

MABP = (HR * SV) * TPR

SymNS (β1-receptors)

Contractility
- SymNS (β1-receptors)

Preload
- Venous compliance
  - SymNS (α1-receptors)
- Plasma volume
  - Water volume
  - Sodium retention
    - Renin-Angio-Aldo

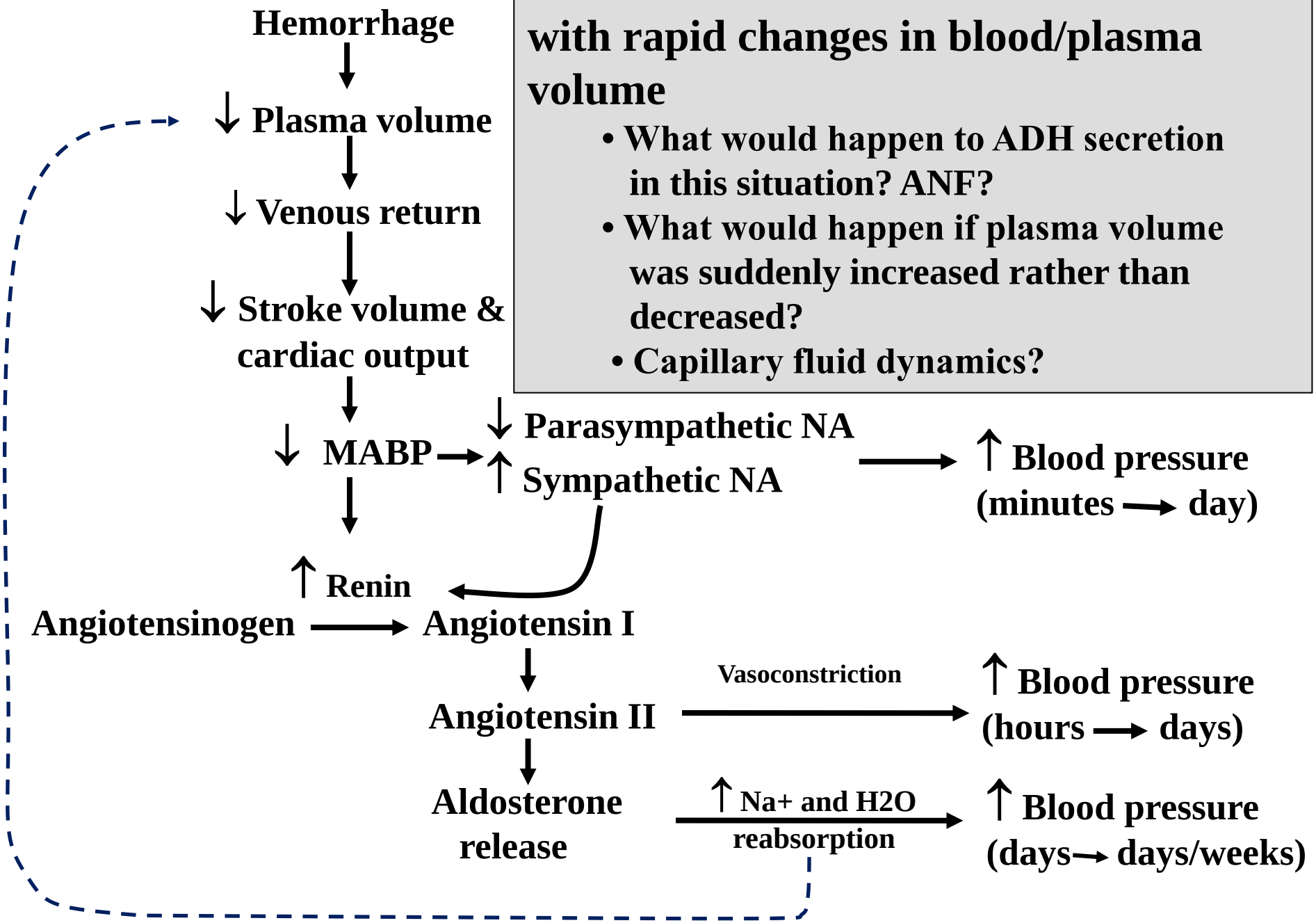
Arterial vaso-constriction
- SymNS (α1 -receptors)
- Angiotensin II (AII receptors)

NITRIC OXIDE DONORS (direct vasodilation)

Nitroglycerin

BLOOD PRESSURE REGULATION with rapid changes in blood/plasma volume

- What would happen to ADH secretion in this situation? ANF?
- What would happen if plasma volume was suddenly increased rather than decreased?
- Capillary fluid dynamics?



I. The BLOOD and its components
II. The CIRCULATIONS
III. HEMODYNAMICS and fundamental pressure, resistance and flow inter-relationships
IV. The HEART as a highly coordinated, well regulated two-step "twin" pump
V. The VASCULATURE as a well regulated system of conduits
VI. BLOOD PRESSURE regulation
VII. Determinates of MYOCARDIAL WORK

Read and study all of chapter 12

Independent studies:

BLOOD components

HEMOSTASIS

LIPOPROTEIN METABOLISM (Google Drive presentation)

More demands placed on the heart

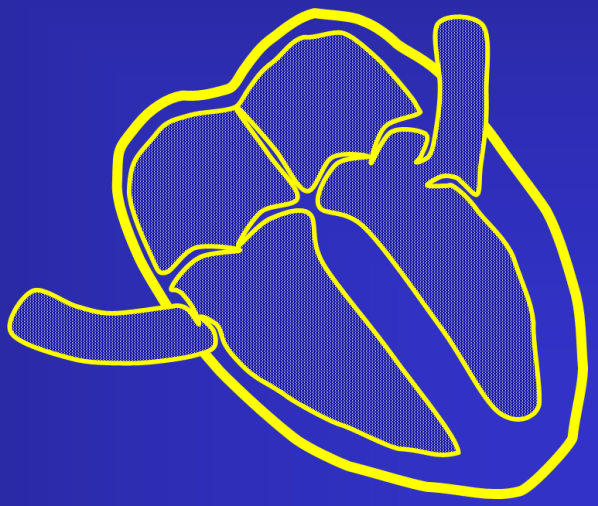
More myocardial work is required

More coronary blood flow is required

But if coronary artery disease is present, blood flow may be limited and ischemia may result (which may lead to angina)

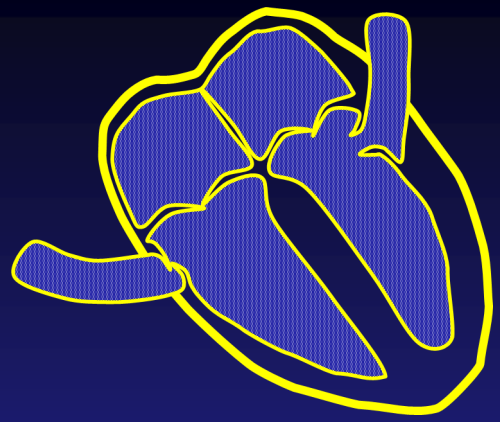
Determinates of Myocardial Work

- **C - contractility (velocity of contraction/relaxation)**
- **R - rate (heart rate)**
- **A - afterload (MABP)**
- **P - preload (ventricular filling)**



Determinates of Myocardial Work

- **C - contractility**
- **R - rate**
- **A - afterload**
- **P - preload**



*MABP = (HR * SV) * TPR*

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